Dual Negative Selection for Anomaly Detection at Cold Start: A Normal-Only Comparison with PyOD

Kishor Datta Gupta Dipankar Dasgupta

October 12, 2026

Abstract

A newly installed sensor system may need an initial anomaly detector before it has accumulated a long operating history or observed a failure. We examine Dual Negative Selection Algorithm (Dual NSA) as a candidate for this cold start, using a small set of accepted normal observations and no real outliers during training. The evidence comes from 17 ODDS tabular datasets, five splits, two feature representations, and six training sizes, with 13 NSA configurations and 43 other PyOD entries. Four central tables compare Dual NSA on every dataset at 100%, 25%, 10%, and 5% of a capped normal training pool. In PCA(2), its mean AUROC is 0.7172, 0.7603, 0.7626, and 0.7264, respectively; its rank against 25 other detectors and itself improves from 13/26 to 10/26, 6/26, and 5/26. At 5%, models learn from 4–30 normals. A neural batch-size sensitivity expands the comparison field and leaves Dual NSA 5/27. These findings support Dual NSA as a competitive starting option under the tested constraints. KDE remains ahead in mean AUROC, average precision declines at the smallest sizes, and dataset-level results vary substantially. Newly commissioned sensors are a proposed application; the experiment does not evaluate chronological sensor streams or deployed alarm performance.

1 Introduction: detecting anomalies before a long history exists

Consider a sensor set that has just begun operation. An operator may be able to approve a short commissioning period as normal, but there may be no examples of equipment faults, unusual environmental conditions, or attacks. The system still needs a first way to flag readings that depart from that initial experience. We call this condition a *normal-only cold start*: a small accepted normal sample is available, while real anomalous training examples are absent.

Negative selection offers a direct way to formulate that task. It learns a representation of self from normal examples and constructs detectors for departures from that self [1]. The Dual NSA implementation evaluated here combines negative detectors with an enclosing model of normal clusters. This gives it two ways to score an unfamiliar observation: matching a negative detector and falling outside the learned normal envelopes. Both can be learned without examples of real failures. Many conventional anomaly detectors also learn from normal-only data, so the practical argument requires a comparison with those alternatives.

Our question is whether Dual NSA remains a useful candidate when very few accepted normal observations are available. We compare it with PyOD detectors [2] under identical normal-only preprocessing and fitting rules. We bring the dataset-level comparisons into the main paper because an aggregate rank alone cannot guide a choice for a new system. The four settings are the full capped normal pool as a reference and 25%, 10%, and 5% retention as increasingly restricted starting samples. Results at 75% and 50%, every other NSA configuration, and the full detector roster remain in the appendix.

The central observation is that Dual NSA becomes more competitive in the two-dimensional PCA

pipeline as normal training availability shrinks. Its AUROC rank improves in a fixed comparison field, and it produces qualified results in every PCA(2) test case. This gives a reason to include it among the initial methods considered for a new sensor set with trusted commissioning data. The case is conditional: KDE is stronger on average, some datasets favor other methods, and a better rank at 5% does not imply better absolute detection than at 10%.

We make three contributions. First, we provide four direct, five-split comparisons on all 17 datasets, with actual normal training counts and explicit comparison-field ranks. Second, we explain what these results support for a normal-only cold start and what remains untested for newly commissioned sensors. Third, we retain the broader 57,120-record experiment, 740 supplemental neural attempts, failures, and audits in an extensive appendix and a reproducible code-and-results repository. Dual NSA was chosen for this focused presentation after inspecting the benchmark; the analysis is exploratory.

2 Dual NSA and the cold start setting

2.1 The model learned from accepted normal observations

The tested NSA implementation belongs to the pinned PyOD contribution described in the appendix [3]. For a point x in the estimator's fitted coordinate system, negative hyperspheres with centers c_j and radii r_j give a signed matching score

$$s_{\text{neg}}(x) = \max_j \{r_j - \|x - c_j\|\}. \tag{1}$$

A positive score indicates membership in at least one negative detector. Dual NSA additionally constructs enclosing self-cluster balls with centers a_k and expanded radii R_k. Its envelope and final scores are

$$s_{\text{env}}(x) = \min_k \{\|x - a_k\| - R_k\}, \qquad s_{\text{dual}}(x) = \max\{s_{\text{neg}}(x), s_{\text{env}}(x)\}. \tag{2}$$

An observation outside every self ball has a positive envelope margin, even when the sampled negative spheres leave a region uncovered. This makes Dual NSA an understandable initial scoring rule: departures from the normal envelopes can be identified without waiting to collect labeled faults. It also makes an assumption that needs checking in the application. A legitimate but previously unseen operating state can lie outside a small learned envelope.

The benchmark evaluates the outcome of this geometry. It does not isolate the envelope term experimentally, measure detector-space coverage, or establish why a particular retained sample performs well. Any explanation linking the rank improvement to self-envelope geometry is a hypothesis for further study.

2.2 What cold start means in this experiment

The training sample is small and clean by construction. Labels are used to remove anomalous rows from the training partition; no anomaly labels are passed to the initial fitting call. This models an operator having accepted a short normal commissioning sample. It does not solve the problem of deciding which initial readings are trustworthy when the system is already faulty.

The proposed sensor use case is a set of newly commissioned sensors, for example several readings describing the same equipment or environmental condition. An initial normal-only detector can supply scores while more accepted operating data are collected. Our data are heterogeneous tabular benchmarks, not a chronological record of such a commissioning process. The sensor interpretation is therefore a motivated application of the tested constraint, rather than a validated deployment result. Sequential dependence, drift, operating-mode changes, and alarm thresholds require their own evaluation.

3 Experimental comparison

3.1 Shared training and test data

We use 17 public ODDS datasets [4]: Arrhythmia, Cardio, Glass, Ionosphere, Letter, Lympho, MNIST, Musk, Optdigits, Pendigits, Pima, Satellite, Satimage-2, Shuttle, Vertebral, Vowels, and WBC. Each of five seeds, 42–46, makes a stratified 60/40 train/test split. All training anomalies are discarded before any preprocessing or model fitting. The normal training pool is capped at 600 observations, and the separate mixed test set is capped at 2,000 while preserving its class mix.

Nested seed-fixed subsets retain 100%, 75%, 50%, 25%, 10%, or 5% of that capped normal pool. Parent row order is preserved, and test rows stay fixed as training availability changes. The four main tables show 100%, 25%, 10%, and 5%. Here 100% means the full *capped normal pool*, not all observations in the original dataset. At 10%, actual training samples contain 8–60 normals; at 5%, they contain 4–30. Every detector receives the same features for a dataset, seed, representation, and retained size. Initial `fit` calls receive normal features without a label array.

We evaluate both original features and PCA(2). Standardization and PCA are fitted anew on retained normals at each size; the held-out test rows use those fitted transformations. Thus the study compares the whole normal-only pipeline as data availability changes. PCA directions are not held fixed. The main dataset tables use PCA(2), where Dual NSA has complete coverage and its strongest comparative standing. Original-feature results are reported in the main discussion and in companion appendix tables.

3.2 Methods displayed and methods attempted

The full grid has 11 NSA strategy families represented by 13 configurations and 43 other registered tabular PyOD entries: $17 \times 5 \times 2 \times 6 \times 56 = 57{,}120$ primary records. Constructors, epochs, ensemble sizes, and the CPU budget are documented in the appendix. This is a comparison of the recorded implementations and settings; it does not describe optimally tuned versions of every detector.

The four main dataset tables display Dual NSA, KDE, KNN, Isolation Forest (IForest), and One-Class SVM (OCSVM). This fixed comparator set covers density estimation, distances, isolation, and a conventional one-class boundary. KDE and KNN also give demanding empirical references near the small-sample settings. These display choices were made after inspection and are descriptive. All four conventional methods have 85/85 qualified outcomes at every size. The rank column includes the wider fixed comparison field, so hiding other columns does not improve Dual NSA's reported position. Full comparisons and AP companion tables are provided in the appendix.

XGBOD and DevNet require labeled anomalies for their intended supervised training in this roster and are unsuitable for the specified input. That restriction does not apply to most PyOD methods. Other failures reflect dimensions, sample-size settings, construction errors, or the computation budget. We mark failed cases *unsuitable under the tested configuration*; the label describes whether a configuration worked, not its detection accuracy or whether a different constructor could work.

3.3 Scores, complete fields, and uncertainty

We report AUROC and average precision (AP); larger values are better. Dataset rows average five qualified split scores. Overall means weight the 17 dataset means equally. A primary score is qualified when the recorded fit plus scoring time is at most 15 seconds. The nominal Python timer sometimes failed to stop native callbacks, so this measured-time rule is a disclosed post hoc enforcement. Raw-success sensitivity results retain the valid over-budget scores. Failed runs remain unscored, and no missing result is replaced by zero.

The primary PCA(2) field contains the same 25 other detectors qualified in all 85 cases at every one of the six retained sizes, plus Dual NSA. Other NSA configurations are not included in this focal denominator. For dataset d, with five-split AUROC mean $A_{d,m}$, the reported Dual rank is

$$r_d = 1 + \sum_{m \in \mathcal{B}} \mathbf{1}\{A_{d,m} > A_{d,\text{dual}} + 10^{-12}\}, \qquad |\mathcal{B}| = 25. \tag{3}$$

Ranks use competition ties with tolerance 10^{-12}. A rank such as 5/26 is a position among 26 methods, not a score percentile or a rank against all 43 attempted PyOD entries. The rank fraction $100r/26$ expresses the position as a percentage of that fixed field. A separate neural training-batch sensitivity adds VAE to the complete PCA(2) field; those ranks use 27 methods and are labeled separately.

Paired differences match successful dataset/seed cells. Descriptive 95% intervals use 5,000 dataset-cluster bootstrap draws and are not reported for fewer than five matched datasets. The comparisons are exploratory and unadjusted for multiplicity. Twelve comparator implementations use query-batch-dependent scoring; the appendix identifies them and supplies an inductive sensitivity. No mixed test features enter initial fitting, but these inference semantics differ from Dual NSA's pointwise scoring.

4 Results: Dual NSA as an initial alternative

4.1 Aggregate standing and four dataset comparisons

Table 1 summarizes the same four settings examined in detail in Tables 2–5. The dataset tables keep every task in the comparison, including tasks where Dual NSA is weak. Bold scores identify a maximum only among the five displayed methods, whereas the Dual rank uses all 25 fixed conventional comparators.

The same four PyOD comparators appear in every dataset table: KDE (density), KNN (distance), IForest (isolation), and OCSVM (one-class boundary). All four were selected after inspecting benchmark results and each has complete six-size qualified coverage. The comparison is descriptive. Bold identifies the highest AUROC among the five displayed methods within that dataset, including ties. Dual's rank uses the full fixed field of 25 other detectors plus Dual (26 total), not just these four comparators. Scores require five measured-$\leq$ *15-second successes; a dash never denotes zero.*

Table 1: Equal-dataset macro summary for PCA(2). Dual and KDE entries pair AUROC/AP; KNN reports AUROC. Dual's AUROC rank and rank percentage appear together: primary field 26 methods, neural-batch sensitivity 27. Percentage is 100 times rank/field size, with lower values better. Each field is fixed across all six sizes; the sensitivity leaves Dual scores unchanged.

Retained	Fit normals	Dual A/P	KDE A/P	KNN AUC	Dual AUC rank	Adapted AUC rank
100%	84–600	0.7172/0.4718	0.7890/0.4817	0.7958	13/26 (50.00%)	13/27 (48.15%)
25%	21–150	0.7603/0.4699	0.7947/0.4711	0.8005	10/26 (38.46%)	10/27 (37.04%)
10%	8–60	0.7626/0.4206	0.7769/0.4373	0.7706	6/26 (23.08%)	6/27 (22.22%)
5%	4–30	0.7264/0.3755	0.7496/0.3949	0.7082	5/26 (19.23%)	5/27 (18.52%)

Table 2: PCA(2), 100 percent retained normals: five-seed mean AUROC on all 17 datasets; n counts retained training normals. Dual's rank uses the fixed 26-method field; rank percentage is 100 times rank/26 (lower is better). Total sums n for one split, reports dataset-macro scores, and ranks those means. Bold marks maxima among displayed methods.

Dataset	n	Dual	KDE	KNN	IForest	OCSVM	Dual rank	Rank %
arrhythmia	231	0.6367	0.6594	**0.6659**	0.6543	0.6620	17/26	65.38%
cardio	600	**0.9424**	0.9376	0.9276	0.9343	0.9217	1/26	3.85%
glass	123	0.7622	0.7976	**0.8762**	0.7902	0.8488	13/26	50.00%
ionosphere	135	0.6933	0.7637	**0.7723**	0.6778	0.7713	11/26	42.31%
letter	600	0.3732	0.5900	**0.6455**	0.4976	0.5769	20/26	76.92%
lympho	84	0.8500	**0.8879**	0.8397	**0.8879**	0.8190	13/26	50.00%
mnist	600	0.6212	**0.7235**	0.6651	0.6924	0.6447	19/26	73.08%
musk	600	0.9995	**0.9999**	0.9998	0.9814	**0.9999**	7/26	26.92%
optdigits	600	0.3492	0.5304	**0.6062**	0.4323	0.5502	23/26	88.46%
pendigits	600	0.9102	0.9812	0.9745	0.9645	**0.9847**	14/26	53.85%
pima	299	0.6350	**0.7104**	0.6930	0.6935	0.6645	15/26	57.69%
satellite	600	0.8165	**0.8431**	0.8339	0.7955	0.8262	8/26	30.77%
satimage-2	600	0.9266	0.9823	**0.9847**	0.9682	0.9775	16/26	61.54%
shuttle	600	0.9787	0.9791	**0.9889**	0.9731	0.9819	10/26	38.46%
vertebral	126	**0.3819**	0.3583	0.3079	0.3599	0.3740	8/26	30.77%
vowels	600	0.4153	0.7244	**0.8196**	0.6759	0.7175	21/26	80.77%
wbc	213	0.9002	**0.9432**	0.9283	0.9291	0.9405	17/26	65.38%
Total	7211	0.7172	0.7890	**0.7958**	0.7593	0.7801	13/26	50.00%

Table 3: PCA(2), 25 percent retained normals: five-seed mean AUROC on all 17 datasets; n counts retained training normals. Dual's rank uses the fixed 26-method field; rank percentage is 100 times rank/26 (lower is better). Total sums n for one split, reports dataset-macro scores, and ranks those means. Bold marks maxima among displayed methods.

Dataset	n	Dual	KDE	KNN	IForest	OCSVM	Dual rank	Rank %
arrhythmia	57	**0.7113**	0.6726	0.6888	0.6877	0.6652	1/26	3.85%
cardio	150	0.9274	**0.9295**	0.9179	0.9240	0.9249	4/26	15.38%
glass	30	**0.9122**	0.8726	0.8305	0.7976	0.9012	1/26	3.85%
ionosphere	33	0.7433	**0.7529**	0.7233	0.6488	0.7482	5/26	19.23%
letter	150	0.5155	0.5991	**0.6263**	0.5009	0.5773	15/26	57.69%
lympho	21	0.7310	0.8328	**0.8879**	0.8569	0.6879	16/26	61.54%
mnist	150	0.6448	**0.7420**	0.6968	0.6908	0.6774	18/26	69.23%
musk	150	0.9937	0.9984	**0.9990**	0.9664	0.9972	9/26	34.62%
optdigits	150	0.2992	0.6090	**0.6610**	0.4738	0.6272	24/26	92.31%
pendigits	150	0.9045	**0.9676**	0.9665	0.9305	0.9653	13/26	50.00%
pima	74	0.6606	**0.7190**	0.7012	0.6994	0.6721	17/26	65.38%
satellite	150	0.8179	**0.8339**	0.8251	0.7769	0.8107	6/26	23.08%
satimage-2	150	0.9664	**0.9767**	0.9744	0.9460	0.9712	9/26	34.62%
shuttle	150	**0.9834**	0.9757	0.9743	0.9575	0.9766	2/26	7.69%
vertebral	31	**0.4327**	0.3917	0.4077	0.3915	0.3847	9/26	34.62%
vowels	150	0.7654	0.7129	**0.8005**	0.6333	0.7328	4/26	15.38%
wbc	53	0.9153	0.9231	**0.9273**	0.9022	0.9196	10/26	38.46%
Total	1799	0.7603	0.7947	**0.8005**	0.7520	0.7788	10/26	38.46%

Table 4: PCA(2), 10 percent retained normals: five-seed mean AUROC on all 17 datasets; n counts retained training normals. Dual's rank uses the fixed 26-method field; rank percentage is 100 times rank/26 (lower is better). Total sums n for one split, reports dataset-macro scores, and ranks those means. Bold marks maxima among displayed methods.

Dataset	n	Dual	KDE	KNN	IForest	OCSVM	Dual rank	Rank %
arrhythmia	23	0.6401	0.6397	**0.7097**	0.6841	0.6310	16/26	61.54%
cardio	60	0.8860	**0.9248**	0.9182	0.9218	0.9043	16/26	61.54%
glass	12	0.8470	0.8616	0.7512	0.7530	**0.8829**	5/26	19.23%
ionosphere	13	0.7478	**0.7515**	0.6548	0.6569	0.7511	5/26	19.23%
letter	60	0.6229	0.6275	**0.6398**	0.4944	0.6352	6/26	23.08%
lympho	8	0.7207	**0.7931**	0.7828	0.7862	0.7466	12/26	46.15%
mnist	60	0.6640	**0.6934**	0.6699	0.6927	0.6556	15/26	57.69%
musk	60	0.9797	**0.9860**	0.9808	0.8723	0.9831	8/26	30.77%
optdigits	60	0.5067	**0.5491**	0.5395	0.4234	0.5373	11/26	42.31%
pendigits	60	0.8600	**0.9439**	0.9173	0.8959	0.9326	15/26	57.69%
pima	29	0.6628	**0.7188**	0.7104	0.6902	0.6656	12/26	46.15%
satellite	60	0.8136	**0.8175**	0.8052	0.7256	0.8066	3/26	11.54%
satimage-2	60	**0.9721**	0.9685	0.9652	0.9034	0.9616	2/26	7.69%
shuttle	60	0.9660	0.9746	0.9677	0.9456	**0.9756**	10/26	38.46%
vertebral	12	**0.4613**	0.4105	0.4401	0.3965	0.3968	8/26	30.77%
vowels	60	**0.7337**	0.6559	0.7306	0.5305	0.6772	1/26	3.85%
wbc	21	0.8799	0.8910	**0.9170**	0.8602	0.8844	11/26	42.31%
Total	718	0.7626	**0.7769**	0.7706	0.7196	0.7663	6/26	23.08%

Table 5: PCA(2), 5 percent retained normals: five-seed mean AUROC on all 17 datasets; n counts retained training normals. Dual's rank uses the fixed 26-method field; rank percentage is 100 times rank/26 (lower is better). Total sums n for one split, reports dataset-macro scores, and ranks those means. Bold marks maxima among displayed methods.

Dataset	n	Dual	KDE	KNN	IForest	OCSVM	Dual rank	Rank %
arrhythmia	11	0.6220	0.6356	0.6546	**0.6567**	0.6337	14/26	53.85%
cardio	30	0.8666	**0.9086**	0.9064	0.9059	0.8971	16/26	61.54%
glass	6	0.7799	0.7835	0.3787	0.6604	**0.8192**	4/26	15.38%
ionosphere	6	0.7147	**0.7350**	0.4874	0.6197	0.7338	5/26	19.23%
letter	30	0.6014	0.5996	0.5797	0.4628	**0.6055**	2/26	7.69%
lympho	4	0.4121	0.5431	**0.7500**	0.6716	0.5362	24/26	92.31%
mnist	30	0.6626	0.7168	0.7264	**0.7387**	0.6595	18/26	69.23%
musk	30	0.9350	**0.9537**	0.9358	0.7970	0.9488	7/26	26.92%
optdigits	30	**0.6226**	0.5938	0.5308	0.4757	0.6070	1/26	3.85%
pendigits	30	0.8256	**0.8954**	0.8634	0.8842	0.8754	17/26	65.38%
pima	14	0.5903	0.6707	**0.6805**	0.6735	0.6240	16/26	61.54%
satellite	30	**0.8185**	0.8181	0.7859	0.6893	0.8097	1/26	3.85%
satimage-2	30	0.9544	0.9543	0.9408	0.8547	**0.9553**	4/26	15.38%
shuttle	30	**0.9616**	0.9575	0.9381	0.9242	0.9615	4/26	15.38%
vertebral	6	0.3853	**0.4407**	0.3081	0.3862	0.4282	19/26	73.08%
vowels	30	**0.7107**	0.6310	0.6538	0.4133	0.6633	1/26	3.85%
wbc	10	0.8856	0.9050	**0.9196**	0.8655	0.8975	12/26	46.15%
Total	357	0.7264	**0.7496**	0.7082	0.6870	0.7445	5/26	19.23%

4.2 What changes when the normal sample is scarce?

Dual NSA's PCA(2) AUROC rises from 0.7172 at the full capped normal pool to 0.7603 at 25% and 0.7626 at 10%, then declines to 0.7264 at 5%. Its rank improves across these settings from 13/26 to 10/26, 6/26, and 5/26. The corresponding 10% and 5% positions are top 23.08% and 19.23% of this fixed field. These improvements give a concrete reason to consider Dual NSA during a normal-only cold start. They do not establish that reducing data always helps it.

KNN provides a useful reference for the changing gap. Its PCA(2) AUROC is 0.7958 at 100% and 0.7706 at 10%; Dual NSA's mean deficit narrows from 0.0786 to 0.0080. The matched change in that relative gap is +0.0707, with descriptive interval [+0.013, +0.136]. KDE remains the strongest mean-AUROC method in the fixed field at both smallest sizes, scoring 0.7769 at 10% and 0.7496 at 5%. Dual-minus-KDE differences are −0.0143 [−0.032, +0.004] and −0.0232 [−0.045, −0.002]. Dual NSA is competitive here, while a density baseline remains a strong starting candidate.

The AP comparison places an important limit on the argument. Dual NSA's PCA(2) AP is 0.4718, 0.4699, 0.4206, and 0.3755 across the four displayed sizes. Relative to 100%, paired AP changes at 10% and 5% are −0.0511 [−0.0990, −0.0084] and −0.0962 [−0.1756, −0.0310]. Paired AUROC changes are +0.0454 [−0.0013, +0.1002] and +0.0092 [−0.0687, +0.0828]. Better ordinal standing at 5% coexists with lower AP and with lower AUROC than at 10%. A system that depends on a small queue of high-priority alerts needs to check precision and its operating threshold separately.

4.3 Dataset variation and the original-feature check

The four dataset tables show why the starting choice must be validated locally. At 10%, Dual NSA's fixed-field rank ranges from 1/26 on Vowels to 16/26 on Arrhythmia. At 5%, it ranges from 1/26 on Optdigits to 24/26 on Lympho. Optdigits has only 0.6226 AUROC despite its first-place relative position; Lympho has 0.4121. Comparative rank and absolute discrimination must both be considered. The results do not support a uniform top-rank claim across datasets.

With original features, Dual NSA achieves 85/85 qualified outcomes at both 10% and 5%. Its AUROC/AP pairs are 0.7792/0.5112 and 0.7900/0.5066, and its AUROC ranks against the fixed 19-other-detector field are 14/20 and 10/20. KDE is ahead, with AUROC 0.8781 and 0.8577. Dual NSA has only 81/85 qualified cases at 100% and 83/85 at 25%, so it has no complete four-setting native rank comparison. On the 81 cases matched to 100%, its AUROC changes at 10% and 5% are +0.0780 [+0.0339, +0.1305] and +0.0903 [+0.0410, +0.1436]. Companion appendix tables retain the incomplete cases and their counts. The especially favorable PCA(2) ranks should not be generalized to arbitrary sensor feature dimensions.

4.4 Does a small-sample repair change the comparison?

The pinned AutoEncoder and VAE use training batches of 32 with incomplete batches discarded. At 5%, every sample is smaller than 32, so the original loops receive no training batch. This is a sample-size implementation failure; it does not establish a need for training outliers. A separate 740-attempt sensitivity uses training batch size $\min(32, n_{\text{fit}})$ while preserving inference batches, splits, preprocessing, epochs, the scoring procedure, and budget. Both methods then qualify in all 85 cases of every new 10% and 5% panel.

The repair adds VAE to the complete six-size PCA(2) comparison field. Dual NSA's four AUROC ranks become 13/27, 10/27, 6/27, and 5/27, corresponding to top 22.22% at 10% and 18.52% at 5%. Its small-sample standing persists when that concrete comparator failure is repaired. A separate primary sensitivity removes query-batch-dependent scorers: the fixed inductive field contains 20 other detectors plus Dual NSA, giving ranks 13/21, 10/21, 6/21, and 5/21. This keeps the same ordinal positions while changing the rank fractions to top 28.57% and 23.81% at the two smallest sizes. The native fixed field does not expand after the neural repair because unrelated historical failures remain. Full outcomes, provenance, and a brief startup timing overlap are disclosed in the appendix.

4.5 How the other NSA mechanisms compare

Table 6 places the other tested NSA configurations beside Dual NSA using the same fixed PCA(2) field. Each NSA is inserted separately into the field, so its rank is directly interpretable against conventional alternatives. Dual NSA leads the complete NSA group at 10% and 5% by both mean AUROC and AP. The table also preserves the three binary configurations that fail to generate valid negative detectors in the tested PCA(2) construction.

All 13 NSA configurations are retained here. Each AUROC rank inserts the focal NSA into the same 25 other PyOD detectors with complete qualified coverage at all six sizes. The 5 percent scores are shown only for configurations with all 85 qualified dataset/seed outcomes; Q preserves every configuration's coverage. All three binary configurations are unsuitable under the tested PCA(2) construction, with 0/85 qualified outcomes. A missing rank or score is not zero.

Table 6: All 13 NSA configurations in PCA(2): AUROC rank against the same fixed 25 other PyOD detectors plus the focal NSA at each retained size. Five-percent AUROC/AP are complete-panel macro means; Q is five-percent qualified coverage out of 85.

NSA configuration	100% rank	25% rank	10% rank	5% rank	5% AUROC	5% AP	5% Q
NSA fixed	22/26	19/26	19/26	19/26	0.5917	0.2200	85/85
NSA variable	15/26	11/26	8/26	10/26	0.6683	0.2879	85/85
NSA coverage	15/26	14/26	9/26	10/26	0.6669	0.2838	85/85
NSA grid	15/26	11/26	8/26	10/26	0.6775	0.3002	85/85
NSA hierarchical	15/26	14/26	12/26	12/26	0.6579	0.2647	85/85
NSA voronoi	15/26	11/26	8/26	8/26	0.6882	0.3153	85/85
NSA deterministic	21/26	17/26	17/26	17/26	0.6066	0.2198	85/85
NSA suppressed	15/26	12/26	9/26	12/26	0.6650	0.2955	85/85
NSA dual	13/26	10/26	6/26	5/26	0.7264	0.3755	85/85
NSA annealed	21/26	19/26	17/26	19/26	0.5966	0.2187	85/85
NSA binary Hamming	—	—	—	—	—	—	0/85
NSA binary contiguous	—	—	—	—	—	—	0/85
NSA binary chunk	—	—	—	—	—	—	0/85

All ten continuous PCA(2) NSA configurations improve their relative AUROC ranks at 10% and 5% compared with 100%. At 10%, all ten increase their absolute AUROC point estimates; at 5%, only three do, and all ten have lower AP. The native-feature family results are much less favorable. The broader family evidence therefore supports a focused argument for Dual NSA under these constraints, with representation and strategy choice remaining central. The appendix provides every strategy, all six sizes, original-feature results, failure reasons, paired contrasts, and alternative rank fields.

5 Discussion: a starting detector for newly commissioned sensors

5.1 When the proposed use case fits

A sensor set that has just begun collecting data can have enough trusted readings to describe an initial normal state while having no real failure examples. Under that condition, Dual NSA can be a good alternative to include in the first detector comparison. The rationale combines its normal-only training rule, pointwise scoring, complete small-sample outcomes, and competitive standing in the tested PCA(2) pipeline. It can supply an initial anomaly score without making collection of labeled faults a prerequisite.

The practical role is an initial candidate whose performance is checked as data become available. KDE and KNN should remain demanding reference methods: neither requires real training outliers in this experiment, and both are strong comparators. The tables give an operator a basis for choosing candidates, rather than a reason to select the same detector for every newly installed system. A small trusted commissioning set must also cover the normal modes that matter. Otherwise, a new but legitimate mode may be flagged simply because it was absent from the starting sample.

For multichannel sensors, the favorable PCA(2) result suggests evaluating a compact normal-only representation when such a representation preserves meaningful operating variation. Dimensional reduction is part of the tested pipeline; we do not show that it is appropriate for every sensor set. The original-feature results provide a useful check against treating the PCA advantage as a property of all

NSA deployments. A single scalar sensor, a strongly time-dependent process, and a high-dimensional multichannel system may need different representations and comparison methods.

5.2 What the experiment establishes

The evidence establishes that the evaluated Dual NSA implementation works with very small normal-only training matrices and can rank near the front of a fixed field in PCA(2). It also establishes that this standing survives a targeted neural batch-size repair. The observation concerns comparative behavior under scarcity. At 5%, the improved rank partly reflects deterioration among competing methods, while Dual NSA itself loses performance relative to 10%.

No experiment here chooses an operational alarm threshold, measures false alarms per hour, or tests a chronological stream after commissioning. AUROC and AP are held-out ranking measures. PyOD's contamination value of 0.1 calibrates internal score thresholds; it does not add training anomalies and does not validate a 10% deployed alarm rate. Thus the benchmark supports a candidate for initial monitoring, with sensor-specific threshold and stream validation still needed before interpreting scores as reliable alarms.

Query-batch-dependent comparators also need a separate interpretation for streaming use. The twelve audited implementations are COF, LOCI, SOS, SOD, ECOD, COPOD, RGraph, SUOD, LMDD, DIF, CD, and ROD. They may use other observations in the mixed query batch when producing scores, while Dual NSA scores pointwise from its fitted state. Their benchmark outcomes are retained, and an inductive sensitivity removes those entries from its field. This distinction concerns inference semantics and does not mean that the normal-only initial fitting rule was violated.

5.3 Limits and next validation

The clean starting data are selected with known benchmark labels. Real commissioning data can be contaminated or incomplete. The random stratified splits do not model temporal drift or the order in which a sensor would encounter operating modes. Training and test caps, fixed constructors, reduced epochs for some models, timing load, and the 15-second qualification limit further bound the comparison. Failures indicate unsuitability under the recorded configuration; the AutoEncoder/VAE repair demonstrates why that scope matters.

A direct sensor study should next use a chronological accepted-normal commissioning period, hold out later normal modes and faults, and compare Dual NSA with the same strong alternatives. It should report calibrated false-alarm rates, precision under the actual anomaly prevalence, detection delay, and sensitivity to contaminated commissioning data. Holding the representation fixed in an ablation and measuring envelope coverage would help distinguish sample-size effects from changes in PCA geometry. These are proposed follow-up evaluations, not outcomes of the present benchmark.

6 Reproducibility and conclusion

The public code-and-results repository provides the executable benchmark and machine-readable evidence.

The tested source is pinned to commit `0d6c901652f82`; its full commit and tree identifiers are recorded in the appendix. Dataset and split hashes identify the inputs. Independent audits reconstructed the retained normal subsets and checked zero fitted outliers. Raw aggregate records, coverage, sanitized logs, recovery evidence, and reporting arithmetic are retained. Per-observation decision vectors were not saved, so the reporting audit cannot independently recompute AUROC or AP from predictions. The

appendix contains the complete experimental report, including both smaller and intermediate training sizes and every attempted method.

Dual NSA is a defensible alternative to consider at a normal-only cold start, particularly in the tested low-dimensional pipeline. With only 4–30 training normals at 5%, it ranks 5/26 in the primary PCA(2) field and 5/27 after the neural batch-size check. That result supports the proposed use case of newly commissioned sensor sets with accepted normal data. Dataset variation, AP losses, native-feature results, and strong density and distance baselines keep the argument specific: Dual NSA can be a useful starting candidate, and the sensor application still needs local, chronological validation.

References

[1] Stephanie Forrest, Alan S. Perelson, Lawrence Allen, and Rajesh Cherukuri. Self-nonself discrimination in a computer. In *Proceedings of the IEEE Computer Society Symposium on Research in Security and Privacy*, pages 202–212, 1994. doi:10.1109/RISP.1994.296580.

[2] Yue Zhao, Zain Nasrullah, and Zheng Li. PyOD: A Python toolbox for scalable outlier detection. *Journal of Machine Learning Research*, 20(96):1–7, 2019. https://www.jmlr.org/papers/v20/19-011.html.

[3] PyOD contributors. Negative-selection detector implementation and strategy guide, pinned source tree associated with pull request 747. Tested commit `0d6c901652f82c813da45875873ef4a8560cf049`. https://github.com/yzhao062/pyod/pull/747. The source snapshot accompanying this paper identifies the precise evaluated implementation.

[4] Stony Brook University. Outlier Detection DataSets (ODDS). http://odds.cs.stonybrook.edu/. The experiment's dataset manifest identifies the source files and SHA-256 hashes used in this study.

A How to read the extended results

These tables preserve the full roster, including methods that could not produce a valid score. Every setting contains 17 datasets and five split seeds, or 85 expected runs per configuration. Training uses only retained normal rows. Anomalies occur only in the held-out test set. The retained percentages refer to a capped normal training pool, rather than to the entire original dataset.

AUROC is the area under the receiver operating characteristic curve; AP is average precision. Larger values are better for both. Each dataset has equal weight in aggregate scores. A score with fewer than 85 completed runs is an available-case description and can change when difficult cases are missing. Such a score is not used in the complete-coverage rankings. A dash denotes an undefined score, never a zero assigned to a failure.

The primary tables qualify a recorded success only when the measured fitting time plus scoring time is at most 15 seconds. This rule was applied during analysis after native callbacks sometimes escaped the nominal Python alarm. The original recorded-success analysis is also provided below. Confidence intervals use 5,000 dataset-cluster bootstrap draws. They are descriptive 95% intervals without correction for the many comparisons in this appendix; they should not be treated as confirmatory evidence of general superiority.

B Pinned implementation and constructor settings

The evaluation uses the pinned PyOD source revision `0d6c901652f82c813da45875873ef4a8560cf049`, with an initially clean source checkout. NSA is expanded into its eleven strategy families; the binary family is evaluated under three matching rules, yielding thirteen configurations. These settings describe the tested implementation, rather than every possible negative selection algorithm.

Table 7: Common NSA constructor defaults in the pinned source. Strategy-specific code uses the relevant parameters. A target of 100 detectors is not a guarantee that the bounded candidate search finds 100 valid detectors.

Parameter	Value
Target detectors	100
Self radius / fixed detector radius	0.1 / 0.1
Maximum candidates	10,000
Sampling margin	0.5
Target coverage / samples / confidence	0.9 / 256 / 0.95
Grid depth / hierarchy levels	4 / 4
Outlier fraction / outlier radius	0.95 / 0.5
Dual clusters / annealing steps	3 / 20
Contamination parameter	0.1
Random state	The split seed (42–46)

For binary Hamming, the matching threshold is the source default of one bit (zero for a single feature, which is not present in these two panels). Contiguous and chunk matching use the same fixed label-free rule on every dataset: $\max(1, \text{round}(0.4d))$, where d is the feature count supplied to the detector. This follows the upstream example's 12-of-30-feature ratio. These rules were not chosen using test anomaly labels.

Other detectors retain the pinned constructor defaults except the declared changes below. Neighbor counts are bounded by the number of retained rows minus one: ABOD and KNN use at most five neighbors; LOF, COF, and SOD use at most twenty. HDBSCAN uses $\min(5, n)$ as its minimum cluster size. LSCP receives a fixed LOF/KNN/HBOS ensemble because its constructor has no default detector list. Available random states are set to the split seed and available worker-count options to one. DeepSVDD receives the supplied feature count. Verbosity options are reduced without changing scoring.

Table 8: Declared reductions in neural detector compute settings, applied before inspecting outcomes. The comparison therefore screens these budgeted implementations and does not claim to reproduce the original papers' full training budgets.

Detector	Setting	Tested	Pinned default
DIF	n_ensemble	10	50
DeepSVDD	epochs	50	100
AnoGAN	epochs / epochs_query	50 / 5	500 / 20
ALAD	epochs	50	200
LUNAR	n_epochs	50	200

PyOD's contamination parameter remains 0.1 where provided. It determines a training-score threshold and does not inject anomaly rows. AUROC and AP are computed from continuous decision scores, so this threshold is not used to select a prediction cutoff in the reported ranking metrics.

C Dataset sizes and retained normal training pools

Table 9: Dataset inventory and numbers of retained fitting normals. The full pool is capped at 600 normals, and the mixed test set is capped at 2,000 rows. All source anomaly labels are excluded from fitting.

Dataset	Rows	Features	Normals	Anomalies	100%	75%	50%	25%
arrhythmia	452	274	386	66	231	173	115	57
cardio	1831	21	1655	176	600	450	300	150
glass	214	9	205	9	123	92	61	30
ionosphere	351	33	225	126	135	101	67	33
letter	1600	32	1500	100	600	450	300	150
lympho	148	18	142	6	84	63	42	21
mnist	7603	100	6903	700	600	450	300	150
musk	3062	166	2965	97	600	450	300	150
optdigits	5216	64	5066	150	600	450	300	150
pendigits	6870	16	6714	156	600	450	300	150
pima	768	8	500	268	299	224	149	74
satellite	6435	36	4399	2036	600	450	300	150
satimage-2	5803	36	5732	71	600	450	300	150
shuttle	49097	9	45586	3511	600	450	300	150
vertebral	240	6	210	30	126	94	63	31
vowels	1456	12	1406	50	600	450	300	150
wbc	378	30	357	21	213	159	106	53

D Sensitivity of the main dataset ranks to the comparison field

The main table uses the same 36 configurations on every dataset. The baseline comparison below contains Dual NSA plus the 26 other PyOD detectors with complete coverage, making a fixed field of 27. The local field instead includes every configuration that completed all five seeds on that particular dataset. Both alternatives retain the same 25%-normal, PCA(2), zero-training-anomaly setting. A field change can alter a rank even though the detector score is unchanged.

Table 10: Dual NSA dataset ranks under three explicitly defined comparison fields. Top position is 100 times rank divided by field size.

Dataset	Fixed rank	Fixed top	Baseline rank	Baseline top	Local rank	Local top
arrhythmia	3/36	Top 8.33%	1/27	Top 3.70%	3/42	Top 7.14%
cardio	11/36	Top 30.56%	4/27	Top 14.81%	11/40	Top 27.50%
glass	3/36	Top 8.33%	1/27	Top 3.70%	3/41	Top 7.32%
ionosphere	8/36	Top 22.22%	5/27	Top 18.52%	8/41	Top 19.51%
letter	17/36	Top 47.22%	16/27	Top 59.26%	18/40	Top 45.00%
lympho	19/36	Top 52.78%	16/27	Top 59.26%	21/40	Top 52.50%
mnist	21/36	Top 58.33%	18/27	Top 66.67%	26/42	Top 61.90%
musk	12/36	Top 33.33%	9/27	Top 33.33%	14/45	Top 31.11%
optdigits	26/36	Top 72.22%	25/27	Top 92.59%	29/39	Top 74.36%
pendigits	16/36	Top 44.44%	13/27	Top 48.15%	17/39	Top 43.59%
pima	18/36	Top 50.00%	18/27	Top 66.67%	22/42	Top 52.38%
satellite	6/36	Top 16.67%	6/27	Top 22.22%	7/44	Top 15.91%
satimage-2	9/36	Top 25.00%	9/27	Top 33.33%	10/40	Top 25.00%
shuttle	2/36	Top 5.56%	2/27	Top 7.41%	2/46	Top 4.35%
vertebral	13/36	Top 36.11%	10/27	Top 37.04%	16/42	Top 38.10%
vowels	7/36	Top 19.44%	4/27	Top 14.81%	7/45	Top 15.56%
wbc	10/36	Top 27.78%	10/27	Top 37.04%	12/47	Top 25.53%

Table 11: Dual NSA split-specific ranks in the fixed field of 36. These ranks show variation that is hidden by a dataset's mean score.

Dataset	Seed 42	Seed 43	Seed 44	Seed 45	Seed 46
arrhythmia	2/36	9/36	9/36	5/36	16/36
cardio	10/36	11/36	19/36	17/36	1/36
glass	13/36	3/36	7/36	5/36	12/36
ionosphere	10/36	13/36	12/36	6/36	5/36
letter	12/36	9/36	18/36	22/36	15/36
lympho	32/36	2/36	21/36	20/36	6/36
mnist	22/36	13/36	17/36	14/36	31/36
musk	16/36	9/36	15/36	1/36	7/36
optdigits	33/36	28/36	34/36	18/36	32/36
pendigits	10/36	20/36	17/36	12/36	25/36
pima	24/36	15/36	17/36	14/36	23/36
satellite	7/36	11/36	4/36	4/36	10/36
satimage-2	11/36	1/36	20/36	1/36	11/36
shuttle	3/36	3/36	1/36	5/36	2/36
vertebral	30/36	11/36	11/36	13/36	25/36
vowels	3/36	6/36	14/36	3/36	11/36
wbc	17/36	14/36	5/36	7/36	18/36

E All detector scores and coverage

Each cell reports AUROC / AP (completed runs out of 85). Scores are available-case dataset macro means. The coverage count is part of the result: two scores with different completed cases are not a controlled comparison.

Table 12: All 56 configurations: Original features. Cell format is AUROC / AP (coverage).

Configuration	100% retained	75% retained	50% retained	25% retained
NSA fixed	0.3597 / 0.1287 (85/85)	0.3504 / 0.1272 (85/85)	0.3460 / 0.1232 (85/85)	0.3297 / 0.1211 (85/85)
NSA variable	0.5931 / 0.3038 (85/85)	0.5821 / 0.2865 (85/85)	0.5531 / 0.2456 (85/85)	0.5025 / 0.2054 (85/85)
NSA coverage	0.5931 / 0.3040 (85/85)	0.5814 / 0.2855 (85/85)	0.5534 / 0.2455 (85/85)	0.5028 / 0.2053 (85/85)
NSA grid	0.5972 / 0.3026 (85/85)	0.5880 / 0.2869 (85/85)	0.5569 / 0.2510 (85/85)	0.5025 / 0.1994 (84/85)
NSA hierarchical	0.2968 / 0.1173 (85/85)	0.3078 / 0.1145 (85/85)	0.2753 / 0.0938 (85/85)	0.2712 / 0.1092 (85/85)
NSA voronoi	— / — (0/85)	— / — (0/85)	— / — (0/85)	— / — (0/85)
NSA deterministic	— / — (0/85)	— / — (0/85)	— / — (0/85)	— / — (0/85)
NSA suppressed	0.5883 / 0.2982 (85/85)	0.5656 / 0.2705 (85/85)	0.5516 / 0.2440 (85/85)	0.4920 / 0.1989 (85/85)
NSA dual	0.7047 / 0.4609 (81/85)	0.7352 / 0.5006 (80/85)	0.7318 / 0.4966 (81/85)	0.7488 / 0.5142 (83/85)
NSA annealed	0.3161 / 0.1120 (85/85)	0.3112 / 0.1058 (85/85)	0.3023 / 0.1070 (85/85)	0.2789 / 0.0988 (85/85)
NSA binary Hamming	0.5259 / 0.1681 (76/85)	0.5226 / 0.1643 (83/85)	0.5288 / 0.1573 (85/85)	0.5172 / 0.1449 (85/85)
NSA binary contiguous	0.5350 / 0.1578 (59/85)	0.5404 / 0.1498 (60/85)	0.5335 / 0.1511 (60/85)	0.5218 / 0.1203 (63/85)
NSA binary chunk	0.5269 / 0.1261 (75/85)	0.5292 / 0.1258 (75/85)	0.5328 / 0.1211 (75/85)	0.5191 / 0.1082 (75/85)
ECOD	0.7855 / 0.4741 (77/85)	0.7824 / 0.4631 (85/85)	0.7783 / 0.4520 (85/85)	0.7738 / 0.4399 (85/85)
ABOD	0.8951 / 0.6279 (85/85)	0.8927 / 0.6118 (85/85)	0.8841 / 0.5849 (85/85)	0.8724 / 0.5576 (85/85)
COPOD	0.7865 / 0.4573 (85/85)	0.7830 / 0.4587 (85/85)	0.7837 / 0.4551 (85/85)	0.7797 / 0.4464 (85/85)
MAD	— / — (0/85)	— / — (0/85)	— / — (0/85)	— / — (0/85)
SOS	0.5868 / 0.2218 (85/85)	0.5883 / 0.2283 (84/85)	0.5897 / 0.2296 (85/85)	0.5884 / 0.2196 (82/85)
QMCD	0.5376 / 0.2370 (85/85)	0.5264 / 0.2173 (84/85)	0.5124 / 0.1981 (85/85)	0.4679 / 0.1766 (85/85)
KDE	0.8970 / 0.6439 (85/85)	0.8945 / 0.6312 (85/85)	0.8932 / 0.6188 (85/85)	0.8933 / 0.5962 (85/85)
Sampling	0.8620 / 0.5595 (85/85)	0.8671 / 0.5543 (85/85)	0.8685 / 0.5545 (85/85)	0.8652 / 0.5464 (85/85)
GMM	0.8654 / 0.5497 (85/85)	0.8615 / 0.5444 (85/85)	0.8637 / 0.5419 (85/85)	0.8612 / 0.5270 (85/85)
PCA	0.8073 / 0.5304 (82/85)	0.8061 / 0.5268 (81/85)	0.7995 / 0.5151 (80/85)	0.8112 / 0.5230 (85/85)
KPCA	0.9023 / 0.6464 (85/85)	0.8986 / 0.6301 (85/85)	0.8967 / 0.6222 (85/85)	0.8925 / 0.6004 (85/85)
MCD	0.8427 / 0.4702 (85/85)	0.8450 / 0.4687 (85/85)	0.8445 / 0.4679 (85/85)	0.8415 / 0.4620 (85/85)
CD	0.8192 / 0.4719 (79/85)	0.8173 / 0.4705 (79/85)	0.8168 / 0.4680 (75/85)	0.8182 / 0.4599 (78/85)

Continued on next page

Table 12 continued

Configuration	100% retained	75% retained	50% retained	25% retained
OCSVM	0.8409 / 0.5674 (85/85)	0.8396 / 0.5612 (85/85)	0.8404 / 0.5632 (85/85)	0.8422 / 0.5559 (85/85)
LMDD	0.7588 / 0.4520 (60/85)	0.7562 / 0.4447 (59/85)	0.7109 / 0.4243 (50/85)	0.7517 / 0.4390 (53/85)
LOF	0.8879 / 0.5786 (85/85)	0.8833 / 0.5781 (85/85)	0.8738 / 0.5648 (85/85)	0.8551 / 0.5477 (85/85)
COF	0.6746 / 0.3429 (85/85)	0.6751 / 0.3395 (85/85)	0.6727 / 0.3404 (85/85)	0.6751 / 0.3407 (81/85)
CBLOF	0.8803 / 0.5702 (84/85)	0.8816 / 0.5695 (83/85)	0.8815 / 0.5712 (85/85)	0.8705 / 0.5522 (75/85)
LOCI	0.6768 / 0.3329 (30/85)	0.6766 / 0.3217 (28/85)	0.7023 / 0.3639 (22/85)	0.7128 / 0.3758 (24/85)
HBOS	0.8133 / 0.4952 (85/85)	0.8130 / 0.4919 (85/85)	0.8130 / 0.4862 (85/85)	0.8119 / 0.4759 (73/85)
HDBSCAN	0.6946 / 0.2392 (85/85)	0.6732 / 0.2225 (85/85)	0.6414 / 0.2021 (85/85)	0.6099 / 0.1703 (85/85)
KNN	0.8973 / 0.6163 (85/85)	0.8937 / 0.6034 (85/85)	0.8896 / 0.5941 (85/85)	0.8782 / 0.5731 (85/85)
SOD	0.7553 / 0.3523 (85/85)	0.7541 / 0.3393 (78/85)	0.7209 / 0.3110 (16/85)	0.7488 / 0.3419 (60/85)
ROD	0.8244 / 0.5105 (42/85)	0.8096 / 0.4694 (35/85)	0.7919 / 0.4636 (31/85)	0.7902 / 0.4756 (38/85)
IForest	0.8533 / 0.5258 (85/85)	0.8527 / 0.5310 (85/85)	0.8478 / 0.5157 (85/85)	0.8368 / 0.4787 (85/85)
INNE	0.8732 / 0.5757 (85/85)	0.8718 / 0.5687 (85/85)	0.8679 / 0.5518 (85/85)	0.8629 / 0.5290 (85/85)
DIF	0.8191 / 0.4428 (85/85)	0.8106 / 0.4243 (85/85)	0.8081 / 0.4195 (85/85)	0.8027 / 0.3970 (72/85)
FeatureBagging	0.8891 / 0.5859 (85/85)	0.8822 / 0.5767 (85/85)	0.8721 / 0.5665 (85/85)	0.8540 / 0.5507 (85/85)
LSCP	0.8924 / 0.5914 (85/85)	0.8868 / 0.5841 (85/85)	0.8784 / 0.5758 (85/85)	0.8528 / 0.5281 (72/85)
LODA	0.7426 / 0.4150 (85/85)	0.7374 / 0.3993 (85/85)	0.7245 / 0.3788 (85/85)	0.7221 / 0.3356 (85/85)
SUOD	0.8503 / 0.5440 (85/85)	0.8451 / 0.5383 (85/85)	0.8436 / 0.5335 (69/85)	0.8308 / 0.5069 (61/85)
XGBOD	— / — (0/85)	— / — (0/85)	— / — (0/85)	— / — (0/85)
AutoEncoder	0.8684 / 0.5702 (85/85)	0.8627 / 0.5625 (85/85)	0.8517 / 0.5511 (85/85)	0.8564 / 0.5614 (57/85)
VAE	0.8299 / 0.5466 (85/85)	0.8234 / 0.5340 (85/85)	0.8141 / 0.5253 (85/85)	0.8317 / 0.5436 (69/85)
SO_GAAL	0.5286 / 0.2693 (85/85)	0.5146 / 0.2353 (85/85)	0.5232 / 0.2235 (85/85)	0.5259 / 0.2439 (85/85)
MO_GAAL	0.5438 / 0.2592 (85/85)	0.5156 / 0.2274 (85/85)	0.5443 / 0.2339 (85/85)	0.5393 / 0.2496 (85/85)
DeepSVDD	0.8651 / 0.5908 (85/85)	0.8546 / 0.5765 (85/85)	0.8545 / 0.5698 (85/85)	0.8395 / 0.5429 (85/85)
AnoGAN	0.7902 / 0.4839 (85/85)	0.7849 / 0.4778 (83/85)	0.7942 / 0.4886 (83/85)	0.8000 / 0.4652 (77/85)
ALAD	0.5409 / 0.1470 (85/85)	0.5485 / 0.1579 (85/85)	0.5489 / 0.1575 (85/85)	0.5579 / 0.1625 (85/85)
AE1SVM	0.8363 / 0.5458 (85/85)	0.8287 / 0.5277 (85/85)	0.8283 / 0.5290 (85/85)	0.8180 / 0.4992 (79/85)
DevNet	— / — (0/85)	— / — (0/85)	— / — (0/85)	— / — (0/85)
RGraph	0.3840 / 0.1682 (26/85)	0.4713 / 0.1978 (21/85)	0.4236 / 0.2126 (21/85)	0.5040 / 0.2547 (31/85)
LUNAR	0.9058 / 0.6768 (85/85)	0.9022 / 0.6401 (85/85)	0.8967 / 0.6285 (85/85)	0.8870 / 0.5962 (85/85)

Table 13: All 56 configurations: PCA to two dimensions. Cell format is AUROC / AP (coverage).

Configuration	100% retained	75% retained	50% retained	25% retained
NSA fixed	0.5827 / 0.3427 (85/85)	0.5923 / 0.3328 (85/85)	0.6202 / 0.3564 (85/85)	0.6299 / 0.3474 (85/85)
NSA variable	0.6928 / 0.4441 (85/85)	0.6982 / 0.4435 (85/85)	0.7139 / 0.4466 (85/85)	0.7240 / 0.4204 (85/85)
NSA coverage	0.6694 / 0.4253 (85/85)	0.6763 / 0.4310 (85/85)	0.7039 / 0.4422 (85/85)	0.6954 / 0.4102 (85/85)
NSA grid	0.6886 / 0.4361 (85/85)	0.6942 / 0.4385 (85/85)	0.7105 / 0.4454 (85/85)	0.7261 / 0.4208 (85/85)
NSA hierarchical	0.6754 / 0.4218 (85/85)	0.6676 / 0.4090 (85/85)	0.6921 / 0.4266 (85/85)	0.6836 / 0.3857 (85/85)
NSA voronoi	0.7043 / 0.4521 (85/85)	0.7164 / 0.4590 (85/85)	0.7315 / 0.4550 (85/85)	0.7432 / 0.4347 (85/85)
NSA deterministic	0.6032 / 0.3594 (85/85)	0.6149 / 0.3586 (85/85)	0.6459 / 0.3928 (85/85)	0.6420 / 0.3450 (85/85)
NSA suppressed	0.6738 / 0.4214 (85/85)	0.6841 / 0.4288 (85/85)	0.7063 / 0.4402 (85/85)	0.7120 / 0.4186 (85/85)
NSA dual	0.7172 / 0.4718 (85/85)	0.7268 / 0.4777 (85/85)	0.7419 / 0.4801 (85/85)	0.7603 / 0.4699 (85/85)
NSA annealed	0.5998 / 0.3672 (85/85)	0.6183 / 0.3715 (85/85)	0.6401 / 0.3835 (85/85)	0.6286 / 0.3403 (85/85)
NSA binary Hamming	— / — (0/85)	— / — (0/85)	— / — (0/85)	— / — (0/85)
NSA binary contiguous	— / — (0/85)	— / — (0/85)	— / — (0/85)	— / — (0/85)
NSA binary chunk	— / — (0/85)	— / — (0/85)	— / — (0/85)	— / — (0/85)
ECOD	0.6201 / 0.2698 (85/85)	0.6197 / 0.2721 (85/85)	0.6176 / 0.2782 (85/85)	0.6199 / 0.2584 (85/85)
ABOD	0.7853 / 0.4831 (85/85)	0.7806 / 0.4832 (85/85)	0.7903 / 0.4860 (85/85)	0.7893 / 0.4706 (85/85)
COPOD	0.6634 / 0.3118 (85/85)	0.6625 / 0.3113 (85/85)	0.6686 / 0.3238 (85/85)	0.6632 / 0.2999 (85/85)
MAD	— / — (0/85)	— / — (0/85)	— / — (0/85)	— / — (0/85)
SOS	0.5326 / 0.1547 (85/85)	0.5402 / 0.1676 (85/85)	0.5350 / 0.1654 (85/85)	0.5280 / 0.1596 (85/85)
QMCD	0.6582 / 0.3382 (85/85)	0.6570 / 0.3381 (85/85)	0.6609 / 0.3288 (85/85)	0.6353 / 0.2828 (85/85)
KDE	0.7890 / 0.4817 (85/85)	0.7913 / 0.4874 (85/85)	0.7956 / 0.4908 (85/85)	0.7947 / 0.4711 (85/85)
Sampling	0.7607 / 0.4333 (85/85)	0.7529 / 0.4388 (85/85)	0.7430 / 0.4191 (85/85)	0.7648 / 0.4329 (85/85)
GMM	0.6345 / 0.3568 (85/85)	0.6330 / 0.3605 (85/85)	0.6370 / 0.3756 (85/85)	0.6495 / 0.3643 (85/85)
PCA	0.6656 / 0.3605 (85/85)	0.6581 / 0.3583 (85/85)	0.6471 / 0.3520 (85/85)	0.6498 / 0.3566 (85/85)
KPCA	0.7862 / 0.4854 (85/85)	0.7892 / 0.4885 (85/85)	0.7980 / 0.4916 (85/85)	0.7937 / 0.4676 (85/85)
MCD	0.7063 / 0.3673 (85/85)	0.6980 / 0.3525 (85/85)	0.7027 / 0.3622 (85/85)	0.7071 / 0.3539 (85/85)
CD	0.5831 / 0.2942 (85/85)	0.5807 / 0.2954 (85/85)	0.5874 / 0.3025 (85/85)	0.5910 / 0.2704 (85/85)
OCSVM	0.7801 / 0.4803 (85/85)	0.7797 / 0.4835 (85/85)	0.7918 / 0.4909 (85/85)	0.7788 / 0.4655 (85/85)
LMDD	0.5580 / 0.2448 (85/85)	0.5717 / 0.2531 (85/85)	0.5898 / 0.2627 (85/85)	0.5774 / 0.2553 (80/85)
LOF	0.7781 / 0.4604 (85/85)	0.7819 / 0.4543 (85/85)	0.7830 / 0.4577 (85/85)	0.7651 / 0.4127 (85/85)
COF	0.6141 / 0.2250 (85/85)	0.6216 / 0.2319 (85/85)	0.6207 / 0.2384 (85/85)	0.6265 / 0.2417 (85/85)
CBLOF	0.7815 / 0.4532 (85/85)	0.7783 / 0.4506 (85/85)	0.7859 / 0.4816 (85/85)	0.7739 / 0.4533 (64/85)

Continued on next page

Table 13 continued

Configuration	100% retained	75% retained	50% retained	25% retained
LOCI	0.5585 / 0.2439 (30/85)	0.5521 / 0.2607 (28/85)	0.5792 / 0.2616 (29/85)	0.4614 / 0.2154 (14/85)
HBOS	0.7581 / 0.3902 (85/85)	0.7488 / 0.3775 (85/85)	0.7594 / 0.3772 (85/85)	0.7562 / 0.3551 (68/85)
HDBSCAN	0.7163 / 0.2391 (85/85)	0.7255 / 0.2265 (85/85)	0.6932 / 0.2004 (85/85)	0.6342 / 0.1791 (85/85)
KNN	0.7958 / 0.4872 (85/85)	0.7993 / 0.4891 (85/85)	0.8051 / 0.4942 (85/85)	0.8005 / 0.4644 (85/85)
SOD	0.5747 / 0.2075 (85/85)	0.5643 / 0.2061 (82/85)	0.5674 / 0.2152 (85/85)	0.5591 / 0.2197 (18/85)
ROD	0.6650 / 0.3133 (85/85)	0.6663 / 0.3204 (85/85)	0.6707 / 0.3200 (85/85)	0.6701 / 0.3153 (72/85)
IForest	0.7593 / 0.4023 (85/85)	0.7571 / 0.3930 (85/85)	0.7598 / 0.3933 (85/85)	0.7520 / 0.3441 (85/85)
INNE	0.7795 / 0.4684 (85/85)	0.7813 / 0.4711 (85/85)	0.7831 / 0.4740 (85/85)	0.7712 / 0.4468 (85/85)
DIF	0.7577 / 0.4216 (85/85)	0.7536 / 0.4321 (85/85)	0.7643 / 0.4348 (85/85)	0.7600 / 0.3986 (74/85)
FeatureBagging	0.7695 / 0.4496 (85/85)	0.7688 / 0.4439 (85/85)	0.7728 / 0.4498 (85/85)	0.7639 / 0.4178 (85/85)
LSCP	0.7893 / 0.4757 (85/85)	0.7888 / 0.4784 (85/85)	0.7934 / 0.4670 (85/85)	0.7933 / 0.4520 (77/85)
LODA	0.7270 / 0.3456 (85/85)	0.7243 / 0.3364 (85/85)	0.7248 / 0.3476 (85/85)	0.7092 / 0.3058 (85/85)
SUOD	0.7576 / 0.3982 (85/85)	0.7600 / 0.4050 (84/85)	0.7620 / 0.4100 (85/85)	0.7608 / 0.3780 (35/85)
XGBOD	— / — (0/85)	— / — (0/85)	— / — (0/85)	— / — (0/85)
AutoEncoder	0.6830 / 0.3647 (85/85)	0.6767 / 0.3495 (85/85)	0.6591 / 0.3211 (85/85)	0.6651 / 0.3316 (60/85)
VAE	0.6718 / 0.3697 (85/85)	0.6690 / 0.3697 (85/85)	0.6623 / 0.3577 (85/85)	0.6461 / 0.3866 (70/85)
SO_GAAL	0.5817 / 0.2066 (85/85)	0.5506 / 0.2070 (85/85)	0.5616 / 0.2154 (85/85)	0.5535 / 0.2083 (85/85)
MO_GAAL	0.5718 / 0.2341 (85/85)	0.5644 / 0.2481 (85/85)	0.5640 / 0.2429 (85/85)	0.5268 / 0.2257 (85/85)
DeepSVDD	0.7368 / 0.3906 (85/85)	0.7434 / 0.3894 (85/85)	0.7448 / 0.4010 (85/85)	0.7238 / 0.3890 (85/85)
AnoGAN	0.6247 / 0.2801 (85/85)	0.6322 / 0.2816 (85/85)	0.6288 / 0.2914 (85/85)	0.6430 / 0.2808 (77/85)
ALAD	0.5051 / 0.1300 (85/85)	0.5233 / 0.1343 (85/85)	0.5282 / 0.1465 (85/85)	0.5193 / 0.1361 (85/85)
AE1SVM	0.6570 / 0.3360 (85/85)	0.6475 / 0.3384 (85/85)	0.6531 / 0.3500 (85/85)	0.6635 / 0.3449 (80/85)
DevNet	— / — (0/85)	— / — (0/85)	— / — (0/85)	— / — (0/85)
RGraph	0.4360 / 0.1545 (35/85)	0.4184 / 0.1541 (34/85)	0.4146 / 0.1242 (45/85)	0.4026 / 0.1524 (32/85)
LUNAR	0.7745 / 0.4576 (85/85)	0.7783 / 0.4632 (85/85)	0.7913 / 0.4759 (85/85)	0.7784 / 0.4449 (85/85)

F Paired changes when the normal training pool shrinks

A change is reduced-size score minus the same configuration's 100%-retained score, using only matching successful dataset/seed cases. Positive changes favor the reduced pool. The pair and dataset counts show how much evidence contributes; a missing CI means fewer than five independent datasets were available.

Table 14: NSA: paired training-size changes on original features. Each interval is a descriptive dataset-cluster 95 percent bootstrap interval.

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
NSA fixed	75%	85/85	17/17	-0.0093	[-0.0236, +0.0024]	-0.0015	[-0.0064, +0.0045]
NSA variable	75%	85/85	17/17	-0.0110	[-0.0166, -0.0055]	-0.0173	[-0.0282, -0.0072]
NSA coverage	75%	85/85	17/17	-0.0117	[-0.0172, -0.0064]	-0.0184	[-0.0290, -0.0098]
NSA grid	75%	85/85	17/17	-0.0092	[-0.0214, +0.0024]	-0.0156	[-0.0290, -0.0042]
NSA hierarchical	75%	85/85	17/17	+0.0110	[-0.0355, +0.0611]	-0.0028	[-0.0228, +0.0161]
NSA voronoi	75%	0/85	0/17	—	[—, —]	—	[—, —]
NSA deterministic	75%	0/85	0/17	—	[—, —]	—	[—, —]
NSA suppressed	75%	85/85	17/17	-0.0228	[-0.0407, -0.0090]	-0.0277	[-0.0503, -0.0098]
NSA dual	75%	80/85	16/17	+0.0354	[+0.0101, +0.0649]	+0.0303	[+0.0059, +0.0685]
NSA annealed	75%	85/85	17/17	-0.0048	[-0.0162, +0.0069]	-0.0062	[-0.0140, +0.0001]
NSA binary Hamming	75%	76/85	16/17	-0.0078	[-0.0220, +0.0041]	-0.0048	[-0.0112, +0.0018]
NSA binary contiguous	75%	59/85	12/17	+0.0076	[-0.0118, +0.0294]	-0.0078	[-0.0191, +0.0015]
NSA binary chunk	75%	75/85	15/17	+0.0023	[-0.0094, +0.0180]	-0.0004	[-0.0054, +0.0045]
NSA fixed	50%	85/85	17/17	-0.0137	[-0.0390, +0.0174]	-0.0055	[-0.0134, +0.0042]
NSA variable	50%	85/85	17/17	-0.0400	[-0.0750, -0.0083]	-0.0582	[-0.1113, -0.0162]
NSA coverage	50%	85/85	17/17	-0.0397	[-0.0750, -0.0090]	-0.0585	[-0.1096, -0.0164]
NSA grid	50%	85/85	17/17	-0.0403	[-0.0787, -0.0105]	-0.0515	[-0.1006, -0.0121]
NSA hierarchical	50%	85/85	17/17	-0.0216	[-0.0662, +0.0230]	-0.0236	[-0.0449, -0.0062]
NSA voronoi	50%	0/85	0/17	—	[—, —]	—	[—, —]
NSA deterministic	50%	0/85	0/17	—	[—, —]	—	[—, —]
NSA suppressed	50%	85/85	17/17	-0.0368	[-0.0669, -0.0110]	-0.0543	[-0.1075, -0.0131]
NSA dual	50%	80/85	16/17	+0.0327	[+0.0011, +0.0691]	+0.0331	[+0.0018, +0.0726]
NSA annealed	50%	85/85	17/17	-0.0138	[-0.0304, +0.0012]	-0.0050	[-0.0217, +0.0059]
NSA binary Hamming	50%	76/85	16/17	-0.0022	[-0.0226, +0.0187]	-0.0141	[-0.0450, +0.0062]

Continued on next page

Table 14 continued

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
NSA binary contiguous	50%	59/85	12/17	-0.0025	[-0.0236, +0.0211]	-0.0067	[-0.0123, -0.0015]
NSA binary chunk	50%	75/85	15/17	+0.0060	[-0.0083, +0.0243]	-0.0050	[-0.0144, +0.0021]
NSA fixed	25%	85/85	17/17	-0.0300	[-0.0638, +0.0097]	-0.0076	[-0.0160, -0.0001]
NSA variable	25%	85/85	17/17	-0.0906	[-0.1755, -0.0182]	-0.0984	[-0.1923, -0.0259]
NSA coverage	25%	85/85	17/17	-0.0904	[-0.1774, -0.0168]	-0.0987	[-0.1911, -0.0243]
NSA grid	25%	84/85	17/17	-0.0939	[-0.1812, -0.0246]	-0.1027	[-0.1964, -0.0352]
NSA hierarchical	25%	85/85	17/17	-0.0256	[-0.0768, +0.0233]	-0.0081	[-0.0338, +0.0141]
NSA voronoi	25%	0/85	0/17	—	[—, —]	—	[—, —]
NSA deterministic	25%	0/85	0/17	—	[—, —]	—	[—, —]
NSA suppressed	25%	85/85	17/17	-0.0964	[-0.1874, -0.0226]	-0.0993	[-0.1936, -0.0252]
NSA dual	25%	80/85	16/17	+0.0550	[+0.0175, +0.0973]	+0.0554	[+0.0133, +0.1094]
NSA annealed	25%	85/85	17/17	-0.0371	[-0.0754, -0.0109]	-0.0132	[-0.0225, -0.0051]
NSA binary Hamming	25%	76/85	16/17	-0.0149	[-0.0368, +0.0032]	-0.0250	[-0.0680, +0.0005]
NSA binary contiguous	25%	59/85	12/17	-0.0165	[-0.0437, +0.0109]	-0.0272	[-0.0661, -0.0050]
NSA binary chunk	25%	75/85	15/17	-0.0078	[-0.0254, +0.0090]	-0.0179	[-0.0408, -0.0024]

Table 15: Other PyOD detectors: paired training-size changes on original features. Each interval is a descriptive dataset-cluster 95 percent bootstrap interval.

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
ECOD	75%	77/85	17/17	-0.0027	[-0.0039, -0.0018]	-0.0102	[-0.0184, -0.0044]
ABOD	75%	85/85	17/17	-0.0024	[-0.0059, +0.0008]	-0.0161	[-0.0272, -0.0061]
COPOD	75%	85/85	17/17	-0.0035	[-0.0084, -0.0004]	+0.0014	[-0.0042, +0.0099]
MAD	75%	0/85	0/17	—	[—, —]	—	[—, —]
SOS	75%	84/85	17/17	+0.0013	[-0.0019, +0.0052]	+0.0065	[+0.0000, +0.0176]
QMCD	75%	84/85	17/17	-0.0114	[-0.0228, -0.0012]	-0.0202	[-0.0397, -0.0058]
KDE	75%	85/85	17/17	-0.0025	[-0.0054, +0.0001]	-0.0127	[-0.0235, -0.0041]
Sampling	75%	85/85	17/17	+0.0051	[-0.0052, +0.0154]	-0.0052	[-0.0325, +0.0175]
GMM	75%	85/85	17/17	-0.0038	[-0.0100, +0.0002]	-0.0053	[-0.0117, +0.0001]
PCA	75%	81/85	17/17	-0.0013	[-0.0022, -0.0005]	-0.0058	[-0.0093, -0.0028]
KPCA	75%	85/85	17/17	-0.0037	[-0.0065, -0.0014]	-0.0163	[-0.0257, -0.0083]
MCD	75%	85/85	17/17	+0.0023	[-0.0036, +0.0088]	-0.0015	[-0.0331, +0.0201]
CD	75%	79/85	16/17	-0.0019	[-0.0108, +0.0072]	-0.0014	[-0.0143, +0.0114]
OCSVM	75%	85/85	17/17	-0.0013	[-0.0032, +0.0001]	-0.0063	[-0.0111, -0.0021]
LMDD	75%	59/85	12/17	-0.0024	[-0.0058, +0.0008]	-0.0069	[-0.0171, -0.0004]
LOF	75%	85/85	17/17	-0.0046	[-0.0106, +0.0007]	-0.0004	[-0.0093, +0.0098]
COF	75%	85/85	17/17	+0.0005	[-0.0005, +0.0016]	-0.0033	[-0.0073, +0.0003]
CBLOF	75%	82/85	17/17	+0.0020	[-0.0080, +0.0129]	-0.0000	[-0.0148, +0.0201]
LOCI	75%	28/85	6/17	+0.0042	[-0.0045, +0.0132]	-0.0077	[-0.0331, +0.0123]
HBOS	75%	85/85	17/17	-0.0003	[-0.0033, +0.0032]	-0.0033	[-0.0074, +0.0006]
HDBSCAN	75%	85/85	17/17	-0.0214	[-0.0434, -0.0006]	-0.0167	[-0.0399, +0.0078]
KNN	75%	85/85	17/17	-0.0036	[-0.0062, -0.0011]	-0.0129	[-0.0214, -0.0052]
SOD	75%	78/85	17/17	-0.0000	[-0.0065, +0.0074]	-0.0090	[-0.0253, +0.0031]
ROD	75%	35/85	8/17	-0.0031	[-0.0099, +0.0015]	-0.0025	[-0.0123, +0.0086]
IForest	75%	85/85	17/17	-0.0005	[-0.0048, +0.0032]	+0.0052	[-0.0067, +0.0232]
INNE	75%	85/85	17/17	-0.0014	[-0.0036, +0.0006]	-0.0070	[-0.0142, -0.0004]
DIF	75%	85/85	17/17	-0.0085	[-0.0173, +0.0001]	-0.0184	[-0.0337, -0.0051]
FeatureBagging	75%	85/85	17/17	-0.0069	[-0.0141, -0.0007]	-0.0092	[-0.0187, +0.0005]
LSCP	75%	85/85	17/17	-0.0055	[-0.0111, -0.0012]	-0.0073	[-0.0160, +0.0005]
LODA	75%	85/85	17/17	-0.0052	[-0.0235, +0.0103]	-0.0157	[-0.0334, -0.0031]

Continued on next page

Table 15 continued

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
SUOD	75%	85/85	17/17	-0.0052	[-0.0094, -0.0015]	-0.0056	[-0.0120, -0.0000]
XGBOD	75%	0/85	0/17	—	[—, —]	—	[—, —]
AutoEncoder	75%	85/85	17/17	-0.0057	[-0.0097, -0.0017]	-0.0077	[-0.0140, -0.0008]
VAE	75%	85/85	17/17	-0.0065	[-0.0130, -0.0012]	-0.0126	[-0.0250, -0.0030]
SO_GAAL	75%	85/85	17/17	-0.0140	[-0.0846, +0.0485]	-0.0341	[-0.0964, +0.0202]
MO_GAAL	75%	85/85	17/17	-0.0282	[-0.0961, +0.0292]	-0.0318	[-0.0840, +0.0125]
DeepSVDD	75%	85/85	17/17	-0.0104	[-0.0197, -0.0018]	-0.0143	[-0.0280, -0.0014]
AnoGAN	75%	83/85	17/17	-0.0039	[-0.0147, +0.0054]	-0.0050	[-0.0247, +0.0095]
ALAD	75%	85/85	17/17	+0.0077	[-0.0111, +0.0301]	+0.0109	[-0.0049, +0.0290]
AE1SVM	75%	85/85	17/17	-0.0076	[-0.0135, -0.0020]	-0.0181	[-0.0281, -0.0091]
DevNet	75%	0/85	0/17	—	[—, —]	—	[—, —]
RGraph	75%	20/85	4/17	+0.0173	[—, —]	-0.0008	[—, —]
LUNAR	75%	85/85	17/17	-0.0036	[-0.0067, -0.0007]	-0.0367	[-0.0616, -0.0153]
ECOD	50%	77/85	17/17	-0.0069	[-0.0103, -0.0038]	-0.0214	[-0.0384, -0.0096]
ABOD	50%	85/85	17/17	-0.0110	[-0.0182, -0.0043]	-0.0430	[-0.0711, -0.0184]
COPOD	50%	85/85	17/17	-0.0029	[-0.0050, -0.0009]	-0.0022	[-0.0077, +0.0048]
MAD	50%	0/85	0/17	—	[—, —]	—	[—, —]
SOS	50%	85/85	17/17	+0.0028	[-0.0003, +0.0081]	+0.0078	[-0.0031, +0.0258]
QMCD	50%	85/85	17/17	-0.0252	[-0.0564, +0.0124]	-0.0389	[-0.0898, -0.0004]
KDE	50%	85/85	17/17	-0.0038	[-0.0101, +0.0030]	-0.0251	[-0.0443, -0.0048]
Sampling	50%	85/85	17/17	+0.0065	[-0.0044, +0.0185]	-0.0050	[-0.0272, +0.0142]
GMM	50%	85/85	17/17	-0.0017	[-0.0106, +0.0088]	-0.0077	[-0.0230, +0.0115]
PCA	50%	80/85	16/17	+0.0025	[-0.0015, +0.0070]	+0.0017	[-0.0110, +0.0161]
KPCA	50%	85/85	17/17	-0.0056	[-0.0102, -0.0011]	-0.0242	[-0.0436, -0.0066]
MCD	50%	85/85	17/17	+0.0018	[-0.0101, +0.0153]	-0.0024	[-0.0354, +0.0265]
CD	50%	75/85	16/17	-0.0024	[-0.0084, +0.0039]	-0.0039	[-0.0121, +0.0031]
OCSVM	50%	85/85	17/17	-0.0005	[-0.0056, +0.0044]	-0.0042	[-0.0159, +0.0079]
LMDD	50%	50/85	10/17	-0.0061	[-0.0181, +0.0060]	-0.0056	[-0.0242, +0.0112]
LOF	50%	85/85	17/17	-0.0141	[-0.0295, -0.0005]	-0.0138	[-0.0406, +0.0141]
COF	50%	85/85	17/17	-0.0019	[-0.0077, +0.0019]	-0.0025	[-0.0113, +0.0032]
CBLOF	50%	84/85	17/17	+0.0013	[-0.0043, +0.0063]	+0.0013	[-0.0155, +0.0184]
LOCI	50%	22/85	5/17	+0.0258	[-0.0031, +0.0729]	+0.0382	[-0.0095, +0.0989]
HBOS	50%	85/85	17/17	-0.0003	[-0.0052, +0.0051]	-0.0090	[-0.0167, -0.0031]

Continued on next page

Table 15 continued

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
HDBSCAN	50%	85/85	17/17	-0.0532	[-0.0957, -0.0197]	-0.0371	[-0.0761, +0.0078]
KNN	50%	85/85	17/17	-0.0077	[-0.0137, -0.0024]	-0.0221	[-0.0426, -0.0026]
SOD	50%	16/85	15/17	-0.0069	[-0.0268, +0.0123]	-0.0011	[-0.0108, +0.0099]
ROD	50%	31/85	7/17	-0.0038	[-0.0163, +0.0044]	+0.0037	[-0.0177, +0.0294]
IForest	50%	85/85	17/17	-0.0055	[-0.0114, +0.0000]	-0.0101	[-0.0207, +0.0000]
INNE	50%	85/85	17/17	-0.0053	[-0.0123, -0.0007]	-0.0238	[-0.0416, -0.0090]
DIF	50%	85/85	17/17	-0.0110	[-0.0196, -0.0037]	-0.0232	[-0.0453, -0.0045]
FeatureBagging	50%	85/85	17/17	-0.0171	[-0.0357, -0.0017]	-0.0193	[-0.0535, +0.0113]
LSCP	50%	85/85	17/17	-0.0140	[-0.0257, -0.0042]	-0.0156	[-0.0338, +0.0036]
LODA	50%	85/85	17/17	-0.0181	[-0.0673, +0.0117]	-0.0362	[-0.0614, -0.0143]
SUOD	50%	69/85	17/17	-0.0080	[-0.0158, -0.0022]	-0.0091	[-0.0260, +0.0048]
XGBOD	50%	0/85	0/17	—	[—, —]	—	[—, —]
AutoEncoder	50%	85/85	17/17	-0.0167	[-0.0288, -0.0057]	-0.0190	[-0.0332, -0.0041]
VAE	50%	85/85	17/17	-0.0159	[-0.0344, -0.0013]	-0.0213	[-0.0407, -0.0060]
SO_GAAL	50%	85/85	17/17	-0.0054	[-0.0321, +0.0208]	-0.0458	[-0.1176, +0.0072]
MO_GAAL	50%	85/85	17/17	+0.0004	[-0.0245, +0.0255]	-0.0253	[-0.0720, +0.0217]
DeepSVDD	50%	85/85	17/17	-0.0106	[-0.0225, -0.0001]	-0.0210	[-0.0419, -0.0017]
AnoGAN	50%	83/85	17/17	+0.0040	[-0.0064, +0.0117]	+0.0059	[-0.0069, +0.0193]
ALAD	50%	85/85	17/17	+0.0080	[-0.0143, +0.0344]	+0.0105	[+0.0001, +0.0228]
AE1SVM	50%	85/85	17/17	-0.0080	[-0.0142, -0.0029]	-0.0168	[-0.0308, -0.0043]
DevNet	50%	0/85	0/17	—	[—, —]	—	[—, —]
RGraph	50%	20/85	4/17	+0.0436	[—, —]	+0.0804	[—, —]
LUNAR	50%	85/85	17/17	-0.0091	[-0.0155, -0.0025]	-0.0483	[-0.0800, -0.0222]
ECOD	25%	77/85	17/17	-0.0111	[-0.0156, -0.0070]	-0.0335	[-0.0609, -0.0153]
ABOD	25%	85/85	17/17	-0.0227	[-0.0385, -0.0079]	-0.0703	[-0.1229, -0.0256]
COPOD	25%	85/85	17/17	-0.0068	[-0.0157, +0.0013]	-0.0109	[-0.0189, -0.0033]
MAD	25%	0/85	0/17	—	[—, —]	—	[—, —]
SOS	25%	82/85	17/17	+0.0013	[-0.0039, +0.0093]	-0.0023	[-0.0065, +0.0012]
QMCD	25%	85/85	17/17	-0.0697	[-0.1392, +0.0073]	-0.0604	[-0.1337, +0.0020]
KDE	25%	85/85	17/17	-0.0037	[-0.0155, +0.0089]	-0.0477	[-0.0838, -0.0125]
Sampling	25%	85/85	17/17	+0.0032	[-0.0077, +0.0144]	-0.0131	[-0.0410, +0.0114]
GMM	25%	85/85	17/17	-0.0042	[-0.0165, +0.0088]	-0.0227	[-0.0459, +0.0057]
PCA	25%	82/85	17/17	+0.0039	[-0.0030, +0.0124]	-0.0086	[-0.0295, +0.0119]

Continued on next page

Table 15 continued

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
KPCA	25%	85/85	17/17	-0.0097	[-0.0184, -0.0022]	-0.0460	[-0.0849, -0.0116]
MCD	25%	85/85	17/17	-0.0012	[-0.0234, +0.0212]	-0.0082	[-0.0480, +0.0303]
CD	25%	78/85	16/17	-0.0010	[-0.0168, +0.0146]	-0.0120	[-0.0320, +0.0075]
OCSVM	25%	85/85	17/17	+0.0013	[-0.0137, +0.0169]	-0.0115	[-0.0289, +0.0063]
LMDD	25%	53/85	12/17	-0.0078	[-0.0202, +0.0041]	-0.0147	[-0.0372, +0.0040]
LOF	25%	85/85	17/17	-0.0328	[-0.0601, -0.0074]	-0.0309	[-0.0736, +0.0103]
COF	25%	81/85	17/17	-0.0004	[-0.0053, +0.0039]	-0.0021	[-0.0121, +0.0086]
CBLOF	25%	74/85	17/17	-0.0080	[-0.0144, -0.0017]	-0.0183	[-0.0396, +0.0026]
LOCI	25%	24/85	6/17	+0.0246	[-0.0152, +0.0826]	+0.0410	[-0.0131, +0.1215]
HBOS	25%	73/85	17/17	-0.0016	[-0.0097, +0.0081]	-0.0203	[-0.0427, -0.0029]
HDBSCAN	25%	85/85	17/17	-0.0847	[-0.1427, -0.0340]	-0.0689	[-0.1120, -0.0300]
KNN	25%	85/85	17/17	-0.0191	[-0.0337, -0.0066]	-0.0431	[-0.0798, -0.0083]
SOD	25%	60/85	17/17	-0.0082	[-0.0211, +0.0019]	-0.0151	[-0.0266, -0.0050]
ROD	25%	35/85	9/17	-0.0001	[-0.0099, +0.0083]	+0.0033	[-0.0179, +0.0287]
IForest	25%	85/85	17/17	-0.0165	[-0.0274, -0.0068]	-0.0472	[-0.0749, -0.0241]
INNE	25%	85/85	17/17	-0.0104	[-0.0221, +0.0000]	-0.0466	[-0.0717, -0.0249]
DIF	25%	72/85	17/17	-0.0126	[-0.0218, -0.0037]	-0.0430	[-0.0793, -0.0154]
FeatureBagging	25%	85/85	17/17	-0.0351	[-0.0629, -0.0095]	-0.0352	[-0.0807, +0.0077]
LSCP	25%	72/85	16/17	-0.0335	[-0.0536, -0.0154]	-0.0486	[-0.0847, -0.0141]
LODA	25%	85/85	17/17	-0.0205	[-0.0721, +0.0256]	-0.0794	[-0.1379, -0.0247]
SUOD	25%	61/85	17/17	-0.0180	[-0.0343, -0.0066]	-0.0361	[-0.0740, -0.0073]
XGBOD	25%	0/85	0/17	—	[—, —]	—	[—, —]
AutoEncoder	25%	57/85	14/17	-0.0399	[-0.0725, -0.0110]	-0.0447	[-0.0859, -0.0135]
VAE	25%	69/85	14/17	-0.0289	[-0.0609, -0.0049]	-0.0386	[-0.0653, -0.0171]
SO_GAAL	25%	85/85	17/17	-0.0028	[-0.0586, +0.0538]	-0.0254	[-0.0931, +0.0339]
MO_GAAL	25%	85/85	17/17	-0.0045	[-0.0634, +0.0516]	-0.0096	[-0.0541, +0.0423]
DeepSVDD	25%	85/85	17/17	-0.0255	[-0.0500, -0.0041]	-0.0478	[-0.0891, -0.0132]
AnoGAN	25%	77/85	17/17	+0.0094	[-0.0027, +0.0215]	-0.0223	[-0.0606, +0.0090]
ALAD	25%	85/85	17/17	+0.0170	[-0.0040, +0.0401]	+0.0155	[+0.0012, +0.0349]
AE1SVM	25%	79/85	16/17	-0.0122	[-0.0248, -0.0017]	-0.0241	[-0.0428, -0.0067]
DevNet	25%	0/85	0/17	—	[—, —]	—	[—, —]
RGraph	25%	26/85	6/17	+0.0759	[-0.0136, +0.1779]	+0.0789	[-0.0055, +0.1920]
LUNAR	25%	85/85	17/17	-0.0188	[-0.0299, -0.0086]	-0.0806	[-0.1337, -0.0341]

Table 16: NSA: paired training-size changes on pca to two dimensions. Each interval is a descriptive dataset-cluster 95 percent bootstrap interval.

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
NSA fixed	75%	85/85	17/17	+0.0096	[-0.0059, +0.0265]	-0.0099	[-0.0306, +0.0072]
NSA variable	75%	85/85	17/17	+0.0054	[-0.0046, +0.0152]	-0.0006	[-0.0160, +0.0137]
NSA coverage	75%	85/85	17/17	+0.0069	[-0.0018, +0.0161]	+0.0057	[-0.0086, +0.0194]
NSA grid	75%	85/85	17/17	+0.0057	[-0.0049, +0.0165]	+0.0024	[-0.0133, +0.0177]
NSA hierarchical	75%	85/85	17/17	-0.0078	[-0.0319, +0.0127]	-0.0128	[-0.0415, +0.0108]
NSA voronoi	75%	85/85	17/17	+0.0122	[+0.0028, +0.0225]	+0.0069	[-0.0047, +0.0201]
NSA deterministic	75%	85/85	17/17	+0.0117	[-0.0028, +0.0266]	-0.0009	[-0.0279, +0.0254]
NSA suppressed	75%	85/85	17/17	+0.0102	[-0.0014, +0.0226]	+0.0075	[-0.0097, +0.0230]
NSA dual	75%	85/85	17/17	+0.0096	[+0.0023, +0.0174]	+0.0060	[-0.0072, +0.0199]
NSA annealed	75%	85/85	17/17	+0.0185	[+0.0037, +0.0388]	+0.0043	[-0.0152, +0.0247]
NSA binary Hamming	75%	0/85	0/17	—	[—, —]	—	[—, —]
NSA binary contiguous	75%	0/85	0/17	—	[—, —]	—	[—, —]
NSA binary chunk	75%	0/85	0/17	—	[—, —]	—	[—, —]
NSA fixed	50%	85/85	17/17	+0.0376	[+0.0067, +0.0745]	+0.0138	[-0.0209, +0.0440]
NSA variable	50%	85/85	17/17	+0.0212	[-0.0045, +0.0478]	+0.0024	[-0.0227, +0.0266]
NSA coverage	50%	85/85	17/17	+0.0345	[+0.0031, +0.0714]	+0.0169	[-0.0131, +0.0466]
NSA grid	50%	85/85	17/17	+0.0219	[-0.0030, +0.0500]	+0.0093	[-0.0131, +0.0334]
NSA hierarchical	50%	85/85	17/17	+0.0167	[-0.0070, +0.0417]	+0.0048	[-0.0289, +0.0402]
NSA voronoi	50%	85/85	17/17	+0.0272	[+0.0047, +0.0547]	+0.0029	[-0.0238, +0.0340]
NSA deterministic	50%	85/85	17/17	+0.0428	[+0.0008, +0.0943]	+0.0334	[-0.0142, +0.0838]
NSA suppressed	50%	85/85	17/17	+0.0324	[+0.0114, +0.0580]	+0.0188	[-0.0029, +0.0435]
NSA dual	50%	85/85	17/17	+0.0247	[+0.0042, +0.0480]	+0.0083	[-0.0097, +0.0317]
NSA annealed	50%	85/85	17/17	+0.0403	[-0.0027, +0.1060]	+0.0163	[-0.0238, +0.0551]
NSA binary Hamming	50%	0/85	0/17	—	[—, —]	—	[—, —]
NSA binary contiguous	50%	0/85	0/17	—	[—, —]	—	[—, —]
NSA binary chunk	50%	0/85	0/17	—	[—, —]	—	[—, —]
NSA fixed	25%	85/85	17/17	+0.0473	[-0.0148, +0.1185]	+0.0047	[-0.0526, +0.0625]
NSA variable	25%	85/85	17/17	+0.0313	[-0.0244, +0.0954]	-0.0237	[-0.0779, +0.0202]

Continued on next page

Table 16 continued

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
NSA coverage	25%	85/85	17/17	+0.0260	[-0.0307, +0.0863]	-0.0152	[-0.0678, +0.0275]
NSA grid	25%	85/85	17/17	+0.0375	[-0.0124, +0.0947]	-0.0153	[-0.0588, +0.0273]
NSA hierarchical	25%	85/85	17/17	+0.0082	[-0.0459, +0.0591]	-0.0361	[-0.0881, +0.0061]
NSA voronoi	25%	85/85	17/17	+0.0389	[-0.0100, +0.0973]	-0.0174	[-0.0587, +0.0232]
NSA deterministic	25%	85/85	17/17	+0.0389	[-0.0253, +0.1147]	-0.0145	[-0.0798, +0.0444]
NSA suppressed	25%	85/85	17/17	+0.0382	[-0.0089, +0.0934]	-0.0028	[-0.0415, +0.0316]
NSA dual	25%	85/85	17/17	+0.0431	[+0.0010, +0.0950]	-0.0019	[-0.0309, +0.0279]
NSA annealed	25%	85/85	17/17	+0.0288	[-0.0388, +0.1111]	-0.0270	[-0.0901, +0.0334]
NSA binary Hamming	25%	0/85	0/17	—	[—, —]	—	[—, —]
NSA binary contiguous	25%	0/85	0/17	—	[—, —]	—	[—, —]
NSA binary chunk	25%	0/85	0/17	—	[—, —]	—	[—, —]

Table 17: Other PyOD detectors: paired training-size changes on pca to two dimensions. Each interval is a descriptive dataset-cluster 95 percent bootstrap interval.

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
ECOD	75%	85/85	17/17	-0.0004	[-0.0050, +0.0048]	+0.0024	[-0.0074, +0.0153]
ABOD	75%	85/85	17/17	-0.0046	[-0.0189, +0.0066]	+0.0001	[-0.0082, +0.0091]
COPOD	75%	85/85	17/17	-0.0009	[-0.0082, +0.0048]	-0.0005	[-0.0107, +0.0101]
MAD	75%	0/85	0/17	—	[—, —]	—	[—, —]
SOS	75%	85/85	17/17	+0.0076	[-0.0035, +0.0230]	+0.0128	[+0.0011, +0.0342]
QMCD	75%	85/85	17/17	-0.0012	[-0.0083, +0.0059]	-0.0000	[-0.0174, +0.0195]
KDE	75%	85/85	17/17	+0.0023	[-0.0020, +0.0078]	+0.0057	[-0.0042, +0.0172]
Sampling	75%	85/85	17/17	-0.0078	[-0.0220, +0.0045]	+0.0054	[-0.0218, +0.0312]
GMM	75%	85/85	17/17	-0.0016	[-0.0090, +0.0060]	+0.0037	[-0.0067, +0.0164]
PCA	75%	85/85	17/17	-0.0074	[-0.0284, +0.0119]	-0.0022	[-0.0151, +0.0110]
KPCA	75%	85/85	17/17	+0.0031	[-0.0049, +0.0096]	+0.0031	[-0.0066, +0.0136]
MCD	75%	85/85	17/17	-0.0083	[-0.0229, +0.0040]	-0.0148	[-0.0405, +0.0043]
CD	75%	85/85	17/17	-0.0024	[-0.0133, +0.0060]	+0.0012	[-0.0069, +0.0102]
OCSVM	75%	85/85	17/17	-0.0003	[-0.0043, +0.0037]	+0.0032	[-0.0040, +0.0119]
LMDD	75%	85/85	17/17	+0.0137	[-0.0028, +0.0301]	+0.0083	[-0.0076, +0.0265]
LOF	75%	85/85	17/17	+0.0038	[-0.0088, +0.0164]	-0.0061	[-0.0174, +0.0048]
COF	75%	85/85	17/17	+0.0075	[-0.0018, +0.0190]	+0.0069	[+0.0002, +0.0150]
CBLOF	75%	85/85	17/17	-0.0031	[-0.0147, +0.0115]	-0.0027	[-0.0221, +0.0167]
LOCI	75%	28/85	6/17	-0.0039	[-0.0178, +0.0086]	+0.0106	[-0.0159, +0.0372]
HBOS	75%	85/85	17/17	-0.0092	[-0.0194, -0.0005]	-0.0127	[-0.0324, +0.0048]
HDBSCAN	75%	85/85	17/17	+0.0092	[-0.0074, +0.0255]	-0.0127	[-0.0382, +0.0112]
KNN	75%	85/85	17/17	+0.0034	[-0.0040, +0.0137]	+0.0018	[-0.0094, +0.0128]
SOD	75%	82/85	17/17	-0.0086	[-0.0310, +0.0107]	-0.0015	[-0.0107, +0.0067]
ROD	75%	85/85	17/17	+0.0013	[-0.0077, +0.0136]	+0.0072	[-0.0082, +0.0270]
IForest	75%	85/85	17/17	-0.0022	[-0.0088, +0.0045]	-0.0093	[-0.0369, +0.0120]
INNE	75%	85/85	17/17	+0.0019	[-0.0093, +0.0133]	+0.0027	[-0.0060, +0.0137]
DIF	75%	85/85	17/17	-0.0041	[-0.0140, +0.0064]	+0.0104	[-0.0024, +0.0264]
FeatureBagging	75%	85/85	17/17	-0.0007	[-0.0122, +0.0118]	-0.0057	[-0.0136, +0.0011]
LSCP	75%	85/85	17/17	-0.0005	[-0.0111, +0.0109]	+0.0027	[-0.0112, +0.0173]
LODA	75%	85/85	17/17	-0.0027	[-0.0116, +0.0062]	-0.0092	[-0.0219, +0.0021]

Continued on next page

Table 17 continued

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
SUOD	75%	84/85	17/17	+0.0002	[-0.0048, +0.0046]	+0.0067	[-0.0093, +0.0255]
XGBOD	75%	0/85	0/17	—	[—, —]	—	[—, —]
AutoEncoder	75%	85/85	17/17	-0.0063	[-0.0255, +0.0134]	-0.0152	[-0.0496, +0.0269]
VAE	75%	85/85	17/17	-0.0028	[-0.0193, +0.0118]	+0.0000	[-0.0195, +0.0183]
SO_GAAL	75%	85/85	17/17	-0.0311	[-0.0860, +0.0122]	+0.0005	[-0.0284, +0.0254]
MO_GAAL	75%	85/85	17/17	-0.0074	[-0.0398, +0.0209]	+0.0140	[-0.0071, +0.0363]
DeepSVDD	75%	85/85	17/17	+0.0066	[-0.0086, +0.0238]	-0.0013	[-0.0161, +0.0147]
AnoGAN	75%	85/85	17/17	+0.0076	[-0.0090, +0.0252]	+0.0015	[-0.0102, +0.0128]
ALAD	75%	85/85	17/17	+0.0182	[+0.0003, +0.0402]	+0.0043	[-0.0039, +0.0126]
AE1SVM	75%	85/85	17/17	-0.0095	[-0.0315, +0.0084]	+0.0024	[-0.0211, +0.0225]
DevNet	75%	0/85	0/17	—	[—, —]	—	[—, —]
RGraph	75%	34/85	7/17	-0.0177	[-0.0456, +0.0020]	-0.0004	[-0.0029, +0.0019]
LUNAR	75%	85/85	17/17	+0.0038	[-0.0089, +0.0185]	+0.0056	[-0.0050, +0.0176]
ECOD	50%	85/85	17/17	-0.0025	[-0.0135, +0.0090]	+0.0084	[-0.0093, +0.0326]
ABOD	50%	85/85	17/17	+0.0050	[-0.0091, +0.0216]	+0.0029	[-0.0090, +0.0192]
COPOD	50%	85/85	17/17	+0.0053	[-0.0086, +0.0215]	+0.0120	[-0.0080, +0.0337]
MAD	50%	0/85	0/17	—	[—, —]	—	[—, —]
SOS	50%	85/85	17/17	+0.0024	[-0.0052, +0.0102]	+0.0107	[-0.0008, +0.0289]
QMCD	50%	85/85	17/17	+0.0027	[-0.0138, +0.0186]	-0.0094	[-0.0450, +0.0231]
KDE	50%	85/85	17/17	+0.0067	[-0.0021, +0.0180]	+0.0090	[-0.0056, +0.0280]
Sampling	50%	85/85	17/17	-0.0176	[-0.0363, +0.0016]	-0.0142	[-0.0386, +0.0110]
GMM	50%	85/85	17/17	+0.0024	[-0.0103, +0.0171]	+0.0188	[+0.0004, +0.0470]
PCA	50%	85/85	17/17	-0.0185	[-0.0388, +0.0005]	-0.0086	[-0.0232, +0.0058]
KPCA	50%	85/85	17/17	+0.0118	[+0.0017, +0.0221]	+0.0062	[-0.0093, +0.0262]
MCD	50%	85/85	17/17	-0.0036	[-0.0150, +0.0080]	-0.0051	[-0.0245, +0.0102]
CD	50%	85/85	17/17	+0.0043	[-0.0122, +0.0200]	+0.0083	[-0.0080, +0.0254]
OCSVM	50%	85/85	17/17	+0.0117	[+0.0007, +0.0249]	+0.0105	[-0.0046, +0.0296]
LMDD	50%	85/85	17/17	+0.0318	[+0.0005, +0.0751]	+0.0180	[+0.0031, +0.0352]
LOF	50%	85/85	17/17	+0.0050	[-0.0096, +0.0177]	-0.0027	[-0.0290, +0.0253]
COF	50%	85/85	17/17	+0.0066	[-0.0059, +0.0227]	+0.0134	[-0.0063, +0.0445]
CBLOF	50%	85/85	17/17	+0.0044	[-0.0124, +0.0243]	+0.0283	[-0.0043, +0.0760]
LOCI	50%	29/85	6/17	+0.0189	[-0.0052, +0.0489]	+0.0138	[+0.0042, +0.0256]
HBOS	50%	85/85	17/17	+0.0013	[-0.0132, +0.0148]	-0.0130	[-0.0405, +0.0128]

Continued on next page

Table 17 continued

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
HDBSCAN	50%	85/85	17/17	-0.0231	[-0.0691, +0.0173]	-0.0388	[-0.0780, -0.0055]
KNN	50%	85/85	17/17	+0.0092	[-0.0027, +0.0221]	+0.0070	[-0.0138, +0.0325]
SOD	50%	85/85	17/17	-0.0073	[-0.0294, +0.0152]	+0.0077	[-0.0241, +0.0421]
ROD	50%	85/85	17/17	+0.0056	[-0.0054, +0.0166]	+0.0067	[-0.0095, +0.0256]
IForest	50%	85/85	17/17	+0.0005	[-0.0075, +0.0089]	-0.0090	[-0.0344, +0.0178]
INNE	50%	85/85	17/17	+0.0036	[-0.0077, +0.0158]	+0.0056	[-0.0107, +0.0286]
DIF	50%	85/85	17/17	+0.0066	[-0.0094, +0.0247]	+0.0132	[-0.0083, +0.0343]
FeatureBagging	50%	85/85	17/17	+0.0033	[-0.0131, +0.0176]	+0.0002	[-0.0191, +0.0187]
LSCP	50%	85/85	17/17	+0.0040	[-0.0080, +0.0164]	-0.0087	[-0.0291, +0.0111]
LODA	50%	85/85	17/17	-0.0022	[-0.0167, +0.0125]	+0.0020	[-0.0151, +0.0180]
SUOD	50%	85/85	17/17	+0.0044	[-0.0020, +0.0114]	+0.0118	[-0.0135, +0.0425]
XGBOD	50%	0/85	0/17	—	[—, —]	—	[—, —]
AutoEncoder	50%	85/85	17/17	-0.0239	[-0.0721, +0.0173]	-0.0436	[-0.1271, +0.0245]
VAE	50%	85/85	17/17	-0.0095	[-0.0268, +0.0072]	-0.0120	[-0.0417, +0.0168]
SO_GAAL	50%	85/85	17/17	-0.0201	[-0.0578, +0.0178]	+0.0088	[-0.0135, +0.0338]
MO_GAAL	50%	85/85	17/17	-0.0078	[-0.0449, +0.0261]	+0.0088	[-0.0049, +0.0240]
DeepSVDD	50%	85/85	17/17	+0.0079	[-0.0186, +0.0369]	+0.0104	[-0.0197, +0.0478]
AnoGAN	50%	85/85	17/17	+0.0041	[-0.0224, +0.0330]	+0.0114	[-0.0087, +0.0299]
ALAD	50%	85/85	17/17	+0.0231	[-0.0019, +0.0536]	+0.0165	[-0.0036, +0.0433]
AE1SVM	50%	85/85	17/17	-0.0039	[-0.0284, +0.0195]	+0.0140	[-0.0121, +0.0433]
DevNet	50%	0/85	0/17	—	[—, —]	—	[—, —]
RGraph	50%	35/85	7/17	-0.0274	[-0.0679, +0.0062]	-0.0003	[-0.0052, +0.0049]
LUNAR	50%	85/85	17/17	+0.0168	[+0.0013, +0.0354]	+0.0182	[-0.0069, +0.0549]
ECOD	25%	85/85	17/17	-0.0002	[-0.0195, +0.0199]	-0.0114	[-0.0324, +0.0079]
ABOD	25%	85/85	17/17	+0.0040	[-0.0087, +0.0179]	-0.0125	[-0.0322, +0.0068]
COPOD	25%	85/85	17/17	-0.0001	[-0.0181, +0.0195]	-0.0118	[-0.0381, +0.0151]
MAD	25%	0/85	0/17	—	[—, —]	—	[—, —]
SOS	25%	85/85	17/17	-0.0047	[-0.0165, +0.0070]	+0.0049	[-0.0019, +0.0150]
QMCD	25%	85/85	17/17	-0.0228	[-0.0710, +0.0205]	-0.0554	[-0.1308, +0.0128]
KDE	25%	85/85	17/17	+0.0057	[-0.0084, +0.0219]	-0.0106	[-0.0334, +0.0083]
Sampling	25%	85/85	17/17	+0.0041	[-0.0181, +0.0261]	-0.0004	[-0.0346, +0.0339]
GMM	25%	85/85	17/17	+0.0150	[-0.0022, +0.0357]	+0.0076	[-0.0119, +0.0251]
PCA	25%	85/85	17/17	-0.0158	[-0.0524, +0.0161]	-0.0040	[-0.0225, +0.0156]

Continued on next page

Table 17 continued

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
KPCA	25%	85/85	17/17	+0.0075	[-0.0140, +0.0291]	-0.0178	[-0.0442, +0.0064]
MCD	25%	85/85	17/17	+0.0008	[-0.0179, +0.0183]	-0.0134	[-0.0392, +0.0082]
CD	25%	85/85	17/17	+0.0079	[-0.0244, +0.0402]	-0.0238	[-0.0683, +0.0126]
OCSVM	25%	85/85	17/17	-0.0013	[-0.0218, +0.0167]	-0.0148	[-0.0378, +0.0056]
LMDD	25%	80/85	17/17	+0.0302	[-0.0121, +0.0813]	+0.0159	[-0.0117, +0.0433]
LOF	25%	85/85	17/17	-0.0129	[-0.0434, +0.0131]	-0.0477	[-0.0879, -0.0115]
COF	25%	85/85	17/17	+0.0124	[-0.0046, +0.0305]	+0.0167	[+0.0018, +0.0372]
CBLOF	25%	64/85	17/17	-0.0037	[-0.0259, +0.0204]	+0.0005	[-0.0283, +0.0306]
LOCI	25%	14/85	5/17	-0.0403	[-0.1277, +0.0144]	-0.0149	[-0.0403, +0.0148]
HBOS	25%	68/85	17/17	+0.0030	[-0.0315, +0.0356]	-0.0303	[-0.0850, +0.0316]
HDBSCAN	25%	85/85	17/17	-0.0821	[-0.1465, -0.0189]	-0.0601	[-0.1069, -0.0164]
KNN	25%	85/85	17/17	+0.0047	[-0.0114, +0.0231]	-0.0228	[-0.0530, +0.0068]
SOD	25%	18/85	15/17	-0.0023	[-0.1078, +0.0875]	+0.0167	[-0.0418, +0.0724]
ROD	25%	72/85	17/17	+0.0069	[-0.0208, +0.0301]	+0.0060	[-0.0212, +0.0336]
IForest	25%	85/85	17/17	-0.0073	[-0.0182, +0.0043]	-0.0582	[-0.1071, -0.0148]
INNE	25%	85/85	17/17	-0.0083	[-0.0354, +0.0140]	-0.0216	[-0.0477, -0.0020]
DIF	25%	74/85	17/17	+0.0006	[-0.0174, +0.0199]	-0.0279	[-0.0712, +0.0101]
FeatureBagging	25%	85/85	17/17	-0.0057	[-0.0358, +0.0210]	-0.0318	[-0.0644, -0.0010]
LSCP	25%	77/85	16/17	+0.0076	[-0.0088, +0.0265]	-0.0305	[-0.0605, -0.0024]
LODA	25%	85/85	17/17	-0.0178	[-0.0514, +0.0122]	-0.0398	[-0.0940, +0.0076]
SUOD	25%	35/85	17/17	-0.0102	[-0.0263, +0.0072]	-0.0360	[-0.0718, -0.0063]
XGBOD	25%	0/85	0/17	—	[—, —]	—	[—, —]
AutoEncoder	25%	60/85	14/17	-0.0308	[-0.0600, -0.0037]	-0.0761	[-0.1648, -0.0040]
VAE	25%	70/85	14/17	-0.0432	[-0.0786, -0.0132]	-0.0232	[-0.0560, +0.0079]
SO_GAAL	25%	85/85	17/17	-0.0282	[-0.0773, +0.0168]	+0.0017	[-0.0143, +0.0184]
MO_GAAL	25%	85/85	17/17	-0.0450	[-0.0757, -0.0158]	-0.0084	[-0.0283, +0.0086]
DeepSVDD	25%	85/85	17/17	-0.0131	[-0.0548, +0.0242]	-0.0016	[-0.0385, +0.0322]
AnoGAN	25%	77/85	17/17	+0.0207	[-0.0151, +0.0591]	-0.0041	[-0.0337, +0.0244]
ALAD	25%	85/85	17/17	+0.0142	[-0.0065, +0.0376]	+0.0061	[-0.0061, +0.0232]
AE1SVM	25%	80/85	16/17	+0.0048	[-0.0168, +0.0250]	+0.0139	[-0.0108, +0.0457]
DevNet	25%	0/85	0/17	—	[—, —]	—	[—, —]
RGraph	25%	32/85	7/17	-0.0321	[-0.0896, +0.0186]	-0.0016	[-0.0120, +0.0081]
LUNAR	25%	85/85	17/17	+0.0039	[-0.0121, +0.0233]	-0.0128	[-0.0252, -0.0007]

G Complete-coverage rankings for both metrics

A configuration enters a panel only if every one of its 85 runs succeeds within the measured time budget. The eligible field can therefore differ across panels. Scores are unrounded when ranked; displayed values are rounded. Aggregate tables break exact ties by configuration name, as in the released analysis. Rank movement alone is not an effect size.

Table 18: Complete-coverage overall rankings: Original features, 100 percent retained normal training pool. All 39 configurations have 85 of 85 qualified runs.

Configuration	AUROC	AUROC rank	Top position	AP	AP rank
LUNAR	0.9058	1/39	2.56%	0.6768	1/39
KPCA	0.9023	2/39	5.13%	0.6464	2/39
KNN	0.8973	3/39	7.69%	0.6163	5/39
KDE	0.8970	4/39	10.26%	0.6439	3/39
ABOD	0.8951	5/39	12.82%	0.6279	4/39
LSCP	0.8924	6/39	15.38%	0.5914	6/39
FeatureBagging	0.8891	7/39	17.95%	0.5859	8/39
LOF	0.8879	8/39	20.51%	0.5786	9/39
INNE	0.8732	9/39	23.08%	0.5757	10/39
AutoEncoder	0.8684	10/39	25.64%	0.5702	11/39
GMM	0.8654	11/39	28.21%	0.5497	14/39
DeepSVDD	0.8651	12/39	30.77%	0.5908	7/39
Sampling	0.8620	13/39	33.33%	0.5595	13/39
IForest	0.8533	14/39	35.90%	0.5258	18/39
SUOD	0.8503	15/39	38.46%	0.5440	17/39
MCD	0.8427	16/39	41.03%	0.4702	21/39
OCSVM	0.8409	17/39	43.59%	0.5674	12/39
AE1SVM	0.8363	18/39	46.15%	0.5458	16/39
VAE	0.8299	19/39	48.72%	0.5466	15/39
DIF	0.8191	20/39	51.28%	0.4428	23/39
HBOS	0.8133	21/39	53.85%	0.4952	19/39
AnoGAN	0.7902	22/39	56.41%	0.4839	20/39
COPOD	0.7865	23/39	58.97%	0.4573	22/39
SOD	0.7553	24/39	61.54%	0.3523	25/39
LODA	0.7426	25/39	64.10%	0.4150	24/39
HDBSCAN	0.6946	26/39	66.67%	0.2392	33/39
COF	0.6746	27/39	69.23%	0.3429	26/39
NSA grid	0.5972	28/39	71.79%	0.3026	29/39
NSA coverage	0.5931	29/39	74.36%	0.3040	27/39
NSA variable	0.5931	30/39	76.92%	0.3038	28/39
NSA suppressed	0.5883	31/39	79.49%	0.2982	30/39
SOS	0.5868	32/39	82.05%	0.2218	35/39
MO_GAAL	0.5438	33/39	84.62%	0.2592	32/39
ALAD	0.5409	34/39	87.18%	0.1470	36/39
QMCD	0.5376	35/39	89.74%	0.2370	34/39
SO_GAAL	0.5286	36/39	92.31%	0.2693	31/39

Continued on next page

Table 18 continued

Configuration	AUROC	AUROC rank	Top position	AP	AP rank
NSA fixed	0.3597	37/39	94.87%	0.1287	37/39
NSA annealed	0.3161	38/39	97.44%	0.1120	39/39
NSA hierarchical	0.2968	39/39	100.00%	0.1173	38/39

Table 19: Complete-coverage overall rankings: Original features, 75 percent retained normal training pool. All 36 configurations have 85 of 85 qualified runs.

Configuration	AUROC	AUROC rank	Top position	AP	AP rank
LUNAR	0.9022	1/36	2.78%	0.6401	1/36
KPCA	0.8986	2/36	5.56%	0.6301	3/36
KDE	0.8945	3/36	8.33%	0.6312	2/36
KNN	0.8937	4/36	11.11%	0.6034	5/36
ABOD	0.8927	5/36	13.89%	0.6118	4/36
LSCP	0.8868	6/36	16.67%	0.5841	6/36
LOF	0.8833	7/36	19.44%	0.5781	7/36
FeatureBagging	0.8822	8/36	22.22%	0.5767	8/36
INNE	0.8718	9/36	25.00%	0.5687	10/36
Sampling	0.8671	10/36	27.78%	0.5543	13/36
AutoEncoder	0.8627	11/36	30.56%	0.5625	11/36
GMM	0.8615	12/36	33.33%	0.5444	14/36
DeepSVDD	0.8546	13/36	36.11%	0.5765	9/36
IForest	0.8527	14/36	38.89%	0.5310	17/36
SUOD	0.8451	15/36	41.67%	0.5383	15/36
MCD	0.8450	16/36	44.44%	0.4687	20/36
OCSVM	0.8396	17/36	47.22%	0.5612	12/36
AE1SVM	0.8287	18/36	50.00%	0.5277	18/36
VAE	0.8234	19/36	52.78%	0.5340	16/36
HBOS	0.8130	20/36	55.56%	0.4919	19/36
DIF	0.8106	21/36	58.33%	0.4243	23/36
COPOD	0.7830	22/36	61.11%	0.4587	22/36
ECOD	0.7824	23/36	63.89%	0.4631	21/36
LODA	0.7374	24/36	66.67%	0.3993	24/36
COF	0.6751	25/36	69.44%	0.3395	25/36
HDBSCAN	0.6732	26/36	72.22%	0.2225	32/36
NSA grid	0.5880	27/36	75.00%	0.2869	26/36
NSA variable	0.5821	28/36	77.78%	0.2865	27/36
NSA coverage	0.5814	29/36	80.56%	0.2855	28/36
NSA suppressed	0.5656	30/36	83.33%	0.2705	29/36
ALAD	0.5485	31/36	86.11%	0.1579	33/36
MO_GAAL	0.5156	32/36	88.89%	0.2274	31/36
SO_GAAL	0.5146	33/36	91.67%	0.2353	30/36
NSA fixed	0.3504	34/36	94.44%	0.1272	34/36
NSA annealed	0.3112	35/36	97.22%	0.1058	36/36

Continued on next page

Table 19 continued

Configuration	AUROC	AUROC rank	Top position	AP	AP rank
NSA hierarchical	0.3078	36/36	100.00%	0.1145	35/36

Table 20: Complete-coverage overall rankings: Original features, 50 percent retained normal training pool. All 39 configurations have 85 of 85 qualified runs.

Configuration	AUROC	AUROC rank	Top position	AP	AP rank
LUNAR	0.8967	1/39	2.56%	0.6285	1/39
KPCA	0.8967	2/39	5.13%	0.6222	2/39
KDE	0.8932	3/39	7.69%	0.6188	3/39
KNN	0.8896	4/39	10.26%	0.5941	4/39
ABOD	0.8841	5/39	12.82%	0.5849	5/39
CBLOF	0.8815	6/39	15.38%	0.5712	7/39
LSCP	0.8784	7/39	17.95%	0.5758	6/39
LOF	0.8738	8/39	20.51%	0.5648	10/39
FeatureBagging	0.8721	9/39	23.08%	0.5665	9/39
Sampling	0.8685	10/39	25.64%	0.5545	12/39
INNE	0.8679	11/39	28.21%	0.5518	13/39
GMM	0.8637	12/39	30.77%	0.5419	15/39
DeepSVDD	0.8545	13/39	33.33%	0.5698	8/39
AutoEncoder	0.8517	14/39	35.90%	0.5511	14/39
IForest	0.8478	15/39	38.46%	0.5157	18/39
MCD	0.8445	16/39	41.03%	0.4679	20/39
OCSVM	0.8404	17/39	43.59%	0.5632	11/39
AE1SVM	0.8283	18/39	46.15%	0.5290	16/39
VAE	0.8141	19/39	48.72%	0.5253	17/39
HBOS	0.8130	20/39	51.28%	0.4862	19/39
DIF	0.8081	21/39	53.85%	0.4195	23/39
COPOD	0.7837	22/39	56.41%	0.4551	21/39
ECOD	0.7783	23/39	58.97%	0.4520	22/39
LODA	0.7245	24/39	61.54%	0.3788	24/39
COF	0.6727	25/39	64.10%	0.3404	25/39
HDBSCAN	0.6414	26/39	66.67%	0.2021	33/39
SOS	0.5897	27/39	69.23%	0.2296	31/39
NSA grid	0.5569	28/39	71.79%	0.2510	26/39
NSA coverage	0.5534	29/39	74.36%	0.2455	28/39
NSA variable	0.5531	30/39	76.92%	0.2456	27/39
NSA suppressed	0.5516	31/39	79.49%	0.2440	29/39
ALAD	0.5489	32/39	82.05%	0.1575	35/39
MO_GAAL	0.5443	33/39	84.62%	0.2339	30/39
NSA binary Hamming	0.5288	34/39	87.18%	0.1573	36/39
SO_GAAL	0.5232	35/39	89.74%	0.2235	32/39
QMCD	0.5124	36/39	92.31%	0.1981	34/39
NSA fixed	0.3460	37/39	94.87%	0.1232	37/39

Continued on next page

Table 20 continued

Configuration	AUROC	AUROC rank	Top position	AP	AP rank
NSA annealed	0.3023	38/39	97.44%	0.1070	38/39
NSA hierarchical	0.2753	39/39	100.00%	0.0938	39/39

Table 21: Complete-coverage overall rankings: Original features, 25 percent retained normal training pool. All 30 configurations have 85 of 85 qualified runs.

Configuration	AUROC	AUROC rank	Top position	AP	AP rank
KDE	0.8933	1/30	3.33%	0.5962	2/30
KPCA	0.8925	2/30	6.67%	0.6004	1/30
LUNAR	0.8870	3/30	10.00%	0.5962	3/30
KNN	0.8782	4/30	13.33%	0.5731	4/30
ABOD	0.8724	5/30	16.67%	0.5576	5/30
Sampling	0.8652	6/30	20.00%	0.5464	9/30
INNE	0.8629	7/30	23.33%	0.5290	11/30
GMM	0.8612	8/30	26.67%	0.5270	12/30
LOF	0.8551	9/30	30.00%	0.5477	8/30
FeatureBagging	0.8540	10/30	33.33%	0.5507	7/30
OCSVM	0.8422	11/30	36.67%	0.5559	6/30
MCD	0.8415	12/30	40.00%	0.4620	15/30
DeepSVDD	0.8395	13/30	43.33%	0.5429	10/30
IForest	0.8368	14/30	46.67%	0.4787	14/30
PCA	0.8112	15/30	50.00%	0.5230	13/30
COPOD	0.7797	16/30	53.33%	0.4464	16/30
ECOD	0.7738	17/30	56.67%	0.4399	17/30
LODA	0.7221	18/30	60.00%	0.3356	18/30
HDBSCAN	0.6099	19/30	63.33%	0.1703	25/30
ALAD	0.5579	20/30	66.67%	0.1625	26/30
MO_GAAL	0.5393	21/30	70.00%	0.2496	19/30
SO_GAAL	0.5259	22/30	73.33%	0.2439	20/30
NSA binary Hamming	0.5172	23/30	76.67%	0.1449	27/30
NSA coverage	0.5028	24/30	80.00%	0.2053	22/30
NSA variable	0.5025	25/30	83.33%	0.2054	21/30
NSA suppressed	0.4920	26/30	86.67%	0.1989	23/30
QMCD	0.4679	27/30	90.00%	0.1766	24/30
NSA fixed	0.3297	28/30	93.33%	0.1211	28/30
NSA annealed	0.2789	29/30	96.67%	0.0988	30/30
NSA hierarchical	0.2712	30/30	100.00%	0.1092	29/30

Table 22: Complete-coverage overall rankings: PCA to two dimensions, 100 percent retained normal training pool. All 48 configurations have 85 of 85 qualified runs.

Configuration	AUROC	AUROC rank	Top position	AP	AP rank
KNN	0.7958	1/48	2.08%	0.4872	1/48
LSCP	0.7893	2/48	4.17%	0.4757	6/48
KDE	0.7890	3/48	6.25%	0.4817	4/48
KPCA	0.7862	4/48	8.33%	0.4854	2/48
ABOD	0.7853	5/48	10.42%	0.4831	3/48
CBLOF	0.7815	6/48	12.50%	0.4532	11/48
OCSVM	0.7801	7/48	14.58%	0.4803	5/48
INNE	0.7795	8/48	16.67%	0.4684	8/48
LOF	0.7781	9/48	18.75%	0.4604	9/48
LUNAR	0.7745	10/48	20.83%	0.4576	10/48
FeatureBagging	0.7695	11/48	22.92%	0.4496	13/48
Sampling	0.7607	12/48	25.00%	0.4333	16/48
IForest	0.7593	13/48	27.08%	0.4023	21/48
HBOS	0.7581	14/48	29.17%	0.3902	24/48
DIF	0.7577	15/48	31.25%	0.4216	19/48
SUOD	0.7576	16/48	33.33%	0.3982	22/48
DeepSVDD	0.7368	17/48	35.42%	0.3906	23/48
LODA	0.7270	18/48	37.50%	0.3456	32/48
NSA dual	0.7172	19/48	39.58%	0.4718	7/48
HDBSCAN	0.7163	20/48	41.67%	0.2391	42/48
MCD	0.7063	21/48	43.75%	0.3673	26/48
NSA voronoi	0.7043	22/48	45.83%	0.4521	12/48
NSA variable	0.6928	23/48	47.92%	0.4441	14/48
NSA grid	0.6886	24/48	50.00%	0.4361	15/48
AutoEncoder	0.6830	25/48	52.08%	0.3647	28/48
NSA hierarchical	0.6754	26/48	54.17%	0.4218	18/48
NSA suppressed	0.6738	27/48	56.25%	0.4214	20/48
VAE	0.6718	28/48	58.33%	0.3697	25/48
NSA coverage	0.6694	29/48	60.42%	0.4253	17/48
PCA	0.6656	30/48	62.50%	0.3605	29/48
ROD	0.6650	31/48	64.58%	0.3133	36/48
COPOD	0.6634	32/48	66.67%	0.3118	37/48
QMCD	0.6582	33/48	68.75%	0.3382	34/48
AE1SVM	0.6570	34/48	70.83%	0.3360	35/48
GMM	0.6345	35/48	72.92%	0.3568	31/48
AnoGAN	0.6247	36/48	75.00%	0.2801	39/48
ECOD	0.6201	37/48	77.08%	0.2698	40/48
COF	0.6141	38/48	79.17%	0.2250	44/48
NSA deterministic	0.6032	39/48	81.25%	0.3594	30/48
NSA annealed	0.5998	40/48	83.33%	0.3672	27/48
CD	0.5831	41/48	85.42%	0.2942	38/48
NSA fixed	0.5827	42/48	87.50%	0.3427	33/48

Continued on next page

Table 22 continued

Configuration	AUROC	AUROC rank	Top position	AP	AP rank
SO_GAAL	0.5817	43/48	89.58%	0.2066	46/48
SOD	0.5747	44/48	91.67%	0.2075	45/48
MO_GAAL	0.5718	45/48	93.75%	0.2341	43/48
LMDD	0.5580	46/48	95.83%	0.2448	41/48
SOS	0.5326	47/48	97.92%	0.1547	47/48
ALAD	0.5051	48/48	100.00%	0.1300	48/48

Table 23: Complete-coverage overall rankings: PCA to two dimensions, 75 percent retained normal training pool. All 46 configurations have 85 of 85 qualified runs.

Configuration	AUROC	AUROC rank	Top position	AP	AP rank
KNN	0.7993	1/46	2.17%	0.4891	1/46
KDE	0.7913	2/46	4.35%	0.4874	3/46
KPCA	0.7892	3/46	6.52%	0.4885	2/46
LSCP	0.7888	4/46	8.70%	0.4784	6/46
LOF	0.7819	5/46	10.87%	0.4543	11/46
INNE	0.7813	6/46	13.04%	0.4711	8/46
ABOD	0.7806	7/46	15.22%	0.4832	5/46
OCSVM	0.7797	8/46	17.39%	0.4835	4/46
CBLOF	0.7783	9/46	19.57%	0.4506	12/46
LUNAR	0.7783	10/46	21.74%	0.4632	9/46
FeatureBagging	0.7688	11/46	23.91%	0.4439	13/46
IForest	0.7571	12/46	26.09%	0.3930	21/46
DIF	0.7536	13/46	28.26%	0.4321	17/46
Sampling	0.7529	14/46	30.43%	0.4388	15/46
HBOS	0.7488	15/46	32.61%	0.3775	23/46
DeepSVDD	0.7434	16/46	34.78%	0.3894	22/46
NSA dual	0.7268	17/46	36.96%	0.4777	7/46
HDBSCAN	0.7255	18/46	39.13%	0.2265	43/46
LODA	0.7243	19/46	41.30%	0.3364	33/46
NSA voronoi	0.7164	20/46	43.48%	0.4590	10/46
NSA variable	0.6982	21/46	45.65%	0.4435	14/46
MCD	0.6980	22/46	47.83%	0.3525	29/46
NSA grid	0.6942	23/46	50.00%	0.4385	16/46
NSA suppressed	0.6841	24/46	52.17%	0.4288	19/46
AutoEncoder	0.6767	25/46	54.35%	0.3495	30/46
NSA coverage	0.6763	26/46	56.52%	0.4310	18/46
VAE	0.6690	27/46	58.70%	0.3697	25/46
NSA hierarchical	0.6676	28/46	60.87%	0.4090	20/46
ROD	0.6663	29/46	63.04%	0.3204	35/46
COPOD	0.6625	30/46	65.22%	0.3113	36/46
PCA	0.6581	31/46	67.39%	0.3583	28/46
QMCD	0.6570	32/46	69.57%	0.3381	32/46

Continued on next page

Table 23 continued

Configuration	AUROC	AUROC rank	Top position	AP	AP rank
AE1SVM	0.6475	33/46	71.74%	0.3384	31/46
GMM	0.6330	34/46	73.91%	0.3605	26/46
AnoGAN	0.6322	35/46	76.09%	0.2816	38/46
COF	0.6216	36/46	78.26%	0.2319	42/46
ECOD	0.6197	37/46	80.43%	0.2721	39/46
NSA annealed	0.6183	38/46	82.61%	0.3715	24/46
NSA deterministic	0.6149	39/46	84.78%	0.3586	27/46
NSA fixed	0.5923	40/46	86.96%	0.3328	34/46
CD	0.5807	41/46	89.13%	0.2954	37/46
LMDD	0.5717	42/46	91.30%	0.2531	40/46
MO_GAAL	0.5644	43/46	93.48%	0.2481	41/46
SO_GAAL	0.5506	44/46	95.65%	0.2070	44/46
SOS	0.5402	45/46	97.83%	0.1676	45/46
ALAD	0.5233	46/46	100.00%	0.1343	46/46

Table 24: Complete-coverage overall rankings: PCA to two dimensions, 50 percent retained normal training pool. All 48 configurations have 85 of 85 qualified runs.

Configuration	AUROC	AUROC rank	Top position	AP	AP rank
KNN	0.8051	1/48	2.08%	0.4942	1/48
KPCA	0.7980	2/48	4.17%	0.4916	2/48
KDE	0.7956	3/48	6.25%	0.4908	4/48
LSCP	0.7934	4/48	8.33%	0.4670	10/48
OCSVM	0.7918	5/48	10.42%	0.4909	3/48
LUNAR	0.7913	6/48	12.50%	0.4759	8/48
ABOD	0.7903	7/48	14.58%	0.4860	5/48
CBLOF	0.7859	8/48	16.67%	0.4816	6/48
INNE	0.7831	9/48	18.75%	0.4740	9/48
LOF	0.7830	10/48	20.83%	0.4577	11/48
FeatureBagging	0.7728	11/48	22.92%	0.4498	13/48
DIF	0.7643	12/48	25.00%	0.4348	18/48
SUOD	0.7620	13/48	27.08%	0.4100	21/48
IForest	0.7598	14/48	29.17%	0.3933	23/48
HBOS	0.7594	15/48	31.25%	0.3772	26/48
DeepSVDD	0.7448	16/48	33.33%	0.4010	22/48
Sampling	0.7430	17/48	35.42%	0.4191	20/48
NSA dual	0.7419	18/48	37.50%	0.4801	7/48
NSA voronoi	0.7315	19/48	39.58%	0.4550	12/48
LODA	0.7248	20/48	41.67%	0.3476	33/48
NSA variable	0.7139	21/48	43.75%	0.4466	14/48
NSA grid	0.7105	22/48	45.83%	0.4454	15/48
NSA suppressed	0.7063	23/48	47.92%	0.4402	17/48
NSA coverage	0.7039	24/48	50.00%	0.4422	16/48

Continued on next page

Table 24 continued

Configuration	AUROC	AUROC rank	Top position	AP	AP rank
MCD	0.7027	25/48	52.08%	0.3622	28/48
HDBSCAN	0.6932	26/48	54.17%	0.2004	46/48
NSA hierarchical	0.6921	27/48	56.25%	0.4266	19/48
ROD	0.6707	28/48	58.33%	0.3200	37/48
COPOD	0.6686	29/48	60.42%	0.3238	35/48
VAE	0.6623	30/48	62.50%	0.3577	29/48
QMCD	0.6609	31/48	64.58%	0.3288	34/48
AutoEncoder	0.6591	32/48	66.67%	0.3211	36/48
AE1SVM	0.6531	33/48	68.75%	0.3500	32/48
PCA	0.6471	34/48	70.83%	0.3520	31/48
NSA deterministic	0.6459	35/48	72.92%	0.3928	24/48
NSA annealed	0.6401	36/48	75.00%	0.3835	25/48
GMM	0.6370	37/48	77.08%	0.3756	27/48
AnoGAN	0.6288	38/48	79.17%	0.2914	39/48
COF	0.6207	39/48	81.25%	0.2384	43/48
NSA fixed	0.6202	40/48	83.33%	0.3564	30/48
ECOD	0.6176	41/48	85.42%	0.2782	40/48
LMDD	0.5898	42/48	87.50%	0.2627	41/48
CD	0.5874	43/48	89.58%	0.3025	38/48
SOD	0.5674	44/48	91.67%	0.2152	45/48
MO_GAAL	0.5640	45/48	93.75%	0.2429	42/48
SO_GAAL	0.5616	46/48	95.83%	0.2154	44/48
SOS	0.5350	47/48	97.92%	0.1654	47/48
ALAD	0.5282	48/48	100.00%	0.1465	48/48

Table 25: Complete-coverage overall rankings: PCA to two dimensions, 25 percent retained normal training pool. All 36 configurations have 85 of 85 qualified runs.

Configuration	AUROC	AUROC rank	Top position	AP	AP rank
KNN	0.8005	1/36	2.78%	0.4644	6/36
KDE	0.7947	2/36	5.56%	0.4711	1/36
KPCA	0.7937	3/36	8.33%	0.4676	4/36
ABOD	0.7893	4/36	11.11%	0.4706	2/36
OCSVM	0.7788	5/36	13.89%	0.4655	5/36
LUNAR	0.7784	6/36	16.67%	0.4449	8/36
INNE	0.7712	7/36	19.44%	0.4468	7/36
LOF	0.7651	8/36	22.22%	0.4127	15/36
Sampling	0.7648	9/36	25.00%	0.4329	10/36
FeatureBagging	0.7639	10/36	27.78%	0.4178	14/36
NSA dual	0.7603	11/36	30.56%	0.4699	3/36
IForest	0.7520	12/36	33.33%	0.3441	24/36
NSA voronoi	0.7432	13/36	36.11%	0.4347	9/36
NSA grid	0.7261	14/36	38.89%	0.4208	11/36

Continued on next page

Table 25 continued

Configuration	AUROC	AUROC rank	Top position	AP	AP rank
NSA variable	0.7240	15/36	41.67%	0.4204	12/36
DeepSVDD	0.7238	16/36	44.44%	0.3890	17/36
NSA suppressed	0.7120	17/36	47.22%	0.4186	13/36
LODA	0.7092	18/36	50.00%	0.3058	26/36
MCD	0.7071	19/36	52.78%	0.3539	21/36
NSA coverage	0.6954	20/36	55.56%	0.4102	16/36
NSA hierarchical	0.6836	21/36	58.33%	0.3857	18/36
COPOD	0.6632	22/36	61.11%	0.2999	27/36
PCA	0.6498	23/36	63.89%	0.3566	20/36
GMM	0.6495	24/36	66.67%	0.3643	19/36
NSA deterministic	0.6420	25/36	69.44%	0.3450	23/36
QMCD	0.6353	26/36	72.22%	0.2828	28/36
HDBSCAN	0.6342	27/36	75.00%	0.1791	34/36
NSA fixed	0.6299	28/36	77.78%	0.3474	22/36
NSA annealed	0.6286	29/36	80.56%	0.3403	25/36
COF	0.6265	30/36	83.33%	0.2417	31/36
ECOD	0.6199	31/36	86.11%	0.2584	30/36
CD	0.5910	32/36	88.89%	0.2704	29/36
SO_GAAL	0.5535	33/36	91.67%	0.2083	33/36
SOS	0.5280	34/36	94.44%	0.1596	35/36
MO_GAAL	0.5268	35/36	97.22%	0.2257	32/36
ALAD	0.5193	36/36	100.00%	0.1361	36/36

H Every NSA configuration against Isolation Forest and KNN

These same-size comparisons use the identical dataset/seed tests. A positive difference favors NSA. They cover both chosen reference detectors at all four retained fractions, including partial coverage. The complete repository contains comparisons against all 43 other detectors, rather than only these two references.

Table 26: All NSA versus IForest/KNN matched contrasts: Original features. Differences are NSA minus reference at the same training size.

NSA configuration	Reference	Retained	Pairs	ΔAUROC	95% CI	ΔAP	95% CI
NSA fixed	KNN	100%	85/85	-0.5376	[-0.6641, -0.4081]	-0.4876	[-0.6277, -0.3497]
NSA fixed	IForest	100%	85/85	-0.4936	[-0.6345, -0.3532]	-0.3971	[-0.5504, -0.2403]
NSA variable	KNN	100%	85/85	-0.3042	[-0.4424, -0.1816]	-0.3124	[-0.4602, -0.1793]
NSA variable	IForest	100%	85/85	-0.2602	[-0.4066, -0.1374]	-0.2220	[-0.3605, -0.0971]
NSA coverage	KNN	100%	85/85	-0.3042	[-0.4424, -0.1851]	-0.3123	[-0.4598, -0.1814]
NSA coverage	IForest	100%	85/85	-0.2602	[-0.4049, -0.1352]	-0.2219	[-0.3545, -0.0950]
NSA grid	KNN	100%	85/85	-0.3001	[-0.4429, -0.1761]	-0.3137	[-0.4583, -0.1753]
NSA grid	IForest	100%	85/85	-0.2561	[-0.3983, -0.1286]	-0.2232	[-0.3588, -0.0926]
NSA hierarchical	KNN	100%	85/85	-0.6005	[-0.7241, -0.4646]	-0.4989	[-0.6351, -0.3566]
NSA hierarchical	IForest	100%	85/85	-0.5565	[-0.6899, -0.4115]	-0.4085	[-0.5531, -0.2590]
NSA voronoi	KNN	100%	0/85	—	[—, —]	—	[—, —]
NSA voronoi	IForest	100%	0/85	—	[—, —]	—	[—, —]
NSA deterministic	KNN	100%	0/85	—	[—, —]	—	[—, —]
NSA deterministic	IForest	100%	0/85	—	[—, —]	—	[—, —]
NSA suppressed	KNN	100%	85/85	-0.3089	[-0.4423, -0.1852]	-0.3180	[-0.4633, -0.1879]
NSA suppressed	IForest	100%	85/85	-0.2649	[-0.4073, -0.1411]	-0.2276	[-0.3609, -0.1022]
NSA dual	KNN	100%	81/85	-0.1928	[-0.2675, -0.1226]	-0.1564	[-0.2300, -0.0923]
NSA dual	IForest	100%	81/85	-0.1489	[-0.2248, -0.0793]	-0.0649	[-0.1504, +0.0375]
NSA annealed	KNN	100%	85/85	-0.5812	[-0.7034, -0.4542]	-0.5043	[-0.6439, -0.3627]
NSA annealed	IForest	100%	85/85	-0.5372	[-0.6667, -0.3959]	-0.4138	[-0.5630, -0.2639]
NSA binary Hamming	KNN	100%	76/85	-0.3977	[-0.4655, -0.3311]	-0.4784	[-0.6027, -0.3518]
NSA binary Hamming	IForest	100%	76/85	-0.3524	[-0.4292, -0.2784]	-0.3831	[-0.5086, -0.2633]
NSA binary contiguous	KNN	100%	59/85	-0.3995	[-0.4564, -0.3423]	-0.5032	[-0.6499, -0.3650]
NSA binary contiguous	IForest	100%	59/85	-0.3608	[-0.4304, -0.2902]	-0.4173	[-0.5487, -0.2869]
NSA binary chunk	KNN	100%	75/85	-0.4088	[-0.4442, -0.3722]	-0.5252	[-0.6455, -0.4048]

Continued on next page

Table 26 continued

NSA configuration	Reference	Retained	Pairs	ΔAUROC	95% CI	ΔAP	95% CI
NSA binary chunk	IForest	100%	75/85	-0.3606	[-0.4104, -0.3062]	-0.4233	[-0.5589, -0.2882]
NSA fixed	KNN	75%	85/85	-0.5433	[-0.6765, -0.4154]	-0.4762	[-0.6197, -0.3284]
NSA fixed	IForest	75%	85/85	-0.5023	[-0.6470, -0.3580]	-0.4038	[-0.5625, -0.2469]
NSA variable	KNN	75%	85/85	-0.3116	[-0.4467, -0.1883]	-0.3169	[-0.4609, -0.1830]
NSA variable	IForest	75%	85/85	-0.2707	[-0.4141, -0.1430]	-0.2445	[-0.3811, -0.1227]
NSA coverage	KNN	75%	85/85	-0.3122	[-0.4532, -0.1910]	-0.3179	[-0.4627, -0.1893]
NSA coverage	IForest	75%	85/85	-0.2713	[-0.4173, -0.1441]	-0.2455	[-0.3798, -0.1223]
NSA grid	KNN	75%	85/85	-0.3056	[-0.4455, -0.1794]	-0.3165	[-0.4704, -0.1768]
NSA grid	IForest	75%	85/85	-0.2647	[-0.4171, -0.1308]	-0.2441	[-0.3813, -0.1103]
NSA hierarchical	KNN	75%	85/85	-0.5859	[-0.7002, -0.4641]	-0.4889	[-0.6210, -0.3500]
NSA hierarchical	IForest	75%	85/85	-0.5450	[-0.6696, -0.4162]	-0.4165	[-0.5540, -0.2784]
NSA voronoi	KNN	75%	0/85	—	[—, —]	—	[—, —]
NSA voronoi	IForest	75%	0/85	—	[—, —]	—	[—, —]
NSA deterministic	KNN	75%	0/85	—	[—, —]	—	[—, —]
NSA deterministic	IForest	75%	0/85	—	[—, —]	—	[—, —]
NSA suppressed	KNN	75%	85/85	-0.3281	[-0.4641, -0.2067]	-0.3329	[-0.4771, -0.2010]
NSA suppressed	IForest	75%	85/85	-0.2872	[-0.4305, -0.1566]	-0.2605	[-0.3953, -0.1338]
NSA dual	KNN	75%	80/85	-0.1563	[-0.2256, -0.0897]	-0.1057	[-0.1656, -0.0553]
NSA dual	IForest	75%	80/85	-0.1168	[-0.1873, -0.0455]	-0.0382	[-0.1246, +0.0565]
NSA annealed	KNN	75%	85/85	-0.5824	[-0.7056, -0.4560]	-0.4976	[-0.6369, -0.3582]
NSA annealed	IForest	75%	85/85	-0.5415	[-0.6774, -0.4062]	-0.4253	[-0.5753, -0.2806]
NSA binary Hamming	KNN	75%	83/85	-0.3709	[-0.4509, -0.2867]	-0.4388	[-0.5736, -0.3074]
NSA binary Hamming	IForest	75%	83/85	-0.3312	[-0.4169, -0.2417]	-0.3666	[-0.4974, -0.2452]
NSA binary contiguous	KNN	75%	60/85	-0.3912	[-0.4428, -0.3380]	-0.5040	[-0.6465, -0.3599]
NSA binary contiguous	IForest	75%	60/85	-0.3575	[-0.4190, -0.2908]	-0.4325	[-0.5661, -0.3006]
NSA binary chunk	KNN	75%	75/85	-0.4027	[-0.4435, -0.3643]	-0.5111	[-0.6332, -0.3885]
NSA binary chunk	IForest	75%	75/85	-0.3587	[-0.4101, -0.3029]	-0.4292	[-0.5592, -0.2951]
NSA fixed	KNN	50%	85/85	-0.5436	[-0.6820, -0.4005]	-0.4709	[-0.6253, -0.3189]
NSA fixed	IForest	50%	85/85	-0.5018	[-0.6498, -0.3488]	-0.3925	[-0.5475, -0.2403]
NSA variable	KNN	50%	85/85	-0.3365	[-0.4768, -0.2130]	-0.3485	[-0.4944, -0.2099]
NSA variable	IForest	50%	85/85	-0.2947	[-0.4467, -0.1633]	-0.2702	[-0.4068, -0.1317]
NSA coverage	KNN	50%	85/85	-0.3362	[-0.4732, -0.2110]	-0.3486	[-0.4990, -0.2117]
NSA coverage	IForest	50%	85/85	-0.2944	[-0.4373, -0.1599]	-0.2702	[-0.4133, -0.1352]

Continued on next page

Table 26 continued

NSA configuration	Reference	Retained	Pairs	ΔAUROC	95% CI	ΔAP	95% CI
NSA grid	KNN	50%	85/85	-0.3327	[-0.4740, -0.2064]	-0.3431	[-0.4922, -0.2021]
NSA grid	IForest	50%	85/85	-0.2909	[-0.4365, -0.1523]	-0.2647	[-0.4050, -0.1229]
NSA hierarchical	KNN	50%	85/85	-0.6143	[-0.7309, -0.4865]	-0.5003	[-0.6402, -0.3606]
NSA hierarchical	IForest	50%	85/85	-0.5725	[-0.6964, -0.4443]	-0.4220	[-0.5698, -0.2753]
NSA voronoi	KNN	50%	0/85	—	[—, —]	—	[—, —]
NSA voronoi	IForest	50%	0/85	—	[—, —]	—	[—, —]
NSA deterministic	KNN	50%	0/85	—	[—, —]	—	[—, —]
NSA deterministic	IForest	50%	0/85	—	[—, —]	—	[—, —]
NSA suppressed	KNN	50%	85/85	-0.3380	[-0.4824, -0.2136]	-0.3501	[-0.5028, -0.2096]
NSA suppressed	IForest	50%	85/85	-0.2962	[-0.4417, -0.1650]	-0.2717	[-0.4183, -0.1395]
NSA dual	KNN	50%	81/85	-0.1579	[-0.2259, -0.0966]	-0.0977	[-0.1441, -0.0570]
NSA dual	IForest	50%	81/85	-0.1166	[-0.1901, -0.0514]	-0.0199	[-0.1062, +0.0846]
NSA annealed	KNN	50%	85/85	-0.5873	[-0.7116, -0.4537]	-0.4871	[-0.6308, -0.3466]
NSA annealed	IForest	50%	85/85	-0.5455	[-0.6763, -0.4031]	-0.4087	[-0.5527, -0.2611]
NSA binary Hamming	KNN	50%	85/85	-0.3608	[-0.4448, -0.2654]	-0.4368	[-0.5744, -0.3025]
NSA binary Hamming	IForest	50%	85/85	-0.3190	[-0.4101, -0.2193]	-0.3584	[-0.4982, -0.2310]
NSA binary contiguous	KNN	50%	60/85	-0.3941	[-0.4456, -0.3404]	-0.4969	[-0.6417, -0.3568]
NSA binary contiguous	IForest	50%	60/85	-0.3604	[-0.4235, -0.2962]	-0.4185	[-0.5466, -0.2903]
NSA binary chunk	KNN	50%	75/85	-0.3946	[-0.4371, -0.3545]	-0.5055	[-0.6346, -0.3742]
NSA binary chunk	IForest	50%	75/85	-0.3498	[-0.4045, -0.2892]	-0.4176	[-0.5582, -0.2820]
NSA fixed	KNN	25%	85/85	-0.5485	[-0.6969, -0.3990]	-0.4520	[-0.6067, -0.2984]
NSA fixed	IForest	25%	85/85	-0.5071	[-0.6602, -0.3487]	-0.3576	[-0.5114, -0.2092]
NSA variable	KNN	25%	85/85	-0.3757	[-0.5235, -0.2309]	-0.3677	[-0.5386, -0.2062]
NSA variable	IForest	25%	85/85	-0.3343	[-0.4915, -0.1845]	-0.2733	[-0.4265, -0.1231]
NSA coverage	KNN	25%	85/85	-0.3754	[-0.5297, -0.2301]	-0.3678	[-0.5326, -0.2072]
NSA coverage	IForest	25%	85/85	-0.3340	[-0.4858, -0.1844]	-0.2734	[-0.4273, -0.1261]
NSA grid	KNN	25%	84/85	-0.3764	[-0.5263, -0.2322]	-0.3740	[-0.5332, -0.2202]
NSA grid	IForest	25%	84/85	-0.3346	[-0.4825, -0.1892]	-0.2794	[-0.4330, -0.1333]
NSA hierarchical	KNN	25%	85/85	-0.6069	[-0.7379, -0.4657]	-0.4639	[-0.6181, -0.3061]
NSA hierarchical	IForest	25%	85/85	-0.5655	[-0.7004, -0.4243]	-0.3694	[-0.5177, -0.2246]
NSA voronoi	KNN	25%	0/85	—	[—, —]	—	[—, —]
NSA voronoi	IForest	25%	0/85	—	[—, —]	—	[—, —]
NSA deterministic	KNN	25%	0/85	—	[—, —]	—	[—, —]

Continued on next page

Table 26 continued

NSA configuration	Reference	Retained	Pairs	ΔAUROC	95% CI	ΔAP	95% CI
NSA deterministic	IForest	25%	0/85	—	[—, —]	—	[—, —]
NSA suppressed	KNN	25%	85/85	-0.3862	[-0.5400, -0.2324]	-0.3742	[-0.5432, -0.2208]
NSA suppressed	IForest	25%	85/85	-0.3448	[-0.5036, -0.1877]	-0.2797	[-0.4350, -0.1289]
NSA dual	KNN	25%	83/85	-0.1295	[-0.1959, -0.0666]	-0.0595	[-0.1023, -0.0202]
NSA dual	IForest	25%	83/85	-0.0876	[-0.1702, -0.0146]	+0.0366	[-0.0539, +0.1564]
NSA annealed	KNN	25%	85/85	-0.5992	[-0.7302, -0.4606]	-0.4743	[-0.6208, -0.3244]
NSA annealed	IForest	25%	85/85	-0.5578	[-0.6915, -0.4144]	-0.3799	[-0.5222, -0.2437]
NSA binary Hamming	KNN	25%	85/85	-0.3610	[-0.4451, -0.2699]	-0.4282	[-0.5694, -0.2991]
NSA binary Hamming	IForest	25%	85/85	-0.3196	[-0.4088, -0.2187]	-0.3338	[-0.4568, -0.2143]
NSA binary contiguous	KNN	25%	63/85	-0.3810	[-0.4264, -0.3367]	-0.4587	[-0.6106, -0.3158]
NSA binary contiguous	IForest	25%	63/85	-0.3340	[-0.3948, -0.2676]	-0.3450	[-0.4732, -0.2208]
NSA binary chunk	KNN	25%	75/85	-0.3952	[-0.4373, -0.3546]	-0.4945	[-0.6324, -0.3534]
NSA binary chunk	IForest	25%	75/85	-0.3521	[-0.4063, -0.2920]	-0.3888	[-0.5277, -0.2574]

Table 27: All NSA versus IForest/KNN matched contrasts: PCA to two dimensions. Differences are NSA minus reference at the same training size.

NSA configuration	Reference	Retained	Pairs	ΔAUROC	95% CI	ΔAP	95% CI
NSA fixed	KNN	100%	85/85	-0.2132	[-0.2927, -0.1377]	-0.1446	[-0.2120, -0.0828]
NSA fixed	IForest	100%	85/85	-0.1766	[-0.2331, -0.1220]	-0.0596	[-0.1083, -0.0130]
NSA variable	KNN	100%	85/85	-0.1031	[-0.1694, -0.0463]	-0.0431	[-0.0842, -0.0047]
NSA variable	IForest	100%	85/85	-0.0665	[-0.1075, -0.0304]	+0.0418	[-0.0011, +0.0902]
NSA coverage	KNN	100%	85/85	-0.1265	[-0.2029, -0.0619]	-0.0619	[-0.1011, -0.0261]
NSA coverage	IForest	100%	85/85	-0.0899	[-0.1402, -0.0468]	+0.0230	[-0.0192, +0.0710]
NSA grid	KNN	100%	85/85	-0.1073	[-0.1727, -0.0481]	-0.0511	[-0.0966, -0.0096]
NSA grid	IForest	100%	85/85	-0.0707	[-0.1141, -0.0333]	+0.0338	[-0.0112, +0.0833]
NSA hierarchical	KNN	100%	85/85	-0.1204	[-0.1940, -0.0540]	-0.0654	[-0.1264, -0.0134]
NSA hierarchical	IForest	100%	85/85	-0.0839	[-0.1343, -0.0388]	+0.0195	[-0.0334, +0.0724]
NSA voronoi	KNN	100%	85/85	-0.0915	[-0.1545, -0.0393]	-0.0351	[-0.0712, -0.0001]
NSA voronoi	IForest	100%	85/85	-0.0550	[-0.0919, -0.0220]	+0.0499	[+0.0083, +0.0956]
NSA deterministic	KNN	100%	85/85	-0.1927	[-0.2719, -0.1150]	-0.1278	[-0.2019, -0.0651]
NSA deterministic	IForest	100%	85/85	-0.1561	[-0.2160, -0.0975]	-0.0428	[-0.1012, +0.0156]
NSA suppressed	KNN	100%	85/85	-0.1220	[-0.1854, -0.0673]	-0.0659	[-0.1065, -0.0283]
NSA suppressed	IForest	100%	85/85	-0.0855	[-0.1249, -0.0498]	+0.0191	[-0.0225, +0.0664]
NSA dual	KNN	100%	85/85	-0.0786	[-0.1407, -0.0283]	-0.0155	[-0.0469, +0.0185]
NSA dual	IForest	100%	85/85	-0.0421	[-0.0781, -0.0144]	+0.0695	[+0.0275, +0.1181]
NSA annealed	KNN	100%	85/85	-0.1960	[-0.2884, -0.1092]	-0.1200	[-0.1881, -0.0589]
NSA annealed	IForest	100%	85/85	-0.1595	[-0.2276, -0.0939]	-0.0351	[-0.0839, +0.0110]
NSA binary Hamming	KNN	100%	0/85	—	[—, —]	—	[—, —]
NSA binary Hamming	IForest	100%	0/85	—	[—, —]	—	[—, —]
NSA binary contiguous	KNN	100%	0/85	—	[—, —]	—	[—, —]
NSA binary contiguous	IForest	100%	0/85	—	[—, —]	—	[—, —]
NSA binary chunk	KNN	100%	0/85	—	[—, —]	—	[—, —]
NSA binary chunk	IForest	100%	0/85	—	[—, —]	—	[—, —]
NSA fixed	KNN	75%	85/85	-0.2070	[-0.2893, -0.1236]	-0.1563	[-0.2411, -0.0858]
NSA fixed	IForest	75%	85/85	-0.1648	[-0.2255, -0.1028]	-0.0602	[-0.1034, -0.0172]
NSA variable	KNN	75%	85/85	-0.1011	[-0.1637, -0.0447]	-0.0456	[-0.0850, -0.0112]
NSA variable	IForest	75%	85/85	-0.0589	[-0.1018, -0.0221]	+0.0505	[+0.0098, +0.0975]

Continued on next page

Table 27 continued

NSA configuration	Reference	Retained	Pairs	ΔAUROC	95% CI	ΔAP	95% CI
NSA coverage	KNN	75%	85/85	-0.1230	[-0.1916, -0.0625]	-0.0581	[-0.0971, -0.0220]
NSA coverage	IForest	75%	85/85	-0.0808	[-0.1243, -0.0414]	+0.0380	[-0.0023, +0.0846]
NSA grid	KNN	75%	85/85	-0.1050	[-0.1693, -0.0474]	-0.0506	[-0.0990, -0.0105]
NSA grid	IForest	75%	85/85	-0.0629	[-0.1049, -0.0246]	+0.0455	[+0.0056, +0.0922]
NSA hierarchical	KNN	75%	85/85	-0.1316	[-0.2023, -0.0650]	-0.0800	[-0.1367, -0.0310]
NSA hierarchical	IForest	75%	85/85	-0.0895	[-0.1421, -0.0422]	+0.0161	[-0.0310, +0.0700]
NSA voronoi	KNN	75%	85/85	-0.0828	[-0.1407, -0.0317]	-0.0301	[-0.0652, +0.0007]
NSA voronoi	IForest	75%	85/85	-0.0407	[-0.0753, -0.0063]	+0.0660	[+0.0229, +0.1146]
NSA deterministic	KNN	75%	85/85	-0.1844	[-0.2614, -0.1082]	-0.1305	[-0.2111, -0.0591]
NSA deterministic	IForest	75%	85/85	-0.1423	[-0.2001, -0.0861]	-0.0344	[-0.0861, +0.0169]
NSA suppressed	KNN	75%	85/85	-0.1152	[-0.1738, -0.0650]	-0.0602	[-0.1121, -0.0193]
NSA suppressed	IForest	75%	85/85	-0.0731	[-0.1072, -0.0399]	+0.0359	[-0.0039, +0.0797]
NSA dual	KNN	75%	85/85	-0.0725	[-0.1272, -0.0252]	-0.0113	[-0.0381, +0.0119]
NSA dual	IForest	75%	85/85	-0.0303	[-0.0625, -0.0021]	+0.0848	[+0.0339, +0.1463]
NSA annealed	KNN	75%	85/85	-0.1809	[-0.2596, -0.1036]	-0.1176	[-0.1868, -0.0523]
NSA annealed	IForest	75%	85/85	-0.1388	[-0.1963, -0.0847]	-0.0215	[-0.0626, +0.0211]
NSA binary Hamming	KNN	75%	0/85	—	[—, —]	—	[—, —]
NSA binary Hamming	IForest	75%	0/85	—	[—, —]	—	[—, —]
NSA binary contiguous	KNN	75%	0/85	—	[—, —]	—	[—, —]
NSA binary contiguous	IForest	75%	0/85	—	[—, —]	—	[—, —]
NSA binary chunk	KNN	75%	0/85	—	[—, —]	—	[—, —]
NSA binary chunk	IForest	75%	0/85	—	[—, —]	—	[—, —]
NSA fixed	KNN	50%	85/85	-0.1848	[-0.2566, -0.1186]	-0.1378	[-0.2173, -0.0724]
NSA fixed	IForest	50%	85/85	-0.1395	[-0.1910, -0.0880]	-0.0369	[-0.0853, +0.0149]
NSA variable	KNN	50%	85/85	-0.0912	[-0.1531, -0.0402]	-0.0477	[-0.0938, -0.0088]
NSA variable	IForest	50%	85/85	-0.0458	[-0.0878, -0.0057]	+0.0533	[+0.0050, +0.1081]
NSA coverage	KNN	50%	85/85	-0.1012	[-0.1719, -0.0430]	-0.0520	[-0.0987, -0.0113]
NSA coverage	IForest	50%	85/85	-0.0558	[-0.1059, -0.0080]	+0.0489	[+0.0002, +0.1044]
NSA grid	KNN	50%	85/85	-0.0946	[-0.1556, -0.0415]	-0.0488	[-0.0958, -0.0090]
NSA grid	IForest	50%	85/85	-0.0493	[-0.0934, -0.0076]	+0.0521	[+0.0050, +0.1089]
NSA hierarchical	KNN	50%	85/85	-0.1130	[-0.1812, -0.0506]	-0.0676	[-0.1326, -0.0122]
NSA hierarchical	IForest	50%	85/85	-0.0676	[-0.1211, -0.0181]	+0.0333	[-0.0311, +0.1015]
NSA voronoi	KNN	50%	85/85	-0.0736	[-0.1253, -0.0306]	-0.0392	[-0.0810, -0.0028]

Continued on next page

Table 27 continued

NSA configuration	Reference	Retained	Pairs	ΔAUROC	95% CI	ΔAP	95% CI
NSA voronoi	IForest	50%	85/85	-0.0283	[-0.0639, +0.0065]	+0.0617	[+0.0097, +0.1189]
NSA deterministic	KNN	50%	85/85	-0.1592	[-0.2296, -0.0917]	-0.1014	[-0.1880, -0.0312]
NSA deterministic	IForest	50%	85/85	-0.1138	[-0.1731, -0.0580]	-0.0005	[-0.0638, +0.0644]
NSA suppressed	KNN	50%	85/85	-0.0988	[-0.1512, -0.0532]	-0.0540	[-0.0985, -0.0161]
NSA suppressed	IForest	50%	85/85	-0.0535	[-0.0888, -0.0201]	+0.0469	[-0.0029, +0.1056]
NSA dual	KNN	50%	85/85	-0.0632	[-0.1162, -0.0205]	-0.0141	[-0.0384, +0.0066]
NSA dual	IForest	50%	85/85	-0.0179	[-0.0489, +0.0117]	+0.0868	[+0.0320, +0.1500]
NSA annealed	KNN	50%	85/85	-0.1650	[-0.2343, -0.0978]	-0.1107	[-0.1885, -0.0494]
NSA annealed	IForest	50%	85/85	-0.1197	[-0.1749, -0.0702]	-0.0098	[-0.0597, +0.0429]
NSA binary Hamming	KNN	50%	0/85	—	[—, —]	—	[—, —]
NSA binary Hamming	IForest	50%	0/85	—	[—, —]	—	[—, —]
NSA binary contiguous	KNN	50%	0/85	—	[—, —]	—	[—, —]
NSA binary contiguous	IForest	50%	0/85	—	[—, —]	—	[—, —]
NSA binary chunk	KNN	50%	0/85	—	[—, —]	—	[—, —]
NSA binary chunk	IForest	50%	0/85	—	[—, —]	—	[—, —]
NSA fixed	KNN	25%	85/85	-0.1706	[-0.2544, -0.0940]	-0.1171	[-0.2154, -0.0352]
NSA fixed	IForest	25%	85/85	-0.1221	[-0.1927, -0.0596]	+0.0033	[-0.0590, +0.0660]
NSA variable	KNN	25%	85/85	-0.0765	[-0.1399, -0.0220]	-0.0440	[-0.1192, +0.0163]
NSA variable	IForest	25%	85/85	-0.0280	[-0.0855, +0.0222]	+0.0763	[+0.0230, +0.1412]
NSA coverage	KNN	25%	85/85	-0.1051	[-0.1721, -0.0459]	-0.0543	[-0.1260, +0.0071]
NSA coverage	IForest	25%	85/85	-0.0566	[-0.1109, -0.0074]	+0.0661	[+0.0129, +0.1327]
NSA grid	KNN	25%	85/85	-0.0744	[-0.1347, -0.0211]	-0.0436	[-0.1175, +0.0164]
NSA grid	IForest	25%	85/85	-0.0259	[-0.0770, +0.0231]	+0.0768	[+0.0195, +0.1443]
NSA hierarchical	KNN	25%	85/85	-0.1169	[-0.1913, -0.0501]	-0.0788	[-0.1703, -0.0105]
NSA hierarchical	IForest	25%	85/85	-0.0684	[-0.1347, -0.0115]	+0.0416	[-0.0162, +0.1060]
NSA voronoi	KNN	25%	85/85	-0.0573	[-0.1020, -0.0159]	-0.0297	[-0.0819, +0.0198]
NSA voronoi	IForest	25%	85/85	-0.0088	[-0.0507, +0.0327]	+0.0907	[+0.0380, +0.1527]
NSA deterministic	KNN	25%	85/85	-0.1585	[-0.2416, -0.0816]	-0.1195	[-0.2299, -0.0286]
NSA deterministic	IForest	25%	85/85	-0.1100	[-0.1808, -0.0475]	+0.0009	[-0.0749, +0.0792]
NSA suppressed	KNN	25%	85/85	-0.0885	[-0.1462, -0.0376]	-0.0459	[-0.1015, +0.0035]
NSA suppressed	IForest	25%	85/85	-0.0400	[-0.0877, +0.0063]	+0.0745	[+0.0208, +0.1354]
NSA dual	KNN	25%	85/85	-0.0402	[-0.0919, -0.0019]	+0.0055	[-0.0163, +0.0311]
NSA dual	IForest	25%	85/85	+0.0083	[-0.0281, +0.0418]	+0.1259	[+0.0596, +0.2064]

Continued on next page

Table 27 continued

NSA configuration	Reference	Retained	Pairs	ΔAUROC	95% CI	ΔAP	95% CI
NSA annealed	KNN	25%	85/85	-0.1719	[-0.2514, -0.0988]	-0.1242	[-0.2199, -0.0425]
NSA annealed	IForest	25%	85/85	-0.1234	[-0.1859, -0.0633]	-0.0038	[-0.0682, +0.0656]
NSA binary Hamming	KNN	25%	0/85	—	[—, —]	—	[—, —]
NSA binary Hamming	IForest	25%	0/85	—	[—, —]	—	[—, —]
NSA binary contiguous	KNN	25%	0/85	—	[—, —]	—	[—, —]
NSA binary contiguous	IForest	25%	0/85	—	[—, —]	—	[—, —]
NSA binary chunk	KNN	25%	0/85	—	[—, —]	—	[—, —]
NSA binary chunk	IForest	25%	0/85	—	[—, —]	—	[—, —]

I Per-dataset scores for every NSA configuration

The following tables show every one of the 13 NSA configurations on each dataset and at each retained size, separately for original features and PCA(2). A cell is the mean over its qualified split seeds and reports AUROC / AP (completed seeds out of five). Thus a low score remains visible, and an unsuccessful setting remains a dash with zero coverage.

Table 28: NSA/fixed: dataset results on original features. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.2751 / 0.1205 (5/5)	0.2737 / 0.1119 (5/5)	0.2448 / 0.1071 (5/5)	0.2307 / 0.0995 (5/5)
cardio	0.4551 / 0.1948 (5/5)	0.4360 / 0.1803 (5/5)	0.4050 / 0.1626 (5/5)	0.3605 / 0.1405 (5/5)
glass	0.3628 / 0.1422 (5/5)	0.4085 / 0.1478 (5/5)	0.5518 / 0.1063 (5/5)	0.5744 / 0.1716 (5/5)
ionosphere	0.2328 / 0.2726 (5/5)	0.2216 / 0.2700 (5/5)	0.1943 / 0.2540 (5/5)	0.1524 / 0.2488 (5/5)
letter	0.5872 / 0.1078 (5/5)	0.5801 / 0.1016 (5/5)	0.5784 / 0.1024 (5/5)	0.5735 / 0.0963 (5/5)
lympho	0.2103 / 0.0350 (5/5)	0.2121 / 0.0350 (5/5)	0.1379 / 0.0328 (5/5)	0.0586 / 0.0274 (5/5)
mnist	0.4172 / 0.1012 (5/5)	0.3992 / 0.0935 (5/5)	0.3954 / 0.0935 (5/5)	0.3910 / 0.0966 (5/5)
musk	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)
optdigits	0.6501 / 0.0848 (5/5)	0.6250 / 0.0871 (5/5)	0.6091 / 0.0793 (5/5)	0.6303 / 0.0783 (5/5)
pendigits	0.6192 / 0.1132 (5/5)	0.6170 / 0.1054 (5/5)	0.6169 / 0.1054 (5/5)	0.6098 / 0.1050 (5/5)
pima	0.4526 / 0.3359 (5/5)	0.4603 / 0.3418 (5/5)	0.4548 / 0.3382 (5/5)	0.4586 / 0.3447 (5/5)
satellite	0.3527 / 0.2737 (5/5)	0.3522 / 0.2706 (5/5)	0.3413 / 0.2650 (5/5)	0.3359 / 0.2634 (5/5)
satimage-2	0.1063 / 0.0080 (5/5)	0.0870 / 0.0080 (5/5)	0.0837 / 0.0077 (5/5)	0.0800 / 0.0073 (5/5)
shuttle	0.1544 / 0.0561 (5/5)	0.0561 / 0.0378 (5/5)	0.0341 / 0.0374 (5/5)	0.0048 / 0.0370 (5/5)
vertebral	0.5353 / 0.1756 (5/5)	0.5437 / 0.2119 (5/5)	0.5456 / 0.2287 (5/5)	0.5095 / 0.1820 (5/5)
vowels	0.5545 / 0.1124 (5/5)	0.5381 / 0.1034 (5/5)	0.5556 / 0.1194 (5/5)	0.5452 / 0.1105 (5/5)
wbc	0.1486 / 0.0372 (5/5)	0.1465 / 0.0397 (5/5)	0.1333 / 0.0384 (5/5)	0.0894 / 0.0330 (5/5)

Table 29: NSA/variable: dataset results on original features. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.4014 / 0.1563 (5/5)	0.3663 / 0.1375 (5/5)	0.2985 / 0.1301 (5/5)	0.2497 / 0.1018 (5/5)
cardio	0.8507 / 0.5414 (5/5)	0.8428 / 0.4986 (5/5)	0.7933 / 0.3874 (5/5)	0.7241 / 0.3129 (5/5)

Continued on next page

Table 29 continued

Dataset	100% retained	75% retained	50% retained	25% retained
glass	0.5707 / 0.1044 (5/5)	0.5701 / 0.1002 (5/5)	0.6537 / 0.1087 (5/5)	0.7293 / 0.1776 (5/5)
ionosphere	0.5045 / 0.5486 (5/5)	0.4787 / 0.5254 (5/5)	0.4514 / 0.5031 (5/5)	0.3502 / 0.4077 (5/5)
letter	0.6774 / 0.1760 (5/5)	0.6811 / 0.1649 (5/5)	0.6880 / 0.1546 (5/5)	0.7020 / 0.1584 (5/5)
lympho	0.8172 / 0.5113 (5/5)	0.7879 / 0.4306 (5/5)	0.6966 / 0.2165 (5/5)	0.3500 / 0.0623 (5/5)
mnist	0.6012 / 0.2148 (5/5)	0.5829 / 0.1856 (5/5)	0.5695 / 0.1724 (5/5)	0.5311 / 0.1550 (5/5)
musk	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)
optdigits	0.8514 / 0.2001 (5/5)	0.8448 / 0.1874 (5/5)	0.8483 / 0.1945 (5/5)	0.8760 / 0.2422 (5/5)
pendigits	0.8447 / 0.2553 (5/5)	0.8394 / 0.2293 (5/5)	0.8516 / 0.2992 (5/5)	0.8700 / 0.2429 (5/5)
pima	0.6530 / 0.5102 (5/5)	0.6383 / 0.4970 (5/5)	0.6641 / 0.5196 (5/5)	0.6852 / 0.5597 (5/5)
satellite	0.3889 / 0.3216 (5/5)	0.3946 / 0.3266 (5/5)	0.3853 / 0.3205 (5/5)	0.3678 / 0.3042 (5/5)
satimage-2	0.1518 / 0.0266 (5/5)	0.1325 / 0.0247 (5/5)	0.1340 / 0.0186 (5/5)	0.1294 / 0.0114 (5/5)
shuttle	0.8914 / 0.8494 (5/5)	0.8816 / 0.8303 (5/5)	0.6589 / 0.5271 (5/5)	0.3874 / 0.2300 (5/5)
vertebral	0.5351 / 0.1562 (5/5)	0.5363 / 0.1779 (5/5)	0.4879 / 0.1390 (5/5)	0.4593 / 0.1250 (5/5)
vowels	0.8097 / 0.4236 (5/5)	0.8059 / 0.3875 (5/5)	0.7983 / 0.3483 (5/5)	0.8072 / 0.3035 (5/5)
wbc	0.5333 / 0.1526 (5/5)	0.5116 / 0.1512 (5/5)	0.4226 / 0.1187 (5/5)	0.3238 / 0.0806 (5/5)

Table 30: NSA/coverage: dataset results on original features. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.4014 / 0.1563 (5/5)	0.3663 / 0.1375 (5/5)	0.2985 / 0.1301 (5/5)	0.2497 / 0.1018 (5/5)
cardio	0.8507 / 0.5414 (5/5)	0.8428 / 0.4986 (5/5)	0.7933 / 0.3874 (5/5)	0.7241 / 0.3129 (5/5)
glass	0.5707 / 0.1044 (5/5)	0.5701 / 0.1002 (5/5)	0.6537 / 0.1087 (5/5)	0.7323 / 0.1778 (5/5)
ionosphere	0.5045 / 0.5486 (5/5)	0.4787 / 0.5254 (5/5)	0.4514 / 0.5031 (5/5)	0.3502 / 0.4077 (5/5)
letter	0.6774 / 0.1760 (5/5)	0.6811 / 0.1649 (5/5)	0.6880 / 0.1546 (5/5)	0.7020 / 0.1584 (5/5)
lympho	0.8172 / 0.5113 (5/5)	0.7879 / 0.4306 (5/5)	0.6966 / 0.2165 (5/5)	0.3500 / 0.0623 (5/5)
mnist	0.6012 / 0.2148 (5/5)	0.5829 / 0.1856 (5/5)	0.5695 / 0.1724 (5/5)	0.5311 / 0.1550 (5/5)
musk	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)

Continued on next page

Table 30 continued

Dataset	100% retained	75% retained	50% retained	25% retained
optdigits	0.8514 / 0.2001 (5/5)	0.8448 / 0.1874 (5/5)	0.8483 / 0.1945 (5/5)	0.8760 / 0.2422 (5/5)
pendigits	0.8447 / 0.2553 (5/5)	0.8394 / 0.2293 (5/5)	0.8516 / 0.2992 (5/5)	0.8700 / 0.2429 (5/5)
pima	0.6530 / 0.5102 (5/5)	0.6383 / 0.4970 (5/5)	0.6641 / 0.5196 (5/5)	0.6852 / 0.5597 (5/5)
satellite	0.3889 / 0.3216 (5/5)	0.3946 / 0.3266 (5/5)	0.3853 / 0.3205 (5/5)	0.3678 / 0.3042 (5/5)
satimage-2	0.1518 / 0.0266 (5/5)	0.1325 / 0.0247 (5/5)	0.1340 / 0.0186 (5/5)	0.1294 / 0.0114 (5/5)
shuttle	0.8914 / 0.8494 (5/5)	0.8816 / 0.8303 (5/5)	0.6589 / 0.5271 (5/5)	0.3874 / 0.2300 (5/5)
vertebral	0.5357 / 0.1586 (5/5)	0.5258 / 0.1604 (5/5)	0.4937 / 0.1379 (5/5)	0.4607 / 0.1231 (5/5)
vowels	0.8097 / 0.4236 (5/5)	0.8059 / 0.3875 (5/5)	0.7983 / 0.3483 (5/5)	0.8072 / 0.3035 (5/5)
wbc	0.5333 / 0.1526 (5/5)	0.5116 / 0.1512 (5/5)	0.4226 / 0.1187 (5/5)	0.3238 / 0.0806 (5/5)

Table 31: NSA/grid: dataset results on original features. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.4110 / 0.1572 (5/5)	0.3741 / 0.1459 (5/5)	0.2979 / 0.1319 (5/5)	0.2507 / 0.1011 (5/5)
cardio	0.8144 / 0.5168 (5/5)	0.8149 / 0.4816 (5/5)	0.7634 / 0.3658 (5/5)	0.7342 / 0.3156 (5/5)
glass	0.5939 / 0.0993 (5/5)	0.6402 / 0.1167 (5/5)	0.6329 / 0.1106 (5/5)	0.6774 / 0.1179 (5/5)
ionosphere	0.5179 / 0.5583 (5/5)	0.4965 / 0.5384 (5/5)	0.4703 / 0.5161 (5/5)	0.3646 / 0.4150 (5/5)
letter	0.6960 / 0.1868 (5/5)	0.6992 / 0.1829 (5/5)	0.6971 / 0.1616 (5/5)	0.7072 / 0.1591 (5/5)
lympho	0.7793 / 0.4235 (5/5)	0.7466 / 0.3434 (5/5)	0.6983 / 0.2070 (5/5)	0.3534 / 0.0609 (5/5)
mnist	0.5882 / 0.1996 (5/5)	0.5836 / 0.1806 (5/5)	0.5622 / 0.1633 (5/5)	0.5277 / 0.1518 (5/5)
musk	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)
optdigits	0.8296 / 0.2465 (5/5)	0.8377 / 0.2421 (5/5)	0.8608 / 0.2574 (5/5)	0.8597 / 0.2415 (5/5)
pendigits	0.8594 / 0.2248 (5/5)	0.8617 / 0.2173 (5/5)	0.8525 / 0.2639 (5/5)	0.8557 / 0.1890 (5/5)
pima	0.6535 / 0.5033 (5/5)	0.6457 / 0.4956 (5/5)	0.6643 / 0.5171 (5/5)	0.6793 / 0.5410 (5/5)
satellite	0.3890 / 0.3249 (5/5)	0.3990 / 0.3339 (5/5)	0.3931 / 0.3278 (5/5)	0.3771 / 0.3089 (5/5)
satimage-2	0.1460 / 0.0255 (5/5)	0.1248 / 0.0241 (5/5)	0.1251 / 0.0173 (5/5)	0.1345 / 0.0119 (5/5)
shuttle	0.9522 / 0.9235 (5/5)	0.8763 / 0.8485 (5/5)	0.6894 / 0.5905 (5/5)	0.4067 / 0.2571 (5/5)

Continued on next page

Table 31 continued

Dataset	100% retained	75% retained	50% retained	25% retained
vertebral	0.5556 / 0.1558 (5/5)	0.5470 / 0.1685 (5/5)	0.5208 / 0.1453 (5/5)	0.4794 / 0.1318 (4/5)
vowels	0.8239 / 0.4275 (5/5)	0.8320 / 0.3968 (5/5)	0.8150 / 0.3625 (5/5)	0.8090 / 0.2923 (5/5)
wbc	0.5427 / 0.1538 (5/5)	0.5170 / 0.1451 (5/5)	0.4238 / 0.1128 (5/5)	0.3259 / 0.0777 (5/5)

Table 32: NSA/hierarchical: dataset results on original features. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.2604 / 0.1050 (5/5)	0.3013 / 0.1222 (5/5)	0.2522 / 0.1142 (5/5)	0.2180 / 0.1056 (5/5)
cardio	0.1615 / 0.0734 (5/5)	0.1573 / 0.0592 (5/5)	0.2597 / 0.0668 (5/5)	0.2105 / 0.0592 (5/5)
glass	0.5787 / 0.1841 (5/5)	0.3841 / 0.0606 (5/5)	0.3128 / 0.0563 (5/5)	0.2768 / 0.0889 (5/5)
ionosphere	0.2229 / 0.2892 (5/5)	0.1508 / 0.2534 (5/5)	0.1441 / 0.2462 (5/5)	0.0985 / 0.2319 (5/5)
letter	0.4673 / 0.0669 (5/5)	0.5131 / 0.0719 (5/5)	0.4824 / 0.0757 (5/5)	0.4565 / 0.0633 (5/5)
lympho	0.0724 / 0.0269 (5/5)	0.3621 / 0.0733 (5/5)	0.2879 / 0.0394 (5/5)	0.0862 / 0.0289 (5/5)
mnist	0.4123 / 0.1083 (5/5)	0.3663 / 0.0806 (5/5)	0.3448 / 0.0733 (5/5)	0.3052 / 0.0665 (5/5)
musk	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)
optdigits	0.5414 / 0.0523 (5/5)	0.5574 / 0.0524 (5/5)	0.5548 / 0.0394 (5/5)	0.6198 / 0.1065 (5/5)
pendigits	0.2809 / 0.0162 (5/5)	0.1746 / 0.0133 (5/5)	0.2858 / 0.0211 (5/5)	0.4617 / 0.0997 (5/5)
pima	0.4973 / 0.3819 (5/5)	0.5469 / 0.4198 (5/5)	0.4365 / 0.3433 (5/5)	0.4955 / 0.3801 (5/5)
satellite	0.2678 / 0.2323 (5/5)	0.2876 / 0.2365 (5/5)	0.2511 / 0.2193 (5/5)	0.2812 / 0.2537 (5/5)
satimage-2	0.0436 / 0.0069 (5/5)	0.0823 / 0.0067 (5/5)	0.0311 / 0.0063 (5/5)	0.0644 / 0.0073 (5/5)
shuttle	0.2233 / 0.1817 (5/5)	0.3358 / 0.2606 (5/5)	0.1307 / 0.0581 (5/5)	0.0461 / 0.0427 (5/5)
vertebral	0.5087 / 0.1642 (5/5)	0.4030 / 0.1265 (5/5)	0.4601 / 0.1551 (5/5)	0.5101 / 0.2080 (5/5)
vowels	0.3315 / 0.0437 (5/5)	0.3865 / 0.0448 (5/5)	0.3289 / 0.0300 (5/5)	0.3658 / 0.0535 (5/5)
wbc	0.1762 / 0.0452 (5/5)	0.2234 / 0.0488 (5/5)	0.1168 / 0.0328 (5/5)	0.1146 / 0.0447 (5/5)

Table 33: NSA/voronoi: dataset results on original features. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)

Continued on next page

Table 33 continued

Dataset	100% retained	75% retained	50% retained	25% retained
cardio	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
glass	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
ionosphere	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
letter	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
lympho	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
mnist	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
musk	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
optdigits	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
pendigits	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
pima	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
satellite	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
satimage-2	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
shuttle	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
vertebral	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
vowels	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
wbc	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)

Table 34: NSA/deterministic: dataset results on original features. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
cardio	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
glass	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
ionosphere	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
letter	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
lympho	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
mnist	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
musk	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
optdigits	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
pendigits	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
pima	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
satellite	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
satimage-2	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
shuttle	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
vertebral	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
vowels	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
wbc	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)

Table 35: NSA/suppressed: dataset results on original features. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.4014 / 0.1563 (5/5)	0.3663 / 0.1375 (5/5)	0.2985 / 0.1301 (5/5)	0.2497 / 0.1018 (5/5)
cardio	0.8740 / 0.6169 (5/5)	0.8416 / 0.5307 (5/5)	0.8211 / 0.4449 (5/5)	0.7682 / 0.3912 (5/5)
glass	0.5915 / 0.1028 (5/5)	0.5848 / 0.1029 (5/5)	0.6073 / 0.0956 (5/5)	0.6518 / 0.1418 (5/5)

Continued on next page

Table 35 continued

Dataset	100% retained	75% retained	50% retained	25% retained
ionosphere	0.5047 / 0.5533 (5/5)	0.4803 / 0.5270 (5/5)	0.4508 / 0.5019 (5/5)	0.3541 / 0.4085 (5/5)
letter	0.6773 / 0.1778 (5/5)	0.6821 / 0.1688 (5/5)	0.6867 / 0.1522 (5/5)	0.7030 / 0.1596 (5/5)
lympho	0.7948 / 0.5177 (5/5)	0.7052 / 0.4053 (5/5)	0.6810 / 0.1931 (5/5)	0.3224 / 0.0540 (5/5)
mnist	0.6011 / 0.2147 (5/5)	0.5797 / 0.1823 (5/5)	0.5663 / 0.1689 (5/5)	0.5247 / 0.1510 (5/5)
musk	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)
optdigits	0.8470 / 0.1909 (5/5)	0.8428 / 0.1921 (5/5)	0.8479 / 0.1936 (5/5)	0.8592 / 0.1903 (5/5)
pendigits	0.8264 / 0.1881 (5/5)	0.8126 / 0.1699 (5/5)	0.8457 / 0.2276 (5/5)	0.8708 / 0.2257 (5/5)
pima	0.6532 / 0.4985 (5/5)	0.6521 / 0.4995 (5/5)	0.6841 / 0.5302 (5/5)	0.7007 / 0.5639 (5/5)
satellite	0.3890 / 0.3214 (5/5)	0.3944 / 0.3263 (5/5)	0.3864 / 0.3209 (5/5)	0.3680 / 0.3016 (5/5)
satimage-2	0.1517 / 0.0266 (5/5)	0.1325 / 0.0247 (5/5)	0.1339 / 0.0186 (5/5)	0.1362 / 0.0114 (5/5)
shuttle	0.8396 / 0.7998 (5/5)	0.7158 / 0.6540 (5/5)	0.6474 / 0.5376 (5/5)	0.2699 / 0.1615 (5/5)
vertebral	0.5397 / 0.1558 (5/5)	0.5244 / 0.1571 (5/5)	0.5196 / 0.1555 (5/5)	0.4627 / 0.1329 (5/5)
vowels	0.7912 / 0.3928 (5/5)	0.8006 / 0.3641 (5/5)	0.7856 / 0.3445 (5/5)	0.7978 / 0.2891 (5/5)
wbc	0.5193 / 0.1401 (5/5)	0.4993 / 0.1399 (5/5)	0.4148 / 0.1162 (5/5)	0.3240 / 0.0806 (5/5)

Table 36: NSA/dual: dataset results on original features. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.4685 / 0.2641 (5/5)	0.5359 / 0.2887 (5/5)	0.6212 / 0.3734 (5/5)	0.6861 / 0.4023 (5/5)
cardio	0.6175 / 0.4529 (5/5)	0.7020 / 0.5372 (5/5)	0.7561 / 0.5655 (5/5)	0.7369 / 0.5337 (5/5)
glass	0.7726 / 0.2231 (5/5)	0.8055 / 0.2362 (5/5)	0.7323 / 0.1773 (5/5)	0.8140 / 0.1820 (5/5)
ionosphere	0.8734 / 0.8968 (5/5)	0.8624 / 0.8897 (5/5)	0.8605 / 0.8872 (5/5)	0.9037 / 0.9189 (5/5)
letter	0.5336 / 0.1160 (5/5)	0.5127 / 0.1172 (5/5)	0.5351 / 0.1051 (5/5)	0.5992 / 0.1306 (5/5)
lympho	0.9207 / 0.6542 (5/5)	0.9224 / 0.6733 (5/5)	0.9259 / 0.6906 (5/5)	0.9759 / 0.7972 (5/5)
mnist	0.7833 / 0.3100 (1/5)	— / — (0/5)	0.7200 / 0.3873 (1/5)	0.6528 / 0.3302 (3/5)
musk	1.0000 / 1.0000 (5/5)	1.0000 / 1.0000 (5/5)	1.0000 / 1.0000 (5/5)	1.0000 / 1.0000 (5/5)
optdigits	0.3844 / 0.0243 (5/5)	0.5289 / 0.0351 (5/5)	0.3934 / 0.0251 (5/5)	0.3510 / 0.0215 (5/5)

Continued on next page

Table 36 continued

Dataset	100% retained	75% retained	50% retained	25% retained
pendigits	0.6119 / 0.0843 (5/5)	0.5947 / 0.0821 (5/5)	0.5622 / 0.0697 (5/5)	0.6431 / 0.0982 (5/5)
pima	0.6075 / 0.4733 (5/5)	0.6393 / 0.5056 (5/5)	0.6310 / 0.5012 (5/5)	0.6673 / 0.5107 (5/5)
satellite	0.6239 / 0.6294 (5/5)	0.6094 / 0.6189 (5/5)	0.6182 / 0.6262 (5/5)	0.6299 / 0.6353 (5/5)
satimage-2	0.9649 / 0.9226 (5/5)	0.9722 / 0.9316 (5/5)	0.9581 / 0.9329 (5/5)	0.9705 / 0.9403 (5/5)
shuttle	0.9963 / 0.9065 (5/5)	0.9961 / 0.9035 (5/5)	0.9944 / 0.8860 (5/5)	0.9964 / 0.8771 (5/5)
vertebral	0.4714 / 0.1396 (5/5)	0.5018 / 0.1409 (5/5)	0.4786 / 0.1400 (5/5)	0.4000 / 0.1168 (5/5)
vowels	0.7023 / 0.2014 (5/5)	0.8788 / 0.4733 (5/5)	0.8716 / 0.4595 (5/5)	0.9355 / 0.5646 (5/5)
wbc	0.6481 / 0.5375 (5/5)	0.7017 / 0.5764 (5/5)	0.7812 / 0.6155 (5/5)	0.7667 / 0.6828 (5/5)

Table 37: NSA/annealed: dataset results on original features. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.2723 / 0.1172 (5/5)	0.2665 / 0.1099 (5/5)	0.2350 / 0.1129 (5/5)	0.2238 / 0.1040 (5/5)
cardio	0.3141 / 0.0780 (5/5)	0.2824 / 0.0746 (5/5)	0.2803 / 0.0707 (5/5)	0.2832 / 0.0672 (5/5)
glass	0.5201 / 0.1969 (5/5)	0.5726 / 0.1615 (5/5)	0.4171 / 0.0810 (5/5)	0.4671 / 0.1352 (5/5)
ionosphere	0.2371 / 0.3010 (5/5)	0.2026 / 0.2633 (5/5)	0.2163 / 0.2851 (5/5)	0.1723 / 0.2599 (5/5)
letter	0.5127 / 0.0690 (5/5)	0.5168 / 0.0716 (5/5)	0.4988 / 0.0685 (5/5)	0.4915 / 0.0694 (5/5)
lympho	0.3569 / 0.0728 (5/5)	0.3552 / 0.0759 (5/5)	0.2879 / 0.0833 (5/5)	0.0690 / 0.0274 (5/5)
mnist	0.3323 / 0.0740 (5/5)	0.3279 / 0.0730 (5/5)	0.3201 / 0.0729 (5/5)	0.2918 / 0.0646 (5/5)
musk	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)	0.0000 / 0.0165 (5/5)
optdigits	0.4705 / 0.0305 (5/5)	0.4739 / 0.0304 (5/5)	0.4840 / 0.0327 (5/5)	0.5157 / 0.0349 (5/5)
pendigits	0.3963 / 0.0198 (5/5)	0.4090 / 0.0211 (5/5)	0.4001 / 0.0208 (5/5)	0.3995 / 0.0197 (5/5)
pima	0.4263 / 0.3285 (5/5)	0.4418 / 0.3359 (5/5)	0.4684 / 0.3525 (5/5)	0.3997 / 0.3093 (5/5)
satellite	0.3092 / 0.2606 (5/5)	0.3219 / 0.2666 (5/5)	0.3182 / 0.2625 (5/5)	0.3176 / 0.2638 (5/5)
satimage-2	0.0671 / 0.0068 (5/5)	0.0569 / 0.0064 (5/5)	0.0648 / 0.0068 (5/5)	0.0621 / 0.0066 (5/5)
shuttle	0.0250 / 0.0405 (5/5)	0.0186 / 0.0414 (5/5)	0.0066 / 0.0370 (5/5)	0.0040 / 0.0370 (5/5)
vertebral	0.5262 / 0.1967 (5/5)	0.4810 / 0.1580 (5/5)	0.5542 / 0.2269 (5/5)	0.5208 / 0.1910 (5/5)

Continued on next page

Table 37 continued

Dataset	100% retained	75% retained	50% retained	25% retained
vowels	0.4578 / 0.0578 (5/5)	0.4099 / 0.0560 (5/5)	0.4397 / 0.0546 (5/5)	0.3870 / 0.0360 (5/5)
wbc	0.1493 / 0.0370 (5/5)	0.1545 / 0.0357 (5/5)	0.1472 / 0.0344 (5/5)	0.1368 / 0.0363 (5/5)

Table 38: NSA/binary/hamming: dataset results on original features. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.5274 / 0.1559 (5/5)	0.5078 / 0.1513 (5/5)	0.5337 / 0.1618 (5/5)	0.5262 / 0.1594 (5/5)
cardio	0.5380 / 0.1139 (5/5)	0.5315 / 0.1090 (5/5)	0.5329 / 0.1104 (5/5)	0.5226 / 0.1066 (5/5)
glass	0.4061 / 0.0481 (5/5)	0.4162 / 0.0562 (5/5)	0.5070 / 0.0819 (5/5)	0.4457 / 0.0571 (5/5)
ionosphere	0.4629 / 0.3731 (5/5)	0.4983 / 0.3998 (5/5)	0.4917 / 0.3933 (5/5)	0.4924 / 0.3911 (5/5)
letter	0.5084 / 0.0760 (5/5)	0.5190 / 0.0788 (5/5)	0.5157 / 0.0775 (5/5)	0.5086 / 0.0723 (5/5)
lympho	0.6078 / 0.1373 (5/5)	0.5233 / 0.1061 (5/5)	0.5147 / 0.0674 (5/5)	0.4750 / 0.0897 (5/5)
mnist	0.5741 / 0.1219 (5/5)	0.5725 / 0.1220 (5/5)	0.5742 / 0.1218 (5/5)	0.5800 / 0.1220 (5/5)
musk	0.4335 / 0.0339 (5/5)	0.4273 / 0.0299 (5/5)	0.3874 / 0.0278 (5/5)	0.3921 / 0.0287 (5/5)
optdigits	0.4969 / 0.0330 (5/5)	0.4922 / 0.0327 (5/5)	0.4872 / 0.0318 (5/5)	0.4978 / 0.0338 (5/5)
pendigits	0.5305 / 0.0265 (5/5)	0.5245 / 0.0265 (5/5)	0.5076 / 0.0257 (5/5)	0.5036 / 0.0249 (5/5)
pima	0.4464 / 0.3343 (1/5)	0.4782 / 0.3459 (4/5)	0.4849 / 0.3452 (5/5)	0.4843 / 0.3422 (5/5)
satellite	0.4013 / 0.2815 (5/5)	0.4027 / 0.2821 (5/5)	0.4026 / 0.2826 (5/5)	0.3968 / 0.2800 (5/5)
satimage-2	0.2851 / 0.0116 (5/5)	0.2837 / 0.0114 (5/5)	0.2957 / 0.0116 (5/5)	0.2763 / 0.0115 (5/5)
shuttle	0.9472 / 0.7725 (5/5)	0.9514 / 0.7477 (5/5)	0.9016 / 0.5657 (5/5)	0.8968 / 0.4518 (5/5)
vertebral	— / — (0/5)	0.5055 / 0.1466 (4/5)	0.6263 / 0.2184 (5/5)	0.6102 / 0.1710 (5/5)
vowels	0.6729 / 0.0727 (5/5)	0.6759 / 0.0655 (5/5)	0.6453 / 0.0669 (5/5)	0.6169 / 0.0495 (5/5)
wbc	0.5760 / 0.0982 (5/5)	0.5736 / 0.0821 (5/5)	0.5809 / 0.0844 (5/5)	0.5670 / 0.0716 (5/5)

Table 39: NSA/binary/rcontiguous: dataset results on original features. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.4954 / 0.1673 (5/5)	0.4980 / 0.1628 (5/5)	0.4854 / 0.1571 (5/5)	0.4664 / 0.1518 (5/5)

Continued on next page

Table 39 continued

Dataset	100% retained	75% retained	50% retained	25% retained
cardio	0.6121 / 0.1465 (5/5)	0.6486 / 0.1553 (5/5)	0.6060 / 0.1297 (5/5)	0.5799 / 0.1185 (5/5)
glass	— / — (0/5)	— / — (0/5)	— / — (0/5)	0.4040 / 0.0511 (1/5)
ionosphere	0.5847 / 0.4472 (5/5)	0.5465 / 0.4144 (5/5)	0.5479 / 0.4238 (5/5)	0.5411 / 0.4208 (5/5)
letter	0.5666 / 0.0775 (5/5)	0.5596 / 0.0781 (5/5)	0.5425 / 0.0713 (5/5)	0.5526 / 0.0741 (5/5)
lympho	0.7483 / 0.3898 (5/5)	0.7517 / 0.3392 (5/5)	0.7690 / 0.3709 (5/5)	0.6905 / 0.1672 (5/5)
mnist	0.5668 / 0.1314 (5/5)	0.5637 / 0.1302 (5/5)	0.5652 / 0.1294 (5/5)	0.5588 / 0.1290 (5/5)
musk	0.4471 / 0.0352 (5/5)	0.5295 / 0.0500 (5/5)	0.5412 / 0.0460 (5/5)	0.4975 / 0.0362 (5/5)
optdigits	0.4702 / 0.0463 (5/5)	0.4657 / 0.0457 (5/5)	0.4690 / 0.0466 (5/5)	0.4404 / 0.0399 (5/5)
pendigits	0.4363 / 0.0258 (4/5)	0.4885 / 0.0286 (5/5)	0.4990 / 0.0253 (5/5)	0.5440 / 0.0293 (5/5)
pima	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
satellite	0.4709 / 0.3210 (5/5)	0.4537 / 0.3109 (5/5)	0.4501 / 0.3081 (5/5)	0.4714 / 0.3211 (5/5)
satimage-2	0.5966 / 0.0226 (5/5)	0.5777 / 0.0238 (5/5)	0.5447 / 0.0221 (5/5)	0.4765 / 0.0197 (5/5)
shuttle	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
vertebral	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
vowels	— / — (0/5)	— / — (0/5)	— / — (0/5)	0.6511 / 0.0622 (2/5)
wbc	0.4253 / 0.0832 (5/5)	0.4021 / 0.0592 (5/5)	0.3815 / 0.0833 (5/5)	0.4309 / 0.0632 (5/5)

Table 40: NSA/binary/rchunk: dataset results on original features. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.5245 / 0.1809 (5/5)	0.5309 / 0.1868 (5/5)	0.5257 / 0.1873 (5/5)	0.5276 / 0.1815 (5/5)
cardio	0.5757 / 0.1342 (5/5)	0.5554 / 0.1144 (5/5)	0.5594 / 0.1183 (5/5)	0.5149 / 0.1012 (5/5)
glass	0.4262 / 0.0552 (5/5)	0.4021 / 0.0494 (5/5)	0.5271 / 0.0584 (5/5)	0.4412 / 0.0458 (5/5)
ionosphere	0.5027 / 0.3694 (5/5)	0.4978 / 0.3735 (5/5)	0.5179 / 0.3827 (5/5)	0.4765 / 0.3576 (5/5)
letter	0.5336 / 0.0730 (5/5)	0.5233 / 0.0774 (5/5)	0.5200 / 0.0695 (5/5)	0.5168 / 0.0681 (5/5)
lympho	0.5552 / 0.0851 (5/5)	0.6483 / 0.1050 (5/5)	0.5474 / 0.0651 (5/5)	0.5405 / 0.0603 (5/5)
mnist	0.4711 / 0.0883 (5/5)	0.4689 / 0.0877 (5/5)	0.4682 / 0.0876 (5/5)	0.4645 / 0.0873 (5/5)
musk	0.4827 / 0.0364 (5/5)	0.4644 / 0.0365 (5/5)	0.4588 / 0.0335 (5/5)	0.4884 / 0.0337 (5/5)
optdigits	0.4405 / 0.0267 (5/5)	0.4362 / 0.0265 (5/5)	0.4528 / 0.0273 (5/5)	0.4419 / 0.0264 (5/5)

Continued on next page

Table 40 continued

Dataset	100% retained	75% retained	50% retained	25% retained
pendigits	0.5302 / 0.0279 (5/5)	0.5456 / 0.0298 (5/5)	0.5724 / 0.0306 (5/5)	0.5790 / 0.0302 (5/5)
pima	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
satellite	0.4516 / 0.3057 (5/5)	0.4690 / 0.3184 (5/5)	0.4683 / 0.3176 (5/5)	0.4788 / 0.3237 (5/5)
satimage-2	0.4932 / 0.0257 (5/5)	0.4726 / 0.0249 (5/5)	0.4555 / 0.0235 (5/5)	0.4386 / 0.0150 (5/5)
shuttle	0.6947 / 0.3104 (5/5)	0.6835 / 0.2918 (5/5)	0.6726 / 0.2536 (5/5)	0.6133 / 0.1477 (5/5)
vertebral	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
vowels	0.6468 / 0.1028 (5/5)	0.6512 / 0.0972 (5/5)	0.6666 / 0.0934 (5/5)	0.6818 / 0.0726 (5/5)
wbc	0.5747 / 0.0701 (5/5)	0.5890 / 0.0670 (5/5)	0.5800 / 0.0682 (5/5)	0.5820 / 0.0726 (5/5)

Table 41: NSA/fixed: dataset results on pca to two dimensions. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.5864 / 0.3586 (5/5)	0.6147 / 0.3637 (5/5)	0.6364 / 0.3695 (5/5)	0.6572 / 0.3868 (5/5)
cardio	0.9145 / 0.6175 (5/5)	0.9191 / 0.6000 (5/5)	0.9310 / 0.6644 (5/5)	0.9286 / 0.6385 (5/5)
glass	0.3909 / 0.1216 (5/5)	0.3604 / 0.1588 (5/5)	0.6213 / 0.1736 (5/5)	0.8646 / 0.3979 (5/5)
ionosphere	0.3435 / 0.3554 (5/5)	0.3620 / 0.3710 (5/5)	0.5302 / 0.4642 (5/5)	0.6238 / 0.4978 (5/5)
letter	0.2960 / 0.0452 (5/5)	0.3180 / 0.0479 (5/5)	0.3062 / 0.0430 (5/5)	0.3317 / 0.0463 (5/5)
lympho	0.7948 / 0.3311 (5/5)	0.8983 / 0.3553 (5/5)	0.8000 / 0.4216 (5/5)	0.8207 / 0.3190 (5/5)
mnist	0.5698 / 0.1418 (5/5)	0.6053 / 0.1800 (5/5)	0.5958 / 0.1688 (5/5)	0.5633 / 0.1296 (5/5)
musk	0.8764 / 0.7165 (5/5)	0.9132 / 0.6919 (5/5)	0.9105 / 0.7641 (5/5)	0.9278 / 0.7730 (5/5)
optdigits	0.2407 / 0.0178 (5/5)	0.2276 / 0.0176 (5/5)	0.2343 / 0.0180 (5/5)	0.2256 / 0.0176 (5/5)
pendigits	0.6891 / 0.1217 (5/5)	0.7264 / 0.1353 (5/5)	0.7743 / 0.2392 (5/5)	0.7415 / 0.2316 (5/5)
pima	0.5651 / 0.4250 (5/5)	0.5397 / 0.4024 (5/5)	0.6121 / 0.4679 (5/5)	0.6335 / 0.4970 (5/5)
satellite	0.6417 / 0.6135 (5/5)	0.6307 / 0.5951 (5/5)	0.5870 / 0.5570 (5/5)	0.5765 / 0.5507 (5/5)
satimage-2	0.7144 / 0.4888 (5/5)	0.6648 / 0.3547 (5/5)	0.6539 / 0.3165 (5/5)	0.4860 / 0.1932 (5/5)
shuttle	0.9628 / 0.9081 (5/5)	0.9656 / 0.8876 (5/5)	0.9708 / 0.8841 (5/5)	0.9561 / 0.7909 (5/5)
vertebral	0.3440 / 0.0997 (5/5)	0.3583 / 0.1059 (5/5)	0.3982 / 0.1100 (5/5)	0.4101 / 0.1335 (5/5)
vowels	0.3248 / 0.0268 (5/5)	0.3322 / 0.0266 (5/5)	0.3425 / 0.0266 (5/5)	0.3413 / 0.0307 (5/5)

Continued on next page

Table 41 continued

Dataset	100% retained	75% retained	50% retained	25% retained
wbc	0.6503 / 0.4361 (5/5)	0.6328 / 0.3634 (5/5)	0.6398 / 0.3704 (5/5)	0.6203 / 0.2711 (5/5)

Table 42: NSA/variable: dataset results on pca to two dimensions. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.6389 / 0.3679 (5/5)	0.6614 / 0.4331 (5/5)	0.6541 / 0.4121 (5/5)	0.7142 / 0.4301 (5/5)
cardio	0.9429 / 0.7453 (5/5)	0.9398 / 0.7276 (5/5)	0.9374 / 0.7120 (5/5)	0.9282 / 0.6643 (5/5)
glass	0.7579 / 0.2142 (5/5)	0.7573 / 0.2276 (5/5)	0.8915 / 0.3408 (5/5)	0.9122 / 0.3375 (5/5)
ionosphere	0.6768 / 0.5575 (5/5)	0.7136 / 0.5710 (5/5)	0.7387 / 0.5958 (5/5)	0.7464 / 0.5930 (5/5)
letter	0.3406 / 0.0539 (5/5)	0.3495 / 0.0545 (5/5)	0.3926 / 0.0584 (5/5)	0.4764 / 0.0903 (5/5)
lympho	0.8466 / 0.4657 (5/5)	0.8328 / 0.4186 (5/5)	0.9069 / 0.4670 (5/5)	0.7190 / 0.3144 (5/5)
mnist	0.6143 / 0.1824 (5/5)	0.6206 / 0.1930 (5/5)	0.6207 / 0.2023 (5/5)	0.6557 / 0.1990 (5/5)
musk	0.9995 / 0.9886 (5/5)	0.9988 / 0.9766 (5/5)	0.9986 / 0.9809 (5/5)	0.9942 / 0.9407 (5/5)
optdigits	0.2812 / 0.0188 (5/5)	0.2599 / 0.0184 (5/5)	0.2398 / 0.0180 (5/5)	0.2809 / 0.0193 (5/5)
pendigits	0.9211 / 0.4013 (5/5)	0.9475 / 0.4226 (5/5)	0.9552 / 0.4375 (5/5)	0.9143 / 0.3863 (5/5)
pima	0.6290 / 0.4845 (5/5)	0.6321 / 0.4864 (5/5)	0.6369 / 0.4925 (5/5)	0.6545 / 0.5084 (5/5)
satellite	0.7903 / 0.7590 (5/5)	0.7974 / 0.7632 (5/5)	0.7714 / 0.7321 (5/5)	0.7689 / 0.7204 (5/5)
satimage-2	0.8601 / 0.7198 (5/5)	0.8215 / 0.6403 (5/5)	0.7865 / 0.5665 (5/5)	0.6232 / 0.3535 (5/5)
shuttle	0.9700 / 0.9398 (5/5)	0.9731 / 0.9168 (5/5)	0.9797 / 0.9064 (5/5)	0.9685 / 0.8864 (5/5)
vertebral	0.3659 / 0.1159 (5/5)	0.3635 / 0.1080 (5/5)	0.3238 / 0.1011 (5/5)	0.4248 / 0.1156 (5/5)
vowels	0.3890 / 0.0427 (5/5)	0.3950 / 0.0397 (5/5)	0.5184 / 0.0700 (5/5)	0.7682 / 0.1350 (5/5)
wbc	0.7528 / 0.4928 (5/5)	0.8049 / 0.5417 (5/5)	0.7842 / 0.4979 (5/5)	0.7585 / 0.4527 (5/5)

Table 43: NSA/coverage: dataset results on pca to two dimensions. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.6371 / 0.3875 (5/5)	0.6442 / 0.4269 (5/5)	0.6498 / 0.4173 (5/5)	0.6908 / 0.4191 (5/5)
cardio	0.9432 / 0.7483 (5/5)	0.9429 / 0.7450 (5/5)	0.9443 / 0.7314 (5/5)	0.9286 / 0.6697 (5/5)

Continued on next page

Table 43 continued

Dataset	100% retained	75% retained	50% retained	25% retained
glass	0.6445 / 0.1592 (5/5)	0.6390 / 0.2192 (5/5)	0.8957 / 0.3359 (5/5)	0.8915 / 0.3467 (5/5)
ionosphere	0.6580 / 0.5438 (5/5)	0.6591 / 0.5353 (5/5)	0.7230 / 0.5937 (5/5)	0.7420 / 0.5855 (5/5)
letter	0.3116 / 0.0468 (5/5)	0.3408 / 0.0532 (5/5)	0.3678 / 0.0539 (5/5)	0.4453 / 0.0663 (5/5)
lympho	0.8086 / 0.3446 (5/5)	0.8328 / 0.3595 (5/5)	0.9069 / 0.4192 (5/5)	0.6190 / 0.2937 (5/5)
mnist	0.6172 / 0.1831 (5/5)	0.6218 / 0.1898 (5/5)	0.6159 / 0.1914 (5/5)	0.6247 / 0.1848 (5/5)
musk	0.9996 / 0.9896 (5/5)	0.9979 / 0.9643 (5/5)	0.9966 / 0.9652 (5/5)	0.9954 / 0.9425 (5/5)
optdigits	0.2879 / 0.0190 (5/5)	0.2795 / 0.0189 (5/5)	0.2198 / 0.0176 (5/5)	0.2575 / 0.0192 (5/5)
pendigits	0.8844 / 0.3615 (5/5)	0.8908 / 0.3754 (5/5)	0.9577 / 0.4584 (5/5)	0.8896 / 0.3394 (5/5)
pima	0.6253 / 0.4813 (5/5)	0.6208 / 0.4817 (5/5)	0.6446 / 0.5027 (5/5)	0.6553 / 0.5145 (5/5)
satellite	0.7364 / 0.7085 (5/5)	0.7516 / 0.7140 (5/5)	0.7625 / 0.7197 (5/5)	0.7488 / 0.7014 (5/5)
satimage-2	0.8432 / 0.7041 (5/5)	0.8170 / 0.6354 (5/5)	0.7814 / 0.5634 (5/5)	0.6275 / 0.3629 (5/5)
shuttle	0.9660 / 0.9196 (5/5)	0.9723 / 0.9165 (5/5)	0.9787 / 0.9075 (5/5)	0.9667 / 0.8819 (5/5)
vertebral	0.3786 / 0.1126 (5/5)	0.3621 / 0.1122 (5/5)	0.3468 / 0.1113 (5/5)	0.3716 / 0.1131 (5/5)
vowels	0.2879 / 0.0260 (5/5)	0.3370 / 0.0283 (5/5)	0.3928 / 0.0415 (5/5)	0.6125 / 0.0814 (5/5)
wbc	0.7498 / 0.4950 (5/5)	0.7872 / 0.5513 (5/5)	0.7823 / 0.4878 (5/5)	0.7554 / 0.4510 (5/5)

Table 44: NSA/grid: dataset results on pca to two dimensions. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.6371 / 0.3669 (5/5)	0.6620 / 0.4202 (5/5)	0.6557 / 0.4119 (5/5)	0.7134 / 0.4342 (5/5)
cardio	0.9428 / 0.7473 (5/5)	0.9393 / 0.7262 (5/5)	0.9375 / 0.7136 (5/5)	0.9274 / 0.6489 (5/5)
glass	0.7567 / 0.1882 (5/5)	0.7591 / 0.2605 (5/5)	0.8927 / 0.3376 (5/5)	0.9183 / 0.3594 (5/5)
ionosphere	0.6766 / 0.5563 (5/5)	0.7119 / 0.5662 (5/5)	0.7386 / 0.5916 (5/5)	0.7447 / 0.5800 (5/5)
letter	0.3386 / 0.0531 (5/5)	0.3505 / 0.0546 (5/5)	0.3918 / 0.0578 (5/5)	0.4762 / 0.0909 (5/5)
lympho	0.8448 / 0.4591 (5/5)	0.8328 / 0.4186 (5/5)	0.9017 / 0.4824 (5/5)	0.7448 / 0.3443 (5/5)
mnist	0.6137 / 0.1819 (5/5)	0.6183 / 0.1929 (5/5)	0.6217 / 0.2038 (5/5)	0.6548 / 0.1977 (5/5)
musk	0.9994 / 0.9876 (5/5)	0.9988 / 0.9762 (5/5)	0.9986 / 0.9818 (5/5)	0.9941 / 0.9411 (5/5)

Continued on next page

Table 44 continued

Dataset	100% retained	75% retained	50% retained	25% retained
optdigits	0.2782 / 0.0187 (5/5)	0.2593 / 0.0184 (5/5)	0.2440 / 0.0181 (5/5)	0.2792 / 0.0192 (5/5)
pendigits	0.9166 / 0.3956 (5/5)	0.9471 / 0.4264 (5/5)	0.9554 / 0.4270 (5/5)	0.9113 / 0.3905 (5/5)
pima	0.6301 / 0.4839 (5/5)	0.6320 / 0.4850 (5/5)	0.6356 / 0.4887 (5/5)	0.6549 / 0.5099 (5/5)
satellite	0.7851 / 0.7543 (5/5)	0.7920 / 0.7558 (5/5)	0.7624 / 0.7204 (5/5)	0.7466 / 0.7021 (5/5)
satimage-2	0.8187 / 0.6361 (5/5)	0.7728 / 0.5604 (5/5)	0.7402 / 0.5409 (5/5)	0.6704 / 0.4045 (5/5)
shuttle	0.9700 / 0.9374 (5/5)	0.9733 / 0.9240 (5/5)	0.9808 / 0.9087 (5/5)	0.9673 / 0.8854 (5/5)
vertebral	0.3667 / 0.1165 (5/5)	0.3579 / 0.1050 (5/5)	0.3262 / 0.1017 (5/5)	0.4312 / 0.1210 (5/5)
vowels	0.3890 / 0.0431 (5/5)	0.3954 / 0.0398 (5/5)	0.5202 / 0.0703 (5/5)	0.7669 / 0.1319 (5/5)
wbc	0.7417 / 0.4875 (5/5)	0.7997 / 0.5237 (5/5)	0.7748 / 0.5159 (5/5)	0.7418 / 0.3932 (5/5)

Table 45: NSA/hierarchical: dataset results on pca to two dimensions. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.6220 / 0.3533 (5/5)	0.6722 / 0.4029 (5/5)	0.6557 / 0.3796 (5/5)	0.6905 / 0.4343 (5/5)
cardio	0.9425 / 0.7449 (5/5)	0.9418 / 0.7320 (5/5)	0.9318 / 0.7013 (5/5)	0.9287 / 0.6209 (5/5)
glass	0.7616 / 0.2543 (5/5)	0.6201 / 0.1788 (5/5)	0.8994 / 0.4502 (5/5)	0.8896 / 0.3077 (5/5)
ionosphere	0.6783 / 0.5524 (5/5)	0.7272 / 0.6027 (5/5)	0.7344 / 0.5956 (5/5)	0.7356 / 0.5679 (5/5)
letter	0.3227 / 0.0493 (5/5)	0.3462 / 0.0538 (5/5)	0.3731 / 0.0524 (5/5)	0.3978 / 0.0708 (5/5)
lympho	0.8828 / 0.4787 (5/5)	0.8379 / 0.3065 (5/5)	0.8517 / 0.3848 (5/5)	0.7207 / 0.3629 (5/5)
mnist	0.6115 / 0.1822 (5/5)	0.6162 / 0.1935 (5/5)	0.6300 / 0.2083 (5/5)	0.6320 / 0.1948 (5/5)
musk	0.9977 / 0.9728 (5/5)	0.9978 / 0.9616 (5/5)	0.9964 / 0.9584 (5/5)	0.9923 / 0.9080 (5/5)
optdigits	0.2587 / 0.0183 (5/5)	0.1992 / 0.0170 (5/5)	0.2276 / 0.0179 (5/5)	0.2181 / 0.0180 (5/5)
pendigits	0.8911 / 0.3567 (5/5)	0.8986 / 0.3530 (5/5)	0.9488 / 0.4751 (5/5)	0.9095 / 0.3698 (5/5)
pima	0.6279 / 0.4882 (5/5)	0.6209 / 0.4884 (5/5)	0.6264 / 0.4861 (5/5)	0.6337 / 0.4775 (5/5)
satellite	0.7040 / 0.6719 (5/5)	0.7232 / 0.6867 (5/5)	0.6903 / 0.6457 (5/5)	0.6787 / 0.6265 (5/5)
satimage-2	0.7946 / 0.6050 (5/5)	0.7449 / 0.4951 (5/5)	0.7169 / 0.4612 (5/5)	0.4998 / 0.2307 (5/5)
shuttle	0.9632 / 0.9029 (5/5)	0.9702 / 0.9074 (5/5)	0.9769 / 0.8870 (5/5)	0.9592 / 0.8197 (5/5)

Continued on next page

Table 45 continued

Dataset	100% retained	75% retained	50% retained	25% retained
vertebral	0.3976 / 0.1287 (5/5)	0.3625 / 0.1082 (5/5)	0.3651 / 0.1124 (5/5)	0.4181 / 0.1197 (5/5)
vowels	0.3661 / 0.0362 (5/5)	0.3665 / 0.0360 (5/5)	0.4496 / 0.0533 (5/5)	0.6214 / 0.0997 (5/5)
wbc	0.6595 / 0.3748 (5/5)	0.7042 / 0.4297 (5/5)	0.6920 / 0.3824 (5/5)	0.6951 / 0.3271 (5/5)

Table 46: NSA/voronoi: dataset results on pca to two dimensions. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.6330 / 0.3771 (5/5)	0.6651 / 0.4296 (5/5)	0.6534 / 0.4081 (5/5)	0.7092 / 0.4264 (5/5)
cardio	0.9400 / 0.7363 (5/5)	0.9398 / 0.7354 (5/5)	0.9360 / 0.7145 (5/5)	0.9259 / 0.6435 (5/5)
glass	0.7659 / 0.1952 (5/5)	0.8232 / 0.2603 (5/5)	0.9000 / 0.3888 (5/5)	0.9238 / 0.3657 (5/5)
ionosphere	0.7138 / 0.5824 (5/5)	0.7284 / 0.5870 (5/5)	0.7441 / 0.6057 (5/5)	0.7503 / 0.5929 (5/5)
letter	0.3560 / 0.0568 (5/5)	0.3616 / 0.0571 (5/5)	0.4170 / 0.0678 (5/5)	0.5302 / 0.0963 (5/5)
lympho	0.8500 / 0.4736 (5/5)	0.8190 / 0.4656 (5/5)	0.8672 / 0.3814 (5/5)	0.7379 / 0.4037 (5/5)
mnist	0.6054 / 0.1745 (5/5)	0.6189 / 0.1838 (5/5)	0.6177 / 0.1954 (5/5)	0.6509 / 0.1900 (5/5)
musk	0.9993 / 0.9852 (5/5)	0.9988 / 0.9751 (5/5)	0.9987 / 0.9809 (5/5)	0.9930 / 0.9381 (5/5)
optdigits	0.2684 / 0.0186 (5/5)	0.2818 / 0.0189 (5/5)	0.2626 / 0.0185 (5/5)	0.3707 / 0.0263 (5/5)
pendigits	0.9303 / 0.4253 (5/5)	0.9614 / 0.4386 (5/5)	0.9614 / 0.4111 (5/5)	0.8894 / 0.3105 (5/5)
pima	0.6301 / 0.4831 (5/5)	0.6481 / 0.4947 (5/5)	0.6388 / 0.4930 (5/5)	0.6601 / 0.5124 (5/5)
satellite	0.8112 / 0.7849 (5/5)	0.8141 / 0.7856 (5/5)	0.8019 / 0.7680 (5/5)	0.7943 / 0.7544 (5/5)
satimage-2	0.8914 / 0.7522 (5/5)	0.8755 / 0.7131 (5/5)	0.8528 / 0.6485 (5/5)	0.7767 / 0.5304 (5/5)
shuttle	0.9723 / 0.9351 (5/5)	0.9760 / 0.9159 (5/5)	0.9830 / 0.9006 (5/5)	0.9316 / 0.8246 (5/5)
vertebral	0.3625 / 0.1183 (5/5)	0.3569 / 0.1082 (5/5)	0.3800 / 0.1136 (5/5)	0.4341 / 0.1252 (5/5)
vowels	0.4303 / 0.0554 (5/5)	0.4779 / 0.0537 (5/5)	0.6167 / 0.0862 (5/5)	0.7876 / 0.1529 (5/5)
wbc	0.8130 / 0.5324 (5/5)	0.8332 / 0.5805 (5/5)	0.8038 / 0.5538 (5/5)	0.7686 / 0.4974 (5/5)

Table 47: NSA/deterministic: dataset results on pca to two dimensions. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.5772 / 0.3446 (5/5)	0.6117 / 0.3876 (5/5)	0.6365 / 0.4110 (5/5)	0.6645 / 0.4118 (5/5)
cardio	0.9476 / 0.7444 (5/5)	0.9425 / 0.7088 (5/5)	0.9414 / 0.7007 (5/5)	0.9389 / 0.6570 (5/5)
glass	0.3732 / 0.0844 (5/5)	0.4128 / 0.0962 (5/5)	0.7317 / 0.2859 (5/5)	0.8707 / 0.2985 (5/5)
ionosphere	0.4406 / 0.4118 (5/5)	0.4810 / 0.4445 (5/5)	0.5858 / 0.5068 (5/5)	0.6688 / 0.5383 (5/5)
letter	0.3248 / 0.0488 (5/5)	0.3323 / 0.0496 (5/5)	0.3711 / 0.0523 (5/5)	0.3798 / 0.0601 (5/5)
lympho	0.8207 / 0.3365 (5/5)	0.8828 / 0.4415 (5/5)	0.8362 / 0.5965 (5/5)	0.7241 / 0.2095 (5/5)
mnist	0.6289 / 0.2006 (5/5)	0.6079 / 0.1724 (5/5)	0.6029 / 0.1803 (5/5)	0.5915 / 0.1740 (5/5)
musk	0.9736 / 0.8890 (5/5)	0.9694 / 0.8311 (5/5)	0.9869 / 0.9016 (5/5)	0.9807 / 0.9025 (5/5)
optdigits	0.2776 / 0.0186 (5/5)	0.2944 / 0.0191 (5/5)	0.2660 / 0.0187 (5/5)	0.2529 / 0.0184 (5/5)
pendigits	0.7294 / 0.1203 (5/5)	0.8047 / 0.2277 (5/5)	0.8957 / 0.3387 (5/5)	0.7897 / 0.2145 (5/5)
pima	0.5368 / 0.4008 (5/5)	0.5229 / 0.3912 (5/5)	0.5486 / 0.4107 (5/5)	0.6266 / 0.4968 (5/5)
satellite	0.5890 / 0.5674 (5/5)	0.6003 / 0.5744 (5/5)	0.5838 / 0.5651 (5/5)	0.5770 / 0.5517 (5/5)
satimage-2	0.6922 / 0.4602 (5/5)	0.6339 / 0.3522 (5/5)	0.5780 / 0.2806 (5/5)	0.4466 / 0.1315 (5/5)
shuttle	0.9604 / 0.9246 (5/5)	0.9724 / 0.9354 (5/5)	0.9720 / 0.9217 (5/5)	0.9592 / 0.8154 (5/5)
vertebral	0.3859 / 0.1135 (5/5)	0.3980 / 0.1121 (5/5)	0.4317 / 0.1210 (5/5)	0.4450 / 0.1476 (5/5)
vowels	0.3697 / 0.0294 (5/5)	0.3641 / 0.0292 (5/5)	0.3879 / 0.0398 (5/5)	0.3883 / 0.0530 (5/5)
wbc	0.6262 / 0.4155 (5/5)	0.6214 / 0.3222 (5/5)	0.6243 / 0.3465 (5/5)	0.6102 / 0.1836 (5/5)

Table 48: NSA/suppressed: dataset results on pca to two dimensions. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.6343 / 0.3439 (5/5)	0.6559 / 0.4137 (5/5)	0.6462 / 0.4016 (5/5)	0.6895 / 0.4226 (5/5)
cardio	0.9188 / 0.6540 (5/5)	0.9204 / 0.6668 (5/5)	0.9184 / 0.6625 (5/5)	0.9031 / 0.6006 (5/5)
glass	0.7134 / 0.1641 (5/5)	0.7232 / 0.2309 (5/5)	0.8378 / 0.3356 (5/5)	0.9000 / 0.3117 (5/5)
ionosphere	0.6905 / 0.5593 (5/5)	0.7006 / 0.5630 (5/5)	0.7248 / 0.5934 (5/5)	0.7210 / 0.5689 (5/5)
letter	0.3393 / 0.0588 (5/5)	0.3463 / 0.0536 (5/5)	0.3962 / 0.0649 (5/5)	0.4911 / 0.0889 (5/5)

Continued on next page

Table 48 continued

Dataset	100% retained	75% retained	50% retained	25% retained
lympho	0.8000 / 0.4457 (5/5)	0.7707 / 0.4663 (5/5)	0.8707 / 0.4311 (5/5)	0.7310 / 0.3785 (5/5)
mnist	0.6061 / 0.1664 (5/5)	0.6149 / 0.1842 (5/5)	0.6106 / 0.1846 (5/5)	0.6383 / 0.1824 (5/5)
musk	0.9814 / 0.9541 (5/5)	0.9722 / 0.9310 (5/5)	0.9870 / 0.9717 (5/5)	0.9714 / 0.9273 (5/5)
optdigits	0.2862 / 0.0189 (5/5)	0.2888 / 0.0190 (5/5)	0.2586 / 0.0185 (5/5)	0.2894 / 0.0203 (5/5)
pendigits	0.8494 / 0.3072 (5/5)	0.8794 / 0.3033 (5/5)	0.8787 / 0.3298 (5/5)	0.8437 / 0.2956 (5/5)
pima	0.6232 / 0.4649 (5/5)	0.6368 / 0.4767 (5/5)	0.6330 / 0.4833 (5/5)	0.6412 / 0.4981 (5/5)
satellite	0.7516 / 0.7145 (5/5)	0.7704 / 0.7307 (5/5)	0.7550 / 0.7171 (5/5)	0.6968 / 0.6692 (5/5)
satimage-2	0.8236 / 0.7138 (5/5)	0.7857 / 0.6217 (5/5)	0.8030 / 0.6245 (5/5)	0.7158 / 0.4995 (5/5)
shuttle	0.9621 / 0.9162 (5/5)	0.9675 / 0.9079 (5/5)	0.9769 / 0.9105 (5/5)	0.9635 / 0.8860 (5/5)
vertebral	0.3446 / 0.1048 (5/5)	0.3375 / 0.1057 (5/5)	0.3573 / 0.1053 (5/5)	0.4804 / 0.1340 (5/5)
vowels	0.3746 / 0.0528 (5/5)	0.4457 / 0.0578 (5/5)	0.5444 / 0.0954 (5/5)	0.7189 / 0.1564 (5/5)
wbc	0.7561 / 0.5239 (5/5)	0.8130 / 0.5578 (5/5)	0.8082 / 0.5530 (5/5)	0.7095 / 0.4760 (5/5)

Table 49: NSA/dual: dataset results on pca to two dimensions. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.6367 / 0.3770 (5/5)	0.6569 / 0.4298 (5/5)	0.6445 / 0.4052 (5/5)	0.7113 / 0.4481 (5/5)
cardio	0.9424 / 0.7414 (5/5)	0.9399 / 0.7335 (5/5)	0.9369 / 0.7233 (5/5)	0.9274 / 0.6582 (5/5)
glass	0.7622 / 0.1988 (5/5)	0.7707 / 0.2608 (5/5)	0.8878 / 0.3606 (5/5)	0.9122 / 0.3174 (5/5)
ionosphere	0.6933 / 0.5713 (5/5)	0.7202 / 0.5915 (5/5)	0.7369 / 0.5928 (5/5)	0.7433 / 0.5771 (5/5)
letter	0.3732 / 0.0559 (5/5)	0.3926 / 0.0590 (5/5)	0.4245 / 0.0610 (5/5)	0.5155 / 0.0917 (5/5)
lympho	0.8500 / 0.5266 (5/5)	0.8362 / 0.4705 (5/5)	0.8948 / 0.4836 (5/5)	0.7310 / 0.4008 (5/5)
mnist	0.6212 / 0.1793 (5/5)	0.6215 / 0.1966 (5/5)	0.6171 / 0.1946 (5/5)	0.6448 / 0.1929 (5/5)
musk	0.9995 / 0.9883 (5/5)	0.9991 / 0.9813 (5/5)	0.9988 / 0.9808 (5/5)	0.9937 / 0.9451 (5/5)
optdigits	0.3492 / 0.0208 (5/5)	0.3477 / 0.0206 (5/5)	0.3053 / 0.0195 (5/5)	0.2992 / 0.0201 (5/5)
pendigits	0.9102 / 0.3953 (5/5)	0.9580 / 0.4417 (5/5)	0.9540 / 0.4263 (5/5)	0.9045 / 0.3766 (5/5)
pima	0.6350 / 0.4888 (5/5)	0.6448 / 0.4922 (5/5)	0.6433 / 0.4962 (5/5)	0.6606 / 0.5134 (5/5)

Continued on next page

Table 49 continued

Dataset	100% retained	75% retained	50% retained	25% retained
satellite	0.8165 / 0.7920 (5/5)	0.8149 / 0.7871 (5/5)	0.8157 / 0.7847 (5/5)	0.8179 / 0.7823 (5/5)
satimage-2	0.9266 / 0.8555 (5/5)	0.9398 / 0.8824 (5/5)	0.9423 / 0.8697 (5/5)	0.9664 / 0.8691 (5/5)
shuttle	0.9787 / 0.9241 (5/5)	0.9846 / 0.9109 (5/5)	0.9886 / 0.8986 (5/5)	0.9834 / 0.8483 (5/5)
vertebral	0.3819 / 0.1200 (5/5)	0.3665 / 0.1084 (5/5)	0.3506 / 0.1056 (5/5)	0.4327 / 0.1421 (5/5)
vowels	0.4153 / 0.0482 (5/5)	0.4477 / 0.0446 (5/5)	0.5490 / 0.0682 (5/5)	0.7654 / 0.1394 (5/5)
wbc	0.9002 / 0.7365 (5/5)	0.9139 / 0.7107 (5/5)	0.9215 / 0.6907 (5/5)	0.9153 / 0.6657 (5/5)

Table 50: NSA/annealed: dataset results on pca to two dimensions. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	0.5894 / 0.3805 (5/5)	0.6210 / 0.3987 (5/5)	0.6303 / 0.3975 (5/5)	0.6681 / 0.4155 (5/5)
cardio	0.9470 / 0.7620 (5/5)	0.9445 / 0.7436 (5/5)	0.9410 / 0.7037 (5/5)	0.9294 / 0.6312 (5/5)
glass	0.2872 / 0.0874 (5/5)	0.4378 / 0.1026 (5/5)	0.7793 / 0.2673 (5/5)	0.8268 / 0.3127 (5/5)
ionosphere	0.3875 / 0.3731 (5/5)	0.4006 / 0.3708 (5/5)	0.4665 / 0.4174 (5/5)	0.6864 / 0.5481 (5/5)
letter	0.3077 / 0.0464 (5/5)	0.3162 / 0.0474 (5/5)	0.3441 / 0.0481 (5/5)	0.3504 / 0.0491 (5/5)
lympho	0.8431 / 0.3534 (5/5)	0.8603 / 0.4113 (5/5)	0.8310 / 0.4925 (5/5)	0.7224 / 0.2360 (5/5)
mnist	0.6086 / 0.1858 (5/5)	0.5984 / 0.1686 (5/5)	0.6071 / 0.1820 (5/5)	0.5435 / 0.1374 (5/5)
musk	0.9395 / 0.8005 (5/5)	0.9404 / 0.7978 (5/5)	0.9425 / 0.8580 (5/5)	0.9712 / 0.8498 (5/5)
optdigits	0.2459 / 0.0182 (5/5)	0.2996 / 0.0193 (5/5)	0.2542 / 0.0184 (5/5)	0.2214 / 0.0176 (5/5)
pendigits	0.7894 / 0.1572 (5/5)	0.7874 / 0.2619 (5/5)	0.8589 / 0.2918 (5/5)	0.7509 / 0.2007 (5/5)
pima	0.5998 / 0.4500 (5/5)	0.6141 / 0.4827 (5/5)	0.6340 / 0.4866 (5/5)	0.6031 / 0.4625 (5/5)
satellite	0.6532 / 0.6320 (5/5)	0.6592 / 0.6326 (5/5)	0.6183 / 0.5969 (5/5)	0.6120 / 0.5884 (5/5)
satimage-2	0.7273 / 0.5242 (5/5)	0.6931 / 0.4352 (5/5)	0.6445 / 0.3284 (5/5)	0.5117 / 0.1729 (5/5)
shuttle	0.9500 / 0.8860 (5/5)	0.9696 / 0.9195 (5/5)	0.9663 / 0.8855 (5/5)	0.9505 / 0.7829 (5/5)
vertebral	0.3946 / 0.1181 (5/5)	0.4081 / 0.1218 (5/5)	0.3732 / 0.1156 (5/5)	0.3831 / 0.1167 (5/5)
vowels	0.2951 / 0.0315 (5/5)	0.3085 / 0.0276 (5/5)	0.3220 / 0.0263 (5/5)	0.3390 / 0.0446 (5/5)
wbc	0.6312 / 0.4366 (5/5)	0.6523 / 0.3739 (5/5)	0.6679 / 0.4041 (5/5)	0.6160 / 0.2183 (5/5)

Table 51: NSA/binary/hamming: dataset results on pca to two dimensions. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
cardio	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
glass	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
ionosphere	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
letter	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
lympho	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
mnist	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
musk	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
optdigits	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
pendigits	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
pima	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
satellite	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
satimage-2	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
shuttle	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
vertebral	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
vowels	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
wbc	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)

Table 52: NSA/binary/rcontiguous: dataset results on pca to two dimensions. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
cardio	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
glass	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
ionosphere	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
letter	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
lympho	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
mnist	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
musk	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
optdigits	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
pendigits	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
pima	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
satellite	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
satimage-2	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
shuttle	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
vertebral	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
vowels	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
wbc	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)

Table 53: NSA/binary/rchunk: dataset results on pca to two dimensions. Cell format is AUROC / AP (completed seeds).

Dataset	100% retained	75% retained	50% retained	25% retained
arrhythmia	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
cardio	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
glass	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)

Continued on next page

Table 53 continued

Dataset	100% retained	75% retained	50% retained	25% retained
ionosphere	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
letter	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
lympho	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
mnist	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
musk	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
optdigits	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
pendigits	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
pima	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
satellite	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
satimage-2	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
shuttle	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
vertebral	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
vowels	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)
wbc	— / — (0/5)	— / — (0/5)	— / — (0/5)	— / — (0/5)

J Failure and applicability accounting

Failures are not assigned a detection score. XGBOD and DevNet require labeled anomalies for their intended supervised training and are inapplicable under this protocol. Geometry restrictions are separate: MAD requires one dimension; the Voronoi NSA configuration requires two or three; deterministic NSA requires two. All other listed configurations were attempted when applicable. Errors and timeouts describe this implementation and compute setting, rather than proving that the detector mathematically requires outlier examples.

Table 54: Terminal outcome totals across all 38,080 configuration runs, after measured-time qualification.

Outcome	Count	Share
success	32392	85.06%
overbudget success	118	0.31%
timeout	1548	4.07%
error	1301	3.42%
build error	1	0.00%
requires labeled outliers	1360	3.57%
requires 1d	680	1.79%
requires 2d or 3d	340	0.89%
requires 2d	340	0.89%

Table 55: Outcome counts for each configuration across 680 expected runs. S=qualified success, O=recorded success over budget, T=timeout, E=error, B=build error, L=requires labeled outliers; the last three columns are dimension restrictions.

Configuration	S	O	T	E	B	L	1D	2/3D	2D
NSA fixed	680	0	0	0	0	0	0	0	0
NSA variable	680	0	0	0	0	0	0	0	0
NSA coverage	680	0	0	0	0	0	0	0	0
NSA grid	679	0	1	0	0	0	0	0	0
NSA hierarchical	680	0	0	0	0	0	0	0	0
NSA voronoi	340	0	0	0	0	0	0	340	0
NSA deterministic	340	0	0	0	0	0	0	0	340
NSA suppressed	680	0	0	0	0	0	0	0	0
NSA dual	665	0	0	15	0	0	0	0	0
NSA annealed	680	0	0	0	0	0	0	0	0
NSA binary Hamming	329	0	0	351	0	0	0	0	0
NSA binary contiguous	242	0	0	438	0	0	0	0	0
NSA binary chunk	300	0	0	380	0	0	0	0	0
ECOD	672	8	0	0	0	0	0	0	0
ABOD	680	0	0	0	0	0	0	0	0
COPOD	680	0	0	0	0	0	0	0	0
MAD	0	0	0	0	0	0	680	0	0
SOS	676	0	2	2	0	0	0	0	0
QMCD	679	0	1	0	0	0	0	0	0
KDE	680	0	0	0	0	0	0	0	0
Sampling	680	0	0	0	0	0	0	0	0
GMM	680	0	0	0	0	0	0	0	0
PCA	668	0	0	12	0	0	0	0	0
KPCA	680	0	0	0	0	0	0	0	0
MCD	680	0	0	0	0	0	0	0	0
CD	651	0	13	16	0	0	0	0	0
OCSVM	680	0	0	0	0	0	0	0	0
LMDD	557	0	123	0	0	0	0	0	0
LOF	680	0	0	0	0	0	0	0	0

Continued on next page

Table 55 continued

Configuration	S	O	T	E	B	L	1D	2/3D	2D
COF	676	0	4	0	0	0	0	0	0
CBLOF	646	4	28	1	1	0	0	0	0
LOCI	205	5	470	0	0	0	0	0	0
HBOS	651	9	20	0	0	0	0	0	0
HDBSCAN	680	0	0	0	0	0	0	0	0
KNN	680	0	0	0	0	0	0	0	0
SOD	509	55	116	0	0	0	0	0	0
ROD	473	8	198	1	0	0	0	0	0
IForest	680	0	0	0	0	0	0	0	0
INNE	680	0	0	0	0	0	0	0	0
DIF	656	24	0	0	0	0	0	0	0
FeatureBagging	680	0	0	0	0	0	0	0	0
LSCP	659	0	11	10	0	0	0	0	0
LODA	680	0	0	0	0	0	0	0	0
SUOD	589	1	89	1	0	0	0	0	0
XGBOD	0	0	0	0	0	680	0	0	0
AutoEncoder	627	4	19	30	0	0	0	0	0
VAE	649	0	0	31	0	0	0	0	0
SO_GAAL	680	0	0	0	0	0	0	0	0
MO_GAAL	680	0	0	0	0	0	0	0	0
DeepSVDD	680	0	0	0	0	0	0	0	0
AnoGAN	660	0	20	0	0	0	0	0	0
ALAD	680	0	0	0	0	0	0	0	0
AE1SVM	669	0	0	11	0	0	0	0	0
DevNet	0	0	0	0	0	680	0	0	0
RGraph	245	0	433	2	0	0	0	0	0
LUNAR	680	0	0	0	0	0	0	0	0

The raw evidence contains one incomplete SOD timeout on Satellite, original features, seed 43, 75% retained. AUROC was partially computed as 0.595705, but AP was absent; neither metric enters the scored tables. One HBOS AP value exceeds one by machine rounding (1.0000000000000002); the raw value is preserved and accepted only within a 10^{-12} numerical tolerance.

Twelve scorers depend on the query batch: CD, COF, COPOD, DIF, ECOD, LMDD, LOCI, RGraph, ROD, SOD, SOS, SUOD. They receive the same fixed mixed test batches as other methods. Their batch dependence means this is not a pure one-row inductive comparison. CD, DIF, LMDD, and ROD were identified by source review as batch dependent even though the active factory metadata underflagged them; the analysis reporting flag was corrected without changing their implementation.

K Sensitivity to the measured time qualification

This section compares all recorded successes with the primary successes that satisfy the 15-second measured fitting-plus-scoring limit. The 118 successful-but-over-budget runs keep their original scores and statuses in raw files. For panels with incomplete coverage, either macro mean is available-case, so any difference can reflect a change in the successful subset.

Table 56: Recorded-success sensitivity: Original features. In each cell the first line gives raw AUROC/AP (raw successes), and the second gives qualified AUROC/AP (qualified successes), both out of 85.

Configuration	100% retained	75% retained	50% retained	25% retained
NSA fixed	0.3597/0.1287 (85)	0.3504/0.1272 (85)	0.3460/0.1232 (85)	0.3297/0.1211 (85)
	0.3597/0.1287 (85)	0.3504/0.1272 (85)	0.3460/0.1232 (85)	0.3297/0.1211 (85)
NSA variable	0.5931/0.3038 (85)	0.5821/0.2865 (85)	0.5531/0.2456 (85)	0.5025/0.2054 (85)
	0.5931/0.3038 (85)	0.5821/0.2865 (85)	0.5531/0.2456 (85)	0.5025/0.2054 (85)
NSA coverage	0.5931/0.3040 (85)	0.5814/0.2855 (85)	0.5534/0.2455 (85)	0.5028/0.2053 (85)
	0.5931/0.3040 (85)	0.5814/0.2855 (85)	0.5534/0.2455 (85)	0.5028/0.2053 (85)
NSA grid	0.5972/0.3026 (85)	0.5880/0.2869 (85)	0.5569/0.2510 (85)	0.5025/0.1994 (84)
	0.5972/0.3026 (85)	0.5880/0.2869 (85)	0.5569/0.2510 (85)	0.5025/0.1994 (84)
NSA hierarchical	0.2968/0.1173 (85)	0.3078/0.1145 (85)	0.2753/0.0938 (85)	0.2712/0.1092 (85)
	0.2968/0.1173 (85)	0.3078/0.1145 (85)	0.2753/0.0938 (85)	0.2712/0.1092 (85)
NSA voronoi	—/— (0)	—/— (0)	—/— (0)	—/— (0)
	—/— (0)	—/— (0)	—/— (0)	—/— (0)
NSA deterministic	—/— (0)	—/— (0)	—/— (0)	—/— (0)
	—/— (0)	—/— (0)	—/— (0)	—/— (0)
NSA suppressed	0.5883/0.2982 (85)	0.5656/0.2705 (85)	0.5516/0.2440 (85)	0.4920/0.1989 (85)
	0.5883/0.2982 (85)	0.5656/0.2705 (85)	0.5516/0.2440 (85)	0.4920/0.1989 (85)
NSA dual	0.7047/0.4609 (81)	0.7352/0.5006 (80)	0.7318/0.4966 (81)	0.7488/0.5142 (83)
	0.7047/0.4609 (81)	0.7352/0.5006 (80)	0.7318/0.4966 (81)	0.7488/0.5142 (83)
NSA annealed	0.3161/0.1120 (85)	0.3112/0.1058 (85)	0.3023/0.1070 (85)	0.2789/0.0988 (85)
	0.3161/0.1120 (85)	0.3112/0.1058 (85)	0.3023/0.1070 (85)	0.2789/0.0988 (85)
NSA binary Hamming	0.5259/0.1681 (76)	0.5226/0.1643 (83)	0.5288/0.1573 (85)	0.5172/0.1449 (85)
	0.5259/0.1681 (76)	0.5226/0.1643 (83)	0.5288/0.1573 (85)	0.5172/0.1449 (85)
NSA binary contiguous	0.5350/0.1578 (59)	0.5404/0.1498 (60)	0.5335/0.1511 (60)	0.5218/0.1203 (63)
	0.5350/0.1578 (59)	0.5404/0.1498 (60)	0.5335/0.1511 (60)	0.5218/0.1203 (63)
NSA binary chunk	0.5269/0.1261 (75)	0.5292/0.1258 (75)	0.5328/0.1211 (75)	0.5191/0.1082 (75)
	0.5269/0.1261 (75)	0.5292/0.1258 (75)	0.5328/0.1211 (75)	0.5191/0.1082 (75)
ECOD	0.7853/0.4727 (85)	0.7824/0.4631 (85)	0.7783/0.4520 (85)	0.7738/0.4399 (85)
	0.7855/0.4741 (77)	0.7824/0.4631 (85)	0.7783/0.4520 (85)	0.7738/0.4399 (85)

Continued on next page

Table 56 continued

Configuration	100% retained	75% retained	50% retained	25% retained
ABOD	0.8951/0.6279 (85)	0.8927/0.6118 (85)	0.8841/0.5849 (85)	0.8724/0.5576 (85)
	0.8951/0.6279 (85)	0.8927/0.6118 (85)	0.8841/0.5849 (85)	0.8724/0.5576 (85)
COPOD	0.7865/0.4573 (85)	0.7830/0.4587 (85)	0.7837/0.4551 (85)	0.7797/0.4464 (85)
	0.7865/0.4573 (85)	0.7830/0.4587 (85)	0.7837/0.4551 (85)	0.7797/0.4464 (85)
MAD	—/— (0)	—/— (0)	—/— (0)	—/— (0)
	—/— (0)	—/— (0)	—/— (0)	—/— (0)
SOS	0.5868/0.2218 (85)	0.5883/0.2283 (84)	0.5897/0.2296 (85)	0.5884/0.2196 (82)
	0.5868/0.2218 (85)	0.5883/0.2283 (84)	0.5897/0.2296 (85)	0.5884/0.2196 (82)
QMCD	0.5376/0.2370 (85)	0.5264/0.2173 (84)	0.5124/0.1981 (85)	0.4679/0.1766 (85)
	0.5376/0.2370 (85)	0.5264/0.2173 (84)	0.5124/0.1981 (85)	0.4679/0.1766 (85)
KDE	0.8970/0.6439 (85)	0.8945/0.6312 (85)	0.8932/0.6188 (85)	0.8933/0.5962 (85)
	0.8970/0.6439 (85)	0.8945/0.6312 (85)	0.8932/0.6188 (85)	0.8933/0.5962 (85)
Sampling	0.8620/0.5595 (85)	0.8671/0.5543 (85)	0.8685/0.5545 (85)	0.8652/0.5464 (85)
	0.8620/0.5595 (85)	0.8671/0.5543 (85)	0.8685/0.5545 (85)	0.8652/0.5464 (85)
GMM	0.8654/0.5497 (85)	0.8615/0.5444 (85)	0.8637/0.5419 (85)	0.8612/0.5270 (85)
	0.8654/0.5497 (85)	0.8615/0.5444 (85)	0.8637/0.5419 (85)	0.8612/0.5270 (85)
PCA	0.8073/0.5304 (82)	0.8061/0.5268 (81)	0.7995/0.5151 (80)	0.8112/0.5230 (85)
	0.8073/0.5304 (82)	0.8061/0.5268 (81)	0.7995/0.5151 (80)	0.8112/0.5230 (85)
KPCA	0.9023/0.6464 (85)	0.8986/0.6301 (85)	0.8967/0.6222 (85)	0.8925/0.6004 (85)
	0.9023/0.6464 (85)	0.8986/0.6301 (85)	0.8967/0.6222 (85)	0.8925/0.6004 (85)
MCD	0.8427/0.4702 (85)	0.8450/0.4687 (85)	0.8445/0.4679 (85)	0.8415/0.4620 (85)
	0.8427/0.4702 (85)	0.8450/0.4687 (85)	0.8445/0.4679 (85)	0.8415/0.4620 (85)
CD	0.8192/0.4719 (79)	0.8173/0.4705 (79)	0.8168/0.4680 (75)	0.8182/0.4599 (78)
	0.8192/0.4719 (79)	0.8173/0.4705 (79)	0.8168/0.4680 (75)	0.8182/0.4599 (78)
OCSVM	0.8409/0.5674 (85)	0.8396/0.5612 (85)	0.8404/0.5632 (85)	0.8422/0.5559 (85)
	0.8409/0.5674 (85)	0.8396/0.5612 (85)	0.8404/0.5632 (85)	0.8422/0.5559 (85)
LMDD	0.7588/0.4520 (60)	0.7562/0.4447 (59)	0.7109/0.4243 (50)	0.7517/0.4390 (53)
	0.7588/0.4520 (60)	0.7562/0.4447 (59)	0.7109/0.4243 (50)	0.7517/0.4390 (53)
LOF	0.8879/0.5786 (85)	0.8833/0.5781 (85)	0.8738/0.5648 (85)	0.8551/0.5477 (85)
	0.8879/0.5786 (85)	0.8833/0.5781 (85)	0.8738/0.5648 (85)	0.8551/0.5477 (85)
COF	0.6746/0.3429 (85)	0.6751/0.3395 (85)	0.6727/0.3404 (85)	0.6751/0.3407 (81)
	0.6746/0.3429 (85)	0.6751/0.3395 (85)	0.6727/0.3404 (85)	0.6751/0.3407 (81)
CBLOF	0.8803/0.5702 (84)	0.8816/0.5695 (83)	0.8815/0.5712 (85)	0.8705/0.5522 (75)
	0.8803/0.5702 (84)	0.8816/0.5695 (83)	0.8815/0.5712 (85)	0.8705/0.5522 (75)
LOCI	0.6768/0.3329 (30)	0.6766/0.3217 (28)	0.7023/0.3639 (22)	0.7161/0.3784 (26)
	0.6768/0.3329 (30)	0.6766/0.3217 (28)	0.7023/0.3639 (22)	0.7128/0.3758 (24)
HBOS	0.8133/0.4952 (85)	0.8130/0.4919 (85)	0.8130/0.4862 (85)	0.8122/0.4751 (75)
	0.8133/0.4952 (85)	0.8130/0.4919 (85)	0.8130/0.4862 (85)	0.8119/0.4759 (73)

Continued on next page

Table 56 continued

Configuration	100% retained	75% retained	50% retained	25% retained
HDBSCAN	0.6946/0.2392 (85)	0.6732/0.2225 (85)	0.6414/0.2021 (85)	0.6099/0.1703 (85)
	0.6946/0.2392 (85)	0.6732/0.2225 (85)	0.6414/0.2021 (85)	0.6099/0.1703 (85)
KNN	0.8973/0.6163 (85)	0.8937/0.6034 (85)	0.8896/0.5941 (85)	0.8782/0.5731 (85)
	0.8973/0.6163 (85)	0.8937/0.6034 (85)	0.8896/0.5941 (85)	0.8782/0.5731 (85)
SOD	0.7553/0.3523 (85)	0.7563/0.3462 (82)	0.7524/0.3576 (46)	0.7485/0.3428 (66)
	0.7553/0.3523 (85)	0.7541/0.3393 (78)	0.7209/0.3110 (16)	0.7488/0.3419 (60)
ROD	0.8244/0.5105 (42)	0.8096/0.4694 (35)	0.7919/0.4636 (31)	0.7895/0.4424 (41)
	0.8244/0.5105 (42)	0.8096/0.4694 (35)	0.7919/0.4636 (31)	0.7902/0.4756 (38)
IForest	0.8533/0.5258 (85)	0.8527/0.5310 (85)	0.8478/0.5157 (85)	0.8368/0.4787 (85)
	0.8533/0.5258 (85)	0.8527/0.5310 (85)	0.8478/0.5157 (85)	0.8368/0.4787 (85)
INNE	0.8732/0.5757 (85)	0.8718/0.5687 (85)	0.8679/0.5518 (85)	0.8629/0.5290 (85)
	0.8732/0.5757 (85)	0.8718/0.5687 (85)	0.8679/0.5518 (85)	0.8629/0.5290 (85)
DIF	0.8191/0.4428 (85)	0.8106/0.4243 (85)	0.8081/0.4195 (85)	0.8060/0.3967 (85)
	0.8191/0.4428 (85)	0.8106/0.4243 (85)	0.8081/0.4195 (85)	0.8027/0.3970 (72)
FeatureBagging	0.8891/0.5859 (85)	0.8822/0.5767 (85)	0.8721/0.5665 (85)	0.8540/0.5507 (85)
	0.8891/0.5859 (85)	0.8822/0.5767 (85)	0.8721/0.5665 (85)	0.8540/0.5507 (85)
LSCP	0.8924/0.5914 (85)	0.8868/0.5841 (85)	0.8784/0.5758 (85)	0.8528/0.5281 (72)
	0.8924/0.5914 (85)	0.8868/0.5841 (85)	0.8784/0.5758 (85)	0.8528/0.5281 (72)
LODA	0.7426/0.4150 (85)	0.7374/0.3993 (85)	0.7245/0.3788 (85)	0.7221/0.3356 (85)
	0.7426/0.4150 (85)	0.7374/0.3993 (85)	0.7245/0.3788 (85)	0.7221/0.3356 (85)
SUOD	0.8503/0.5440 (85)	0.8451/0.5383 (85)	0.8436/0.5335 (69)	0.8305/0.5069 (62)
	0.8503/0.5440 (85)	0.8451/0.5383 (85)	0.8436/0.5335 (69)	0.8308/0.5069 (61)
XGBOD	—/— (0)	—/— (0)	—/— (0)	—/— (0)
	—/— (0)	—/— (0)	—/— (0)	—/— (0)
AutoEncoder	0.8684/0.5702 (85)	0.8627/0.5625 (85)	0.8517/0.5511 (85)	0.8575/0.5628 (58)
	0.8684/0.5702 (85)	0.8627/0.5625 (85)	0.8517/0.5511 (85)	0.8564/0.5614 (57)
VAE	0.8299/0.5466 (85)	0.8234/0.5340 (85)	0.8141/0.5253 (85)	0.8317/0.5436 (69)
	0.8299/0.5466 (85)	0.8234/0.5340 (85)	0.8141/0.5253 (85)	0.8317/0.5436 (69)
SO_GAAL	0.5286/0.2693 (85)	0.5146/0.2353 (85)	0.5232/0.2235 (85)	0.5259/0.2439 (85)
	0.5286/0.2693 (85)	0.5146/0.2353 (85)	0.5232/0.2235 (85)	0.5259/0.2439 (85)
MO_GAAL	0.5438/0.2592 (85)	0.5156/0.2274 (85)	0.5443/0.2339 (85)	0.5393/0.2496 (85)
	0.5438/0.2592 (85)	0.5156/0.2274 (85)	0.5443/0.2339 (85)	0.5393/0.2496 (85)
DeepSVDD	0.8651/0.5908 (85)	0.8546/0.5765 (85)	0.8545/0.5698 (85)	0.8395/0.5429 (85)
	0.8651/0.5908 (85)	0.8546/0.5765 (85)	0.8545/0.5698 (85)	0.8395/0.5429 (85)
AnoGAN	0.7902/0.4839 (85)	0.7849/0.4778 (83)	0.7942/0.4886 (83)	0.8000/0.4652 (77)
	0.7902/0.4839 (85)	0.7849/0.4778 (83)	0.7942/0.4886 (83)	0.8000/0.4652 (77)
ALAD	0.5409/0.1470 (85)	0.5485/0.1579 (85)	0.5489/0.1575 (85)	0.5579/0.1625 (85)
	0.5409/0.1470 (85)	0.5485/0.1579 (85)	0.5489/0.1575 (85)	0.5579/0.1625 (85)

Continued on next page

Table 56 continued

Configuration	100% retained	75% retained	50% retained	25% retained
AE1SVM	0.8363/0.5458 (85)	0.8287/0.5277 (85)	0.8283/0.5290 (85)	0.8180/0.4992 (79)
	0.8363/0.5458 (85)	0.8287/0.5277 (85)	0.8283/0.5290 (85)	0.8180/0.4992 (79)
DevNet	—/— (0)	—/— (0)	—/— (0)	—/— (0)
	—/— (0)	—/— (0)	—/— (0)	—/— (0)
RGraph	0.3840/0.1682 (26)	0.4713/0.1978 (21)	0.4236/0.2126 (21)	0.5040/0.2547 (31)
	0.3840/0.1682 (26)	0.4713/0.1978 (21)	0.4236/0.2126 (21)	0.5040/0.2547 (31)
LUNAR	0.9058/0.6768 (85)	0.9022/0.6401 (85)	0.8967/0.6285 (85)	0.8870/0.5962 (85)
	0.9058/0.6768 (85)	0.9022/0.6401 (85)	0.8967/0.6285 (85)	0.8870/0.5962 (85)

Table 57: Recorded-success sensitivity: PCA to two dimensions. In each cell the first line gives raw AUROC/AP (raw successes), and the second gives qualified AUROC/AP (qualified successes), both out of 85.

Configuration	100% retained	75% retained	50% retained	25% retained
NSA fixed	0.5827/0.3427 (85)	0.5923/0.3328 (85)	0.6202/0.3564 (85)	0.6299/0.3474 (85)
	0.5827/0.3427 (85)	0.5923/0.3328 (85)	0.6202/0.3564 (85)	0.6299/0.3474 (85)
NSA variable	0.6928/0.4441 (85)	0.6982/0.4435 (85)	0.7139/0.4466 (85)	0.7240/0.4204 (85)
	0.6928/0.4441 (85)	0.6982/0.4435 (85)	0.7139/0.4466 (85)	0.7240/0.4204 (85)
NSA coverage	0.6694/0.4253 (85)	0.6763/0.4310 (85)	0.7039/0.4422 (85)	0.6954/0.4102 (85)
	0.6694/0.4253 (85)	0.6763/0.4310 (85)	0.7039/0.4422 (85)	0.6954/0.4102 (85)
NSA grid	0.6886/0.4361 (85)	0.6942/0.4385 (85)	0.7105/0.4454 (85)	0.7261/0.4208 (85)
	0.6886/0.4361 (85)	0.6942/0.4385 (85)	0.7105/0.4454 (85)	0.7261/0.4208 (85)
NSA hierarchical	0.6754/0.4218 (85)	0.6676/0.4090 (85)	0.6921/0.4266 (85)	0.6836/0.3857 (85)
	0.6754/0.4218 (85)	0.6676/0.4090 (85)	0.6921/0.4266 (85)	0.6836/0.3857 (85)
NSA voronoi	0.7043/0.4521 (85)	0.7164/0.4590 (85)	0.7315/0.4550 (85)	0.7432/0.4347 (85)
	0.7043/0.4521 (85)	0.7164/0.4590 (85)	0.7315/0.4550 (85)	0.7432/0.4347 (85)
NSA deterministic	0.6032/0.3594 (85)	0.6149/0.3586 (85)	0.6459/0.3928 (85)	0.6420/0.3450 (85)
	0.6032/0.3594 (85)	0.6149/0.3586 (85)	0.6459/0.3928 (85)	0.6420/0.3450 (85)
NSA suppressed	0.6738/0.4214 (85)	0.6841/0.4288 (85)	0.7063/0.4402 (85)	0.7120/0.4186 (85)
	0.6738/0.4214 (85)	0.6841/0.4288 (85)	0.7063/0.4402 (85)	0.7120/0.4186 (85)
NSA dual	0.7172/0.4718 (85)	0.7268/0.4777 (85)	0.7419/0.4801 (85)	0.7603/0.4699 (85)
	0.7172/0.4718 (85)	0.7268/0.4777 (85)	0.7419/0.4801 (85)	0.7603/0.4699 (85)
NSA annealed	0.5998/0.3672 (85)	0.6183/0.3715 (85)	0.6401/0.3835 (85)	0.6286/0.3403 (85)
	0.5998/0.3672 (85)	0.6183/0.3715 (85)	0.6401/0.3835 (85)	0.6286/0.3403 (85)
NSA binary Hamming	—/— (0)	—/— (0)	—/— (0)	—/— (0)
	—/— (0)	—/— (0)	—/— (0)	—/— (0)
NSA binary contiguous	—/— (0)	—/— (0)	—/— (0)	—/— (0)
	—/— (0)	—/— (0)	—/— (0)	—/— (0)
NSA binary chunk	—/— (0)	—/— (0)	—/— (0)	—/— (0)
	—/— (0)	—/— (0)	—/— (0)	—/— (0)
ECOD	0.6201/0.2698 (85)	0.6197/0.2721 (85)	0.6176/0.2782 (85)	0.6199/0.2584 (85)
	0.6201/0.2698 (85)	0.6197/0.2721 (85)	0.6176/0.2782 (85)	0.6199/0.2584 (85)
ABOD	0.7853/0.4831 (85)	0.7806/0.4832 (85)	0.7903/0.4860 (85)	0.7893/0.4706 (85)
	0.7853/0.4831 (85)	0.7806/0.4832 (85)	0.7903/0.4860 (85)	0.7893/0.4706 (85)
COPOD	0.6634/0.3118 (85)	0.6625/0.3113 (85)	0.6686/0.3238 (85)	0.6632/0.2999 (85)
	0.6634/0.3118 (85)	0.6625/0.3113 (85)	0.6686/0.3238 (85)	0.6632/0.2999 (85)
MAD	—/— (0)	—/— (0)	—/— (0)	—/— (0)
	—/— (0)	—/— (0)	—/— (0)	—/— (0)

Continued on next page

Table 57 continued

Configuration	100% retained	75% retained	50% retained	25% retained
SOS	0.5326/0.1547 (85)	0.5402/0.1676 (85)	0.5350/0.1654 (85)	0.5280/0.1596 (85)
	0.5326/0.1547 (85)	0.5402/0.1676 (85)	0.5350/0.1654 (85)	0.5280/0.1596 (85)
QMCD	0.6582/0.3382 (85)	0.6570/0.3381 (85)	0.6609/0.3288 (85)	0.6353/0.2828 (85)
	0.6582/0.3382 (85)	0.6570/0.3381 (85)	0.6609/0.3288 (85)	0.6353/0.2828 (85)
KDE	0.7890/0.4817 (85)	0.7913/0.4874 (85)	0.7956/0.4908 (85)	0.7947/0.4711 (85)
	0.7890/0.4817 (85)	0.7913/0.4874 (85)	0.7956/0.4908 (85)	0.7947/0.4711 (85)
Sampling	0.7607/0.4333 (85)	0.7529/0.4388 (85)	0.7430/0.4191 (85)	0.7648/0.4329 (85)
	0.7607/0.4333 (85)	0.7529/0.4388 (85)	0.7430/0.4191 (85)	0.7648/0.4329 (85)
GMM	0.6345/0.3568 (85)	0.6330/0.3605 (85)	0.6370/0.3756 (85)	0.6495/0.3643 (85)
	0.6345/0.3568 (85)	0.6330/0.3605 (85)	0.6370/0.3756 (85)	0.6495/0.3643 (85)
PCA	0.6656/0.3605 (85)	0.6581/0.3583 (85)	0.6471/0.3520 (85)	0.6498/0.3566 (85)
	0.6656/0.3605 (85)	0.6581/0.3583 (85)	0.6471/0.3520 (85)	0.6498/0.3566 (85)
KPCA	0.7862/0.4854 (85)	0.7892/0.4885 (85)	0.7980/0.4916 (85)	0.7937/0.4676 (85)
	0.7862/0.4854 (85)	0.7892/0.4885 (85)	0.7980/0.4916 (85)	0.7937/0.4676 (85)
MCD	0.7063/0.3673 (85)	0.6980/0.3525 (85)	0.7027/0.3622 (85)	0.7071/0.3539 (85)
	0.7063/0.3673 (85)	0.6980/0.3525 (85)	0.7027/0.3622 (85)	0.7071/0.3539 (85)
CD	0.5831/0.2942 (85)	0.5807/0.2954 (85)	0.5874/0.3025 (85)	0.5910/0.2704 (85)
	0.5831/0.2942 (85)	0.5807/0.2954 (85)	0.5874/0.3025 (85)	0.5910/0.2704 (85)
OCSVM	0.7801/0.4803 (85)	0.7797/0.4835 (85)	0.7918/0.4909 (85)	0.7788/0.4655 (85)
	0.7801/0.4803 (85)	0.7797/0.4835 (85)	0.7918/0.4909 (85)	0.7788/0.4655 (85)
LMDD	0.5580/0.2448 (85)	0.5717/0.2531 (85)	0.5898/0.2627 (85)	0.5774/0.2553 (80)
	0.5580/0.2448 (85)	0.5717/0.2531 (85)	0.5898/0.2627 (85)	0.5774/0.2553 (80)
LOF	0.7781/0.4604 (85)	0.7819/0.4543 (85)	0.7830/0.4577 (85)	0.7651/0.4127 (85)
	0.7781/0.4604 (85)	0.7819/0.4543 (85)	0.7830/0.4577 (85)	0.7651/0.4127 (85)
COF	0.6141/0.2250 (85)	0.6216/0.2319 (85)	0.6207/0.2384 (85)	0.6265/0.2417 (85)
	0.6141/0.2250 (85)	0.6216/0.2319 (85)	0.6207/0.2384 (85)	0.6265/0.2417 (85)
CBLOF	0.7815/0.4532 (85)	0.7783/0.4506 (85)	0.7859/0.4816 (85)	0.7743/0.4512 (68)
	0.7815/0.4532 (85)	0.7783/0.4506 (85)	0.7859/0.4816 (85)	0.7739/0.4533 (64)
LOCI	0.5585/0.2439 (30)	0.5521/0.2607 (28)	0.5792/0.2616 (29)	0.5296/0.2561 (17)
	0.5585/0.2439 (30)	0.5521/0.2607 (28)	0.5792/0.2616 (29)	0.4614/0.2154 (14)
HBOS	0.7581/0.3902 (85)	0.7488/0.3775 (85)	0.7594/0.3772 (85)	0.7615/0.3580 (75)
	0.7581/0.3902 (85)	0.7488/0.3775 (85)	0.7594/0.3772 (85)	0.7562/0.3551 (68)
HDBSCAN	0.7163/0.2391 (85)	0.7255/0.2265 (85)	0.6932/0.2004 (85)	0.6342/0.1791 (85)
	0.7163/0.2391 (85)	0.7255/0.2265 (85)	0.6932/0.2004 (85)	0.6342/0.1791 (85)
KNN	0.7958/0.4872 (85)	0.7993/0.4891 (85)	0.8051/0.4942 (85)	0.8005/0.4644 (85)
	0.7958/0.4872 (85)	0.7993/0.4891 (85)	0.8051/0.4942 (85)	0.8005/0.4644 (85)
SOD	0.5747/0.2075 (85)	0.5649/0.2059 (83)	0.5674/0.2152 (85)	0.5993/0.2481 (32)
	0.5747/0.2075 (85)	0.5643/0.2061 (82)	0.5674/0.2152 (85)	0.5591/0.2197 (18)

Continued on next page

Table 57 continued

Configuration	100% retained	75% retained	50% retained	25% retained
ROD	0.6650/0.3133 (85)	0.6663/0.3204 (85)	0.6707/0.3200 (85)	0.6697/0.3142 (77)
	0.6650/0.3133 (85)	0.6663/0.3204 (85)	0.6707/0.3200 (85)	0.6701/0.3153 (72)
IForest	0.7593/0.4023 (85)	0.7571/0.3930 (85)	0.7598/0.3933 (85)	0.7520/0.3441 (85)
	0.7593/0.4023 (85)	0.7571/0.3930 (85)	0.7598/0.3933 (85)	0.7520/0.3441 (85)
INNE	0.7795/0.4684 (85)	0.7813/0.4711 (85)	0.7831/0.4740 (85)	0.7712/0.4468 (85)
	0.7795/0.4684 (85)	0.7813/0.4711 (85)	0.7831/0.4740 (85)	0.7712/0.4468 (85)
DIF	0.7577/0.4216 (85)	0.7536/0.4321 (85)	0.7643/0.4348 (85)	0.7583/0.3942 (85)
	0.7577/0.4216 (85)	0.7536/0.4321 (85)	0.7643/0.4348 (85)	0.7600/0.3986 (74)
FeatureBagging	0.7695/0.4496 (85)	0.7688/0.4439 (85)	0.7728/0.4498 (85)	0.7639/0.4178 (85)
	0.7695/0.4496 (85)	0.7688/0.4439 (85)	0.7728/0.4498 (85)	0.7639/0.4178 (85)
LSCP	0.7893/0.4757 (85)	0.7888/0.4784 (85)	0.7934/0.4670 (85)	0.7933/0.4520 (77)
	0.7893/0.4757 (85)	0.7888/0.4784 (85)	0.7934/0.4670 (85)	0.7933/0.4520 (77)
LODA	0.7270/0.3456 (85)	0.7243/0.3364 (85)	0.7248/0.3476 (85)	0.7092/0.3058 (85)
	0.7270/0.3456 (85)	0.7243/0.3364 (85)	0.7248/0.3476 (85)	0.7092/0.3058 (85)
SUOD	0.7576/0.3982 (85)	0.7600/0.4050 (84)	0.7620/0.4100 (85)	0.7608/0.3780 (35)
	0.7576/0.3982 (85)	0.7600/0.4050 (84)	0.7620/0.4100 (85)	0.7608/0.3780 (35)
XGBOD	—/— (0)	—/— (0)	—/— (0)	—/— (0)
	—/— (0)	—/— (0)	—/— (0)	—/— (0)
AutoEncoder	0.6830/0.3647 (85)	0.6767/0.3495 (85)	0.6591/0.3211 (85)	0.6639/0.3325 (63)
	0.6830/0.3647 (85)	0.6767/0.3495 (85)	0.6591/0.3211 (85)	0.6651/0.3316 (60)
VAE	0.6718/0.3697 (85)	0.6690/0.3697 (85)	0.6623/0.3577 (85)	0.6461/0.3866 (70)
	0.6718/0.3697 (85)	0.6690/0.3697 (85)	0.6623/0.3577 (85)	0.6461/0.3866 (70)
SO_GAAL	0.5817/0.2066 (85)	0.5506/0.2070 (85)	0.5616/0.2154 (85)	0.5535/0.2083 (85)
	0.5817/0.2066 (85)	0.5506/0.2070 (85)	0.5616/0.2154 (85)	0.5535/0.2083 (85)
MO_GAAL	0.5718/0.2341 (85)	0.5644/0.2481 (85)	0.5640/0.2429 (85)	0.5268/0.2257 (85)
	0.5718/0.2341 (85)	0.5644/0.2481 (85)	0.5640/0.2429 (85)	0.5268/0.2257 (85)
DeepSVDD	0.7368/0.3906 (85)	0.7434/0.3894 (85)	0.7448/0.4010 (85)	0.7238/0.3890 (85)
	0.7368/0.3906 (85)	0.7434/0.3894 (85)	0.7448/0.4010 (85)	0.7238/0.3890 (85)
AnoGAN	0.6247/0.2801 (85)	0.6322/0.2816 (85)	0.6288/0.2914 (85)	0.6430/0.2808 (77)
	0.6247/0.2801 (85)	0.6322/0.2816 (85)	0.6288/0.2914 (85)	0.6430/0.2808 (77)
ALAD	0.5051/0.1300 (85)	0.5233/0.1343 (85)	0.5282/0.1465 (85)	0.5193/0.1361 (85)
	0.5051/0.1300 (85)	0.5233/0.1343 (85)	0.5282/0.1465 (85)	0.5193/0.1361 (85)
AE1SVM	0.6570/0.3360 (85)	0.6475/0.3384 (85)	0.6531/0.3500 (85)	0.6635/0.3449 (80)
	0.6570/0.3360 (85)	0.6475/0.3384 (85)	0.6531/0.3500 (85)	0.6635/0.3449 (80)
DevNet	—/— (0)	—/— (0)	—/— (0)	—/— (0)
	—/— (0)	—/— (0)	—/— (0)	—/— (0)
RGraph	0.4360/0.1545 (35)	0.4184/0.1541 (34)	0.4146/0.1242 (45)	0.4026/0.1524 (32)
	0.4360/0.1545 (35)	0.4184/0.1541 (34)	0.4146/0.1242 (45)	0.4026/0.1524 (32)

Continued on next page

Table 57 continued

Configuration	100% retained	75% retained	50% retained	25% retained
LUNAR	0.7745/0.4576 (85) 0.7745/0.4576 (85)	0.7783/0.4632 (85) 0.7783/0.4632 (85)	0.7913/0.4759 (85) 0.7913/0.4759 (85)	0.7784/0.4449 (85) 0.7784/0.4449 (85)

Table 58: Configurations with successful recorded runs outside the 15-second measured budget. The maximum and mean are measured fitting-plus-scoring seconds over those successful runs only.

Configuration	Over-budget runs	Maximum seconds	Mean seconds
ECOD	8	16.47	16.03
CBLOF	4	21.19	19.11
LOCI	5	56.92	33.83
HBOS	9	21.91	20.47
SOD	55	65.27	23.47
ROD	8	250.75	47.89
DIF	24	24.47	18.73
SUOD	1	31.20	31.20
AutoEncoder	4	20.70	18.25

L Watchdog interventions and recovered processes

Six LOCI processes stopped making progress despite the nominal Python alarm. Each was terminated only after its raw record file had remained unchanged for at least 600 seconds. The intervention snapshot and hashes were retained, a timeout with no score was recorded, and remaining configurations resumed. This is distinct from assigning a score to an interrupted method.

Table 59: All external LOCI watchdog interventions. Age is seconds since the raw file last changed before action.

Dataset	Seed	Features	Retained	No-progress age	Signal
letter	46	native	25%	913.1	SIGTERM
mnist	46	native	25%	839.4	SIGTERM
satimage-2	46	native	25%	820.7	SIGTERM
satellite	42	pca2	25%	626.3	SIGTERM
vowels	42	pca2	25%	627.1	SIGTERM
optdigits	43	pca2	25%	641.3	SIGTERM

Two runner processes exited before completing their cells. The observed state was captured in sidecars, and the same pinned commands resumed the uncompleted roster. All 56 configuration records are present in both recovered cells. Saved logs do not establish a particular detector as the cause of either exit, so no such attribution is made.

Table 60: Recovered runner cells. Counts refer to completed roster records before recovery and in the final raw evidence.

Cell	Before	Final
glass pca2 seed43 retain25	36	56
pendigits pca2 seed43 retain25	30	56

The source dataset hashes, cell identity/order checks, row labels, transform fitting rules, complete method roster, and all intervention snapshots were audited. The publication generator additionally reconstructs every displayed method mean and aggregate rank from the 38,080 raw records and reproduces the paired bootstrap intervals used above. This verifies the reported values; it does not remove limits from dataset selection, capped sample sizes, two-dimensional projection, detector settings, compute budget, or post-hoc inspection of many results.

M Companion dataset comparisons for the cold start tables

The following AP and original-feature tables accompany the four main PCA(2) AUROC comparisons. The comparator set is unchanged. Incomplete native Dual NSA means disclose qualified split counts and are not used to claim a complete four-setting ranking improvement. *PCA(2) average precision (AP) by dataset. The same four comparators, five-seed requirement, fixed 26-method rank field, and within-displayed-set bolding rule apply as in the main AUROC tables. AP rank is computed independently of AUROC rank.*

Table 61: PCA(2), 100 percent retained normals: five-seed mean AP for Dual NSA and the four unchanged PyOD comparators on all 17 datasets. The rank is Dual's AP rank against the full fixed 26-configuration field.

Dataset	n	Dual	KDE	KNN	IForest	OCSVM	Dual AP rank
arrhythmia	231	0.3770	**0.3864**	0.3860	0.3562	0.3800	11/26
cardio	600	**0.7414**	0.7159	0.6997	0.6830	0.7178	1/26
glass	123	0.1988	0.1838	**0.2461**	0.1300	0.2236	11/26
ionosphere	135	0.5713	0.6084	**0.6243**	0.4567	0.6232	7/26
letter	600	0.0559	0.0847	**0.1083**	0.0647	0.0785	19/26
lympho	84	**0.5266**	0.4109	0.3428	0.3629	0.3881	1/26
mnist	600	0.1793	0.2197	0.1979	**0.2346**	0.1966	19/26
musk	600	0.9883	0.9962	0.9942	0.6950	**0.9962**	7/26
optdigits	600	0.0208	0.0283	**0.0355**	0.0237	0.0303	23/26
pendigits	600	0.3953	0.5848	0.5684	0.4263	**0.6035**	11/26
pima	299	0.4888	**0.5448**	0.5333	0.5198	0.5144	15/26
satellite	600	0.7920	0.7924	**0.7925**	0.7211	0.7880	4/26
satimage-2	600	0.8555	0.8703	**0.8720**	0.6343	0.8711	12/26
shuttle	600	**0.9241**	0.8515	0.9036	0.7406	0.8637	2/26
vertebral	126	**0.1200**	0.1031	0.0971	0.1054	0.1086	7/26
vowels	600	0.0482	0.0696	**0.1477**	0.0563	0.0636	16/26
wbc	213	0.7365	**0.7387**	0.7332	0.6281	0.7186	2/26

Table 62: PCA(2), 25 percent retained normals: five-seed mean AP for Dual NSA and the four unchanged PyOD comparators on all 17 datasets. The rank is Dual's AP rank against the full fixed 26-configuration field.

Dataset	n	Dual	KDE	KNN	IForest	OCSVM	Dual AP rank
arrhythmia	57	0.4481	0.4297	**0.4671**	0.4562	0.4159	9/26
cardio	150	0.6582	0.6631	0.6368	0.5669	**0.6686**	6/26
glass	30	**0.3174**	0.2394	0.1663	0.1378	0.2949	1/26
ionosphere	33	0.5771	0.5847	0.5041	0.4381	**0.5932**	5/26
letter	150	0.0917	0.0908	**0.1016**	0.0649	0.0856	5/26
lympho	21	0.4008	0.4393	**0.4580**	0.3299	0.3374	8/26
mnist	150	0.1929	0.2266	0.2141	**0.2335**	0.2087	14/26
musk	150	0.9451	**0.9782**	0.9777	0.5052	0.9738	8/26
optdigits	150	0.0201	0.0346	**0.0435**	0.0255	0.0394	24/26
pendigits	150	0.3766	0.4373	0.4311	0.2569	**0.4491**	5/26
pima	74	0.5134	**0.5425**	0.5230	0.5232	0.5146	12/26
satellite	150	0.7823	**0.7857**	0.7780	0.6974	0.7767	3/26
satimage-2	150	**0.8691**	0.8578	0.8469	0.2983	0.8576	4/26
shuttle	150	**0.8483**	0.8181	0.7978	0.6307	0.8249	3/26
vertebral	31	**0.1421**	0.1098	0.1149	0.1073	0.1121	8/26
vowels	150	**0.1394**	0.0672	0.1311	0.0495	0.0729	1/26
wbc	53	0.6657	**0.7047**	0.7037	0.5276	0.6884	12/26

Table 63: PCA(2), 10 percent retained normals: five-seed mean AP for Dual NSA and the four unchanged PyOD comparators on all 17 datasets. The rank is Dual's AP rank against the full fixed 26-configuration field.

Dataset	n	Dual	KDE	KNN	IForest	OCSVM	Dual AP rank
arrhythmia	23	0.3116	0.3180	**0.4218**	0.3746	0.3064	19/26
cardio	60	0.5817	0.6412	**0.6577**	0.5457	0.6333	13/26
glass	12	0.2364	0.2551	0.1303	0.1363	**0.2756**	6/26
ionosphere	13	0.5792	0.5904	0.4598	0.4749	**0.5910**	6/26
letter	60	**0.1049**	0.1031	0.0992	0.0656	0.1021	4/26
lympho	8	0.2885	0.2467	0.2773	**0.3118**	0.2247	4/26
mnist	60	0.2105	0.2194	0.2120	**0.2226**	0.2116	15/26
musk	60	0.8122	0.8777	0.8533	0.3864	**0.8843**	7/26
optdigits	60	0.0307	**0.0366**	0.0344	0.0242	0.0339	10/26
pendigits	60	0.2298	**0.3798**	0.3057	0.2195	0.3771	7/26
pima	29	0.5040	**0.5549**	0.5426	0.5319	0.5151	12/26
satellite	60	**0.7728**	0.7711	0.7433	0.6009	0.7676	1/26
satimage-2	60	**0.8509**	0.8263	0.8283	0.2254	0.8252	5/26
shuttle	60	0.7471	0.7881	0.7357	0.5875	**0.7917**	9/26
vertebral	12	0.1229	0.1128	0.1281	**0.1285**	0.1094	12/26
vowels	60	**0.1234**	0.0667	0.0846	0.0382	0.0775	1/26
wbc	21	0.6439	0.6471	**0.6794**	0.3211	0.5756	10/26

Table 64: PCA(2), 5 percent retained normals: five-seed mean AP for Dual NSA and the four unchanged PyOD comparators on all 17 datasets. The rank is Dual's AP rank against the full fixed 26-configuration field.

Dataset	n	Dual	KDE	KNN	IForest	OCSVM	Dual AP rank
arrhythmia	11	0.3307	0.3265	**0.3942**	0.3900	0.3188	12/26
cardio	30	0.5725	**0.5969**	0.5922	0.5485	0.5949	11/26
glass	6	0.1717	0.1902	0.0806	0.1111	**0.2082**	7/26
ionosphere	6	0.5790	**0.5838**	0.4240	0.4807	0.5809	4/26
letter	30	**0.1007**	0.0944	0.0819	0.0653	0.0960	2/26
lympho	4	0.0465	0.1778	**0.2295**	0.2106	0.1774	23/26
mnist	30	0.1920	0.2172	0.2371	**0.2440**	0.1969	20/26
musk	30	0.5974	**0.7466**	0.6902	0.3071	0.7406	6/26
optdigits	30	**0.0474**	0.0406	0.0301	0.0256	0.0453	1/26
pendigits	30	0.1608	**0.2316**	0.1282	0.1361	0.2248	11/26
pima	14	0.4427	0.4994	0.4992	**0.5185**	0.4619	13/26
satellite	30	**0.7658**	0.7620	0.7129	0.5126	0.7567	1/26
satimage-2	30	**0.7986**	0.7287	0.7656	0.2113	0.7230	5/26
shuttle	30	0.7142	0.7139	0.6405	0.4994	**0.7307**	5/26
vertebral	6	0.1125	**0.1201**	0.0988	0.1097	0.1184	17/26
vowels	30	**0.1065**	0.0572	0.0587	0.0291	0.0707	1/26
wbc	10	0.6450	0.6268	**0.6548**	0.3754	0.5091	3/26

Table 65: Original features, 100 percent retained normals: dataset-level AUROC/AP pairs for Dual NSA and four fixed comparators. Within the five shown, maxima are bolded separately for AUROC and AP. Q counts strictly qualified Dual seeds; a dash means fewer than five and is not zero. Dual ranks use the fixed field of 19 other detectors plus Dual (20 total). At 100 and 25 percent, Dual lacks complete overall coverage, so its local dataset ranks do not yield a complete-coverage aggregate rank.

Dataset	n	Dual A/P	Q	KDE A/P	KNN A/P	IForest A/P	OCSVM A/P	Dual ranks A/P
arrhythmia	231	0.4685/0.2641	5/5	0.7712/0.4382	0.8228/0.4840	**0.8337**/**0.5618**	0.8212/0.4838	20/20; 18/20
cardio	600	0.6175/0.4529	5/5	0.9513/0.7217	0.9334/0.6637	0.9532/0.7020	**0.9662**/**0.7541**	19/20; 17/20
glass	123	0.7726/0.2231	5/5	0.8220/0.2599	**0.8524**/**0.2665**	0.7591/0.1519	0.6494/0.2116	11/20; 10/20
ionosphere	135	0.8734/0.8968	5/5	0.9797/0.9701	**0.9849**/**0.9777**	0.9234/0.8853	0.9696/0.9627	14/20; 13/20
letter	600	0.5336/0.1160	5/5	**0.9075**/**0.3839**	0.8576/0.2772	0.6430/0.0953	0.6156/0.1361	17/20; 13/20
lympho	84	0.9207/0.6542	5/5	0.9603/0.7654	0.9759/0.7936	**0.9966**/**0.9500**	0.9810/0.8086	15/20; 15/20
mnist	600	—	1/5	**0.9326**/**0.5904**	0.9300/0.5772	0.8661/0.4096	0.9118/0.5368	—
musk	600	**1.0000**/**1.0000**	5/5	**1.0000**/**1.0000**	**1.0000**/**1.0000**	0.9646/0.4296	**1.0000**/**1.0000**	1/20; 1/20
optdigits	600	0.3844/0.0243	5/5	**0.9442**/**0.1930**	0.8972/0.1157	0.8410/0.0977	0.6347/0.0362	18/20; 20/20
pendigits	600	0.6119/0.0843	5/5	**0.9961**/**0.8307**	0.9883/0.5841	0.9676/0.3445	0.9657/0.3427	19/20; 17/20
pima	299	0.6075/0.4733	5/5	**0.7599**/**0.5854**	0.7486/0.5765	0.7361/0.5746	0.7096/0.5479	17/20; 16/20
satellite	600	0.6239/0.6294	5/5	**0.8770**/**0.8316**	0.8716/0.8217	0.8070/0.7796	0.7644/0.7555	19/20; 16/20
satimage-2	600	0.9649/0.9226	5/5	0.9978/0.9457	**0.9986**/**0.9620**	0.9926/0.8980	0.9966/0.9547	16/20; 8/20
shuttle	600	0.9963/0.9065	5/5	**0.9965**/0.8929	0.9964/0.8785	0.9951/**0.9705**	0.9963/0.8885	10/20; 5/20
vertebral	126	0.4714/0.1396	5/5	0.4502/0.1249	0.4698/0.1295	0.4571/0.1223	**0.5185**/**0.1423**	9/20; 9/20
vowels	600	0.7023/0.2014	5/5	0.9653/**0.7251**	**0.9711**/0.6479	0.8093/0.2145	0.8340/0.3508	16/20; 13/20
wbc	213	0.6481/0.5375	5/5	0.9377/0.6879	0.9552/0.7206	0.9606/**0.7518**	**0.9608**/0.7341	17/20; 16/20

Table 66: Original features, 25 percent retained normals: dataset-level AUROC/AP pairs for Dual NSA and four fixed comparators. Within the five shown, maxima are bolded separately for AUROC and AP. Q counts strictly qualified Dual seeds; a dash means fewer than five and is not zero. Dual ranks use the fixed field of 19 other detectors plus Dual (20 total). At 100 and 25 percent, Dual lacks complete overall coverage, so its local dataset ranks do not yield a complete-coverage aggregate rank.

Dataset	n	Dual A/P	Q	KDE A/P	KNN A/P	IForest A/P	OCSVM A/P	Dual ranks A/P
arrhythmia	57	0.6861/0.4023	5/5	0.8283/0.5532	0.8298/**0.5568**	**0.8318**/0.4985	0.8275/0.5566	16/20; 16/20
cardio	150	0.7369/0.5337	5/5	0.9253/0.6217	0.9253/0.6176	0.9352/0.6314	**0.9561**/**0.6898**	17/20; 14/20
glass	30	0.8140/0.1820	5/5	**0.8640**/**0.2180**	0.7659/0.1406	0.7128/0.1038	0.7433/0.1596	4/20; 5/20
ionosphere	33	0.9037/0.9189	5/5	**0.9762**/**0.9666**	0.9488/0.9298	0.8639/0.7735	0.9665/0.9577	13/20; 10/20
letter	150	0.5992/0.1306	5/5	**0.8451**/**0.2697**	0.7919/0.1949	0.6360/0.0965	0.6217/0.1333	14/20; 13/20
lympho	21	0.9759/0.7972	5/5	0.9707/0.7033	**0.9914**/**0.8667**	0.9828/0.8622	**0.9914**/**0.8667**	9/20; 10/20
mnist	150	—	3/5	**0.9100**/0.4866	0.9088/**0.4883**	0.8694/0.4232	0.9004/0.4598	—
musk	150	**1.0000**/**1.0000**	5/5	0.9996/0.9874	**1.0000**/**1.0000**	0.9035/0.2182	1.0000/0.9999	1/20; 1/20
optdigits	150	0.3510/0.0215	5/5	**0.9140**/**0.1425**	0.8342/0.0783	0.8630/0.1133	0.6654/0.0396	18/20; 18/20
pendigits	150	0.6431/0.0982	5/5	**0.9868**/**0.5937**	0.9670/0.3641	0.9568/0.3010	0.9578/0.3127	19/20; 14/20
pima	74	0.6673/0.5107	5/5	**0.7560**/**0.5912**	0.7528/0.5789	0.7343/0.5668	0.7079/0.5506	14/20; 15/20
satellite	150	0.6299/0.6353	5/5	**0.8708**/**0.8226**	0.8644/0.8118	0.8029/0.7711	0.7640/0.7569	19/20; 16/20
satimage-2	150	0.9705/0.9403	5/5	0.9972/0.9408	**0.9983**/**0.9637**	0.9913/0.8713	0.9958/0.9538	16/20; 6/20
shuttle	150	**0.9964**/0.8771	5/5	0.9955/0.8539	0.9955/0.8567	0.9898/**0.9092**	0.9955/0.8562	2/20; 5/20
vertebral	31	0.4000/0.1168	5/5	0.4498/0.1204	**0.4617**/**0.1229**	0.4240/0.1155	0.4403/0.1172	18/20; 14/20
vowels	150	0.9355/0.5646	5/5	**0.9560**/**0.5704**	0.9465/0.4848	0.7796/0.1884	0.8330/0.3442	7/20; 4/20
wbc	53	0.7667/0.6828	5/5	0.9413/0.6938	0.9465/0.6870	0.9479/0.6935	**0.9505**/**0.6956**	16/20; 11/20

Table 67: Original features, 10 percent retained normals: dataset-level AUROC/AP pairs for Dual NSA and four fixed comparators. Within the five shown, maxima are bolded separately for AUROC and AP. Q counts strictly qualified Dual seeds; a dash means fewer than five and is not zero. Dual ranks use the fixed field of 19 other detectors plus Dual (20 total). At 100 and 25 percent, Dual lacks complete overall coverage, so its local dataset ranks do not yield a complete-coverage aggregate rank.

Dataset	n	Dual A/P	Q	KDE A/P	KNN A/P	IForest A/P	OCSVM A/P	Dual ranks A/P
arrhythmia	23	0.7755/0.4616	5/5	**0.8299**/0.5509	0.8286/**0.5560**	0.8284/0.4690	0.8290/0.5412	15/20; 14/20
cardio	60	0.9081/**0.6820**	5/5	0.9218/0.6207	0.9373/0.6525	0.9271/0.6152	**0.9570**/0.6812	14/20; 4/20
glass	12	0.7378/0.1447	5/5	**0.8171**/**0.1650**	0.7329/0.1354	0.7213/0.1052	0.7659/0.1442	6/20; 9/20
ionosphere	13	0.9262/0.8993	5/5	**0.9467**/0.9223	0.9090/0.8674	0.7976/0.6358	0.9461/**0.9228**	8/20; 8/20
letter	60	0.5589/0.1161	5/5	**0.7935**/**0.2139**	0.7348/0.1532	0.5924/0.0880	0.6192/0.1281	13/20; 11/20
lympho	8	**0.9862**/**0.8133**	5/5	0.9103/0.6661	0.9828/0.7800	0.8983/0.3899	0.9810/0.7889	2/20; 2/20
mnist	60	0.8086/0.3928	5/5	**0.8971**/**0.4588**	0.8920/0.4483	0.8285/0.3273	0.8934/0.4209	13/20; 10/20
musk	60	1.0000/0.9988	5/5	**1.0000**/**1.0000**	1.0000/0.9999	0.7623/0.0779	1.0000/0.9997	8/20; 8/20
optdigits	60	0.3639/0.0212	5/5	**0.8658**/**0.0954**	0.7296/0.0488	0.8175/0.0886	0.6866/0.0431	19/20; 20/20
pendigits	60	0.6806/0.1179	5/5	**0.9684**/**0.3725**	0.9222/0.1975	0.9304/0.2342	0.9369/0.2468	16/20; 13/20
pima	29	0.6509/0.4889	5/5	0.7380/**0.5630**	**0.7422**/0.5551	0.7185/0.5417	0.6900/0.5269	16/20; 16/20
satellite	60	0.6963/0.6913	5/5	**0.8645**/**0.8143**	0.8509/0.7962	0.7647/0.7269	0.7626/0.7541	14/20; 14/20
satimage-2	60	0.9769/0.9408	5/5	**0.9983**/0.9538	0.9978/**0.9591**	0.9832/0.8082	0.9962/0.9549	14/20; 8/20
shuttle	60	**0.9962**/0.8727	5/5	0.9952/0.8506	0.9954/0.8562	0.9868/**0.8944**	0.9953/0.8517	1/20; 4/20
vertebral	12	0.4603/0.1266	5/5	**0.5091**/0.1363	0.4687/0.1221	0.4875/**0.1642**	0.4837/0.1267	17/20; 14/20
vowels	60	0.8334/0.2571	5/5	**0.9348**/**0.4892**	0.8885/0.2882	0.6756/0.0843	0.8159/0.2907	9/20; 9/20
wbc	21	0.8868/0.6653	5/5	**0.9373**/**0.6810**	0.9363/0.6750	0.9153/0.4754	0.9330/0.6775	14/20; 8/20

Table 68: Original features, 5 percent retained normals: dataset-level AUROC/AP pairs for Dual NSA and four fixed comparators. Within the five shown, maxima are bolded separately for AUROC and AP. Q counts strictly qualified Dual seeds; a dash means fewer than five and is not zero. Dual ranks use the fixed field of 19 other detectors plus Dual (20 total). At 100 and 25 percent, Dual lacks complete overall coverage, so its local dataset ranks do not yield a complete-coverage aggregate rank.

Dataset	n	Dual A/P	Q	KDE A/P	KNN A/P	IForest A/P	OCSVM A/P	Dual ranks A/P
arrhythmia	11	0.7804/0.4754	5/5	0.8219/0.5289	**0.8276**/**0.5360**	0.7998/0.4239	0.8234/0.5129	14/20; 13/20
cardio	30	0.9060/0.6542	5/5	0.9212/0.5930	0.9429/0.6604	0.9024/0.5630	**0.9574**/**0.6627**	12/20; 5/20
glass	6	0.7018/0.1023	5/5	**0.7738**/**0.1296**	0.5445/0.0813	0.6921/0.1003	0.7247/0.1085	8/20; 11/20
ionosphere	6	0.9050/0.8524	5/5	**0.9224**/**0.8742**	0.8876/0.8391	0.7474/0.5720	0.9213/0.8660	6/20; 6/20
letter	30	0.5741/0.1313	5/5	**0.7447**/**0.1804**	0.6387/0.1172	0.5471/0.0764	0.6067/0.1202	11/20; 7/20
lympho	4	**0.9793**/**0.8508**	5/5	0.9345/0.6738	0.9500/0.5950	0.8267/0.3797	0.9517/0.8077	2/20; 2/20
mnist	30	0.8140/0.3842	5/5	0.8780/**0.4227**	**0.8808**/0.4189	0.8256/0.3047	0.8730/0.3764	14/20; 8/20
musk	30	0.9983/0.9345	5/5	**0.9987**/**0.9478**	0.9983/0.9316	0.6507/0.0456	0.9984/0.9369	7/20; 6/20
optdigits	30	0.4443/0.0249	5/5	**0.8055**/0.0675	0.6456/0.0374	0.7793/**0.0729**	0.6636/0.0398	18/20; 18/20
pendigits	30	0.7701/0.1710	5/5	**0.9341**/0.2121	0.8699/0.1319	0.9236/**0.2204**	0.9307/0.1932	14/20; 9/20
pima	14	0.6516/0.4929	5/5	0.7088/0.5343	**0.7187**/0.5335	0.7029/**0.5437**	0.6769/0.5111	13/20; 11/20
satellite	30	0.7518/0.7275	5/5	**0.8654**/**0.8137**	0.8290/0.7768	0.7431/0.6617	0.7675/0.7546	11/20; 11/20
satimage-2	30	0.9695/0.9331	5/5	**0.9976**/0.9361	0.9915/0.9456	0.9672/0.5751	0.9945/**0.9478**	12/20; 6/20
shuttle	30	**0.9967**/0.8899	5/5	0.9959/0.8715	0.9958/0.8734	0.9788/0.8033	0.9961/**0.8913**	2/20; 5/20
vertebral	6	0.4250/0.1141	5/5	0.4369/0.1157	0.3593/0.1028	**0.5134**/**0.1912**	0.4202/0.1105	10/20; 10/20
vowels	30	0.8456/0.2589	5/5	**0.9114**/**0.3802**	0.7915/0.1573	0.5981/0.0507	0.8127/0.2454	7/20; 7/20
wbc	10	0.9165/0.6147	5/5	**0.9306**/**0.6223**	0.9049/0.6148	0.8853/0.3930	0.9189/0.6062	7/20; 7/20

N Additional summaries from the full benchmark

These summaries retain the wider analyses while the main discussion focuses on Dual NSA and normal-only cold start. Their captions define the different comparison fields; ranks with different denominators must not be compared as if they came from one field.

Table 69: Dual NSA on each dataset with 25 percent of the normal training pool retained and PCA to two dimensions. Scores average five split seeds. The common field contains 36 configurations with all 85 runs completed within the measured time budget. Top position is 100 times rank divided by 36; smaller is better. Split range gives the smallest and largest rank across the five seeds.

Dataset	AUROC	AP	Rank	Top position	Split range
arrhythmia	0.7113	0.4481	3/36	Top 8.33%	2–16
cardio	0.9274	0.6582	11/36	Top 30.56%	1–19
glass	0.9122	0.3174	3/36	Top 8.33%	3–13
ionosphere	0.7433	0.5771	8/36	Top 22.22%	5–13
letter	0.5155	0.0917	17/36	Top 47.22%	9–22
lympho	0.7310	0.4008	19/36	Top 52.78%	2–32
mnist	0.6448	0.1929	21/36	Top 58.33%	13–31
musk	0.9937	0.9451	12/36	Top 33.33%	1–16
optdigits	0.2992	0.0201	26/36	Top 72.22%	18–34
pendigits	0.9045	0.3766	16/36	Top 44.44%	10–25
pima	0.6606	0.5134	18/36	Top 50.00%	14–24
satellite	0.8179	0.7823	6/36	Top 16.67%	4–11
satimage-2	0.9664	0.8691	9/36	Top 25.00%	1–20
shuttle	0.9834	0.8483	2/36	Top 5.56%	1–5
vertebral	0.4327	0.1421	13/36	Top 36.11%	11–30
vowels	0.7654	0.1394	7/36	Top 19.44%	3–14
wbc	0.9153	0.6657	10/36	Top 27.78%	5–18

Table 70: Dual NSA as normal training availability changes in PCA(2). All displayed scores have complete coverage. KNN is the highest mean-AUROC fully observed comparator in each of these four panels.

Retained	Dual AUROC	Dual AP	KNN AUROC	Dual rank
100%	0.7172	0.4718	0.7958	19/48
75%	0.7268	0.4777	0.7993	17/46
50%	0.7419	0.4801	0.8051	18/48
25%	0.7603	0.4699	0.8005	11/36

N.1 Five and ten percent: an extension with the same training rule

The extension retains 10% and 5% of exactly the same capped normal pools. It adds 19,040 configuration records; the six-size comparison contains 57,120 records. The actual sample sizes are 8–60 normals at 10% and 4–30 normals at 5%. Benchmark preprocessing and initial model-fitting calls use retained normals only. Some comparator scoring implementations also use the mixed query batch, as audited separately. The additional tables use a fixed set of other PyOD detectors with all 85 runs qualified at

Table 71: Highest observed mean-AUROC NSA and other PyOD configuration with complete coverage in each setting. Leaders are selected descriptively from test outcomes; the rank field includes NSA and other configurations.

Features	Retained	NSA leader	AUROC	Rank	Other leader	AUROC
Original	100%	NSA grid	0.5972	28/39	LUNAR	0.9058
Original	75%	NSA grid	0.5880	27/36	LUNAR	0.9022
Original	50%	NSA grid	0.5569	28/39	LUNAR	0.8967
Original	25%	NSA binary Hamming	0.5172	23/30	KDE	0.8933
PCA(2)	100%	NSA dual	0.7172	19/48	KNN	0.7958
PCA(2)	75%	NSA dual	0.7268	17/46	KNN	0.7993
PCA(2)	50%	NSA dual	0.7419	18/48	KNN	0.8051
PCA(2)	25%	NSA dual	0.7603	11/36	KNN	0.8005

every one of the six sizes. Each NSA is ranked against that set plus itself; other NSA configurations do not enlarge its denominator. A dash marks insufficient complete coverage, rather than poor performance assigned an artificial score.

We mark an unsuccessful method–dataset–seed combination as *unsuitable under the tested configuration*. A method with zero qualified outcomes is unsuitable throughout that panel; a partly successful method is unsuitable in the reported number of cases. This functional label covers missing required inputs, dimension restrictions, construction or runtime errors, and the 15-second computation limit. It does not describe detection accuracy or establish that a different constructor could not work. The appendix lists every method's coverage and failure reasons. The separate neural training-batch check shows how a specific small-sample failure can be repaired.

Table 72: Highest-score other PyOD detector in each new setting, selected separately for AUROC and AP within the fixed six-size-qualified baseline field. Every displayed score has 85 qualified cases; field count is the number of other detectors before inserting a focal NSA. Leader selection is descriptive.

Features	Retained	Metric	Other detector	Score	Field count
Original features	10%	AUROC	KDE	0.8781	19
Original features	10%	AP	KDE	0.5620	19
Original features	5%	AUROC	KDE	0.8577	19
Original features	5%	AP	KDE	0.5237	19
PCA(2)	10%	AUROC	KDE	0.7769	25
PCA(2)	10%	AP	KDE	0.4373	25
PCA(2)	5%	AUROC	KDE	0.7496	25
PCA(2)	5%	AP	KDE	0.3949	25

Table 73: Every NSA configuration at 10 percent retained normal training data, PCA(2). AUROC and AP are dataset macro means; incomplete configurations have available-case scores but no overall rank. Rank and top position compare each complete NSA separately with the same six-size-qualified other PyOD detectors for that representation.

NSA configuration	Qualified	AUROC	AP	AUROC rank	Top position	Suitability
NSA fixed	85/85	0.6132	0.2789	19/26	73.08%	Works in 85/85
NSA variable	85/85	0.7273	0.3638	8/26	30.77%	Works in 85/85
NSA coverage	85/85	0.7110	0.3571	9/26	34.62%	Works in 85/85
NSA grid	85/85	0.7243	0.3668	8/26	30.77%	Works in 85/85
NSA hierarchical	85/85	0.6891	0.3316	12/26	46.15%	Works in 85/85
NSA voronoi	85/85	0.7361	0.3689	8/26	30.77%	Works in 85/85
NSA deterministic	85/85	0.6278	0.2893	17/26	65.38%	Works in 85/85
NSA suppressed	85/85	0.7173	0.3547	9/26	34.62%	Works in 85/85
NSA dual	85/85	0.7626	0.4206	6/26	23.08%	Works in 85/85
NSA annealed	85/85	0.6241	0.2869	17/26	65.38%	Works in 85/85
NSA binary Hamming	0/85	—	—	—	—	Unsuitable (0/85)
NSA binary contiguous	0/85	—	—	—	—	Unsuitable (0/85)
NSA binary chunk	0/85	—	—	—	—	Unsuitable (0/85)

Table 74: Every NSA configuration at 5 percent retained normal training data, PCA(2). AUROC and AP are dataset macro means; incomplete configurations have available-case scores but no overall rank. Rank and top position compare each complete NSA separately with the same six-size-qualified other PyOD detectors for that representation.

NSA configuration	Qualified	AUROC	AP	AUROC rank	Top position	Suitability
NSA fixed	85/85	0.5917	0.2200	19/26	73.08%	Works in 85/85
NSA variable	85/85	0.6683	0.2879	10/26	38.46%	Works in 85/85
NSA coverage	85/85	0.6669	0.2838	10/26	38.46%	Works in 85/85
NSA grid	85/85	0.6775	0.3002	10/26	38.46%	Works in 85/85
NSA hierarchical	85/85	0.6579	0.2647	12/26	46.15%	Works in 85/85
NSA voronoi	85/85	0.6882	0.3153	8/26	30.77%	Works in 85/85
NSA deterministic	85/85	0.6066	0.2198	17/26	65.38%	Works in 85/85
NSA suppressed	85/85	0.6650	0.2955	12/26	46.15%	Works in 85/85
NSA dual	85/85	0.7264	0.3755	5/26	19.23%	Works in 85/85
NSA annealed	85/85	0.5966	0.2187	19/26	73.08%	Works in 85/85
NSA binary Hamming	0/85	—	—	—	—	Unsuitable (0/85)
NSA binary contiguous	0/85	—	—	—	—	Unsuitable (0/85)
NSA binary chunk	0/85	—	—	—	—	Unsuitable (0/85)

Table 75: Every NSA configuration at 10 percent retained normal training data, Original features. AUROC and AP are dataset macro means; incomplete configurations have available-case scores but no overall rank. Rank and top position compare each complete NSA separately with the same six-size-qualified other PyOD detectors for that representation.

NSA configuration	Qualified	AUROC	AP	AUROC rank	Top position	Suitability
NSA fixed	85/85	0.2982	0.1093	20/20	100.00%	Works in 85/85
NSA variable	85/85	0.4350	0.1620	20/20	100.00%	Works in 85/85
NSA coverage	85/85	0.4333	0.1620	20/20	100.00%	Works in 85/85
NSA grid	85/85	0.4323	0.1573	20/20	100.00%	Works in 85/85
NSA hierarchical	85/85	0.2452	0.1031	20/20	100.00%	Works in 85/85
NSA voronoi	0/85	—	—	—	—	Unsuitable (0/85)
NSA deterministic	0/85	—	—	—	—	Unsuitable (0/85)
NSA suppressed	85/85	0.4366	0.1614	20/20	100.00%	Works in 85/85
NSA dual	85/85	0.7792	0.5112	14/20	70.00%	Works in 85/85
NSA annealed	85/85	0.2641	0.1040	20/20	100.00%	Works in 85/85
NSA binary Hamming	85/85	0.5024	0.1361	18/20	90.00%	Works in 85/85
NSA binary contiguous	75/85	0.5317	0.1321	—	—	Unsuitable in 10/85
NSA binary chunk	78/85	0.5350	0.1290	—	—	Unsuitable in 7/85

Table 76: Every NSA configuration at 5 percent retained normal training data, Original features. AUROC and AP are dataset macro means; incomplete configurations have available-case scores but no overall rank. Rank and top position compare each complete NSA separately with the same six-size-qualified other PyOD detectors for that representation.

NSA configuration	Qualified	AUROC	AP	AUROC rank	Top position	Suitability
NSA fixed	85/85	0.2951	0.1173	20/20	100.00%	Works in 85/85
NSA variable	85/85	0.3844	0.1509	20/20	100.00%	Works in 85/85
NSA coverage	85/85	0.3819	0.1463	20/20	100.00%	Works in 85/85
NSA grid	85/85	0.3823	0.1417	20/20	100.00%	Works in 85/85
NSA hierarchical	85/85	0.3089	0.1193	20/20	100.00%	Works in 85/85
NSA voronoi	0/85	—	—	—	—	Unsuitable (0/85)
NSA deterministic	0/85	—	—	—	—	Unsuitable (0/85)
NSA suppressed	85/85	0.3803	0.1483	20/20	100.00%	Works in 85/85
NSA dual	85/85	0.7900	0.5066	10/20	50.00%	Works in 85/85
NSA annealed	85/85	0.2564	0.0924	20/20	100.00%	Works in 85/85
NSA binary Hamming	85/85	0.5036	0.1234	17/20	85.00%	Works in 85/85
NSA binary contiguous	75/85	0.5437	0.1339	—	—	Unsuitable in 10/85
NSA binary chunk	84/85	0.5223	0.1186	—	—	Unsuitable in 1/85

Table 77: Dual NSA across all six retained sizes, PCA(2). The primary field holds the other-detector roster fixed. The final column removes the twelve implementations identified as batch dependent from the comparison field; it does not change NSA's scores.

Retained	AUROC	AP	PyOD AUC rank	Top position	Pointwise AUC rank
100%	0.7172	0.4718	13/26	50.00%	13/21
75%	0.7268	0.4777	12/26	46.15%	12/21
50%	0.7419	0.4801	12/26	46.15%	12/21
25%	0.7603	0.4699	10/26	38.46%	10/21
10%	0.7626	0.4206	6/26	23.08%	6/21
5%	0.7264	0.3755	5/26	19.23%	5/21

Table 78: Dual NSA on every dataset with only 10 or 5 percent of the capped normal pool. Each rank uses the same six-size-qualified other-detector field plus Dual NSA. Scores average five successful seeds; a missing value means fewer than five qualified seeds.

Dataset	10% AUROC	10% AUC rank	10% top	5% AUROC	5% AUC rank	5% top
arrhythmia	0.6401	16/26	61.54%	0.6220	14/26	53.85%
cardio	0.8860	16/26	61.54%	0.8666	16/26	61.54%
glass	0.8470	5/26	19.23%	0.7799	4/26	15.38%
ionosphere	0.7478	5/26	19.23%	0.7147	5/26	19.23%
letter	0.6229	6/26	23.08%	0.6014	2/26	7.69%
lympho	0.7207	12/26	46.15%	0.4121	24/26	92.31%
mnist	0.6640	15/26	57.69%	0.6626	18/26	69.23%
musk	0.9797	8/26	30.77%	0.9350	7/26	26.92%
optdigits	0.5067	11/26	42.31%	0.6226	1/26	3.85%
pendigits	0.8600	15/26	57.69%	0.8256	17/26	65.38%
pima	0.6628	12/26	46.15%	0.5903	16/26	61.54%
satellite	0.8136	3/26	11.54%	0.8185	1/26	3.85%
satimage-2	0.9721	2/26	7.69%	0.9544	4/26	15.38%
shuttle	0.9660	10/26	38.46%	0.9616	4/26	15.38%
vertebral	0.4613	8/26	30.77%	0.3853	19/26	73.08%
vowels	0.7337	1/26	3.85%	0.7107	1/26	3.85%
wbc	0.8799	11/26	42.31%	0.8856	12/26	46.15%

A higher NSA score with fewer normals does not by itself establish a relative advantage. We also measure the change in the NSA-minus-comparator gap from 100% retention. Both methods and both sizes must succeed in the same dataset/seed case. Positive differences indicate that NSA gains relative to that unchanged comparator; the intervals remain exploratory and unadjusted.

Table 79: Dual NSA's relative AUROC change versus fixed comparators in PCA(2). The difference is (NSA minus comparator at the reduced size) minus (NSA minus the same comparator at 100 percent); each matched case requires four successful outcomes.

Comparator	Retained	Matched	Δ gap	95% CI
KNN	10%	85/85	+0.0707	[+0.0132, +0.1355]
IForest	10%	85/85	+0.0851	[+0.0349, +0.1461]
KNN	5%	85/85	+0.0968	[-0.0005, +0.1970]
IForest	5%	85/85	+0.0815	[+0.0097, +0.1640]

Table 80: Dual NSA's relative AP change versus fixed comparators in PCA(2). The difference is (NSA minus comparator at the reduced size) minus (NSA minus the same comparator at 100 percent); each matched case requires four successful outcomes.

Comparator	Retained	Matched	Δ gap	95% CI
KNN	10%	85/85	+0.0129	[-0.0306, +0.0566]
IForest	10%	85/85	+0.0455	[-0.0086, +0.1091]
KNN	5%	85/85	+0.0193	[-0.0465, +0.0769]
IForest	5%	85/85	+0.0252	[-0.0414, +0.0897]

N.2 A training-batch check for neural comparators

The pinned AutoEncoder and VAE constructors use training batches of 32 and discard incomplete batches. Every 5% sample contains fewer than 32 normals, so their fixed-configuration training loops receive no batch and fail. This is a sample-size implementation limit; it is not a requirement for anomalous training examples. We therefore ran a separate sensitivity with training batch size $\min(32, n_{\text{fit}})$, while preserving inference batches of 32, the splits, preprocessing, epochs, score definitions, and time budget. The 740 supplemental method attempts cover both methods in every new 10% and 5% cell, plus the 30 historical 25% cells with fewer than 32 normals. Training-forward counts verify that successful runs actually trained. The training batch is reduced in 540 attempts; the other 200 attempts at 10% retain the original batch of 32. The remaining 1,300 earlier AutoEncoder/VAE records are reused with explicit lineage. These supplementary results do not replace the primary fixed-configuration experiment.

An independent source-and-split audit overlapped the startup of the first eight supplemental cells for about four seconds. All sixteen method attempts succeeded in less than 5.1 seconds against the 15-second limit, so their eligibility did not depend on a near-threshold timing decision. Their original records and a timing provenance note are retained. Runtime conclusions remain conditional on machine load.

Table 81: AutoEncoder and VAE with a training batch capped by the normal sample. Scores are available-case dataset means; compare ranks only when coverage is 85/85.

Features	Retained	Method	Original runs	Sensitivity runs	AUROC	AP
Original features	10%	AutoEncoder	50/85	85/85	0.8119	0.5071

Continued on next page

Table 81 continued

Features	Retained	Method	Original runs	Sensitivity runs	AUROC	AP
Original features	10%	VAE	50/85	85/85	0.8021	0.4992
Original features	5%	AutoEncoder	0/85	85/85	0.7952	0.4945
Original features	5%	VAE	0/85	85/85	0.7863	0.4860
PCA(2)	10%	AutoEncoder	50/85	85/85	0.6306	0.3183
PCA(2)	10%	VAE	50/85	85/85	0.6297	0.3295
PCA(2)	5%	AutoEncoder	0/85	85/85	0.6400	0.3151
PCA(2)	5%	VAE	0/85	85/85	0.6243	0.3205

Table 82: Dual NSA ranks after the AutoEncoder/VAE training-batch sensitivity. Each field is fixed across all six sizes within its analysis. Denominators include one focal NSA and other PyOD detectors only. Missing ranks indicate incomplete NSA coverage.

Features	Retained	Dual runs	Original AUROC rank	Adapted AUROC rank	Original AP rank	Adapted AP rank
Original features	100%	81/85	—	—	—	—
Original features	75%	80/85	—	—	—	—
Original features	50%	81/85	—	—	—	—
Original features	25%	83/85	—	—	—	—
Original features	10%	85/85	14/20	14/20	7/20	7/20
Original features	5%	85/85	10/20	10/20	4/20	4/20
PCA(2)	100%	85/85	13/26	13/27	6/26	6/27
PCA(2)	75%	85/85	12/26	12/27	6/26	6/27
PCA(2)	50%	85/85	12/26	12/27	6/26	6/27
PCA(2)	25%	85/85	10/26	10/27	3/26	3/27
PCA(2)	10%	85/85	6/26	6/27	6/26	6/27
PCA(2)	5%	85/85	5/26	5/27	5/26	5/27

O Additional extreme-retention experiments

The earlier appendix sections preserve the four-size experiment. The following tables add the 10% and 5% runs and the six-size fields. Complete coverage remains 85 dataset/seed cases per feature/size panel. All 56 configurations remain in the outcome denominator, including XGBOD and DevNet, whose intended supervised training requires labeled outliers. Small-sample fit errors, geometry restrictions, and timing exclusions are distinct from a requirement for training outliers.

Table 83: Exact six-size comparison fields. A field contains other detectors complete at every size; an NSA rank adds that focal configuration to the count. These selected complete fields do not imply a rank against every one of the 43 attempted detectors.

Features	Field	Count	Other detectors
Original features	Strict other detectors	19	ABOD, COPOD, KDE, GMM, KPCA, MCD, OCSVM, LOF, HDBSCAN, KNN, IForest, INNE, FeatureBagging, LODA, SO_GAAL, MO_GAAL, DeepSVDD, ALAD, LUNAR
Original features	Pointwise subset	18	ABOD, KDE, GMM, KPCA, MCD, OCSVM, LOF, HDBSCAN, KNN, IForest, INNE, FeatureBagging, LODA, SO_GAAL, MO_GAAL, DeepSVDD, ALAD, LUNAR
Original features	Raw other detectors	21	ECOD, ABOD, COPOD, KDE, GMM, KPCA, MCD, OCSVM, LOF, HDBSCAN, KNN, IForest, INNE, DIF, FeatureBagging, LODA, SO_GAAL, MO_GAAL, DeepSVDD, ALAD, LUNAR
PCA(2)	Strict other detectors	25	ECOD, ABOD, COPOD, SOS, QMCD, KDE, GMM, PCA, KPCA, MCD, CD, OCSVM, LOF, COF, HDBSCAN, KNN, IForest, INNE, FeatureBagging, LODA, SO_GAAL, MO_GAAL, DeepSVDD, ALAD, LUNAR
PCA(2)	Pointwise subset	20	ABOD, QMCD, KDE, GMM, PCA, KPCA, MCD, OCSVM, LOF, HDBSCAN, KNN, IForest, INNE, FeatureBagging, LODA, SO_GAAL, MO_GAAL, DeepSVDD, ALAD, LUNAR
PCA(2)	Raw other detectors	26	ECOD, ABOD, COPOD, SOS, QMCD, KDE, GMM, PCA, KPCA, MCD, CD, OCSVM, LOF, COF, HDBSCAN, KNN, IForest, INNE, DIF, FeatureBagging, LODA, SO_GAAL, MO_GAAL, DeepSVDD, ALAD, LUNAR

Table 84: Actual normal fitting rows at all six retained sizes; ranges cover seeds 42–46.

Dataset	100%	75%	50%	25%	10%	5%
arrhythmia	231	173	115	57	23	11
cardio	600	450	300	150	60	30
glass	123	92	61	30	12	6
ionosphere	135	101	67	33	13	6
letter	600	450	300	150	60	30
lympho	84	63	42	21	8	4
mnist	600	450	300	150	60	30
musk	600	450	300	150	60	30
optdigits	600	450	300	150	60	30

Continued on next page

Table 84 continued

Dataset	100%	75%	50%	25%	10%	5%
pendigits	600	450	300	150	60	30
pima	299	224	149	74	29	14
satellite	600	450	300	150	60	30
satimage-2	600	450	300	150	60	30
shuttle	600	450	300	150	60	30
vertebral	126	94	63	31	12	6
vowels	600	450	300	150	60	30
wbc	213	159	106	53	21	10

Table 85: Suitability at 10 and 5 percent retained normal training data, Original features. Unsuitable labels refer to the stated constructor, dimensionality, normal-only input, and measured 15-second budget. Partial coverage is reported by case count; raw successful scores over budget are retained separately.

Configuration	10% runs	10% suitability	10% reasons	5% runs	5% suitability	5% reasons
NSA fixed	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA variable	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA coverage	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA grid	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA hierarchical	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA voronoi	0/85	Unsuitable (0/85)	2D/3D required: 85	0/85	Unsuitable (0/85)	2D/3D required: 85
NSA deterministic	0/85	Unsuitable (0/85)	2D required: 85	0/85	Unsuitable (0/85)	2D required: 85
NSA suppressed	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA dual	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA annealed	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA binary Hamming	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA binary contiguous	75/85	Unsuitable in 10/85	runtime error: 10	75/85	Unsuitable in 10/85	runtime error: 10
NSA binary chunk	78/85	Unsuitable in 7/85	runtime error: 7	84/85	Unsuitable in 1/85	runtime error: 1
ECOD	77/85	Unsuitable in 8/85	over budget: 8	85/85	Works in 85/85	none
ABOD	85/85	Works in 85/85	none	85/85	Works in 85/85	none
COPOD	85/85	Works in 85/85	none	85/85	Works in 85/85	none
MAD	0/85	Unsuitable (0/85)	1D required: 85	0/85	Unsuitable (0/85)	1D required: 85
SOS	84/85	Unsuitable in 1/85	runtime error: 1	85/85	Works in 85/85	none
QMCD	85/85	Works in 85/85	none	85/85	Works in 85/85	none
KDE	85/85	Works in 85/85	none	85/85	Works in 85/85	none
Sampling	65/85	Unsuitable in 20/85	runtime error: 20	50/85	Unsuitable in 35/85	runtime error: 35
GMM	85/85	Works in 85/85	none	85/85	Works in 85/85	none
PCA	85/85	Works in 85/85	none	85/85	Works in 85/85	none
KPCA	85/85	Works in 85/85	none	85/85	Works in 85/85	none
MCD	85/85	Works in 85/85	none	85/85	Works in 85/85	none
CD	78/85	Unsuitable in 7/85	runtime error: 6; timeout: 1	79/85	Unsuitable in 6/85	runtime error: 6
OCSVM	85/85	Works in 85/85	none	85/85	Works in 85/85	none
LMDD	52/85	Unsuitable in 33/85	timeout: 33	55/85	Unsuitable in 30/85	timeout: 30

Continued on next page

Table 85 continued

Configuration	10% runs	10% suitability	10% reasons	5% runs	5% suitability	5% reasons
LOF	85/85	Works in 85/85	none	85/85	Works in 85/85	none
COF	85/85	Works in 85/85	none	85/85	Works in 85/85	none
CBLOF	80/85	Unsuitable in 5/85	runtime error: 5	65/85	Unsuitable in 20/85	runtime error: 20
LOCI	29/85	Unsuitable in 56/85	timeout: 56	30/85	Unsuitable in 55/85	timeout: 55
HBOS	85/85	Works in 85/85	none	85/85	Works in 85/85	none
HDBSCAN	85/85	Works in 85/85	none	85/85	Works in 85/85	none
KNN	85/85	Works in 85/85	none	85/85	Works in 85/85	none
SOD	58/85	Unsuitable in 27/85	construction error: 5; over budget: 9; timeout: 13	55/85	Unsuitable in 30/85	construction error: 30
ROD	39/85	Unsuitable in 46/85	over budget: 1; timeout: 45	42/85	Unsuitable in 43/85	runtime error: 1; over budget: 1; timeout: 41
IForest	85/85	Works in 85/85	none	85/85	Works in 85/85	none
INNE	85/85	Works in 85/85	none	85/85	Works in 85/85	none
DIF	85/85	Works in 85/85	none	85/85	Works in 85/85	none
FeatureBagging	85/85	Works in 85/85	none	85/85	Works in 85/85	none
LSCP	50/85	Unsuitable in 35/85	runtime error: 35	50/85	Unsuitable in 35/85	runtime error: 35
LODA	85/85	Works in 85/85	none	85/85	Works in 85/85	none
SUOD	85/85	Works in 85/85	none	84/85	Unsuitable in 1/85	runtime error: 1
XGBOD	0/85	Unsuitable (0/85)	labeled outliers required: 85	0/85	Unsuitable (0/85)	labeled outliers required: 85
AutoEncoder	50/85	Unsuitable in 35/85	runtime error: 35	0/85	Unsuitable (0/85)	runtime error: 85
VAE	50/85	Unsuitable in 35/85	runtime error: 35	0/85	Unsuitable (0/85)	runtime error: 85
SO_GAAL	85/85	Works in 85/85	none	85/85	Works in 85/85	none
MO_GAAL	85/85	Works in 85/85	none	85/85	Works in 85/85	none
DeepSVDD	85/85	Works in 85/85	none	85/85	Works in 85/85	none
AnoGAN	85/85	Works in 85/85	none	85/85	Works in 85/85	none
ALAD	85/85	Works in 85/85	none	85/85	Works in 85/85	none
AE1SVM	85/85	Works in 85/85	none	85/85	Works in 85/85	none
DevNet	0/85	Unsuitable (0/85)	labeled outliers required: 85	0/85	Unsuitable (0/85)	labeled outliers required: 85
RGraph	35/85	Unsuitable in 50/85	timeout: 50	54/85	Unsuitable in 31/85	timeout: 31
LUNAR	85/85	Works in 85/85	none	85/85	Works in 85/85	none

Table 86: Suitability at 10 and 5 percent retained normal training data, PCA(2). Unsuitable labels refer to the stated constructor, dimensionality, normal-only input, and measured 15-second budget. Partial coverage is reported by case count; raw successful scores over budget are retained separately.

Configuration	10% runs	10% suitability	10% reasons	5% runs	5% suitability	5% reasons
NSA fixed	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA variable	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA coverage	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA grid	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA hierarchical	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA voronoi	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA deterministic	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA suppressed	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA dual	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA annealed	85/85	Works in 85/85	none	85/85	Works in 85/85	none
NSA binary Hamming	0/85	Unsuitable (0/85)	runtime error: 85	0/85	Unsuitable (0/85)	runtime error: 85
NSA binary contiguous	0/85	Unsuitable (0/85)	runtime error: 85	0/85	Unsuitable (0/85)	runtime error: 85
NSA binary chunk	0/85	Unsuitable (0/85)	runtime error: 85	0/85	Unsuitable (0/85)	runtime error: 85
ECOD	85/85	Works in 85/85	none	85/85	Works in 85/85	none
ABOD	85/85	Works in 85/85	none	85/85	Works in 85/85	none
COPOD	85/85	Works in 85/85	none	85/85	Works in 85/85	none
MAD	0/85	Unsuitable (0/85)	1D required: 85	0/85	Unsuitable (0/85)	1D required: 85
SOS	85/85	Works in 85/85	none	85/85	Works in 85/85	none
QMCD	85/85	Works in 85/85	none	85/85	Works in 85/85	none
KDE	85/85	Works in 85/85	none	85/85	Works in 85/85	none
Sampling	65/85	Unsuitable in 20/85	runtime error: 20	50/85	Unsuitable in 35/85	runtime error: 35
GMM	85/85	Works in 85/85	none	85/85	Works in 85/85	none
PCA	85/85	Works in 85/85	none	85/85	Works in 85/85	none
KPCA	85/85	Works in 85/85	none	85/85	Works in 85/85	none
MCD	85/85	Works in 85/85	none	85/85	Works in 85/85	none
CD	85/85	Works in 85/85	none	85/85	Works in 85/85	none
OCSVM	85/85	Works in 85/85	none	85/85	Works in 85/85	none
LMDD	85/85	Works in 85/85	none	85/85	Works in 85/85	none

Continued on next page

Table 86 continued

Configuration	10% runs	10% suitability	10% reasons	5% runs	5% suitability	5% reasons
LOF	85/85	Works in 85/85	none	85/85	Works in 85/85	none
COF	85/85	Works in 85/85	none	85/85	Works in 85/85	none
CBLOF	80/85	Unsuitable in 5/85	runtime error: 5	65/85	Unsuitable in 20/85	runtime error: 20
LOCI	28/85	Unsuitable in 57/85	timeout: 57	30/85	Unsuitable in 55/85	timeout: 55
HBOS	85/85	Works in 85/85	none	85/85	Works in 85/85	none
HDBSCAN	85/85	Works in 85/85	none	85/85	Works in 85/85	none
KNN	85/85	Works in 85/85	none	85/85	Works in 85/85	none
SOD	63/85	Unsuitable in 22/85	construction error: 5; over budget: 5; timeout: 12	55/85	Unsuitable in 30/85	construction error: 30
ROD	85/85	Works in 85/85	none	85/85	Works in 85/85	none
IForest	85/85	Works in 85/85	none	85/85	Works in 85/85	none
INNE	85/85	Works in 85/85	none	85/85	Works in 85/85	none
DIF	83/85	Unsuitable in 2/85	over budget: 2	85/85	Works in 85/85	none
FeatureBagging	85/85	Works in 85/85	none	85/85	Works in 85/85	none
LSCP	50/85	Unsuitable in 35/85	runtime error: 35	50/85	Unsuitable in 35/85	runtime error: 35
LODA	85/85	Works in 85/85	none	85/85	Works in 85/85	none
SUOD	84/85	Unsuitable in 1/85	runtime error: 1	85/85	Works in 85/85	none
XGBOD	0/85	Unsuitable (0/85)	labeled outliers required: 85	0/85	Unsuitable (0/85)	labeled outliers required: 85
AutoEncoder	50/85	Unsuitable in 35/85	runtime error: 35	0/85	Unsuitable (0/85)	runtime error: 85
VAE	50/85	Unsuitable in 35/85	runtime error: 35	0/85	Unsuitable (0/85)	runtime error: 85
SO_GAAL	85/85	Works in 85/85	none	85/85	Works in 85/85	none
MO_GAAL	85/85	Works in 85/85	none	85/85	Works in 85/85	none
DeepSVDD	85/85	Works in 85/85	none	85/85	Works in 85/85	none
AnoGAN	85/85	Works in 85/85	none	85/85	Works in 85/85	none
ALAD	85/85	Works in 85/85	none	85/85	Works in 85/85	none
AE1SVM	85/85	Works in 85/85	none	85/85	Works in 85/85	none
DevNet	0/85	Unsuitable (0/85)	labeled outliers required: 85	0/85	Unsuitable (0/85)	labeled outliers required: 85
RGraph	53/85	Unsuitable in 32/85	timeout: 32	85/85	Works in 85/85	none
LUNAR	85/85	Works in 85/85	none	85/85	Works in 85/85	none

Table 87: Every detector at 10 and 5 percent, Original features. Incomplete means are available-case. Q is strict qualified coverage, R is raw-success coverage.

Configuration	10% AUROC	10% AP	10% Q	10% R	5% AUROC	5% AP	5% Q	5% R
NSA fixed	0.2982	0.1093	85/85	85/85	0.2951	0.1173	85/85	85/85
NSA variable	0.4350	0.1620	85/85	85/85	0.3844	0.1509	85/85	85/85
NSA coverage	0.4333	0.1620	85/85	85/85	0.3819	0.1463	85/85	85/85
NSA grid	0.4323	0.1573	85/85	85/85	0.3823	0.1417	85/85	85/85
NSA hierarchical	0.2452	0.1031	85/85	85/85	0.3089	0.1193	85/85	85/85
NSA voronoi	—	—	0/85	0/85	—	—	0/85	0/85
NSA deterministic	—	—	0/85	0/85	—	—	0/85	0/85
NSA suppressed	0.4366	0.1614	85/85	85/85	0.3803	0.1483	85/85	85/85
NSA dual	0.7792	0.5112	85/85	85/85	0.7900	0.5066	85/85	85/85
NSA annealed	0.2641	0.1040	85/85	85/85	0.2564	0.0924	85/85	85/85
NSA binary Hamming	0.5024	0.1361	85/85	85/85	0.5036	0.1234	85/85	85/85
NSA binary contiguous	0.5317	0.1321	75/85	75/85	0.5437	0.1339	75/85	75/85
NSA binary chunk	0.5350	0.1290	78/85	78/85	0.5223	0.1186	84/85	84/85
ECOD	0.7729	0.4329	77/85	85/85	0.7686	0.4268	85/85	85/85
ABOD	0.8536	0.5352	85/85	85/85	0.8227	0.4993	85/85	85/85
COPOD	0.7776	0.4474	85/85	85/85	0.7762	0.4436	85/85	85/85
MAD	—	—	0/85	0/85	—	—	0/85	0/85
SOS	0.5916	0.2208	84/85	84/85	0.5928	0.2208	85/85	85/85
QMCD	0.4502	0.1812	85/85	85/85	0.3821	0.1485	85/85	85/85
KDE	0.8781	0.5620	85/85	85/85	0.8577	0.5237	85/85	85/85
Sampling	0.8732	0.5472	65/85	65/85	0.8850	0.5191	50/85	50/85
GMM	0.8420	0.5067	85/85	85/85	0.8244	0.4648	85/85	85/85
PCA	0.8061	0.5068	85/85	85/85	0.7907	0.4861	85/85	85/85
KPCA	0.8766	0.5579	85/85	85/85	0.8544	0.5214	85/85	85/85
MCD	0.8178	0.4461	85/85	85/85	0.7862	0.4051	85/85	85/85
CD	0.8216	0.4776	78/85	78/85	0.8162	0.4575	79/85	79/85
OCSVM	0.8407	0.5353	85/85	85/85	0.8258	0.5112	85/85	85/85
LMDD	0.7443	0.4517	52/85	52/85	0.7284	0.4163	55/85	55/85
LOF	0.7931	0.5063	85/85	85/85	0.7739	0.4874	85/85	85/85

Continued on next page

Table 87 continued

Configuration	10% AUROC	10% AP	10% Q	10% R	5% AUROC	5% AP	5% Q	5% R
COF	0.6657	0.3202	85/85	85/85	0.6500	0.2984	85/85	85/85
CBLOF	0.8580	0.5274	80/85	80/85	0.8652	0.5216	65/85	65/85
LOCI	0.7087	0.3674	29/85	29/85	0.6819	0.3476	30/85	30/85
HBOS	0.8043	0.4543	85/85	85/85	0.7848	0.4265	85/85	85/85
HDBSCAN	0.5446	0.1398	85/85	85/85	0.5303	0.1303	85/85	85/85
KNN	0.8558	0.5347	85/85	85/85	0.8104	0.4913	85/85	85/85
SOD	0.7325	0.3226	58/85	67/85	0.7238	0.2796	55/85	55/85
ROD	0.8054	0.4483	39/85	40/85	0.7826	0.4245	42/85	43/85
IForest	0.8021	0.3957	85/85	85/85	0.7696	0.3516	85/85	85/85
INNE	0.8379	0.4534	85/85	85/85	0.7993	0.3844	85/85	85/85
DIF	0.7636	0.3574	85/85	85/85	0.7334	0.3109	85/85	85/85
FeatureBagging	0.8002	0.5099	85/85	85/85	0.7809	0.4873	85/85	85/85
LSCP	0.8724	0.5188	50/85	50/85	0.8487	0.4971	50/85	50/85
LODA	0.6620	0.1956	85/85	85/85	0.4868	0.1312	85/85	85/85
SUOD	0.8190	0.4654	85/85	85/85	0.7981	0.4334	84/85	84/85
XGBOD	—	—	0/85	0/85	—	—	0/85	0/85
AutoEncoder	0.8347	0.5080	50/85	50/85	—	—	0/85	0/85
VAE	0.8194	0.4915	50/85	50/85	—	—	0/85	0/85
SO_GAAL	0.4595	0.2057	85/85	85/85	0.4980	0.2088	85/85	85/85
MO_GAAL	0.4879	0.2013	85/85	85/85	0.4889	0.2088	85/85	85/85
DeepSVDD	0.8089	0.5069	85/85	85/85	0.7927	0.4683	85/85	85/85
AnoGAN	0.7961	0.4628	85/85	85/85	0.7894	0.4754	85/85	85/85
ALAD	0.5461	0.1544	85/85	85/85	0.5458	0.1551	85/85	85/85
AE1SVM	0.8143	0.5101	85/85	85/85	0.7921	0.4924	85/85	85/85
DevNet	—	—	0/85	0/85	—	—	0/85	0/85
RGraph	0.5518	0.2457	35/85	35/85	0.5585	0.2186	54/85	54/85
LUNAR	0.8703	0.5564	85/85	85/85	0.8343	0.5021	85/85	85/85

Table 88: All NSA AUROC ranks against the fixed other-detector field at every size, Original features. Parentheses give coverage; blank ranks prevent a partial-case advantage.

Configuration	100%	75%	50%	25%	10%	5%
NSA fixed	20/20 (85/85)	20/20 (85/85)	20/20 (85/85)	20/20 (85/85)	20/20 (85/85)	20/20 (85/85)
NSA variable	17/20 (85/85)	17/20 (85/85)	17/20 (85/85)	20/20 (85/85)	20/20 (85/85)	20/20 (85/85)
NSA coverage	17/20 (85/85)	17/20 (85/85)	17/20 (85/85)	20/20 (85/85)	20/20 (85/85)	20/20 (85/85)
NSA grid	17/20 (85/85)	17/20 (85/85)	17/20 (85/85)	— (84/85)	20/20 (85/85)	20/20 (85/85)
NSA hierarchical	20/20 (85/85)	20/20 (85/85)	20/20 (85/85)	20/20 (85/85)	20/20 (85/85)	20/20 (85/85)
NSA voronoi	— (0/85)	— (0/85)	— (0/85)	— (0/85)	— (0/85)	— (0/85)
NSA deterministic	— (0/85)	— (0/85)	— (0/85)	— (0/85)	— (0/85)	— (0/85)
NSA suppressed	17/20 (85/85)	17/20 (85/85)	17/20 (85/85)	20/20 (85/85)	20/20 (85/85)	20/20 (85/85)
NSA dual	— (81/85)	— (80/85)	— (81/85)	— (83/85)	14/20 (85/85)	10/20 (85/85)
NSA annealed	20/20 (85/85)	20/20 (85/85)	20/20 (85/85)	20/20 (85/85)	20/20 (85/85)	20/20 (85/85)
NSA binary Hamming	— (76/85)	— (83/85)	19/20 (85/85)	20/20 (85/85)	18/20 (85/85)	17/20 (85/85)
NSA binary contiguous	— (59/85)	— (60/85)	— (60/85)	— (63/85)	— (75/85)	— (75/85)
NSA binary chunk	— (75/85)	— (75/85)	— (75/85)	— (75/85)	— (78/85)	— (84/85)

Table 89: NSA AUROC rank sensitivity, Original features. The raw comparator field requires all six sizes complete, and the focal NSA requires all 85 raw successes in its own panel. Strict uses the measured-time field; pointwise removes batch-dependent scorers. Different denominators identify different fields, so movement between columns is not a score improvement.

NSA configuration	10% strict	10% raw	10% pointwise	5% strict	5% raw	5% pointwise
NSA fixed	20/20	22/22	19/19	20/20	22/22	19/19
NSA variable	20/20	22/22	19/19	20/20	22/22	19/19
NSA coverage	20/20	22/22	19/19	20/20	22/22	19/19
NSA grid	20/20	22/22	19/19	20/20	22/22	19/19
NSA hierarchical	20/20	22/22	19/19	20/20	22/22	19/19
NSA voronoi	—	—	—	—	—	—
NSA deterministic	—	—	—	—	—	—
NSA suppressed	20/20	22/22	19/19	20/20	22/22	19/19
NSA dual	14/20	14/22	14/19	10/20	10/22	10/19
NSA annealed	20/20	22/22	19/19	20/20	22/22	19/19
NSA binary Hamming	18/20	20/22	17/19	17/20	19/22	16/19
NSA binary contiguous	—	—	—	—	—	—
NSA binary chunk	—	—	—	—	—	—

Table 90: NSA AP rank sensitivity, Original features. The raw comparator field requires all six sizes complete, and the focal NSA requires all 85 raw successes in its own panel. Strict uses the measured-time field; pointwise removes batch-dependent scorers. Different denominators identify different fields, so movement between columns is not a score improvement.

NSA configuration	10% strict	10% raw	10% pointwise	5% strict	5% raw	5% pointwise
NSA fixed	20/20	22/22	19/19	20/20	22/22	19/19
NSA variable	18/20	20/22	17/19	18/20	20/22	17/19
NSA coverage	18/20	20/22	17/19	18/20	20/22	17/19
NSA grid	18/20	20/22	17/19	18/20	20/22	17/19
NSA hierarchical	20/20	22/22	19/19	20/20	22/22	19/19
NSA voronoi	—	—	—	—	—	—
NSA deterministic	—	—	—	—	—	—
NSA suppressed	18/20	20/22	17/19	18/20	20/22	17/19
NSA dual	7/20	7/22	7/19	4/20	4/22	4/19
NSA annealed	20/20	22/22	19/19	20/20	22/22	19/19
NSA binary Hamming	20/20	22/22	19/19	20/20	22/22	19/19
NSA binary contiguous	—	—	—	—	—	—
NSA binary chunk	—	—	—	—	—	—

Table 91: Panel-qualified detector ranking, Original features, 10 percent. The field changes across sizes; this is a sensitivity result, not the primary cross-size field.

Configuration	AUROC	AUROC rank	Top position	AP
KDE	0.8781	1/36	2.78%	0.5620
KPCA	0.8766	2/36	5.56%	0.5579
LUNAR	0.8703	3/36	8.33%	0.5564
KNN	0.8558	4/36	11.11%	0.5347
ABOD	0.8536	5/36	13.89%	0.5352
GMM	0.8420	6/36	16.67%	0.5067
OCSVM	0.8407	7/36	19.44%	0.5353
INNE	0.8379	8/36	22.22%	0.4534
SUOD	0.8190	9/36	25.00%	0.4654
MCD	0.8178	10/36	27.78%	0.4461
AE1SVM	0.8143	11/36	30.56%	0.5101
DeepSVDD	0.8089	12/36	33.33%	0.5069
PCA	0.8061	13/36	36.11%	0.5068
HBOS	0.8043	14/36	38.89%	0.4543
IForest	0.8021	15/36	41.67%	0.3957
FeatureBagging	0.8002	16/36	44.44%	0.5099
AnoGAN	0.7961	17/36	47.22%	0.4628
LOF	0.7931	18/36	50.00%	0.5063
NSA dual	0.7792	19/36	52.78%	0.5112
COPOD	0.7776	20/36	55.56%	0.4474
DIF	0.7636	21/36	58.33%	0.3574
COF	0.6657	22/36	61.11%	0.3202
LODA	0.6620	23/36	63.89%	0.1956
ALAD	0.5461	24/36	66.67%	0.1544
HDBSCAN	0.5446	25/36	69.44%	0.1398
NSA binary Hamming	0.5024	26/36	72.22%	0.1361
MO_GAAL	0.4879	27/36	75.00%	0.2013
SO_GAAL	0.4595	28/36	77.78%	0.2057
QMCD	0.4502	29/36	80.56%	0.1812
NSA suppressed	0.4366	30/36	83.33%	0.1614
NSA variable	0.4350	31/36	86.11%	0.1620
NSA coverage	0.4333	32/36	88.89%	0.1620
NSA grid	0.4323	33/36	91.67%	0.1573
NSA fixed	0.2982	34/36	94.44%	0.1093
NSA annealed	0.2641	35/36	97.22%	0.1040
NSA hierarchical	0.2452	36/36	100.00%	0.1031

Table 92: Panel-qualified detector ranking, Original features, 5 percent. The field changes across sizes; this is a sensitivity result, not the primary cross-size field.

Configuration	AUROC	AUROC rank	Top position	AP
KDE	0.8577	1/37	2.70%	0.5237
KPCA	0.8544	2/37	5.41%	0.5214
LUNAR	0.8343	3/37	8.11%	0.5021
OCSVM	0.8258	4/37	10.81%	0.5112
GMM	0.8244	5/37	13.51%	0.4648
ABOD	0.8227	6/37	16.22%	0.4993
KNN	0.8104	7/37	18.92%	0.4913
INNE	0.7993	8/37	21.62%	0.3844
DeepSVDD	0.7927	9/37	24.32%	0.4683
AE1SVM	0.7921	10/37	27.03%	0.4924
PCA	0.7907	11/37	29.73%	0.4861
NSA dual	0.7900	12/37	32.43%	0.5066
AnoGAN	0.7894	13/37	35.14%	0.4754
MCD	0.7862	14/37	37.84%	0.4051
HBOS	0.7848	15/37	40.54%	0.4265
FeatureBagging	0.7809	16/37	43.24%	0.4873
COPOD	0.7762	17/37	45.95%	0.4436
LOF	0.7739	18/37	48.65%	0.4874
IForest	0.7696	19/37	51.35%	0.3516
ECOD	0.7686	20/37	54.05%	0.4268
DIF	0.7334	21/37	56.76%	0.3109
COF	0.6500	22/37	59.46%	0.2984
SOS	0.5928	23/37	62.16%	0.2208
ALAD	0.5458	24/37	64.86%	0.1551
HDBSCAN	0.5303	25/37	67.57%	0.1303
NSA binary Hamming	0.5036	26/37	70.27%	0.1234
SO_GAAL	0.4980	27/37	72.97%	0.2088
MO_GAAL	0.4889	28/37	75.68%	0.2088
LODA	0.4868	29/37	78.38%	0.1312
NSA variable	0.3844	30/37	81.08%	0.1509
NSA grid	0.3823	31/37	83.78%	0.1417
QMCD	0.3821	32/37	86.49%	0.1485
NSA coverage	0.3819	33/37	89.19%	0.1463
NSA suppressed	0.3803	34/37	91.89%	0.1483
NSA hierarchical	0.3089	35/37	94.59%	0.1193
NSA fixed	0.2951	36/37	97.30%	0.1173
NSA annealed	0.2564	37/37	100.00%	0.0924

Table 93: Every configuration's paired change from full normal-pool retention, Original features. Positive favors fewer normals. Intervals are descriptive dataset-cluster bootstrap intervals, unadjusted for multiple comparisons.

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
NSA fixed	10%	85/85	17/17	-0.0615	[-0.0986, -0.0230]	-0.0194	[-0.0352, -0.0043]
NSA variable	10%	85/85	17/17	-0.1581	[-0.2739, -0.0578]	-0.1418	[-0.2484, -0.0520]
NSA coverage	10%	85/85	17/17	-0.1598	[-0.2714, -0.0605]	-0.1420	[-0.2465, -0.0501]
NSA grid	10%	85/85	17/17	-0.1649	[-0.2771, -0.0660]	-0.1452	[-0.2560, -0.0603]
NSA hierarchical	10%	85/85	17/17	-0.0516	[-0.1108, -0.0009]	-0.0142	[-0.0438, +0.0140]
NSA voronoi	10%	0/85	0/17	—	[—, —]	—	[—, —]
NSA deterministic	10%	0/85	0/17	—	[—, —]	—	[—, —]
NSA suppressed	10%	85/85	17/17	-0.1518	[-0.2595, -0.0614]	-0.1369	[-0.2349, -0.0565]
NSA dual	10%	81/85	17/17	+0.0780	[+0.0339, +0.1305]	+0.0532	[+0.0139, +0.0954]
NSA annealed	10%	85/85	17/17	-0.0520	[-0.0954, -0.0166]	-0.0080	[-0.0211, +0.0053]
NSA binary Hamming	10%	76/85	16/17	-0.0240	[-0.0511, +0.0001]	-0.0325	[-0.0896, +0.0016]
NSA binary contiguous	10%	59/85	12/17	-0.0122	[-0.0634, +0.0463]	-0.0209	[-0.0682, +0.0120]
NSA binary chunk	10%	75/85	15/17	+0.0052	[-0.0235, +0.0348]	-0.0151	[-0.0313, -0.0025]
ECOD	10%	69/85	17/17	-0.0140	[-0.0204, -0.0083]	-0.0433	[-0.0759, -0.0195]
ABOD	10%	85/85	17/17	-0.0415	[-0.0685, -0.0189]	-0.0927	[-0.1532, -0.0420]
COPOD	10%	85/85	17/17	-0.0090	[-0.0169, -0.0015]	-0.0098	[-0.0205, +0.0015]
MAD	10%	0/85	0/17	—	[—, —]	—	[—, —]
SOS	10%	84/85	17/17	+0.0047	[-0.0059, +0.0183]	-0.0012	[-0.0087, +0.0063]
QMCD	10%	85/85	17/17	-0.0874	[-0.2297, +0.0645]	-0.0557	[-0.1399, +0.0261]
KDE	10%	85/85	17/17	-0.0189	[-0.0379, +0.0008]	-0.0819	[-0.1448, -0.0282]
Sampling	10%	65/85	13/17	-0.0076	[-0.0184, +0.0030]	-0.0253	[-0.0553, +0.0013]
GMM	10%	85/85	17/17	-0.0233	[-0.0512, +0.0019]	-0.0430	[-0.0784, -0.0036]
PCA	10%	82/85	17/17	-0.0010	[-0.0097, +0.0092]	-0.0230	[-0.0418, -0.0042]
KPCA	10%	85/85	17/17	-0.0257	[-0.0419, -0.0125]	-0.0886	[-0.1501, -0.0373]
MCD	10%	85/85	17/17	-0.0249	[-0.0482, -0.0001]	-0.0241	[-0.0693, +0.0158]
CD	10%	77/85	16/17	+0.0003	[-0.0209, +0.0228]	+0.0037	[-0.0193, +0.0279]
OCSVM	10%	85/85	17/17	-0.0002	[-0.0145, +0.0187]	-0.0322	[-0.0527, -0.0127]
LMDD	10%	52/85	11/17	+0.0019	[-0.0132, +0.0187]	-0.0196	[-0.0575, +0.0122]

Continued on next page

Table 93 continued

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
LOF	10%	85/85	17/17	-0.0948	[-0.1502, -0.0456]	-0.0723	[-0.1319, -0.0179]
COF	10%	85/85	17/17	-0.0089	[-0.0254, +0.0056]	-0.0227	[-0.0595, +0.0005]
CBLOF	10%	79/85	16/17	-0.0164	[-0.0303, -0.0031]	-0.0280	[-0.0499, -0.0065]
LOCI	10%	29/85	6/17	+0.0336	[-0.0112, +0.0981]	+0.0362	[-0.0001, +0.0953]
HBOS	10%	85/85	17/17	-0.0090	[-0.0251, +0.0096]	-0.0409	[-0.0827, -0.0112]
HDBSCAN	10%	85/85	17/17	-0.1500	[-0.2207, -0.0837]	-0.0994	[-0.1551, -0.0506]
KNN	10%	85/85	17/17	-0.0415	[-0.0677, -0.0181]	-0.0815	[-0.1428, -0.0298]
SOD	10%	58/85	16/17	-0.0101	[-0.0272, +0.0050]	-0.0107	[-0.0249, +0.0007]
ROD	10%	38/85	8/17	-0.0071	[-0.0127, -0.0013]	-0.0226	[-0.0323, -0.0129]
IForest	10%	85/85	17/17	-0.0512	[-0.0794, -0.0267]	-0.1302	[-0.2039, -0.0669]
INNE	10%	85/85	17/17	-0.0353	[-0.0604, -0.0160]	-0.1223	[-0.2096, -0.0627]
DIF	10%	85/85	17/17	-0.0555	[-0.0804, -0.0299]	-0.0854	[-0.1220, -0.0510]
FeatureBagging	10%	85/85	17/17	-0.0889	[-0.1425, -0.0387]	-0.0759	[-0.1414, -0.0173]
LSCP	10%	50/85	10/17	-0.0656	[-0.1180, -0.0186]	-0.0904	[-0.1632, -0.0297]
LODA	10%	85/85	17/17	-0.0806	[-0.1674, +0.0004]	-0.2194	[-0.3652, -0.0919]
SUOD	10%	85/85	17/17	-0.0313	[-0.0519, -0.0147]	-0.0786	[-0.1246, -0.0393]
XGBOD	10%	0/85	0/17	—	[—, —]	—	[—, —]
AutoEncoder	10%	50/85	10/17	-0.0779	[-0.1384, -0.0215]	-0.0798	[-0.1481, -0.0262]
VAE	10%	50/85	10/17	-0.0455	[-0.0931, -0.0097]	-0.0660	[-0.1029, -0.0345]
SO_GAAL	10%	85/85	17/17	-0.0691	[-0.1231, -0.0150]	-0.0636	[-0.1317, -0.0046]
MO_GAAL	10%	85/85	17/17	-0.0560	[-0.1027, -0.0090]	-0.0579	[-0.0983, -0.0177]
DeepSVDD	10%	85/85	17/17	-0.0561	[-0.0940, -0.0220]	-0.0838	[-0.1481, -0.0270]
AnoGAN	10%	85/85	17/17	+0.0059	[-0.0088, +0.0197]	-0.0211	[-0.0538, +0.0080]
ALAD	10%	85/85	17/17	+0.0052	[-0.0264, +0.0377]	+0.0074	[-0.0020, +0.0178]
AE1SVM	10%	85/85	17/17	-0.0219	[-0.0351, -0.0099]	-0.0357	[-0.0589, -0.0119]
DevNet	10%	0/85	0/17	—	[—, —]	—	[—, —]
RGraph	10%	26/85	6/17	+0.1433	[+0.0153, +0.2920]	+0.0758	[+0.0017, +0.2119]
LUNAR	10%	85/85	17/17	-0.0354	[-0.0536, -0.0181]	-0.1204	[-0.2024, -0.0511]
NSA fixed	5%	85/85	17/17	-0.0645	[-0.1099, -0.0169]	-0.0113	[-0.0293, +0.0067]
NSA variable	5%	85/85	17/17	-0.2087	[-0.3503, -0.0876]	-0.1529	[-0.2749, -0.0475]
NSA coverage	5%	85/85	17/17	-0.2112	[-0.3489, -0.0904]	-0.1577	[-0.2777, -0.0589]
NSA grid	5%	85/85	17/17	-0.2150	[-0.3484, -0.1012]	-0.1609	[-0.2840, -0.0588]
NSA hierarchical	5%	85/85	17/17	+0.0121	[-0.0489, +0.0717]	+0.0019	[-0.0232, +0.0258]

Continued on next page

Table 93 continued

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
NSA voronoi	5%	0/85	0/17	—	[—, —]	—	[—, —]
NSA deterministic	5%	0/85	0/17	—	[—, —]	—	[—, —]
NSA suppressed	5%	85/85	17/17	-0.2080	[-0.3382, -0.0917]	-0.1500	[-0.2605, -0.0482]
NSA dual	5%	81/85	17/17	+0.0903	[+0.0410, +0.1436]	+0.0493	[+0.0060, +0.0949]
NSA annealed	5%	85/85	17/17	-0.0597	[-0.1040, -0.0217]	-0.0196	[-0.0380, -0.0045]
NSA binary Hamming	5%	76/85	16/17	-0.0234	[-0.0634, +0.0100]	-0.0458	[-0.1250, -0.0003]
NSA binary contiguous	5%	59/85	12/17	-0.0177	[-0.0725, +0.0472]	-0.0180	[-0.0551, +0.0109]
NSA binary chunk	5%	75/85	15/17	-0.0018	[-0.0299, +0.0292]	-0.0234	[-0.0506, -0.0050]
ECOD	5%	77/85	17/17	-0.0162	[-0.0228, -0.0104]	-0.0466	[-0.0799, -0.0206]
ABOD	5%	85/85	17/17	-0.0725	[-0.1113, -0.0391]	-0.1285	[-0.2007, -0.0668]
COPOD	5%	85/85	17/17	-0.0103	[-0.0185, -0.0025]	-0.0137	[-0.0246, -0.0022]
MAD	5%	0/85	0/17	—	[—, —]	—	[—, —]
SOS	5%	85/85	17/17	+0.0059	[-0.0064, +0.0234]	-0.0010	[-0.0163, +0.0173]
QMCD	5%	85/85	17/17	-0.1556	[-0.3417, +0.0553]	-0.0885	[-0.1870, +0.0084]
KDE	5%	85/85	17/17	-0.0393	[-0.0637, -0.0176]	-0.1202	[-0.2047, -0.0548]
Sampling	5%	50/85	10/17	-0.0134	[-0.0309, +0.0032]	-0.0538	[-0.1025, -0.0123]
GMM	5%	85/85	17/17	-0.0409	[-0.0728, -0.0098]	-0.0849	[-0.1318, -0.0364]
PCA	5%	82/85	17/17	-0.0162	[-0.0282, -0.0040]	-0.0433	[-0.0653, -0.0213]
KPCA	5%	85/85	17/17	-0.0479	[-0.0718, -0.0273]	-0.1250	[-0.2007, -0.0600]
MCD	5%	85/85	17/17	-0.0565	[-0.0937, -0.0225]	-0.0651	[-0.1366, -0.0114]
CD	5%	79/85	16/17	-0.0030	[-0.0303, +0.0264]	-0.0144	[-0.0514, +0.0255]
OCSVM	5%	85/85	17/17	-0.0151	[-0.0316, +0.0018]	-0.0562	[-0.0842, -0.0294]
LMDD	5%	55/85	12/17	-0.0305	[-0.0710, +0.0030]	-0.0357	[-0.0717, -0.0048]
LOF	5%	85/85	17/17	-0.1140	[-0.1868, -0.0507]	-0.0912	[-0.1580, -0.0345]
COF	5%	85/85	17/17	-0.0245	[-0.0540, -0.0022]	-0.0444	[-0.0892, -0.0101]
CBLOF	5%	64/85	13/17	-0.0317	[-0.0520, -0.0129]	-0.0562	[-0.0898, -0.0246]
LOCI	5%	30/85	6/17	+0.0051	[-0.0279, +0.0396]	+0.0147	[-0.0235, +0.0644]
HBOS	5%	85/85	17/17	-0.0285	[-0.0484, -0.0089]	-0.0687	[-0.1193, -0.0280]
HDBSCAN	5%	85/85	17/17	-0.1644	[-0.2372, -0.0873]	-0.1089	[-0.1648, -0.0591]
KNN	5%	85/85	17/17	-0.0869	[-0.1347, -0.0445]	-0.1249	[-0.1961, -0.0628]
SOD	5%	55/85	11/17	-0.0027	[-0.0211, +0.0130]	-0.0038	[-0.0174, +0.0077]
ROD	5%	39/85	8/17	-0.0248	[-0.0367, -0.0142]	-0.0448	[-0.0705, -0.0226]

Continued on next page

Table 93 continued

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
IForest	5%	85/85	17/17	-0.0837	[-0.1267, -0.0466]	-0.1742	[-0.2559, -0.1017]
INNE	5%	85/85	17/17	-0.0739	[-0.1126, -0.0399]	-0.1912	[-0.2859, -0.1128]
DIF	5%	85/85	17/17	-0.0858	[-0.1248, -0.0474]	-0.1318	[-0.1914, -0.0759]
FeatureBagging	5%	85/85	17/17	-0.1082	[-0.1785, -0.0506]	-0.0985	[-0.1672, -0.0371]
LSCP	5%	50/85	10/17	-0.0893	[-0.1596, -0.0279]	-0.1121	[-0.2054, -0.0373]
LODA	5%	85/85	17/17	-0.2559	[-0.3908, -0.1209]	-0.2837	[-0.4331, -0.1375]
SUOD	5%	84/85	17/17	-0.0517	[-0.0790, -0.0290]	-0.1098	[-0.1683, -0.0603]
XGBOD	5%	0/85	0/17	—	[—, —]	—	[—, —]
AutoEncoder	5%	0/85	0/17	—	[—, —]	—	[—, —]
VAE	5%	0/85	0/17	—	[—, —]	—	[—, —]
SO_GAAL	5%	85/85	17/17	-0.0306	[-0.1048, +0.0463]	-0.0605	[-0.1428, +0.0039]
MO_GAAL	5%	85/85	17/17	-0.0550	[-0.1300, +0.0294]	-0.0504	[-0.1059, +0.0008]
DeepSVDD	5%	85/85	17/17	-0.0724	[-0.1168, -0.0335]	-0.1225	[-0.1932, -0.0630]
AnoGAN	5%	85/85	17/17	-0.0008	[-0.0194, +0.0156]	-0.0085	[-0.0275, +0.0099]
ALAD	5%	85/85	17/17	+0.0050	[-0.0170, +0.0279]	+0.0080	[-0.0002, +0.0163]
AE1SVM	5%	85/85	17/17	-0.0442	[-0.0640, -0.0256]	-0.0534	[-0.0824, -0.0243]
DevNet	5%	0/85	0/17	—	[—, —]	—	[—, —]
RGraph	5%	26/85	6/17	+0.1779	[-0.0047, +0.3734]	+0.0606	[-0.0556, +0.2132]
LUNAR	5%	85/85	17/17	-0.0715	[-0.1008, -0.0451]	-0.1746	[-0.2699, -0.0930]

Table 94: All NSA configurations versus IForest and KNN, Original features, new sizes. Differences are NSA minus comparator on matched successful cases.

NSA configuration	Reference	Retained	Pairs	ΔAUROC	95% CI	ΔAP	95% CI
NSA fixed	KNN	10%	85/85	-0.5577	[-0.7131, -0.4007]	-0.4255	[-0.5820, -0.2629]
NSA fixed	IForest	10%	85/85	-0.5039	[-0.6516, -0.3501]	-0.2864	[-0.4166, -0.1637]
NSA variable	KNN	10%	85/85	-0.4209	[-0.5902, -0.2474]	-0.3727	[-0.5464, -0.2088]
NSA variable	IForest	10%	85/85	-0.3671	[-0.5309, -0.2043]	-0.2336	[-0.3705, -0.0993]
NSA coverage	KNN	10%	85/85	-0.4225	[-0.5930, -0.2523]	-0.3728	[-0.5438, -0.1991]
NSA coverage	IForest	10%	85/85	-0.3688	[-0.5344, -0.2091]	-0.2337	[-0.3687, -0.1055]
NSA grid	KNN	10%	85/85	-0.4235	[-0.6019, -0.2537]	-0.3774	[-0.5472, -0.2128]
NSA grid	IForest	10%	85/85	-0.3698	[-0.5368, -0.2069]	-0.2383	[-0.3790, -0.1111]
NSA hierarchical	KNN	10%	85/85	-0.6106	[-0.7514, -0.4658]	-0.4316	[-0.5871, -0.2768]
NSA hierarchical	IForest	10%	85/85	-0.5569	[-0.6862, -0.4224]	-0.2925	[-0.4217, -0.1708]
NSA voronoi	KNN	10%	0/85	—	[—, —]	—	[—, —]
NSA voronoi	IForest	10%	0/85	—	[—, —]	—	[—, —]
NSA deterministic	KNN	10%	0/85	—	[—, —]	—	[—, —]
NSA deterministic	IForest	10%	0/85	—	[—, —]	—	[—, —]
NSA suppressed	KNN	10%	85/85	-0.4192	[-0.5942, -0.2501]	-0.3734	[-0.5461, -0.2028]
NSA suppressed	IForest	10%	85/85	-0.3655	[-0.5204, -0.2047]	-0.2343	[-0.3644, -0.1093]
NSA dual	KNN	10%	85/85	-0.0766	[-0.1300, -0.0326]	-0.0235	[-0.0447, -0.0041]
NSA dual	IForest	10%	85/85	-0.0229	[-0.1001, +0.0437]	+0.1156	[+0.0173, +0.2496]
NSA annealed	KNN	10%	85/85	-0.5917	[-0.7325, -0.4455]	-0.4308	[-0.5929, -0.2678]
NSA annealed	IForest	10%	85/85	-0.5380	[-0.6703, -0.4020]	-0.2917	[-0.4246, -0.1647]
NSA binary Hamming	KNN	10%	85/85	-0.3535	[-0.4290, -0.2776]	-0.3986	[-0.5378, -0.2664]
NSA binary Hamming	IForest	10%	85/85	-0.2997	[-0.3744, -0.2262]	-0.2595	[-0.3667, -0.1605]
NSA binary contiguous	KNN	10%	75/85	-0.3575	[-0.4057, -0.3093]	-0.4288	[-0.5772, -0.2863]
NSA binary contiguous	IForest	10%	75/85	-0.2969	[-0.3588, -0.2308]	-0.2693	[-0.3937, -0.1520]
NSA binary chunk	KNN	10%	78/85	-0.3151	[-0.3936, -0.2287]	-0.4047	[-0.5500, -0.2614]
NSA binary chunk	IForest	10%	78/85	-0.2559	[-0.3362, -0.1649]	-0.2626	[-0.3765, -0.1561]
NSA fixed	KNN	5%	85/85	-0.5153	[-0.6879, -0.3315]	-0.3740	[-0.5417, -0.2053]
NSA fixed	IForest	5%	85/85	-0.4745	[-0.6228, -0.3217]	-0.2343	[-0.3490, -0.1217]

Continued on next page

Table 94 continued

NSA configuration	Reference	Retained	Pairs	ΔAUROC	95% CI	ΔAP	95% CI
NSA variable	KNN	5%	85/85	-0.4260	[-0.6264, -0.2239]	-0.3405	[-0.5194, -0.1664]
NSA variable	IForest	5%	85/85	-0.3852	[-0.5595, -0.2136]	-0.2007	[-0.3304, -0.0784]
NSA coverage	KNN	5%	85/85	-0.4285	[-0.6263, -0.2372]	-0.3451	[-0.5127, -0.1684]
NSA coverage	IForest	5%	85/85	-0.3877	[-0.5657, -0.2145]	-0.2053	[-0.3332, -0.0813]
NSA grid	KNN	5%	85/85	-0.4281	[-0.6226, -0.2333]	-0.3496	[-0.5265, -0.1760]
NSA grid	IForest	5%	85/85	-0.3873	[-0.5584, -0.2165]	-0.2099	[-0.3302, -0.0930]
NSA hierarchical	KNN	5%	85/85	-0.5015	[-0.6739, -0.3086]	-0.3721	[-0.5396, -0.2101]
NSA hierarchical	IForest	5%	85/85	-0.4607	[-0.6241, -0.2982]	-0.2324	[-0.3481, -0.1235]
NSA voronoi	KNN	5%	0/85	—	[—, —]	—	[—, —]
NSA voronoi	IForest	5%	0/85	—	[—, —]	—	[—, —]
NSA deterministic	KNN	5%	0/85	—	[—, —]	—	[—, —]
NSA deterministic	IForest	5%	0/85	—	[—, —]	—	[—, —]
NSA suppressed	KNN	5%	85/85	-0.4301	[-0.6272, -0.2346]	-0.3431	[-0.5214, -0.1638]
NSA suppressed	IForest	5%	85/85	-0.3893	[-0.5605, -0.2128]	-0.2034	[-0.3327, -0.0850]
NSA dual	KNN	5%	85/85	-0.0204	[-0.0591, +0.0156]	+0.0153	[-0.0131, +0.0533]
NSA dual	IForest	5%	85/85	+0.0204	[-0.0468, +0.0906]	+0.1550	[+0.0579, +0.2756]
NSA annealed	KNN	5%	85/85	-0.5540	[-0.7118, -0.3868]	-0.3990	[-0.5473, -0.2494]
NSA annealed	IForest	5%	85/85	-0.5132	[-0.6502, -0.3708]	-0.2592	[-0.3650, -0.1618]
NSA binary Hamming	KNN	5%	85/85	-0.3068	[-0.3941, -0.2150]	-0.3679	[-0.5123, -0.2286]
NSA binary Hamming	IForest	5%	85/85	-0.2660	[-0.3414, -0.1978]	-0.2282	[-0.3218, -0.1407]
NSA binary contiguous	KNN	5%	75/85	-0.3029	[-0.3853, -0.2181]	-0.3806	[-0.5358, -0.2323]
NSA binary contiguous	IForest	5%	75/85	-0.2474	[-0.3196, -0.1702]	-0.2156	[-0.3213, -0.1202]
NSA binary chunk	KNN	5%	84/85	-0.2873	[-0.3700, -0.1949]	-0.3725	[-0.5211, -0.2326]
NSA binary chunk	IForest	5%	84/85	-0.2453	[-0.3187, -0.1685]	-0.2310	[-0.3306, -0.1405]

Table 95: NSA/fixed, Original features: dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.1887	0.0913	5/5	20/20	0.2249	0.0961	5/5	20/20
cardio	0.2675	0.0905	5/5	20/20	0.1947	0.0881	5/5	20/20
glass	0.4677	0.0817	5/5	19/20	0.5378	0.2245	5/5	19/20
ionosphere	0.0861	0.2192	5/5	20/20	0.0939	0.2242	5/5	20/20
letter	0.5722	0.1083	5/5	13/20	0.5770	0.1012	5/5	11/20
lympho	0.0121	0.0254	5/5	20/20	0.0431	0.0262	5/5	20/20
mnist	0.2755	0.0754	5/5	20/20	0.2373	0.0666	5/5	20/20
musk	0.0000	0.0165	5/5	20/20	0.0015	0.0165	5/5	20/20
optdigits	0.5791	0.0453	5/5	15/20	0.5806	0.0494	5/5	14/20
pendigits	0.5553	0.0761	5/5	19/20	0.5105	0.0880	5/5	20/20
pima	0.4837	0.3713	5/5	18/20	0.4541	0.3469	5/5	18/20
satellite	0.3315	0.2624	5/5	20/20	0.3176	0.2524	5/5	20/20
satimage-2	0.0697	0.0069	5/5	20/20	0.0645	0.0065	5/5	20/20
shuttle	0.0046	0.0370	5/5	20/20	0.0038	0.0370	5/5	20/20
vertebral	0.5567	0.2012	5/5	4/20	0.5786	0.2084	5/5	3/20
vowels	0.5346	0.1164	5/5	17/20	0.5126	0.1310	5/5	14/20
wbc	0.0837	0.0330	5/5	18/20	0.0845	0.0319	5/5	20/20

Table 96: NSA/variable, Original features: dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.1971	0.0933	5/5	20/20	0.2347	0.1069	5/5	20/20
cardio	0.5410	0.1917	5/5	20/20	0.3397	0.0955	5/5	20/20
glass	0.6488	0.2178	5/5	13/20	0.6707	0.3369	5/5	9/20
ionosphere	0.1919	0.2551	5/5	20/20	0.1127	0.2258	5/5	20/20
letter	0.6921	0.1736	5/5	9/20	0.6648	0.1680	5/5	8/20
lympho	0.1138	0.0298	5/5	20/20	0.0293	0.0257	5/5	20/20
mnist	0.3226	0.0791	5/5	20/20	0.2958	0.0750	5/5	20/20
musk	0.0001	0.0165	5/5	20/20	0.0014	0.0165	5/5	20/20
optdigits	0.8473	0.1736	5/5	3/20	0.8108	0.1190	5/5	1/20
pendigits	0.8220	0.1439	5/5	13/20	0.7157	0.1244	5/5	16/20
pima	0.6896	0.5277	5/5	12/20	0.6256	0.4641	5/5	15/20
satellite	0.3505	0.2830	5/5	20/20	0.3584	0.2971	5/5	20/20
satimage-2	0.1096	0.0084	5/5	20/20	0.0857	0.0084	5/5	20/20
shuttle	0.2643	0.1273	5/5	20/20	0.0743	0.0392	5/5	20/20
vertebral	0.5841	0.1828	5/5	3/20	0.5728	0.2011	5/5	4/20
vowels	0.7873	0.2011	5/5	10/20	0.8145	0.2275	5/5	7/20
wbc	0.2321	0.0497	5/5	18/20	0.1273	0.0338	5/5	20/20

Table 97: NSA/coverage, Original features: dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.1971	0.0933	5/5	20/20	0.2347	0.1069	5/5	20/20
cardio	0.5410	0.1917	5/5	20/20	0.3397	0.0955	5/5	20/20
glass	0.6335	0.2032	5/5	13/20	0.6390	0.3004	5/5	9/20
ionosphere	0.1919	0.2551	5/5	20/20	0.1127	0.2258	5/5	20/20
letter	0.6921	0.1736	5/5	9/20	0.6648	0.1680	5/5	8/20
lympho	0.1190	0.0320	5/5	20/20	0.0293	0.0257	5/5	20/20
mnist	0.3226	0.0791	5/5	20/20	0.2958	0.0750	5/5	20/20
musk	0.0001	0.0165	5/5	20/20	0.0014	0.0165	5/5	20/20
optdigits	0.8473	0.1736	5/5	3/20	0.8108	0.1190	5/5	1/20
pendigits	0.8220	0.1439	5/5	13/20	0.7157	0.1244	5/5	16/20
pima	0.6901	0.5292	5/5	11/20	0.6286	0.4639	5/5	15/20
satellite	0.3505	0.2830	5/5	20/20	0.3584	0.2971	5/5	20/20
satimage-2	0.1096	0.0084	5/5	20/20	0.0857	0.0084	5/5	20/20
shuttle	0.2643	0.1273	5/5	20/20	0.0743	0.0392	5/5	20/20
vertebral	0.5653	0.1924	5/5	4/20	0.5595	0.1597	5/5	4/20
vowels	0.7873	0.2011	5/5	10/20	0.8145	0.2275	5/5	7/20
wbc	0.2321	0.0497	5/5	18/20	0.1273	0.0338	5/5	20/20

Table 98: NSA/grid, Original features: dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.1946	0.0925	5/5	20/20	0.2340	0.1108	5/5	20/20
cardio	0.5305	0.1884	5/5	20/20	0.3275	0.0894	5/5	20/20
glass	0.6341	0.1610	5/5	13/20	0.6354	0.2694	5/5	9/20
ionosphere	0.1987	0.2585	5/5	20/20	0.1156	0.2261	5/5	20/20
letter	0.6908	0.1738	5/5	9/20	0.6776	0.1766	5/5	7/20
lympho	0.1034	0.0298	5/5	20/20	0.0517	0.0263	5/5	20/20
mnist	0.3362	0.0862	5/5	20/20	0.3000	0.0765	5/5	20/20
musk	0.0000	0.0165	5/5	20/20	0.0014	0.0165	5/5	20/20
optdigits	0.8189	0.1514	5/5	4/20	0.7977	0.1137	5/5	3/20
pendigits	0.8277	0.1131	5/5	13/20	0.7238	0.0660	5/5	16/20
pima	0.6849	0.5289	5/5	12/20	0.6427	0.4921	5/5	15/20
satellite	0.3570	0.2842	5/5	20/20	0.3640	0.3001	5/5	20/20
satimage-2	0.1105	0.0084	5/5	20/20	0.0861	0.0078	5/5	20/20
shuttle	0.2850	0.1452	5/5	20/20	0.0787	0.0516	5/5	20/20
vertebral	0.5496	0.1716	5/5	4/20	0.5611	0.1938	5/5	4/20
vowels	0.7921	0.2153	5/5	10/20	0.7702	0.1570	5/5	10/20
wbc	0.2352	0.0500	5/5	18/20	0.1309	0.0353	5/5	20/20

Table 99: NSA/hierarchical, Original features: dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.1866	0.0940	5/5	20/20	0.2365	0.1081	5/5	20/20
cardio	0.2908	0.1646	5/5	20/20	0.2555	0.1213	5/5	20/20
glass	0.1994	0.0394	5/5	20/20	0.5220	0.2171	5/5	19/20
ionosphere	0.0928	0.2231	5/5	20/20	0.1010	0.2217	5/5	20/20
letter	0.4607	0.0798	5/5	18/20	0.5251	0.0944	5/5	13/20
lympho	0.0897	0.0336	5/5	20/20	0.0362	0.0260	5/5	20/20
mnist	0.2644	0.0706	5/5	20/20	0.2155	0.0636	5/5	20/20
musk	0.0000	0.0165	5/5	20/20	0.0014	0.0165	5/5	20/20
optdigits	0.5360	0.0456	5/5	17/20	0.7273	0.0844	5/5	7/20
pendigits	0.3338	0.0426	5/5	20/20	0.3932	0.0208	5/5	20/20
pima	0.4523	0.3527	5/5	18/20	0.6281	0.4794	5/5	15/20
satellite	0.2521	0.2272	5/5	20/20	0.2075	0.2075	5/5	20/20
satimage-2	0.0126	0.0063	5/5	20/20	0.0471	0.0064	5/5	20/20
shuttle	0.0197	0.0385	5/5	20/20	0.0078	0.0370	5/5	20/20
vertebral	0.5403	0.2528	5/5	4/20	0.6054	0.1807	5/5	2/20
vowels	0.3758	0.0347	5/5	18/20	0.6114	0.1083	5/5	11/20
wbc	0.0613	0.0312	5/5	18/20	0.1304	0.0342	5/5	20/20

Table 100: NSA/voronoi, Original features: dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	—	—	0/5	—	—	—	0/5	—
cardio	—	—	0/5	—	—	—	0/5	—
glass	—	—	0/5	—	—	—	0/5	—
ionosphere	—	—	0/5	—	—	—	0/5	—
letter	—	—	0/5	—	—	—	0/5	—
lympho	—	—	0/5	—	—	—	0/5	—
mnist	—	—	0/5	—	—	—	0/5	—
musk	—	—	0/5	—	—	—	0/5	—
optdigits	—	—	0/5	—	—	—	0/5	—
pendigits	—	—	0/5	—	—	—	0/5	—
pima	—	—	0/5	—	—	—	0/5	—
satellite	—	—	0/5	—	—	—	0/5	—
satimage-2	—	—	0/5	—	—	—	0/5	—
shuttle	—	—	0/5	—	—	—	0/5	—
vertebral	—	—	0/5	—	—	—	0/5	—
vowels	—	—	0/5	—	—	—	0/5	—
wbc	—	—	0/5	—	—	—	0/5	—

Table 101: NSA/deterministic, Original features: dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	—	—	0/5	—	—	—	0/5	—
cardio	—	—	0/5	—	—	—	0/5	—
glass	—	—	0/5	—	—	—	0/5	—
ionosphere	—	—	0/5	—	—	—	0/5	—
letter	—	—	0/5	—	—	—	0/5	—
lympho	—	—	0/5	—	—	—	0/5	—
mnist	—	—	0/5	—	—	—	0/5	—
musk	—	—	0/5	—	—	—	0/5	—
optdigits	—	—	0/5	—	—	—	0/5	—
pendigits	—	—	0/5	—	—	—	0/5	—
pima	—	—	0/5	—	—	—	0/5	—
satellite	—	—	0/5	—	—	—	0/5	—
satimage-2	—	—	0/5	—	—	—	0/5	—
shuttle	—	—	0/5	—	—	—	0/5	—
vertebral	—	—	0/5	—	—	—	0/5	—
vowels	—	—	0/5	—	—	—	0/5	—
wbc	—	—	0/5	—	—	—	0/5	—

Table 102: NSA/suppressed, Original features: dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.1971	0.0933	5/5	20/20	0.2347	0.1069	5/5	20/20
cardio	0.5985	0.2611	5/5	17/20	0.3326	0.0923	5/5	20/20
glass	0.6579	0.1760	5/5	12/20	0.6415	0.2762	5/5	9/20
ionosphere	0.1922	0.2565	5/5	20/20	0.1129	0.2267	5/5	20/20
letter	0.6795	0.1731	5/5	9/20	0.6665	0.1627	5/5	7/20
lympho	0.1069	0.0304	5/5	20/20	0.0121	0.0254	5/5	20/20
mnist	0.3300	0.0829	5/5	20/20	0.2870	0.0721	5/5	20/20
musk	0.0001	0.0165	5/5	20/20	0.0014	0.0165	5/5	20/20
optdigits	0.8541	0.1741	5/5	3/20	0.8141	0.1253	5/5	1/20
pendigits	0.7917	0.1002	5/5	15/20	0.6908	0.1319	5/5	16/20
pima	0.6767	0.4971	5/5	14/20	0.6215	0.4567	5/5	15/20
satellite	0.3503	0.2832	5/5	20/20	0.3578	0.2974	5/5	20/20
satimage-2	0.1130	0.0084	5/5	20/20	0.0856	0.0084	5/5	20/20
shuttle	0.2814	0.1690	5/5	20/20	0.1187	0.0637	5/5	20/20
vertebral	0.5629	0.1698	5/5	4/20	0.5429	0.2022	5/5	5/20
vowels	0.7989	0.2026	5/5	10/20	0.8243	0.2224	5/5	7/20
wbc	0.2306	0.0490	5/5	18/20	0.1208	0.0335	5/5	20/20

Table 103: NSA/dual, Original features: dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.7755	0.4616	5/5	15/20	0.7804	0.4754	5/5	14/20
cardio	0.9081	0.6820	5/5	14/20	0.9060	0.6542	5/5	12/20
glass	0.7378	0.1447	5/5	6/20	0.7018	0.1023	5/5	8/20
ionosphere	0.9262	0.8993	5/5	8/20	0.9050	0.8524	5/5	6/20
letter	0.5589	0.1161	5/5	13/20	0.5741	0.1313	5/5	11/20
lympho	0.9862	0.8133	5/5	2/20	0.9793	0.8508	5/5	2/20
mnist	0.8086	0.3928	5/5	13/20	0.8140	0.3842	5/5	14/20
musk	1.0000	0.9988	5/5	8/20	0.9983	0.9345	5/5	7/20
optdigits	0.3639	0.0212	5/5	19/20	0.4443	0.0249	5/5	18/20
pendigits	0.6806	0.1179	5/5	16/20	0.7701	0.1710	5/5	14/20
pima	0.6509	0.4889	5/5	16/20	0.6516	0.4929	5/5	13/20
satellite	0.6963	0.6913	5/5	14/20	0.7518	0.7275	5/5	11/20
satimage-2	0.9769	0.9408	5/5	14/20	0.9695	0.9331	5/5	12/20
shuttle	0.9962	0.8727	5/5	1/20	0.9967	0.8899	5/5	2/20
vertebral	0.4603	0.1266	5/5	17/20	0.4250	0.1141	5/5	10/20
vowels	0.8334	0.2571	5/5	9/20	0.8456	0.2589	5/5	7/20
wbc	0.8868	0.6653	5/5	14/20	0.9165	0.6147	5/5	7/20

Table 104: NSA/annealed, Original features: dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.1905	0.0936	5/5	20/20	0.2253	0.0941	5/5	20/20
cardio	0.2119	0.0743	5/5	20/20	0.1851	0.0611	5/5	20/20
glass	0.4933	0.2500	5/5	19/20	0.4384	0.0663	5/5	20/20
ionosphere	0.1064	0.2344	5/5	20/20	0.0920	0.2211	5/5	20/20
letter	0.4958	0.0681	5/5	17/20	0.5021	0.0686	5/5	16/20
lympho	0.0552	0.0266	5/5	20/20	0.0310	0.0258	5/5	20/20
mnist	0.1940	0.0559	5/5	20/20	0.1801	0.0548	5/5	20/20
musk	0.0001	0.0165	5/5	20/20	0.0012	0.0165	5/5	20/20
optdigits	0.5218	0.0339	5/5	17/20	0.4794	0.0300	5/5	18/20
pendigits	0.3543	0.0176	5/5	20/20	0.4000	0.0500	5/5	20/20
pima	0.5054	0.3694	5/5	18/20	0.4426	0.3347	5/5	18/20
satellite	0.3060	0.2565	5/5	20/20	0.3097	0.2589	5/5	20/20
satimage-2	0.0509	0.0064	5/5	20/20	0.0475	0.0064	5/5	20/20
shuttle	0.0044	0.0370	5/5	20/20	0.0037	0.0370	5/5	20/20
vertebral	0.4556	0.1574	5/5	17/20	0.5784	0.1826	5/5	3/20
vowels	0.4286	0.0367	5/5	18/20	0.3469	0.0298	5/5	18/20
wbc	0.1151	0.0336	5/5	18/20	0.0953	0.0329	5/5	20/20

Table 105: NSA/binary/hamming, Original features: dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.5274	0.1662	5/5	17/20	0.5266	0.1677	5/5	18/20
cardio	0.5298	0.1074	5/5	20/20	0.5204	0.1054	5/5	19/20
glass	0.3091	0.0471	5/5	20/20	0.4524	0.0563	5/5	20/20
ionosphere	0.4806	0.3870	5/5	20/20	0.4508	0.3525	5/5	19/20
letter	0.5264	0.0729	5/5	17/20	0.5043	0.0716	5/5	16/20
lympho	0.4845	0.0655	5/5	15/20	0.4948	0.0514	5/5	18/20
mnist	0.5743	0.1209	5/5	17/20	0.5580	0.1158	5/5	17/20
musk	0.4706	0.0578	5/5	19/20	0.4923	0.0415	5/5	19/20
optdigits	0.5033	0.0332	5/5	17/20	0.4751	0.0310	5/5	18/20
pendigits	0.5067	0.0256	5/5	20/20	0.5699	0.0332	5/5	19/20
pima	0.5136	0.3549	5/5	17/20	0.5010	0.3502	5/5	17/20
satellite	0.4106	0.2838	5/5	20/20	0.4339	0.2922	5/5	20/20
satimage-2	0.2865	0.0115	5/5	20/20	0.2935	0.0113	5/5	20/20
shuttle	0.8670	0.3461	5/5	15/20	0.7137	0.1683	5/5	15/20
vertebral	0.4617	0.1220	5/5	16/20	0.5053	0.1321	5/5	7/20
vowels	0.5375	0.0406	5/5	16/20	0.5324	0.0413	5/5	13/20
wbc	0.5505	0.0716	5/5	15/20	0.5373	0.0760	5/5	15/20

Table 106: NSA/binary/rcontiguous, Original features: dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.4645	0.1520	5/5	20/20	0.4622	0.1533	5/5	19/20
cardio	0.5512	0.1164	5/5	20/20	0.5931	0.1488	5/5	16/20
glass	0.3360	0.0590	5/5	20/20	0.6140	0.1053	5/5	12/20
ionosphere	0.5065	0.3884	5/5	19/20	0.4725	0.3649	5/5	19/20
letter	0.5165	0.0703	5/5	17/20	0.5120	0.0687	5/5	15/20
lympho	0.6216	0.1470	5/5	15/20	0.6190	0.1980	5/5	17/20
mnist	0.5631	0.1267	5/5	17/20	0.5557	0.1263	5/5	17/20
musk	0.4896	0.0598	5/5	19/20	0.4612	0.0381	5/5	19/20
optdigits	0.4830	0.0516	5/5	17/20	0.4718	0.0620	5/5	18/20
pendigits	0.6580	0.0831	5/5	17/20	0.6652	0.0580	5/5	17/20
pima	—	—	0/5	—	—	—	0/5	—
satellite	0.4570	0.3112	5/5	20/20	0.4569	0.3081	5/5	20/20
satimage-2	0.4431	0.0165	5/5	20/20	0.4022	0.0135	5/5	20/20
shuttle	0.6904	0.1763	5/5	16/20	0.6256	0.1509	5/5	15/20
vertebral	—	—	0/5	—	—	—	0/5	—
vowels	0.6850	0.1053	5/5	13/20	0.6951	0.0783	5/5	10/20
wbc	0.5100	0.1174	5/5	17/20	0.5497	0.1342	5/5	15/20

Table 107: NSA/binary/rchunk, Original features: dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.5365	0.1787	5/5	16/20	0.5243	0.1711	5/5	18/20
cardio	0.5150	0.1032	5/5	20/20	0.5065	0.1001	5/5	19/20
glass	0.5213	0.0517	5/5	18/20	0.5866	0.0564	5/5	14/20
ionosphere	0.4263	0.3408	5/5	20/20	0.4395	0.3406	5/5	19/20
letter	0.5146	0.0713	5/5	17/20	0.4915	0.0653	5/5	17/20
lympho	0.5190	0.0581	5/5	15/20	0.5707	0.0467	5/5	17/20
mnist	0.4561	0.0863	5/5	20/20	0.4648	0.0881	5/5	18/20
musk	0.4865	0.0335	5/5	19/20	0.5063	0.0388	5/5	19/20
optdigits	0.4440	0.0263	5/5	18/20	0.4623	0.0273	5/5	18/20
pendigits	0.6383	0.0346	5/5	17/20	0.5515	0.0276	5/5	19/20
pima	—	—	2/5	—	0.4896	0.3471	5/5	17/20
satellite	0.4942	0.3300	5/5	20/20	0.4892	0.3272	5/5	20/20
satimage-2	0.4111	0.0147	5/5	20/20	0.3992	0.0139	5/5	20/20
shuttle	0.7478	0.2055	5/5	16/20	0.6894	0.1261	5/5	15/20
vertebral	—	—	1/5	—	—	—	4/5	—
vowels	0.7192	0.0649	5/5	11/20	0.6625	0.0512	5/5	10/20
wbc	0.5512	0.0658	5/5	15/20	0.5317	0.0608	5/5	16/20

Table 108: Every detector at 10 and 5 percent, PCA(2). Incomplete means are available-case. Q is strict qualified coverage, R is raw-success coverage.

Configuration	10% AUROC	10% AP	10% Q	10% R	5% AUROC	5% AP	5% Q	5% R
NSA fixed	0.6132	0.2789	85/85	85/85	0.5917	0.2200	85/85	85/85
NSA variable	0.7273	0.3638	85/85	85/85	0.6683	0.2879	85/85	85/85
NSA coverage	0.7110	0.3571	85/85	85/85	0.6669	0.2838	85/85	85/85
NSA grid	0.7243	0.3668	85/85	85/85	0.6775	0.3002	85/85	85/85
NSA hierarchical	0.6891	0.3316	85/85	85/85	0.6579	0.2647	85/85	85/85
NSA voronoi	0.7361	0.3689	85/85	85/85	0.6882	0.3153	85/85	85/85
NSA deterministic	0.6278	0.2893	85/85	85/85	0.6066	0.2198	85/85	85/85
NSA suppressed	0.7173	0.3547	85/85	85/85	0.6650	0.2955	85/85	85/85
NSA dual	0.7626	0.4206	85/85	85/85	0.7264	0.3755	85/85	85/85
NSA annealed	0.6241	0.2869	85/85	85/85	0.5966	0.2187	85/85	85/85
NSA binary Hamming	—	—	0/85	0/85	—	—	0/85	0/85
NSA binary contiguous	—	—	0/85	0/85	—	—	0/85	0/85
NSA binary chunk	—	—	0/85	0/85	—	—	0/85	0/85
ECOD	0.6162	0.2412	85/85	85/85	0.6172	0.2507	85/85	85/85
ABOD	0.7751	0.4221	85/85	85/85	0.7437	0.3791	85/85	85/85
COPOD	0.6611	0.2884	85/85	85/85	0.6529	0.2899	85/85	85/85
MAD	—	—	0/85	0/85	—	—	0/85	0/85
SOS	0.5252	0.1472	85/85	85/85	0.5295	0.1529	85/85	85/85
QMCD	0.6058	0.2582	85/85	85/85	0.5224	0.2009	85/85	85/85
KDE	0.7769	0.4373	85/85	85/85	0.7496	0.3949	85/85	85/85
Sampling	0.7742	0.4442	65/85	65/85	0.7676	0.3969	50/85	50/85
GMM	0.6418	0.3487	85/85	85/85	0.6403	0.3479	85/85	85/85
PCA	0.6504	0.3413	85/85	85/85	0.6657	0.3562	85/85	85/85
KPCA	0.7764	0.4290	85/85	85/85	0.7440	0.3837	85/85	85/85
MCD	0.7053	0.3462	85/85	85/85	0.6805	0.3242	85/85	85/85
CD	0.5877	0.2623	85/85	85/85	0.5975	0.2643	85/85	85/85
OCSVM	0.7663	0.4295	85/85	85/85	0.7445	0.3855	85/85	85/85
LMDD	0.5874	0.2547	85/85	85/85	0.5800	0.2384	85/85	85/85
LOF	0.6729	0.3576	85/85	85/85	0.6207	0.3030	85/85	85/85

Continued on next page

Table 108 continued

Configuration	10% AUROC	10% AP	10% Q	10% R	5% AUROC	5% AP	5% Q	5% R
COF	0.6139	0.2165	85/85	85/85	0.6023	0.2040	85/85	85/85
CBLOF	0.7665	0.4357	80/85	80/85	0.7703	0.4210	65/85	65/85
LOCI	0.5645	0.2412	28/85	28/85	0.5263	0.2354	30/85	30/85
HBOS	0.7331	0.2969	85/85	85/85	0.6720	0.2230	85/85	85/85
HDBSCAN	0.5928	0.1505	85/85	85/85	0.5650	0.1351	85/85	85/85
KNN	0.7706	0.4232	85/85	85/85	0.7082	0.3717	85/85	85/85
SOD	0.5624	0.2075	63/85	68/85	0.5845	0.2026	55/85	55/85
ROD	0.6634	0.2968	85/85	85/85	0.6248	0.2797	85/85	85/85
IForest	0.7196	0.3056	85/85	85/85	0.6870	0.2809	85/85	85/85
INNE	0.7452	0.4062	85/85	85/85	0.7139	0.3470	85/85	85/85
DIF	0.7250	0.3661	83/85	85/85	0.6941	0.3325	85/85	85/85
FeatureBagging	0.6948	0.3703	85/85	85/85	0.6355	0.3239	85/85	85/85
LSCP	0.7968	0.4374	50/85	50/85	0.7475	0.3439	50/85	50/85
LODA	0.6645	0.2562	85/85	85/85	0.5771	0.1873	85/85	85/85
SUOD	0.7402	0.3490	84/85	84/85	0.6915	0.2966	85/85	85/85
XGBOD	—	—	0/85	0/85	—	—	0/85	0/85
AutoEncoder	0.6253	0.3012	50/85	50/85	—	—	0/85	0/85
VAE	0.6255	0.3211	50/85	50/85	—	—	0/85	0/85
SO_GAAL	0.5688	0.2370	85/85	85/85	0.5571	0.2157	85/85	85/85
MO_GAAL	0.5610	0.2541	85/85	85/85	0.5434	0.2227	85/85	85/85
DeepSVDD	0.7010	0.3433	85/85	85/85	0.6667	0.3067	85/85	85/85
AnoGAN	0.6425	0.2731	85/85	85/85	0.6493	0.2950	85/85	85/85
ALAD	0.5045	0.1276	85/85	85/85	0.5133	0.1373	85/85	85/85
AE1SVM	0.6548	0.3471	85/85	85/85	0.6558	0.3537	85/85	85/85
DevNet	—	—	0/85	0/85	—	—	0/85	0/85
RGraph	0.4038	0.1112	53/85	53/85	0.3872	0.0982	85/85	85/85
LUNAR	0.7459	0.4038	85/85	85/85	0.7106	0.3666	85/85	85/85

Table 109: All NSA AUROC ranks against the fixed other-detector field at every size, PCA(2). Parentheses give coverage; blank ranks prevent a partial-case advantage.

Configuration	100%	75%	50%	25%	10%	5%
NSA fixed	22/26 (85/85)	21/26 (85/85)	20/26 (85/85)	19/26 (85/85)	19/26 (85/85)	19/26 (85/85)
NSA variable	15/26 (85/85)	14/26 (85/85)	13/26 (85/85)	11/26 (85/85)	8/26 (85/85)	10/26 (85/85)
NSA coverage	15/26 (85/85)	15/26 (85/85)	13/26 (85/85)	14/26 (85/85)	9/26 (85/85)	10/26 (85/85)
NSA grid	15/26 (85/85)	15/26 (85/85)	13/26 (85/85)	11/26 (85/85)	8/26 (85/85)	10/26 (85/85)
NSA hierarchical	15/26 (85/85)	15/26 (85/85)	15/26 (85/85)	14/26 (85/85)	12/26 (85/85)	12/26 (85/85)
NSA voronoi	15/26 (85/85)	14/26 (85/85)	12/26 (85/85)	11/26 (85/85)	8/26 (85/85)	8/26 (85/85)
NSA deterministic	21/26 (85/85)	21/26 (85/85)	18/26 (85/85)	17/26 (85/85)	17/26 (85/85)	17/26 (85/85)
NSA suppressed	15/26 (85/85)	15/26 (85/85)	13/26 (85/85)	12/26 (85/85)	9/26 (85/85)	12/26 (85/85)
NSA dual	13/26 (85/85)	12/26 (85/85)	12/26 (85/85)	10/26 (85/85)	6/26 (85/85)	5/26 (85/85)
NSA annealed	21/26 (85/85)	21/26 (85/85)	18/26 (85/85)	19/26 (85/85)	17/26 (85/85)	19/26 (85/85)
NSA binary Hamming	— (0/85)	— (0/85)	— (0/85)	— (0/85)	— (0/85)	— (0/85)
NSA binary contiguous	— (0/85)	— (0/85)	— (0/85)	— (0/85)	— (0/85)	— (0/85)
NSA binary chunk	— (0/85)	— (0/85)	— (0/85)	— (0/85)	— (0/85)	— (0/85)

Table 110: NSA AUROC rank sensitivity, PCA(2). The raw comparator field requires all six sizes complete, and the focal NSA requires all 85 raw successes in its own panel. Strict uses the measured-time field; pointwise removes batch-dependent scorers. Different denominators identify different fields, so movement between columns is not a score improvement.

NSA configuration	10% strict	10% raw	10% pointwise	5% strict	5% raw	5% pointwise
NSA fixed	19/26	20/27	16/21	19/26	20/27	15/21
NSA variable	8/26	8/27	8/21	10/26	11/27	10/21
NSA coverage	9/26	10/27	9/21	10/26	11/27	10/21
NSA grid	8/26	8/27	8/21	10/26	11/27	10/21
NSA hierarchical	12/26	13/27	12/21	12/26	13/27	12/21
NSA voronoi	8/26	8/27	8/21	8/26	9/27	8/21
NSA deterministic	17/26	18/27	16/21	17/26	18/27	15/21
NSA suppressed	9/26	10/27	9/21	12/26	13/27	12/21
NSA dual	6/26	6/27	6/21	5/26	5/27	5/21
NSA annealed	17/26	18/27	16/21	19/26	20/27	15/21
NSA binary Hamming	—	—	—	—	—	—
NSA binary contiguous	—	—	—	—	—	—
NSA binary chunk	—	—	—	—	—	—

Table 111: NSA AP rank sensitivity, PCA(2). The raw comparator field requires all six sizes complete, and the focal NSA requires all 85 raw successes in its own panel. Strict uses the measured-time field; pointwise removes batch-dependent scorers. Different denominators identify different fields, so movement between columns is not a score improvement.

NSA configuration	10% strict	10% raw	10% pointwise	5% strict	5% raw	5% pointwise
NSA fixed	16/26	17/27	15/21	19/26	20/27	16/21
NSA variable	9/26	10/27	9/21	15/26	16/27	14/21
NSA coverage	10/26	11/27	10/21	15/26	16/27	14/21
NSA grid	9/26	9/27	9/21	14/26	15/27	14/21
NSA hierarchical	14/26	15/27	14/21	16/26	17/27	15/21
NSA voronoi	9/26	9/27	9/21	12/26	13/27	12/21
NSA deterministic	15/26	16/27	15/21	19/26	20/27	16/21
NSA suppressed	10/26	11/27	10/21	14/26	15/27	14/21
NSA dual	6/26	6/27	6/21	5/26	5/27	5/21
NSA annealed	16/26	17/27	15/21	19/26	20/27	16/21
NSA binary Hamming	—	—	—	—	—	—
NSA binary contiguous	—	—	—	—	—	—
NSA binary chunk	—	—	—	—	—	—

Table 112: Panel-qualified detector ranking, PCA(2), 10 percent. The field changes across sizes; this is a sensitivity result, not the primary cross-size field.

Configuration	AUROC	AUROC rank	Top position	AP
KDE	0.7769	1/40	2.50%	0.4373
KPCA	0.7764	2/40	5.00%	0.4290
ABOD	0.7751	3/40	7.50%	0.4221
KNN	0.7706	4/40	10.00%	0.4232
OCSVM	0.7663	5/40	12.50%	0.4295
NSA dual	0.7626	6/40	15.00%	0.4206
LUNAR	0.7459	7/40	17.50%	0.4038
INNE	0.7452	8/40	20.00%	0.4062
NSA voronoi	0.7361	9/40	22.50%	0.3689
HBOS	0.7331	10/40	25.00%	0.2969
NSA variable	0.7273	11/40	27.50%	0.3638
NSA grid	0.7243	12/40	30.00%	0.3668
IForest	0.7196	13/40	32.50%	0.3056
NSA suppressed	0.7173	14/40	35.00%	0.3547
NSA coverage	0.7110	15/40	37.50%	0.3571
MCD	0.7053	16/40	40.00%	0.3462
DeepSVDD	0.7010	17/40	42.50%	0.3433
FeatureBagging	0.6948	18/40	45.00%	0.3703
NSA hierarchical	0.6891	19/40	47.50%	0.3316
LOF	0.6729	20/40	50.00%	0.3576
LODA	0.6645	21/40	52.50%	0.2562
ROD	0.6634	22/40	55.00%	0.2968
COPOD	0.6611	23/40	57.50%	0.2884
AE1SVM	0.6548	24/40	60.00%	0.3471
PCA	0.6504	25/40	62.50%	0.3413
AnoGAN	0.6425	26/40	65.00%	0.2731
GMM	0.6418	27/40	67.50%	0.3487
NSA deterministic	0.6278	28/40	70.00%	0.2893
NSA annealed	0.6241	29/40	72.50%	0.2869
ECOD	0.6162	30/40	75.00%	0.2412
COF	0.6139	31/40	77.50%	0.2165
NSA fixed	0.6132	32/40	80.00%	0.2789
QMCD	0.6058	33/40	82.50%	0.2582
HDBSCAN	0.5928	34/40	85.00%	0.1505
CD	0.5877	35/40	87.50%	0.2623
LMDD	0.5874	36/40	90.00%	0.2547
SO_GAAL	0.5688	37/40	92.50%	0.2370
MO_GAAL	0.5610	38/40	95.00%	0.2541
SOS	0.5252	39/40	97.50%	0.1472
ALAD	0.5045	40/40	100.00%	0.1276

Table 113: Panel-qualified detector ranking, PCA(2), 5 percent. The field changes across sizes; this is a sensitivity result, not the primary cross-size field.

Configuration	AUROC	AUROC rank	Top position	AP
KDE	0.7496	1/43	2.33%	0.3949
OCSVM	0.7445	2/43	4.65%	0.3855
KPCA	0.7440	3/43	6.98%	0.3837
ABOD	0.7437	4/43	9.30%	0.3791
NSA dual	0.7264	5/43	11.63%	0.3755
INNE	0.7139	6/43	13.95%	0.3470
LUNAR	0.7106	7/43	16.28%	0.3666
KNN	0.7082	8/43	18.60%	0.3717
DIF	0.6941	9/43	20.93%	0.3325
SUOD	0.6915	10/43	23.26%	0.2966
NSA voronoi	0.6882	11/43	25.58%	0.3153
IForest	0.6870	12/43	27.91%	0.2809
MCD	0.6805	13/43	30.23%	0.3242
NSA grid	0.6775	14/43	32.56%	0.3002
HBOS	0.6720	15/43	34.88%	0.2230
NSA variable	0.6683	16/43	37.21%	0.2879
NSA coverage	0.6669	17/43	39.53%	0.2838
DeepSVDD	0.6667	18/43	41.86%	0.3067
PCA	0.6657	19/43	44.19%	0.3562
NSA suppressed	0.6650	20/43	46.51%	0.2955
NSA hierarchical	0.6579	21/43	48.84%	0.2647
AE1SVM	0.6558	22/43	51.16%	0.3537
COPOD	0.6529	23/43	53.49%	0.2899
AnoGAN	0.6493	24/43	55.81%	0.2950
GMM	0.6403	25/43	58.14%	0.3479
FeatureBagging	0.6355	26/43	60.47%	0.3239
ROD	0.6248	27/43	62.79%	0.2797
LOF	0.6207	28/43	65.12%	0.3030
ECOD	0.6172	29/43	67.44%	0.2507
NSA deterministic	0.6066	30/43	69.77%	0.2198
COF	0.6023	31/43	72.09%	0.2040
CD	0.5975	32/43	74.42%	0.2643
NSA annealed	0.5966	33/43	76.74%	0.2187
NSA fixed	0.5917	34/43	79.07%	0.2200
LMDD	0.5800	35/43	81.40%	0.2384
LODA	0.5771	36/43	83.72%	0.1873
HDBSCAN	0.5650	37/43	86.05%	0.1351
SO_GAAL	0.5571	38/43	88.37%	0.2157
MO_GAAL	0.5434	39/43	90.70%	0.2227
SOS	0.5295	40/43	93.02%	0.1529
QMCD	0.5224	41/43	95.35%	0.2009
ALAD	0.5133	42/43	97.67%	0.1373

Continued on next page

Table 113 continued

Configuration	AUROC	AUROC rank	Top position	AP
RGraph	0.3872	43/43	100.00%	0.0982

Table 114: Every configuration's paired change from full normal-pool retention, PCA(2). Positive favors fewer normals. Intervals are descriptive dataset-cluster bootstrap intervals, unadjusted for multiple comparisons.

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
NSA fixed	10%	85/85	17/17	+0.0305	[-0.0475, +0.1125]	-0.0638	[-0.1453, +0.0138]
NSA variable	10%	85/85	17/17	+0.0345	[-0.0302, +0.1072]	-0.0803	[-0.1535, -0.0176]
NSA coverage	10%	85/85	17/17	+0.0417	[-0.0279, +0.1205]	-0.0683	[-0.1434, -0.0041]
NSA grid	10%	85/85	17/17	+0.0357	[-0.0279, +0.1040]	-0.0693	[-0.1318, -0.0071]
NSA hierarchical	10%	85/85	17/17	+0.0137	[-0.0629, +0.0924]	-0.0902	[-0.1577, -0.0271]
NSA voronoi	10%	85/85	17/17	+0.0318	[-0.0309, +0.0974]	-0.0833	[-0.1505, -0.0172]
NSA deterministic	10%	85/85	17/17	+0.0247	[-0.0453, +0.1015]	-0.0702	[-0.1543, +0.0098]
NSA suppressed	10%	85/85	17/17	+0.0435	[-0.0227, +0.1164]	-0.0667	[-0.1434, +0.0003]
NSA dual	10%	85/85	17/17	+0.0454	[-0.0013, +0.1002]	-0.0511	[-0.0990, -0.0084]
NSA annealed	10%	85/85	17/17	+0.0243	[-0.0554, +0.1176]	-0.0803	[-0.1677, +0.0015]
NSA binary Hamming	10%	0/85	0/17	—	[—, —]	—	[—, —]
NSA binary contiguous	10%	0/85	0/17	—	[—, —]	—	[—, —]
NSA binary chunk	10%	0/85	0/17	—	[—, —]	—	[—, —]
ECOD	10%	85/85	17/17	-0.0039	[-0.0383, +0.0270]	-0.0285	[-0.0568, -0.0035]
ABOD	10%	85/85	17/17	-0.0102	[-0.0243, +0.0049]	-0.0610	[-0.0958, -0.0320]
COPOD	10%	85/85	17/17	-0.0023	[-0.0321, +0.0239]	-0.0234	[-0.0665, +0.0125]
MAD	10%	0/85	0/17	—	[—, —]	—	[—, —]
SOS	10%	85/85	17/17	-0.0075	[-0.0325, +0.0109]	-0.0075	[-0.0378, +0.0114]
QMCD	10%	85/85	17/17	-0.0524	[-0.1218, +0.0126]	-0.0800	[-0.1798, +0.0097]
KDE	10%	85/85	17/17	-0.0120	[-0.0304, +0.0066]	-0.0444	[-0.0783, -0.0134]
Sampling	10%	65/85	13/17	-0.0079	[-0.0292, +0.0130]	-0.0223	[-0.0579, +0.0092]
GMM	10%	85/85	17/17	+0.0072	[-0.0254, +0.0386]	-0.0080	[-0.0409, +0.0220]
PCA	10%	85/85	17/17	-0.0151	[-0.0456, +0.0157]	-0.0192	[-0.0474, +0.0031]
KPCA	10%	85/85	17/17	-0.0097	[-0.0303, +0.0143]	-0.0564	[-0.1016, -0.0160]
MCD	10%	85/85	17/17	-0.0010	[-0.0253, +0.0212]	-0.0211	[-0.0580, +0.0099]
CD	10%	85/85	17/17	+0.0046	[-0.0426, +0.0465]	-0.0319	[-0.0824, +0.0088]
OCSVM	10%	85/85	17/17	-0.0137	[-0.0286, +0.0018]	-0.0508	[-0.0875, -0.0181]
LMDD	10%	85/85	17/17	+0.0294	[-0.0096, +0.0737]	+0.0099	[-0.0109, +0.0310]

Continued on next page

Table 114 continued

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
LOF	10%	85/85	17/17	-0.1052	[-0.1762, -0.0407]	-0.1028	[-0.1677, -0.0431]
COF	10%	85/85	17/17	-0.0002	[-0.0237, +0.0219]	-0.0085	[-0.0386, +0.0172]
CBLOF	10%	80/85	16/17	-0.0147	[-0.0402, +0.0136]	-0.0258	[-0.0595, +0.0008]
LOCI	10%	28/85	6/17	+0.0142	[-0.0647, +0.0814]	-0.0018	[-0.0427, +0.0364]
HBOS	10%	85/85	17/17	-0.0250	[-0.0559, +0.0037]	-0.0933	[-0.1599, -0.0323]
HDBSCAN	10%	85/85	17/17	-0.1235	[-0.1893, -0.0586]	-0.0887	[-0.1359, -0.0486]
KNN	10%	85/85	17/17	-0.0252	[-0.0518, +0.0042]	-0.0641	[-0.1038, -0.0296]
SOD	10%	63/85	16/17	-0.0027	[-0.0225, +0.0178]	-0.0012	[-0.0231, +0.0193]
ROD	10%	85/85	17/17	-0.0016	[-0.0352, +0.0305]	-0.0165	[-0.0551, +0.0148]
IForest	10%	85/85	17/17	-0.0397	[-0.0620, -0.0164]	-0.0967	[-0.1645, -0.0365]
INNE	10%	85/85	17/17	-0.0342	[-0.0712, -0.0010]	-0.0622	[-0.1095, -0.0235]
DIF	10%	83/85	17/17	-0.0329	[-0.0655, -0.0015]	-0.0572	[-0.1105, -0.0131]
FeatureBagging	10%	85/85	17/17	-0.0747	[-0.1394, -0.0189]	-0.0793	[-0.1272, -0.0330]
LSCP	10%	50/85	10/17	-0.0365	[-0.0616, -0.0140]	-0.0753	[-0.1424, -0.0249]
LODA	10%	85/85	17/17	-0.0625	[-0.1362, +0.0020]	-0.0894	[-0.1733, -0.0052]
SUOD	10%	84/85	17/17	-0.0180	[-0.0449, +0.0066]	-0.0494	[-0.1011, +0.0016]
XGBOD	10%	0/85	0/17	—	[—, —]	—	[—, —]
AutoEncoder	10%	50/85	10/17	-0.0769	[-0.1286, -0.0387]	-0.0768	[-0.1500, -0.0170]
VAE	10%	50/85	10/17	-0.0681	[-0.1149, -0.0258]	-0.0559	[-0.1047, -0.0129]
SO_GAAL	10%	85/85	17/17	-0.0129	[-0.0796, +0.0410]	+0.0304	[+0.0029, +0.0598]
MO_GAAL	10%	85/85	17/17	-0.0108	[-0.0470, +0.0215]	+0.0200	[-0.0090, +0.0504]
DeepSVDD	10%	85/85	17/17	-0.0358	[-0.0817, +0.0090]	-0.0474	[-0.1060, +0.0022]
AnoGAN	10%	85/85	17/17	+0.0178	[-0.0215, +0.0552]	-0.0070	[-0.0542, +0.0358]
ALAD	10%	85/85	17/17	-0.0006	[-0.0204, +0.0209]	-0.0024	[-0.0116, +0.0062]
AE1SVM	10%	85/85	17/17	-0.0022	[-0.0428, +0.0399]	+0.0111	[-0.0218, +0.0496]
DevNet	10%	0/85	0/17	—	[—, —]	—	[—, —]
RGraph	10%	35/85	7/17	-0.0443	[-0.0870, +0.0087]	-0.0067	[-0.0192, +0.0072]
LUNAR	10%	85/85	17/17	-0.0286	[-0.0520, -0.0065]	-0.0538	[-0.0879, -0.0240]
NSA fixed	5%	85/85	17/17	+0.0090	[-0.0930, +0.0996]	-0.1227	[-0.2258, -0.0276]
NSA variable	5%	85/85	17/17	-0.0245	[-0.1100, +0.0686]	-0.1562	[-0.2570, -0.0637]
NSA coverage	5%	85/85	17/17	-0.0025	[-0.0955, +0.0985]	-0.1416	[-0.2489, -0.0481]
NSA grid	5%	85/85	17/17	-0.0110	[-0.1032, +0.0775]	-0.1359	[-0.2261, -0.0525]
NSA hierarchical	5%	85/85	17/17	-0.0175	[-0.0929, +0.0642]	-0.1571	[-0.2569, -0.0726]

Continued on next page

Table 114 continued

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
NSA voronoi	5%	85/85	17/17	-0.0160	[-0.0991, +0.0692]	-0.1368	[-0.2259, -0.0522]
NSA deterministic	5%	85/85	17/17	+0.0035	[-0.0747, +0.0771]	-0.1396	[-0.2454, -0.0378]
NSA suppressed	5%	85/85	17/17	-0.0088	[-0.0843, +0.0723]	-0.1259	[-0.2147, -0.0459]
NSA dual	5%	85/85	17/17	+0.0092	[-0.0687, +0.0828]	-0.0962	[-0.1756, -0.0310]
NSA annealed	5%	85/85	17/17	-0.0032	[-0.0824, +0.0781]	-0.1485	[-0.2542, -0.0492]
NSA binary Hamming	5%	0/85	0/17	—	[—, —]	—	[—, —]
NSA binary contiguous	5%	0/85	0/17	—	[—, —]	—	[—, —]
NSA binary chunk	5%	0/85	0/17	—	[—, —]	—	[—, —]
ECOD	5%	85/85	17/17	-0.0029	[-0.0416, +0.0310]	-0.0191	[-0.0472, +0.0062]
ABOD	5%	85/85	17/17	-0.0416	[-0.0703, -0.0157]	-0.1040	[-0.1621, -0.0543]
COPOD	5%	85/85	17/17	-0.0105	[-0.0673, +0.0308]	-0.0219	[-0.0693, +0.0202]
MAD	5%	0/85	0/17	—	[—, —]	—	[—, —]
SOS	5%	85/85	17/17	-0.0031	[-0.0187, +0.0116]	-0.0018	[-0.0282, +0.0179]
QMCD	5%	85/85	17/17	-0.1357	[-0.2454, -0.0307]	-0.1372	[-0.2449, -0.0366]
KDE	5%	85/85	17/17	-0.0394	[-0.0870, -0.0045]	-0.0868	[-0.1394, -0.0415]
Sampling	5%	50/85	10/17	-0.0215	[-0.0565, +0.0072]	-0.0522	[-0.1042, -0.0095]
GMM	5%	85/85	17/17	+0.0058	[-0.0272, +0.0389]	-0.0089	[-0.0412, +0.0233]
PCA	5%	85/85	17/17	+0.0002	[-0.0481, +0.0510]	-0.0043	[-0.0446, +0.0360]
KPCA	5%	85/85	17/17	-0.0421	[-0.0900, -0.0037]	-0.1016	[-0.1605, -0.0466]
MCD	5%	85/85	17/17	-0.0258	[-0.0517, -0.0010]	-0.0431	[-0.0807, -0.0086]
CD	5%	85/85	17/17	+0.0143	[-0.0229, +0.0520]	-0.0299	[-0.0847, +0.0171]
OCSVM	5%	85/85	17/17	-0.0356	[-0.0739, -0.0060]	-0.0948	[-0.1496, -0.0454]
LMDD	5%	85/85	17/17	+0.0220	[-0.0185, +0.0620]	-0.0064	[-0.0556, +0.0342]
LOF	5%	85/85	17/17	-0.1573	[-0.2565, -0.0692]	-0.1574	[-0.2826, -0.0694]
COF	5%	85/85	17/17	-0.0118	[-0.0435, +0.0189]	-0.0210	[-0.0531, +0.0090]
CBLOF	5%	65/85	13/17	-0.0391	[-0.0641, -0.0176]	-0.0717	[-0.1128, -0.0378]
LOCI	5%	30/85	6/17	-0.0321	[-0.1177, +0.0388]	-0.0085	[-0.0399, +0.0284]
HBOS	5%	85/85	17/17	-0.0861	[-0.1511, -0.0305]	-0.1672	[-0.2418, -0.0944]
HDBSCAN	5%	85/85	17/17	-0.1513	[-0.2154, -0.0822]	-0.1041	[-0.1573, -0.0571]
KNN	5%	85/85	17/17	-0.0876	[-0.1514, -0.0360]	-0.1155	[-0.1782, -0.0605]
SOD	5%	55/85	11/17	+0.0239	[-0.0137, +0.0738]	+0.0202	[-0.0154, +0.0634]
ROD	5%	85/85	17/17	-0.0402	[-0.1027, +0.0108]	-0.0336	[-0.0781, +0.0017]

Continued on next page

Table 114 continued

Configuration	Retained	Pairs	Datasets	ΔAUROC	95% CI	ΔAP	95% CI
IForest	5%	85/85	17/17	-0.0723	[-0.1141, -0.0330]	-0.1214	[-0.1923, -0.0542]
INNE	5%	85/85	17/17	-0.0656	[-0.1139, -0.0196]	-0.1214	[-0.1795, -0.0679]
DIF	5%	85/85	17/17	-0.0635	[-0.1097, -0.0189]	-0.0891	[-0.1663, -0.0290]
FeatureBagging	5%	85/85	17/17	-0.1340	[-0.2197, -0.0574]	-0.1257	[-0.2308, -0.0517]
LSCP	5%	50/85	10/17	-0.0858	[-0.1509, -0.0326]	-0.1688	[-0.3052, -0.0653]
LODA	5%	85/85	17/17	-0.1499	[-0.2620, -0.0499]	-0.1583	[-0.2498, -0.0720]
SUOD	5%	85/85	17/17	-0.0660	[-0.1239, -0.0164]	-0.1016	[-0.1672, -0.0411]
XGBOD	5%	0/85	0/17	—	[—, —]	—	[—, —]
AutoEncoder	5%	0/85	0/17	—	[—, —]	—	[—, —]
VAE	5%	0/85	0/17	—	[—, —]	—	[—, —]
SO_GAAL	5%	85/85	17/17	-0.0247	[-0.0596, +0.0092]	+0.0091	[-0.0180, +0.0344]
MO_GAAL	5%	85/85	17/17	-0.0284	[-0.0721, +0.0202]	-0.0115	[-0.0528, +0.0231]
DeepSVDD	5%	85/85	17/17	-0.0702	[-0.1309, -0.0101]	-0.0839	[-0.1596, -0.0192]
AnoGAN	5%	85/85	17/17	+0.0246	[-0.0109, +0.0579]	+0.0149	[-0.0304, +0.0594]
ALAD	5%	85/85	17/17	+0.0082	[-0.0041, +0.0226]	+0.0073	[-0.0019, +0.0202]
AE1SVM	5%	85/85	17/17	-0.0012	[-0.0601, +0.0596]	+0.0177	[-0.0357, +0.0748]
DevNet	5%	0/85	0/17	—	[—, —]	—	[—, —]
RGraph	5%	35/85	7/17	-0.0320	[-0.0872, +0.0284]	-0.0059	[-0.0214, +0.0114]
LUNAR	5%	85/85	17/17	-0.0639	[-0.1107, -0.0210]	-0.0910	[-0.1490, -0.0451]

Table 115: All NSA configurations versus IForest and KNN, PCA(2), new sizes. Differences are NSA minus comparator on matched successful cases.

NSA configuration	Reference	Retained	Pairs	ΔAUROC	95% CI	ΔAP	95% CI
NSA fixed	KNN	10%	85/85	-0.1574	[-0.2437, -0.0816]	-0.1443	[-0.2526, -0.0566]
NSA fixed	IForest	10%	85/85	-0.1064	[-0.1792, -0.0439]	-0.0267	[-0.0795, +0.0299]
NSA variable	KNN	10%	85/85	-0.0433	[-0.1039, +0.0060]	-0.0594	[-0.1570, +0.0160]
NSA variable	IForest	10%	85/85	+0.0077	[-0.0530, +0.0620]	+0.0582	[+0.0095, +0.1175]
NSA coverage	KNN	10%	85/85	-0.0596	[-0.1190, -0.0067]	-0.0661	[-0.1609, +0.0110]
NSA coverage	IForest	10%	85/85	-0.0085	[-0.0687, +0.0436]	+0.0515	[+0.0037, +0.1093]
NSA grid	KNN	10%	85/85	-0.0463	[-0.1050, +0.0035]	-0.0564	[-0.1494, +0.0232]
NSA grid	IForest	10%	85/85	+0.0047	[-0.0569, +0.0589]	+0.0612	[+0.0097, +0.1248]
NSA hierarchical	KNN	10%	85/85	-0.0815	[-0.1556, -0.0182]	-0.0915	[-0.1909, -0.0078]
NSA hierarchical	IForest	10%	85/85	-0.0305	[-0.1049, +0.0395]	+0.0260	[-0.0224, +0.0822]
NSA voronoi	KNN	10%	85/85	-0.0345	[-0.0810, +0.0067]	-0.0543	[-0.1287, +0.0163]
NSA voronoi	IForest	10%	85/85	+0.0165	[-0.0372, +0.0666]	+0.0633	[+0.0025, +0.1298]
NSA deterministic	KNN	10%	85/85	-0.1428	[-0.2249, -0.0720]	-0.1339	[-0.2436, -0.0424]
NSA deterministic	IForest	10%	85/85	-0.0917	[-0.1658, -0.0281]	-0.0163	[-0.0712, +0.0480]
NSA suppressed	KNN	10%	85/85	-0.0533	[-0.1076, -0.0021]	-0.0684	[-0.1621, +0.0108]
NSA suppressed	IForest	10%	85/85	-0.0023	[-0.0621, +0.0543]	+0.0491	[-0.0066, +0.1149]
NSA dual	KNN	10%	85/85	-0.0080	[-0.0294, +0.0151]	-0.0025	[-0.0297, +0.0256]
NSA dual	IForest	10%	85/85	+0.0430	[+0.0077, +0.0771]	+0.1150	[+0.0364, +0.2053]
NSA annealed	KNN	10%	85/85	-0.1465	[-0.2285, -0.0730]	-0.1362	[-0.2498, -0.0447]
NSA annealed	IForest	10%	85/85	-0.0954	[-0.1604, -0.0347]	-0.0187	[-0.0711, +0.0363]
NSA binary Hamming	KNN	10%	0/85	—	[—, —]	—	[—, —]
NSA binary Hamming	IForest	10%	0/85	—	[—, —]	—	[—, —]
NSA binary contiguous	KNN	10%	0/85	—	[—, —]	—	[—, —]
NSA binary contiguous	IForest	10%	0/85	—	[—, —]	—	[—, —]
NSA binary chunk	KNN	10%	0/85	—	[—, —]	—	[—, —]
NSA binary chunk	IForest	10%	0/85	—	[—, —]	—	[—, —]
NSA fixed	KNN	5%	85/85	-0.1166	[-0.2159, -0.0181]	-0.1517	[-0.2643, -0.0527]
NSA fixed	IForest	5%	85/85	-0.0954	[-0.1745, -0.0239]	-0.0609	[-0.1088, -0.0178]

Continued on next page

Table 115 continued

NSA configuration	Reference	Retained	Pairs	ΔAUROC	95% CI	ΔAP	95% CI
NSA variable	KNN	5%	85/85	-0.0399	[-0.1187, +0.0407]	-0.0838	[-0.1778, -0.0032]
NSA variable	IForest	5%	85/85	-0.0187	[-0.0936, +0.0546]	+0.0070	[-0.0342, +0.0492]
NSA coverage	KNN	5%	85/85	-0.0413	[-0.1271, +0.0388]	-0.0879	[-0.1941, -0.0035]
NSA coverage	IForest	5%	85/85	-0.0201	[-0.1012, +0.0524]	+0.0029	[-0.0422, +0.0479]
NSA grid	KNN	5%	85/85	-0.0307	[-0.1129, +0.0519]	-0.0715	[-0.1676, +0.0092]
NSA grid	IForest	5%	85/85	-0.0095	[-0.0823, +0.0637]	+0.0193	[-0.0291, +0.0678]
NSA hierarchical	KNN	5%	85/85	-0.0503	[-0.1384, +0.0303]	-0.1070	[-0.2074, -0.0214]
NSA hierarchical	IForest	5%	85/85	-0.0291	[-0.1078, +0.0392]	-0.0162	[-0.0582, +0.0206]
NSA voronoi	KNN	5%	85/85	-0.0200	[-0.0988, +0.0584]	-0.0564	[-0.1355, +0.0195]
NSA voronoi	IForest	5%	85/85	+0.0012	[-0.0698, +0.0703]	+0.0345	[-0.0220, +0.0943]
NSA deterministic	KNN	5%	85/85	-0.1016	[-0.2036, -0.0125]	-0.1518	[-0.2710, -0.0536]
NSA deterministic	IForest	5%	85/85	-0.0804	[-0.1674, -0.0039]	-0.0610	[-0.1157, -0.0094]
NSA suppressed	KNN	5%	85/85	-0.0432	[-0.1231, +0.0359]	-0.0762	[-0.1797, +0.0068]
NSA suppressed	IForest	5%	85/85	-0.0220	[-0.0936, +0.0459]	+0.0146	[-0.0371, +0.0717]
NSA dual	KNN	5%	85/85	+0.0182	[-0.0496, +0.0903]	+0.0039	[-0.0343, +0.0383]
NSA dual	IForest	5%	85/85	+0.0394	[-0.0207, +0.0961]	+0.0947	[+0.0166, +0.1804]
NSA annealed	KNN	5%	85/85	-0.1116	[-0.2067, -0.0182]	-0.1530	[-0.2680, -0.0524]
NSA annealed	IForest	5%	85/85	-0.0904	[-0.1688, -0.0207]	-0.0621	[-0.1141, -0.0154]
NSA binary Hamming	KNN	5%	0/85	—	[—, —]	—	[—, —]
NSA binary Hamming	IForest	5%	0/85	—	[—, —]	—	[—, —]
NSA binary contiguous	KNN	5%	0/85	—	[—, —]	—	[—, —]
NSA binary contiguous	IForest	5%	0/85	—	[—, —]	—	[—, —]
NSA binary chunk	KNN	5%	0/85	—	[—, —]	—	[—, —]
NSA binary chunk	IForest	5%	0/85	—	[—, —]	—	[—, —]

Table 116: NSA/fixed, PCA(2): dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.6071	0.2592	5/5	23/26	0.6489	0.2854	5/5	8/26
cardio	0.9037	0.5182	5/5	13/26	0.8812	0.4762	5/5	14/26
glass	0.8000	0.1913	5/5	7/26	0.6006	0.1174	5/5	14/26
ionosphere	0.7024	0.5444	5/5	8/26	0.6709	0.5348	5/5	6/26
letter	0.4455	0.0658	5/5	18/26	0.5220	0.0845	5/5	13/26
lympho	0.5983	0.0872	5/5	19/26	0.3241	0.0558	5/5	24/26
mnist	0.6189	0.1815	5/5	20/26	0.6534	0.1815	5/5	21/26
musk	0.9376	0.6691	5/5	11/26	0.8828	0.3442	5/5	7/26
optdigits	0.1969	0.0179	5/5	26/26	0.3634	0.0214	5/5	24/26
pendigits	0.7726	0.2407	5/5	17/26	0.8248	0.1109	5/5	17/26
pima	0.6459	0.5114	5/5	15/26	0.5947	0.4241	5/5	14/26
satellite	0.5698	0.5238	5/5	20/26	0.5592	0.4703	5/5	20/26
satimage-2	0.4627	0.0654	5/5	25/26	0.4681	0.0256	5/5	25/26
shuttle	0.9174	0.6081	5/5	16/26	0.7189	0.3134	5/5	21/26
vertebral	0.4512	0.1378	5/5	9/26	0.4986	0.1623	5/5	6/26
vowels	0.3310	0.0324	5/5	24/26	0.4399	0.0574	5/5	15/26
wbc	0.4627	0.0869	5/5	25/26	0.4068	0.0742	5/5	24/26

Table 117: NSA/variable, PCA(2): dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.6438	0.3212	5/5	15/26	0.6177	0.3274	5/5	14/26
cardio	0.8860	0.5606	5/5	16/26	0.8694	0.5161	5/5	16/26
glass	0.8555	0.2541	5/5	4/26	0.6463	0.1758	5/5	12/26
ionosphere	0.7522	0.5871	5/5	1/26	0.7258	0.5964	5/5	4/26
letter	0.6470	0.1193	5/5	3/26	0.6027	0.0967	5/5	2/26
lympho	0.8293	0.3529	5/5	1/26	0.5017	0.1708	5/5	23/26
mnist	0.6512	0.2032	5/5	18/26	0.6729	0.2053	5/5	17/26
musk	0.9798	0.8032	5/5	8/26	0.9361	0.5072	5/5	6/26
optdigits	0.4768	0.0290	5/5	12/26	0.6120	0.0438	5/5	1/26
pendigits	0.8738	0.2091	5/5	11/26	0.8221	0.1314	5/5	17/26
pima	0.6676	0.5163	5/5	9/26	0.6003	0.4619	5/5	14/26
satellite	0.7286	0.6794	5/5	9/26	0.6919	0.6253	5/5	9/26
satimage-2	0.6475	0.2269	5/5	21/26	0.7270	0.1702	5/5	17/26
shuttle	0.9452	0.8159	5/5	15/26	0.7535	0.4625	5/5	21/26
vertebral	0.4635	0.1291	5/5	8/26	0.4278	0.1413	5/5	15/26
vowels	0.7349	0.1379	5/5	1/26	0.7085	0.0897	5/5	1/26
wbc	0.5811	0.2393	5/5	21/26	0.4451	0.1723	5/5	24/26

Table 118: NSA/coverage, PCA(2): dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.6548	0.3258	5/5	14/26	0.6204	0.3266	5/5	14/26
cardio	0.8876	0.5577	5/5	16/26	0.8704	0.5079	5/5	16/26
glass	0.8726	0.2708	5/5	2/26	0.6409	0.2198	5/5	12/26
ionosphere	0.7336	0.5712	5/5	6/26	0.7199	0.5985	5/5	4/26
letter	0.6028	0.1057	5/5	8/26	0.6010	0.1000	5/5	2/26
lympho	0.7621	0.3150	5/5	7/26	0.5069	0.1715	5/5	23/26
mnist	0.6279	0.2006	5/5	20/26	0.6757	0.2103	5/5	17/26
musk	0.9798	0.7902	5/5	8/26	0.9334	0.4937	5/5	7/26
optdigits	0.4064	0.0262	5/5	20/26	0.6177	0.0440	5/5	1/26
pendigits	0.8434	0.1948	5/5	15/26	0.8321	0.1509	5/5	17/26
pima	0.6585	0.5142	5/5	13/26	0.5947	0.4364	5/5	14/26
satellite	0.7192	0.6736	5/5	10/26	0.6761	0.6045	5/5	11/26
satimage-2	0.6426	0.2299	5/5	21/26	0.7162	0.0975	5/5	17/26
shuttle	0.9454	0.8159	5/5	15/26	0.7602	0.4640	5/5	21/26
vertebral	0.4653	0.1335	5/5	8/26	0.4504	0.1457	5/5	10/26
vowels	0.7267	0.1105	5/5	3/26	0.7003	0.0888	5/5	1/26
wbc	0.5589	0.2342	5/5	22/26	0.4210	0.1639	5/5	24/26

Table 119: NSA/grid, PCA(2): dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.6327	0.3086	5/5	21/26	0.6163	0.3086	5/5	15/26
cardio	0.8834	0.5525	5/5	16/26	0.8594	0.5120	5/5	17/26
glass	0.8591	0.3201	5/5	4/26	0.7055	0.2495	5/5	7/26
ionosphere	0.7515	0.5886	5/5	2/26	0.7300	0.6153	5/5	4/26
letter	0.6445	0.1181	5/5	3/26	0.5994	0.0936	5/5	3/26
lympho	0.8069	0.3463	5/5	2/26	0.4000	0.0608	5/5	24/26
mnist	0.6432	0.1959	5/5	18/26	0.6667	0.1940	5/5	18/26
musk	0.9787	0.8132	5/5	8/26	0.9367	0.5278	5/5	6/26
optdigits	0.4769	0.0289	5/5	12/26	0.6125	0.0442	5/5	1/26
pendigits	0.8748	0.2145	5/5	11/26	0.8305	0.1788	5/5	17/26
pima	0.6677	0.5251	5/5	9/26	0.6052	0.4695	5/5	13/26
satellite	0.7061	0.6546	5/5	12/26	0.6931	0.6140	5/5	9/26
satimage-2	0.6632	0.2820	5/5	21/26	0.7375	0.1823	5/5	17/26
shuttle	0.9456	0.8001	5/5	14/26	0.8593	0.6123	5/5	18/26
vertebral	0.4764	0.1301	5/5	8/26	0.4222	0.1336	5/5	15/26
vowels	0.7345	0.1321	5/5	1/26	0.7082	0.0979	5/5	1/26
wbc	0.5679	0.2245	5/5	22/26	0.5359	0.2093	5/5	21/26

Table 120: NSA/hierarchical, PCA(2): dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.6454	0.3047	5/5	14/26	0.6187	0.3121	5/5	14/26
cardio	0.8894	0.4994	5/5	16/26	0.8332	0.4386	5/5	18/26
glass	0.8780	0.3008	5/5	2/26	0.6537	0.1171	5/5	11/26
ionosphere	0.7520	0.5995	5/5	1/26	0.6870	0.5608	5/5	5/26
letter	0.5964	0.1028	5/5	10/26	0.6027	0.1047	5/5	2/26
lympho	0.6483	0.2429	5/5	16/26	0.7259	0.3077	5/5	3/26
mnist	0.6609	0.2053	5/5	16/26	0.6749	0.1838	5/5	17/26
musk	0.9730	0.7234	5/5	8/26	0.8946	0.3668	5/5	7/26
optdigits	0.4120	0.0285	5/5	20/26	0.5467	0.0404	5/5	6/26
pendigits	0.8419	0.2181	5/5	15/26	0.8424	0.1363	5/5	16/26
pima	0.6417	0.4791	5/5	15/26	0.6022	0.4435	5/5	14/26
satellite	0.6527	0.5962	5/5	14/26	0.6245	0.5618	5/5	15/26
satimage-2	0.5341	0.2125	5/5	23/26	0.5658	0.1896	5/5	23/26
shuttle	0.9228	0.7031	5/5	15/26	0.7853	0.4213	5/5	21/26
vertebral	0.4474	0.1291	5/5	10/26	0.4260	0.1135	5/5	15/26
vowels	0.7425	0.1341	5/5	1/26	0.6706	0.0774	5/5	4/26
wbc	0.4766	0.1581	5/5	25/26	0.4300	0.1245	5/5	24/26

Table 121: NSA/voronoi, PCA(2): dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.6238	0.2835	5/5	23/26	0.5958	0.2735	5/5	18/26
cardio	0.8735	0.5502	5/5	17/26	0.8392	0.4783	5/5	17/26
glass	0.8927	0.3510	5/5	1/26	0.7165	0.2733	5/5	6/26
ionosphere	0.7599	0.6184	5/5	1/26	0.7316	0.6132	5/5	4/26
letter	0.6380	0.1115	5/5	4/26	0.5979	0.0960	5/5	4/26
lympho	0.7552	0.1929	5/5	7/26	0.5276	0.0751	5/5	21/26
mnist	0.6535	0.1980	5/5	18/26	0.6480	0.1907	5/5	21/26
musk	0.9790	0.8435	5/5	8/26	0.9440	0.6488	5/5	6/26
optdigits	0.5327	0.0332	5/5	10/26	0.6207	0.0470	5/5	1/26
pendigits	0.8435	0.2238	5/5	15/26	0.8086	0.1802	5/5	18/26
pima	0.6667	0.5130	5/5	9/26	0.5891	0.4491	5/5	16/26
satellite	0.7558	0.7046	5/5	8/26	0.7563	0.6855	5/5	8/26
satimage-2	0.8159	0.4221	5/5	18/26	0.8270	0.3191	5/5	17/26
shuttle	0.9134	0.7005	5/5	17/26	0.8159	0.5183	5/5	19/26
vertebral	0.4702	0.1360	5/5	8/26	0.4673	0.1522	5/5	9/26
vowels	0.7280	0.1246	5/5	3/26	0.7040	0.1090	5/5	1/26
wbc	0.6120	0.2640	5/5	20/26	0.5106	0.2513	5/5	22/26

Table 122: NSA/deterministic, PCA(2): dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.6349	0.2742	5/5	19/26	0.5774	0.2187	5/5	19/26
cardio	0.8938	0.4425	5/5	16/26	0.8321	0.3568	5/5	18/26
glass	0.7896	0.2495	5/5	7/26	0.5482	0.0848	5/5	19/26
ionosphere	0.7200	0.5672	5/5	6/26	0.6434	0.5040	5/5	7/26
letter	0.4351	0.0598	5/5	19/26	0.5625	0.0917	5/5	8/26
lympho	0.6534	0.1705	5/5	16/26	0.6966	0.2424	5/5	6/26
mnist	0.6243	0.1926	5/5	20/26	0.6386	0.1811	5/5	21/26
musk	0.9762	0.7451	5/5	8/26	0.9448	0.4160	5/5	6/26
optdigits	0.2199	0.0174	5/5	26/26	0.4310	0.0359	5/5	21/26
pendigits	0.8358	0.2042	5/5	15/26	0.7952	0.1043	5/5	19/26
pima	0.6361	0.4928	5/5	16/26	0.5969	0.4281	5/5	14/26
satellite	0.5642	0.5253	5/5	20/26	0.5543	0.4486	5/5	20/26
satimage-2	0.4171	0.0322	5/5	25/26	0.4258	0.0227	5/5	25/26
shuttle	0.9115	0.5915	5/5	17/26	0.7092	0.2762	5/5	21/26
vertebral	0.4052	0.1193	5/5	18/26	0.5085	0.1713	5/5	5/26
vowels	0.4524	0.1022	5/5	20/26	0.5097	0.0801	5/5	13/26
wbc	0.5033	0.1311	5/5	24/26	0.3384	0.0745	5/5	24/26

Table 123: NSA/suppressed, PCA(2): dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.6014	0.2937	5/5	23/26	0.6285	0.3293	5/5	13/26
cardio	0.8547	0.5263	5/5	19/26	0.8470	0.4991	5/5	17/26
glass	0.8799	0.2950	5/5	2/26	0.6677	0.1987	5/5	9/26
ionosphere	0.7536	0.6130	5/5	1/26	0.7294	0.6001	5/5	4/26
letter	0.6142	0.1029	5/5	8/26	0.5781	0.0920	5/5	7/26
lympho	0.7259	0.1832	5/5	11/26	0.5517	0.1879	5/5	18/26
mnist	0.6418	0.1968	5/5	18/26	0.6414	0.1869	5/5	21/26
musk	0.9694	0.8296	5/5	8/26	0.9342	0.6376	5/5	7/26
optdigits	0.5671	0.0388	5/5	2/26	0.5743	0.0420	5/5	5/26
pendigits	0.8056	0.2434	5/5	16/26	0.8061	0.1758	5/5	18/26
pima	0.6442	0.5052	5/5	15/26	0.5910	0.4351	5/5	14/26
satellite	0.7167	0.6772	5/5	10/26	0.6636	0.5935	5/5	11/26
satimage-2	0.6942	0.2016	5/5	20/26	0.7250	0.1526	5/5	17/26
shuttle	0.9283	0.8001	5/5	15/26	0.7856	0.5190	5/5	21/26
vertebral	0.4746	0.1375	5/5	8/26	0.4329	0.1298	5/5	13/26
vowels	0.7320	0.1314	5/5	1/26	0.6890	0.0965	5/5	2/26
wbc	0.5908	0.2543	5/5	21/26	0.4604	0.1471	5/5	24/26

Table 124: NSA/dual, PCA(2): dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.6401	0.3116	5/5	16/26	0.6220	0.3307	5/5	14/26
cardio	0.8860	0.5817	5/5	16/26	0.8666	0.5725	5/5	16/26
glass	0.8470	0.2364	5/5	5/26	0.7799	0.1717	5/5	4/26
ionosphere	0.7478	0.5792	5/5	5/26	0.7147	0.5790	5/5	5/26
letter	0.6229	0.1049	5/5	6/26	0.6014	0.1007	5/5	2/26
lympho	0.7207	0.2885	5/5	12/26	0.4121	0.0465	5/5	24/26
mnist	0.6640	0.2105	5/5	15/26	0.6626	0.1920	5/5	18/26
musk	0.9797	0.8122	5/5	8/26	0.9350	0.5974	5/5	7/26
optdigits	0.5067	0.0307	5/5	11/26	0.6226	0.0474	5/5	1/26
pendigits	0.8600	0.2298	5/5	15/26	0.8256	0.1608	5/5	17/26
pima	0.6628	0.5040	5/5	12/26	0.5903	0.4427	5/5	16/26
satellite	0.8136	0.7728	5/5	3/26	0.8185	0.7658	5/5	1/26
satimage-2	0.9721	0.8509	5/5	2/26	0.9544	0.7986	5/5	4/26
shuttle	0.9660	0.7471	5/5	10/26	0.9616	0.7142	5/5	4/26
vertebral	0.4613	0.1229	5/5	8/26	0.3853	0.1125	5/5	19/26
vowels	0.7337	0.1234	5/5	1/26	0.7107	0.1065	5/5	1/26
wbc	0.8799	0.6439	5/5	11/26	0.8856	0.6450	5/5	12/26

Table 125: NSA/annealed, PCA(2): dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	0.6307	0.2846	5/5	22/26	0.6024	0.2204	5/5	18/26
cardio	0.9003	0.4787	5/5	13/26	0.8616	0.4076	5/5	17/26
glass	0.7957	0.2402	5/5	7/26	0.5976	0.1721	5/5	14/26
ionosphere	0.7489	0.6258	5/5	4/26	0.6226	0.4736	5/5	7/26
letter	0.4277	0.0640	5/5	19/26	0.4810	0.0723	5/5	14/26
lympho	0.6052	0.1135	5/5	19/26	0.6293	0.1771	5/5	12/26
mnist	0.6096	0.1699	5/5	21/26	0.6512	0.1864	5/5	21/26
musk	0.9197	0.6608	5/5	11/26	0.9187	0.3726	5/5	7/26
optdigits	0.2376	0.0180	5/5	26/26	0.3183	0.0207	5/5	26/26
pendigits	0.8136	0.2278	5/5	16/26	0.8086	0.1176	5/5	18/26
pima	0.6306	0.4901	5/5	16/26	0.5815	0.4117	5/5	16/26
satellite	0.5999	0.5329	5/5	18/26	0.5845	0.4783	5/5	19/26
satimage-2	0.4475	0.0497	5/5	25/26	0.4663	0.0221	5/5	25/26
shuttle	0.9174	0.5910	5/5	16/26	0.7043	0.2736	5/5	21/26
vertebral	0.4315	0.1255	5/5	12/26	0.5129	0.1672	5/5	5/26
vowels	0.3750	0.0499	5/5	23/26	0.4483	0.0738	5/5	15/26
wbc	0.5196	0.1552	5/5	23/26	0.3531	0.0712	5/5	24/26

Table 126: NSA/binary/hamming, PCA(2): dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	—	—	0/5	—	—	—	0/5	—
cardio	—	—	0/5	—	—	—	0/5	—
glass	—	—	0/5	—	—	—	0/5	—
ionosphere	—	—	0/5	—	—	—	0/5	—
letter	—	—	0/5	—	—	—	0/5	—
lympho	—	—	0/5	—	—	—	0/5	—
mnist	—	—	0/5	—	—	—	0/5	—
musk	—	—	0/5	—	—	—	0/5	—
optdigits	—	—	0/5	—	—	—	0/5	—
pendigits	—	—	0/5	—	—	—	0/5	—
pima	—	—	0/5	—	—	—	0/5	—
satellite	—	—	0/5	—	—	—	0/5	—
satimage-2	—	—	0/5	—	—	—	0/5	—
shuttle	—	—	0/5	—	—	—	0/5	—
vertebral	—	—	0/5	—	—	—	0/5	—
vowels	—	—	0/5	—	—	—	0/5	—
wbc	—	—	0/5	—	—	—	0/5	—

Table 127: NSA/binary/rcontiguous, PCA(2): dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	—	—	0/5	—	—	—	0/5	—
cardio	—	—	0/5	—	—	—	0/5	—
glass	—	—	0/5	—	—	—	0/5	—
ionosphere	—	—	0/5	—	—	—	0/5	—
letter	—	—	0/5	—	—	—	0/5	—
lympho	—	—	0/5	—	—	—	0/5	—
mnist	—	—	0/5	—	—	—	0/5	—
musk	—	—	0/5	—	—	—	0/5	—
optdigits	—	—	0/5	—	—	—	0/5	—
pendigits	—	—	0/5	—	—	—	0/5	—
pima	—	—	0/5	—	—	—	0/5	—
satellite	—	—	0/5	—	—	—	0/5	—
satimage-2	—	—	0/5	—	—	—	0/5	—
shuttle	—	—	0/5	—	—	—	0/5	—
vertebral	—	—	0/5	—	—	—	0/5	—
vowels	—	—	0/5	—	—	—	0/5	—
wbc	—	—	0/5	—	—	—	0/5	—

Table 128: NSA/binary/rchunk, PCA(2): dataset scores and fixed other-detector ranks at new sizes. Scores and ranks require all five successful seeds; Q reports qualified coverage.

Dataset	10% AUROC	10% AP	10% Q	10% rank	5% AUROC	5% AP	5% Q	5% rank
arrhythmia	—	—	0/5	—	—	—	0/5	—
cardio	—	—	0/5	—	—	—	0/5	—
glass	—	—	0/5	—	—	—	0/5	—
ionosphere	—	—	0/5	—	—	—	0/5	—
letter	—	—	0/5	—	—	—	0/5	—
lympho	—	—	0/5	—	—	—	0/5	—
mnist	—	—	0/5	—	—	—	0/5	—
musk	—	—	0/5	—	—	—	0/5	—
optdigits	—	—	0/5	—	—	—	0/5	—
pendigits	—	—	0/5	—	—	—	0/5	—
pima	—	—	0/5	—	—	—	0/5	—
satellite	—	—	0/5	—	—	—	0/5	—
satimage-2	—	—	0/5	—	—	—	0/5	—
shuttle	—	—	0/5	—	—	—	0/5	—
vertebral	—	—	0/5	—	—	—	0/5	—
vowels	—	—	0/5	—	—	—	0/5	—
wbc	—	—	0/5	—	—	—	0/5	—

P Every new-size NSA versus other-detector contrast

These tables include all 2,236 new same-size comparisons. Differences are NSA minus the named other detector, using only matched qualified dataset/seed cases. P gives matched cases out of 85. Each CI resamples datasets as clusters and is descriptive, with no multiplicity adjustment. Missing intervals mean fewer than five matched datasets. A positive available-case contrast with partial coverage does not establish a global ranking advantage.

Table 129: NSA/fixed versus all 43 other registered detectors, Original features, 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	77	-0.4849	[-0.6590, -0.3079]	-0.3261	[-0.4749, -0.1852]	85	-0.4735	[-0.6585, -0.2956]	-0.3094	[-0.4624, -0.1662]
ABOD	85	-0.5554	[-0.7080, -0.4015]	-0.4259	[-0.5791, -0.2771]	85	-0.5275	[-0.6938, -0.3588]	-0.3820	[-0.5397, -0.2275]
COPOD	85	-0.4794	[-0.6545, -0.2953]	-0.3381	[-0.5021, -0.1779]	85	-0.4811	[-0.6644, -0.2956]	-0.3262	[-0.4995, -0.1632]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	84	-0.2934	[-0.4186, -0.1762]	-0.1115	[-0.2037, -0.0405]	85	-0.2976	[-0.4209, -0.1840]	-0.1035	[-0.1997, -0.0250]
QMCD	85	-0.1521	[-0.2479, -0.0698]	-0.0720	[-0.1258, -0.0292]	85	-0.0869	[-0.2033, -0.0059]	-0.0311	[-0.0820, +0.0136]
KDE	85	-0.5800	[-0.7230, -0.4392]	-0.4527	[-0.5985, -0.3070]	85	-0.5626	[-0.7119, -0.4035]	-0.4064	[-0.5626, -0.2479]
Sampling	65	-0.5697	[-0.7360, -0.4117]	-0.4449	[-0.6216, -0.2774]	50	-0.5850	[-0.7716, -0.3896]	-0.4354	[-0.6555, -0.2180]
GMM	85	-0.5439	[-0.7013, -0.3856]	-0.3974	[-0.5358, -0.2575]	85	-0.5293	[-0.6878, -0.3629]	-0.3475	[-0.4896, -0.2079]
PCA	85	-0.5079	[-0.6806, -0.3270]	-0.3975	[-0.5654, -0.2356]	85	-0.4956	[-0.6780, -0.3103]	-0.3687	[-0.5300, -0.2022]
KPCA	85	-0.5784	[-0.7254, -0.4291]	-0.4486	[-0.5933, -0.3002]	85	-0.5592	[-0.7198, -0.3980]	-0.4040	[-0.5616, -0.2462]
MCD	85	-0.5196	[-0.6779, -0.3630]	-0.3368	[-0.4851, -0.2022]	85	-0.4911	[-0.6475, -0.3271]	-0.2878	[-0.4251, -0.1573]
CD	78	-0.5197	[-0.6936, -0.3476]	-0.3679	[-0.5132, -0.2288]	79	-0.5168	[-0.6925, -0.3332]	-0.3387	[-0.4865, -0.1926]
OCSVM	85	-0.5425	[-0.7064, -0.3733]	-0.4260	[-0.5859, -0.2682]	85	-0.5306	[-0.6982, -0.3549]	-0.3939	[-0.5605, -0.2321]
LMDD	52	-0.4483	[-0.6778, -0.2205]	-0.3267	[-0.5146, -0.1500]	55	-0.4288	[-0.6432, -0.2132]	-0.2888	[-0.4742, -0.1123]
LOF	85	-0.4949	[-0.6709, -0.3100]	-0.3970	[-0.5663, -0.2263]	85	-0.4788	[-0.6673, -0.2895]	-0.3701	[-0.5405, -0.2012]
COF	85	-0.3675	[-0.5074, -0.2269]	-0.2109	[-0.3114, -0.1194]	85	-0.3549	[-0.4920, -0.2165]	-0.1811	[-0.2670, -0.0963]
CBLOF	80	-0.5419	[-0.6952, -0.3875]	-0.4128	[-0.5725, -0.2552]	65	-0.5757	[-0.7441, -0.4181]	-0.4207	[-0.6043, -0.2526]
LOCI	29	-0.4741	[-0.6729, -0.2526]	-0.2586	[-0.4185, -0.0987]	30	-0.4214	[-0.6473, -0.1752]	-0.2124	[-0.3864, -0.0250]
HBOS	85	-0.5061	[-0.6627, -0.3486]	-0.3450	[-0.4976, -0.1986]	85	-0.4897	[-0.6505, -0.3300]	-0.3091	[-0.4629, -0.1650]
HDBSCAN	85	-0.2465	[-0.3645, -0.1291]	-0.0305	[-0.0711, +0.0056]	85	-0.2352	[-0.3376, -0.1299]	-0.0129	[-0.0509, +0.0241]
KNN	85	-0.5577	[-0.7131, -0.4007]	-0.4255	[-0.5820, -0.2629]	85	-0.5153	[-0.6879, -0.3315]	-0.3740	[-0.5417, -0.2053]
SOD	58	-0.4098	[-0.5612, -0.2584]	-0.2045	[-0.3068, -0.1151]	55	-0.4097	[-0.5529, -0.2658]	-0.1721	[-0.2307, -0.1103]

Continued on next page

Table 129 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ROD	39	-0.4444	[-0.7035, -0.1966]	-0.3226	[-0.5753, -0.1072]	42	-0.4448	[-0.6947, -0.2005]	-0.2974	[-0.5352, -0.0892]
IForest	85	-0.5039	[-0.6516, -0.3501]	-0.2864	[-0.4166, -0.1637]	85	-0.4745	[-0.6228, -0.3217]	-0.2343	[-0.3490, -0.1217]
INNE	85	-0.5397	[-0.6931, -0.3853]	-0.3441	[-0.4956, -0.1976]	85	-0.5042	[-0.6566, -0.3467]	-0.2671	[-0.4067, -0.1375]
DIF	85	-0.4654	[-0.6193, -0.3134]	-0.2481	[-0.3644, -0.1382]	85	-0.4382	[-0.5895, -0.2887]	-0.1936	[-0.2922, -0.1028]
FeatureBagging	85	-0.5021	[-0.6781, -0.3225]	-0.4006	[-0.5658, -0.2399]	85	-0.4858	[-0.6729, -0.2974]	-0.3700	[-0.5319, -0.1974]
LSCP	50	-0.5534	[-0.7606, -0.3523]	-0.4353	[-0.6627, -0.2190]	50	-0.5486	[-0.7534, -0.3331]	-0.4135	[-0.6362, -0.1939]
LODA	85	-0.3639	[-0.4899, -0.2400]	-0.0863	[-0.1321, -0.0435]	85	-0.1917	[-0.2868, -0.1005]	-0.0139	[-0.0371, +0.0095]
SUOD	85	-0.5208	[-0.6811, -0.3574]	-0.3561	[-0.5085, -0.2065]	84	-0.5028	[-0.6713, -0.3298]	-0.3160	[-0.4630, -0.1712]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	-0.5157	[-0.7389, -0.2926]	-0.4245	[-0.6568, -0.2041]	0	—	[—, —]	—	[—, —]
VAE	50	-0.5004	[-0.7331, -0.2705]	-0.4080	[-0.6386, -0.1828]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	-0.1614	[-0.2995, -0.0242]	-0.0964	[-0.1521, -0.0383]	85	-0.2029	[-0.3522, -0.0494]	-0.0915	[-0.1536, -0.0322]
MO_GAAL	85	-0.1897	[-0.3215, -0.0547]	-0.0920	[-0.1483, -0.0372]	85	-0.1937	[-0.3367, -0.0505]	-0.0915	[-0.1615, -0.0237]
DeepSVDD	85	-0.5108	[-0.6800, -0.3411]	-0.3976	[-0.5543, -0.2489]	85	-0.4975	[-0.6667, -0.3232]	-0.3509	[-0.5115, -0.1959]
AnoGAN	85	-0.4979	[-0.6701, -0.3193]	-0.3535	[-0.4950, -0.2125]	85	-0.4943	[-0.6669, -0.3088]	-0.3581	[-0.5096, -0.1975]
ALAD	85	-0.2480	[-0.3570, -0.1414]	-0.0452	[-0.0828, -0.0115]	85	-0.2507	[-0.3615, -0.1413]	-0.0377	[-0.0793, +0.0026]
AE1SVM	85	-0.5161	[-0.6890, -0.3327]	-0.4008	[-0.5660, -0.2429]	85	-0.4969	[-0.6764, -0.3116]	-0.3750	[-0.5423, -0.2098]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	35	-0.2834	[-0.5262, -0.0363]	-0.0995	[-0.2474, +0.0295]	54	-0.2975	[-0.4482, -0.1526]	-0.0999	[-0.2323, +0.0025]
LUNAR	85	-0.5722	[-0.7191, -0.4236]	-0.4471	[-0.6050, -0.2979]	85	-0.5392	[-0.6945, -0.3770]	-0.3848	[-0.5517, -0.2187]

Table 130: NSA/variable versus all 43 other registered detectors, Original features, 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	77	-0.3418	[-0.5358, -0.1523]	-0.2700	[-0.4283, -0.1212]	85	-0.3843	[-0.5870, -0.1791]	-0.2759	[-0.4371, -0.1160]
ABOD	85	-0.4186	[-0.5934, -0.2421]	-0.3731	[-0.5429, -0.2082]	85	-0.4383	[-0.6235, -0.2490]	-0.3485	[-0.5154, -0.1852]
COPOD	85	-0.3426	[-0.5372, -0.1477]	-0.2854	[-0.4620, -0.1208]	85	-0.3919	[-0.6008, -0.1807]	-0.2927	[-0.4706, -0.1191]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	84	-0.1558	[-0.3049, -0.0122]	-0.0576	[-0.1598, +0.0226]	85	-0.2084	[-0.3603, -0.0584]	-0.0699	[-0.1767, +0.0174]
QMCD	85	-0.0153	[-0.1454, +0.1027]	-0.0192	[-0.0895, +0.0369]	85	+0.0023	[-0.1281, +0.0977]	+0.0024	[-0.0566, +0.0577]
KDE	85	-0.4432	[-0.6044, -0.2869]	-0.4000	[-0.5570, -0.2382]	85	-0.4734	[-0.6530, -0.2957]	-0.3729	[-0.5355, -0.2025]
Sampling	65	-0.4228	[-0.6128, -0.2353]	-0.3881	[-0.5818, -0.2053]	50	-0.4689	[-0.6969, -0.2361]	-0.4020	[-0.6333, -0.1697]
GMM	85	-0.4071	[-0.5846, -0.2327]	-0.3447	[-0.4975, -0.1913]	85	-0.4401	[-0.6201, -0.2531]	-0.3139	[-0.4717, -0.1577]
PCA	85	-0.3711	[-0.5621, -0.1737]	-0.3447	[-0.5232, -0.1768]	85	-0.4064	[-0.6032, -0.1922]	-0.3352	[-0.5183, -0.1632]
KPCA	85	-0.4416	[-0.6073, -0.2746]	-0.3958	[-0.5564, -0.2367]	85	-0.4700	[-0.6459, -0.2907]	-0.3705	[-0.5482, -0.1996]
MCD	85	-0.3828	[-0.5605, -0.2065]	-0.2841	[-0.4360, -0.1322]	85	-0.4019	[-0.5851, -0.2172]	-0.2542	[-0.3995, -0.1053]
CD	78	-0.3735	[-0.5652, -0.1814]	-0.3115	[-0.4717, -0.1632]	79	-0.4223	[-0.6230, -0.2198]	-0.3038	[-0.4627, -0.1496]
OCSVM	85	-0.4057	[-0.5945, -0.2210]	-0.3732	[-0.5488, -0.1987]	85	-0.4414	[-0.6257, -0.2471]	-0.3604	[-0.5383, -0.1803]
LMDD	52	-0.2998	[-0.5355, -0.0653]	-0.2723	[-0.4743, -0.0722]	55	-0.3357	[-0.5705, -0.1020]	-0.2543	[-0.4535, -0.0566]
LOF	85	-0.3581	[-0.5578, -0.1643]	-0.3443	[-0.5209, -0.1659]	85	-0.3895	[-0.5998, -0.1688]	-0.3365	[-0.5206, -0.1526]
COF	85	-0.2307	[-0.3862, -0.0653]	-0.1582	[-0.2635, -0.0557]	85	-0.2657	[-0.4274, -0.1013]	-0.1476	[-0.2436, -0.0549]
CBLOF	80	-0.4029	[-0.5729, -0.2338]	-0.3571	[-0.5311, -0.1900]	65	-0.4692	[-0.6637, -0.2696]	-0.3850	[-0.5764, -0.1912]
LOCI	29	-0.3786	[-0.5769, -0.1682]	-0.2289	[-0.4017, -0.0467]	30	-0.3906	[-0.6373, -0.1367]	-0.1926	[-0.3854, +0.0296]
HBOS	85	-0.3693	[-0.5480, -0.1975]	-0.2923	[-0.4639, -0.1328]	85	-0.4004	[-0.5858, -0.2093]	-0.2756	[-0.4407, -0.1181]
HDBSCAN	85	-0.1097	[-0.2482, +0.0278]	+0.0222	[-0.0283, +0.0730]	85	-0.1459	[-0.2807, -0.0121]	+0.0206	[-0.0274, +0.0737]
KNN	85	-0.4209	[-0.5902, -0.2474]	-0.3727	[-0.5464, -0.2088]	85	-0.4260	[-0.6264, -0.2239]	-0.3405	[-0.5194, -0.1664]
SOD	58	-0.2772	[-0.4461, -0.1107]	-0.1520	[-0.2616, -0.0541]	55	-0.2886	[-0.4702, -0.1113]	-0.1310	[-0.1936, -0.0572]
ROD	39	-0.2508	[-0.5143, -0.0221]	-0.2465	[-0.5008, -0.0220]	42	-0.3282	[-0.6117, -0.0600]	-0.2598	[-0.5043, -0.0318]
IForest	85	-0.3671	[-0.5309, -0.2043]	-0.2336	[-0.3705, -0.0993]	85	-0.3852	[-0.5595, -0.2136]	-0.2007	[-0.3304, -0.0784]
INNE	85	-0.4029	[-0.5740, -0.2325]	-0.2913	[-0.4502, -0.1403]	85	-0.4150	[-0.5931, -0.2417]	-0.2336	[-0.3782, -0.0944]
DIF	85	-0.3286	[-0.5045, -0.1573]	-0.1954	[-0.3183, -0.0788]	85	-0.3490	[-0.5203, -0.1699]	-0.1601	[-0.2697, -0.0528]
FeatureBagging	85	-0.3653	[-0.5531, -0.1716]	-0.3479	[-0.5152, -0.1818]	85	-0.3965	[-0.6080, -0.1803]	-0.3364	[-0.5166, -0.1577]

Continued on next page

Table 130 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
LSCP	50	-0.3987	[-0.6440, -0.1574]	-0.3790	[-0.6203, -0.1480]	50	-0.4325	[-0.6908, -0.1766]	-0.3801	[-0.6221, -0.1299]
LODA	85	-0.2271	[-0.3745, -0.0860]	-0.0336	[-0.0851, +0.0179]	85	-0.1024	[-0.2137, +0.0091]	+0.0196	[-0.0122, +0.0567]
SUOD	85	-0.3840	[-0.5583, -0.2052]	-0.3034	[-0.4659, -0.1395]	84	-0.4138	[-0.6088, -0.2267]	-0.2823	[-0.4452, -0.1230]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	-0.3610	[-0.6200, -0.1135]	-0.3682	[-0.6089, -0.1265]	0	—	[—, —]	—	[—, —]
VAE	50	-0.3458	[-0.6167, -0.0889]	-0.3517	[-0.6024, -0.1147]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	-0.0246	[-0.1945, +0.1459]	-0.0437	[-0.1132, +0.0294]	85	-0.1136	[-0.2967, +0.0735]	-0.0579	[-0.1331, +0.0179]
MO_GAAL	85	-0.0529	[-0.2168, +0.1145]	-0.0392	[-0.1097, +0.0318]	85	-0.1045	[-0.2778, +0.0750]	-0.0579	[-0.1383, +0.0266]
DeepSVDD	85	-0.3740	[-0.5607, -0.1839]	-0.3449	[-0.5093, -0.1805]	85	-0.4083	[-0.6092, -0.2031]	-0.3174	[-0.4856, -0.1472]
AnoGAN	85	-0.3611	[-0.5471, -0.1657]	-0.3008	[-0.4503, -0.1476]	85	-0.4050	[-0.6070, -0.1961]	-0.3245	[-0.4944, -0.1518]
ALAD	85	-0.1112	[-0.2412, +0.0198]	+0.0076	[-0.0429, +0.0564]	85	-0.1615	[-0.3014, -0.0190]	-0.0042	[-0.0562, +0.0511]
AE1SVM	85	-0.3794	[-0.5674, -0.1863]	-0.3481	[-0.5234, -0.1764]	85	-0.4077	[-0.6153, -0.1973]	-0.3415	[-0.5266, -0.1661]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	35	-0.1722	[-0.4504, +0.1045]	-0.0519	[-0.2246, +0.1122]	54	-0.2100	[-0.3951, -0.0345]	-0.0658	[-0.2124, +0.0481]
LUNAR	85	-0.4354	[-0.5941, -0.2729]	-0.3944	[-0.5588, -0.2330]	85	-0.4499	[-0.6280, -0.2750]	-0.3513	[-0.5339, -0.1773]

Table 131: NSA/coverage versus all 43 other registered detectors, Original features, 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	77	-0.3421	[-0.5308, -0.1544]	-0.2696	[-0.4261, -0.1170]	85	-0.3868	[-0.5836, -0.1857]	-0.2805	[-0.4423, -0.1240]
ABOD	85	-0.4203	[-0.5958, -0.2522]	-0.3732	[-0.5384, -0.2068]	85	-0.4408	[-0.6280, -0.2476]	-0.3530	[-0.5230, -0.1850]
COPOD	85	-0.3443	[-0.5423, -0.1487]	-0.2855	[-0.4575, -0.1177]	85	-0.3943	[-0.5996, -0.1866]	-0.2973	[-0.4773, -0.1255]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	84	-0.1575	[-0.3035, -0.0164]	-0.0577	[-0.1563, +0.0267]	85	-0.2109	[-0.3591, -0.0635]	-0.0745	[-0.1742, +0.0080]
QMCD	85	-0.0170	[-0.1493, +0.1010]	-0.0193	[-0.0877, +0.0365]	85	-0.0002	[-0.1264, +0.0947]	-0.0022	[-0.0615, +0.0506]
KDE	85	-0.4448	[-0.6053, -0.2880]	-0.4000	[-0.5595, -0.2475]	85	-0.4758	[-0.6505, -0.3079]	-0.3774	[-0.5402, -0.2126]
Sampling	65	-0.4228	[-0.6165, -0.2321]	-0.3879	[-0.5748, -0.2011]	50	-0.4689	[-0.6961, -0.2296]	-0.4020	[-0.6400, -0.1744]
GMM	85	-0.4087	[-0.5801, -0.2350]	-0.3448	[-0.4986, -0.1925]	85	-0.4425	[-0.6253, -0.2641]	-0.3185	[-0.4759, -0.1606]
PCA	85	-0.3728	[-0.5624, -0.1796]	-0.3448	[-0.5264, -0.1821]	85	-0.4088	[-0.6143, -0.2019]	-0.3398	[-0.5170, -0.1664]
KPCA	85	-0.4433	[-0.6034, -0.2866]	-0.3959	[-0.5566, -0.2343]	85	-0.4725	[-0.6482, -0.2919]	-0.3751	[-0.5542, -0.2092]
MCD	85	-0.3845	[-0.5585, -0.2110]	-0.2842	[-0.4339, -0.1433]	85	-0.4044	[-0.5865, -0.2235]	-0.2588	[-0.4063, -0.1189]
CD	78	-0.3753	[-0.5654, -0.1896]	-0.3115	[-0.4698, -0.1594]	79	-0.4250	[-0.6281, -0.2188]	-0.3087	[-0.4661, -0.1560]
OCSVM	85	-0.4074	[-0.5823, -0.2261]	-0.3733	[-0.5506, -0.2025]	85	-0.4439	[-0.6379, -0.2592]	-0.3650	[-0.5403, -0.1890]
LMDD	52	-0.3024	[-0.5403, -0.0731]	-0.2724	[-0.4721, -0.0740]	55	-0.3392	[-0.5719, -0.1109]	-0.2608	[-0.4569, -0.0663]
LOF	85	-0.3598	[-0.5570, -0.1623]	-0.3443	[-0.5232, -0.1667]	85	-0.3920	[-0.6034, -0.1659]	-0.3411	[-0.5181, -0.1624]
COF	85	-0.2324	[-0.3907, -0.0710]	-0.1582	[-0.2691, -0.0535]	85	-0.2681	[-0.4321, -0.1053]	-0.1522	[-0.2447, -0.0629]
CBLOF	80	-0.4050	[-0.5734, -0.2436]	-0.3573	[-0.5337, -0.1916]	65	-0.4689	[-0.6714, -0.2698]	-0.3850	[-0.5841, -0.1994]
LOCI	29	-0.3834	[-0.5803, -0.1823]	-0.2294	[-0.4010, -0.0565]	30	-0.3981	[-0.6373, -0.1590]	-0.2056	[-0.3861, -0.0103]
HBOS	85	-0.3710	[-0.5387, -0.1981]	-0.2924	[-0.4515, -0.1359]	85	-0.4029	[-0.5908, -0.2206]	-0.2802	[-0.4474, -0.1232]
HDBSCAN	85	-0.1113	[-0.2532, +0.0252]	+0.0222	[-0.0293, +0.0690]	85	-0.1484	[-0.2795, -0.0090]	+0.0160	[-0.0318, +0.0648]
KNN	85	-0.4225	[-0.5930, -0.2523]	-0.3728	[-0.5438, -0.1991]	85	-0.4285	[-0.6263, -0.2372]	-0.3451	[-0.5127, -0.1684]
SOD	58	-0.2790	[-0.4381, -0.1176]	-0.1520	[-0.2596, -0.0514]	55	-0.2883	[-0.4710, -0.1035]	-0.1310	[-0.1930, -0.0614]
ROD	39	-0.2543	[-0.4990, -0.0279]	-0.2467	[-0.5053, -0.0120]	42	-0.3324	[-0.6038, -0.0694]	-0.2676	[-0.5027, -0.0439]
IForest	85	-0.3688	[-0.5344, -0.2091]	-0.2337	[-0.3687, -0.1055]	85	-0.3877	[-0.5657, -0.2145]	-0.2053	[-0.3332, -0.0813]
INNE	85	-0.4046	[-0.5726, -0.2388]	-0.2914	[-0.4518, -0.1397]	85	-0.4175	[-0.5962, -0.2411]	-0.2382	[-0.3760, -0.1031]
DIF	85	-0.3303	[-0.5029, -0.1538]	-0.1954	[-0.3216, -0.0737]	85	-0.3515	[-0.5233, -0.1669]	-0.1646	[-0.2750, -0.0618]
FeatureBagging	85	-0.3669	[-0.5554, -0.1763]	-0.3480	[-0.5321, -0.1675]	85	-0.3990	[-0.6102, -0.1785]	-0.3410	[-0.5212, -0.1641]
LSCP	50	-0.3987	[-0.6369, -0.1656]	-0.3790	[-0.6170, -0.1404]	50	-0.4325	[-0.6907, -0.1685]	-0.3801	[-0.6202, -0.1361]

Continued on next page

Table 131 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
LODA	85	-0.2287	[-0.3728, -0.0900]	-0.0336	[-0.0852, +0.0162]	85	-0.1049	[-0.2212, +0.0053]	+0.0151	[-0.0160, +0.0495]
SUOD	85	-0.3857	[-0.5647, -0.2074]	-0.3034	[-0.4659, -0.1417]	84	-0.4162	[-0.6115, -0.2291]	-0.2869	[-0.4477, -0.1335]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	-0.3610	[-0.6074, -0.1000]	-0.3682	[-0.6195, -0.1375]	0	—	[—, —]	—	[—, —]
VAE	50	-0.3458	[-0.6042, -0.0878]	-0.3517	[-0.5935, -0.1130]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	-0.0263	[-0.1881, +0.1392]	-0.0438	[-0.1135, +0.0292]	85	-0.1161	[-0.3030, +0.0769]	-0.0625	[-0.1399, +0.0149]
MO_GAAL	85	-0.0546	[-0.2134, +0.1078]	-0.0393	[-0.1095, +0.0340]	85	-0.1070	[-0.2832, +0.0722]	-0.0625	[-0.1448, +0.0207]
DeepSVDD	85	-0.3757	[-0.5561, -0.1890]	-0.3450	[-0.5113, -0.1789]	85	-0.4108	[-0.6037, -0.2006]	-0.3220	[-0.4884, -0.1535]
AnoGAN	85	-0.3628	[-0.5523, -0.1698]	-0.3008	[-0.4553, -0.1511]	85	-0.4075	[-0.6039, -0.2024]	-0.3291	[-0.4960, -0.1647]
ALAD	85	-0.1128	[-0.2403, +0.0162]	+0.0075	[-0.0415, +0.0538]	85	-0.1639	[-0.3023, -0.0230]	-0.0088	[-0.0586, +0.0451]
AE1SVM	85	-0.3810	[-0.5678, -0.1934]	-0.3482	[-0.5246, -0.1771]	85	-0.4102	[-0.6110, -0.2010]	-0.3461	[-0.5248, -0.1663]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	35	-0.1762	[-0.4523, +0.1009]	-0.0521	[-0.2212, +0.1099]	54	-0.2128	[-0.3998, -0.0434]	-0.0710	[-0.2107, +0.0416]
LUNAR	85	-0.4371	[-0.5940, -0.2764]	-0.3944	[-0.5552, -0.2308]	85	-0.4524	[-0.6281, -0.2678]	-0.3559	[-0.5298, -0.1906]

Table 132: NSA/grid versus all 43 other registered detectors, Original features, 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	77	-0.3449	[-0.5333, -0.1613]	-0.2765	[-0.4326, -0.1286]	85	-0.3864	[-0.5834, -0.1799]	-0.2851	[-0.4471, -0.1339]
ABOD	85	-0.4213	[-0.5939, -0.2536]	-0.3778	[-0.5409, -0.2145]	85	-0.4404	[-0.6180, -0.2451]	-0.3576	[-0.5243, -0.1994]
COPOD	85	-0.3453	[-0.5355, -0.1505]	-0.2901	[-0.4610, -0.1318]	85	-0.3940	[-0.5992, -0.1859]	-0.3019	[-0.4774, -0.1348]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	84	-0.1560	[-0.2984, -0.0165]	-0.0619	[-0.1573, +0.0184]	85	-0.2105	[-0.3555, -0.0708]	-0.0791	[-0.1789, +0.0005]
QMCD	85	-0.0179	[-0.1536, +0.0996]	-0.0239	[-0.0861, +0.0321]	85	+0.0002	[-0.1301, +0.0945]	-0.0068	[-0.0627, +0.0407]
KDE	85	-0.4458	[-0.5981, -0.2913]	-0.4046	[-0.5577, -0.2529]	85	-0.4755	[-0.6445, -0.2977]	-0.3820	[-0.5438, -0.2225]
Sampling	65	-0.4222	[-0.6013, -0.2382]	-0.3892	[-0.5762, -0.2034]	50	-0.4723	[-0.7029, -0.2341]	-0.4136	[-0.6434, -0.1971]
GMM	85	-0.4097	[-0.5775, -0.2354]	-0.3494	[-0.4970, -0.2059]	85	-0.4422	[-0.6244, -0.2563]	-0.3231	[-0.4782, -0.1748]
PCA	85	-0.3737	[-0.5628, -0.1799]	-0.3494	[-0.5208, -0.1768]	85	-0.4085	[-0.6168, -0.1999]	-0.3444	[-0.5175, -0.1719]
KPCA	85	-0.4442	[-0.5996, -0.2862]	-0.4005	[-0.5571, -0.2489]	85	-0.4721	[-0.6471, -0.2945]	-0.3797	[-0.5460, -0.2155]
MCD	85	-0.3855	[-0.5561, -0.2115]	-0.2888	[-0.4387, -0.1437]	85	-0.4040	[-0.5867, -0.2231]	-0.2634	[-0.4066, -0.1255]
CD	78	-0.3759	[-0.5669, -0.1933]	-0.3164	[-0.4654, -0.1698]	79	-0.4244	[-0.6211, -0.2275]	-0.3138	[-0.4721, -0.1600]
OCSVM	85	-0.4084	[-0.5848, -0.2312]	-0.3779	[-0.5496, -0.2107]	85	-0.4435	[-0.6353, -0.2492]	-0.3695	[-0.5426, -0.1982]
LMDD	52	-0.3129	[-0.5554, -0.0792]	-0.2777	[-0.4778, -0.0820]	55	-0.3228	[-0.5580, -0.0914]	-0.2609	[-0.4541, -0.0788]
LOF	85	-0.3608	[-0.5587, -0.1685]	-0.3490	[-0.5245, -0.1762]	85	-0.3916	[-0.5964, -0.1755]	-0.3457	[-0.5301, -0.1702]
COF	85	-0.2334	[-0.3936, -0.0712]	-0.1628	[-0.2711, -0.0656]	85	-0.2678	[-0.4277, -0.1111]	-0.1567	[-0.2497, -0.0672]
CBLOF	80	-0.4051	[-0.5756, -0.2404]	-0.3621	[-0.5327, -0.1901]	65	-0.4703	[-0.6610, -0.2689]	-0.3913	[-0.5812, -0.2058]
LOCI	29	-0.3873	[-0.5961, -0.1868]	-0.2398	[-0.3990, -0.0789]	30	-0.3938	[-0.6197, -0.1560]	-0.2040	[-0.3820, -0.0151]
HBOS	85	-0.3719	[-0.5399, -0.2071]	-0.2970	[-0.4633, -0.1420]	85	-0.4025	[-0.5910, -0.2125]	-0.2848	[-0.4490, -0.1305]
HDBSCAN	85	-0.1123	[-0.2533, +0.0248]	+0.0175	[-0.0313, +0.0647]	85	-0.1480	[-0.2782, -0.0118]	+0.0114	[-0.0323, +0.0562]
KNN	85	-0.4235	[-0.6019, -0.2537]	-0.3774	[-0.5472, -0.2128]	85	-0.4281	[-0.6226, -0.2333]	-0.3496	[-0.5265, -0.1760]
SOD	58	-0.2816	[-0.4488, -0.1185]	-0.1570	[-0.2604, -0.0615]	55	-0.2901	[-0.4735, -0.1132]	-0.1390	[-0.2024, -0.0684]
ROD	39	-0.2554	[-0.5042, -0.0270]	-0.2552	[-0.5096, -0.0323]	42	-0.3313	[-0.5962, -0.0792]	-0.2765	[-0.5073, -0.0519]
IForest	85	-0.3698	[-0.5368, -0.2069]	-0.2383	[-0.3790, -0.1111]	85	-0.3873	[-0.5584, -0.2165]	-0.2099	[-0.3302, -0.0930]
INNE	85	-0.4056	[-0.5673, -0.2365]	-0.2960	[-0.4467, -0.1510]	85	-0.4171	[-0.5885, -0.2457]	-0.2427	[-0.3877, -0.1050]
DIF	85	-0.3313	[-0.5118, -0.1586]	-0.2001	[-0.3213, -0.0834]	85	-0.3511	[-0.5213, -0.1772]	-0.1692	[-0.2731, -0.0677]
FeatureBagging	85	-0.3679	[-0.5607, -0.1771]	-0.3526	[-0.5258, -0.1713]	85	-0.3986	[-0.6025, -0.1872]	-0.3456	[-0.5257, -0.1723]
LSCP	50	-0.3975	[-0.6354, -0.1637]	-0.3806	[-0.6262, -0.1559]	50	-0.4360	[-0.6944, -0.1779]	-0.3916	[-0.6226, -0.1494]

Continued on next page

Table 132 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
LODA	85	-0.2297	[-0.3681, -0.0970]	-0.0382	[-0.0880, +0.0116]	85	-0.1045	[-0.2187, +0.0036]	+0.0105	[-0.0149, +0.0381]
SUOD	85	-0.3866	[-0.5628, -0.2135]	-0.3080	[-0.4689, -0.1575]	84	-0.4158	[-0.5978, -0.2203]	-0.2914	[-0.4407, -0.1358]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	-0.3598	[-0.6069, -0.1081]	-0.3698	[-0.6116, -0.1359]	0	—	[—, —]	—	[—, —]
VAE	50	-0.3446	[-0.6045, -0.0937]	-0.3532	[-0.5971, -0.1160]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	-0.0272	[-0.1891, +0.1343]	-0.0484	[-0.1162, +0.0228]	85	-0.1157	[-0.3001, +0.0624]	-0.0671	[-0.1385, +0.0040]
MO_GAAL	85	-0.0556	[-0.2136, +0.1023]	-0.0439	[-0.1128, +0.0247]	85	-0.1066	[-0.2769, +0.0680]	-0.0671	[-0.1440, +0.0095]
DeepSVDD	85	-0.3766	[-0.5605, -0.1889]	-0.3496	[-0.5129, -0.1865]	85	-0.4104	[-0.6060, -0.2090]	-0.3266	[-0.4896, -0.1660]
AnoGAN	85	-0.3637	[-0.5520, -0.1756]	-0.3055	[-0.4532, -0.1578]	85	-0.4071	[-0.6150, -0.2036]	-0.3337	[-0.4965, -0.1717]
ALAD	85	-0.1138	[-0.2400, +0.0141]	+0.0029	[-0.0422, +0.0492]	85	-0.1636	[-0.2977, -0.0244]	-0.0134	[-0.0599, +0.0329]
AE1SVM	85	-0.3820	[-0.5720, -0.2000]	-0.3528	[-0.5273, -0.1832]	85	-0.4098	[-0.6092, -0.2073]	-0.3507	[-0.5288, -0.1692]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	35	-0.1803	[-0.4418, +0.0855]	-0.0610	[-0.2261, +0.0978]	54	-0.2058	[-0.3921, -0.0323]	-0.0720	[-0.2134, +0.0367]
LUNAR	85	-0.4380	[-0.5977, -0.2807]	-0.3990	[-0.5657, -0.2440]	85	-0.4520	[-0.6263, -0.2710]	-0.3604	[-0.5228, -0.1961]

Table 133: NSA/hierarchical versus all 43 other registered detectors, Original features, 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	77	-0.5265	[-0.6849, -0.3657]	-0.3281	[-0.4747, -0.1940]	85	-0.4597	[-0.6493, -0.2653]	-0.3075	[-0.4621, -0.1613]
ABOD	85	-0.6084	[-0.7420, -0.4584]	-0.4320	[-0.5861, -0.2809]	85	-0.5137	[-0.6865, -0.3366]	-0.3801	[-0.5422, -0.2222]
COPOD	85	-0.5324	[-0.6918, -0.3684]	-0.3443	[-0.5061, -0.1903]	85	-0.4673	[-0.6573, -0.2699]	-0.3243	[-0.4920, -0.1634]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	84	-0.3461	[-0.4521, -0.2442]	-0.1176	[-0.2093, -0.0413]	85	-0.2838	[-0.4164, -0.1599]	-0.1015	[-0.1955, -0.0231]
QMCD	85	-0.2051	[-0.3066, -0.1078]	-0.0781	[-0.1331, -0.0311]	85	-0.0732	[-0.1915, +0.0114]	-0.0292	[-0.0796, +0.0173]
KDE	85	-0.6329	[-0.7572, -0.4926]	-0.4589	[-0.6064, -0.3134]	85	-0.5488	[-0.7125, -0.3867]	-0.4045	[-0.5627, -0.2447]
Sampling	65	-0.6235	[-0.7664, -0.4784]	-0.4546	[-0.6196, -0.2944]	50	-0.5858	[-0.7840, -0.3714]	-0.4431	[-0.6568, -0.2346]
GMM	85	-0.5968	[-0.7356, -0.4526]	-0.4036	[-0.5461, -0.2599]	85	-0.5155	[-0.6874, -0.3381]	-0.3455	[-0.4956, -0.2022]
PCA	85	-0.5609	[-0.7113, -0.3917]	-0.4036	[-0.5652, -0.2455]	85	-0.4818	[-0.6699, -0.2814]	-0.3668	[-0.5349, -0.1990]
KPCA	85	-0.6314	[-0.7573, -0.4932]	-0.4547	[-0.6004, -0.3069]	85	-0.5455	[-0.7087, -0.3722]	-0.4021	[-0.5642, -0.2486]
MCD	85	-0.5726	[-0.7105, -0.4272]	-0.3430	[-0.4816, -0.2109]	85	-0.4773	[-0.6474, -0.3024]	-0.2858	[-0.4271, -0.1541]
CD	78	-0.5738	[-0.7254, -0.4166]	-0.3738	[-0.5163, -0.2336]	79	-0.5041	[-0.6915, -0.3129]	-0.3381	[-0.4856, -0.1952]
OCSVM	85	-0.5955	[-0.7415, -0.4423]	-0.4321	[-0.5889, -0.2753]	85	-0.5169	[-0.6901, -0.3336]	-0.3920	[-0.5570, -0.2241]
LMDD	52	-0.4923	[-0.6836, -0.2825]	-0.3293	[-0.5146, -0.1554]	55	-0.3808	[-0.6063, -0.1675]	-0.2777	[-0.4603, -0.0992]
LOF	85	-0.5479	[-0.7006, -0.3860]	-0.4032	[-0.5642, -0.2397]	85	-0.4650	[-0.6639, -0.2589]	-0.3682	[-0.5370, -0.2034]
COF	85	-0.4205	[-0.5462, -0.2885]	-0.2170	[-0.3202, -0.1216]	85	-0.3411	[-0.4839, -0.1949]	-0.1792	[-0.2739, -0.0917]
CBLOF	80	-0.6030	[-0.7428, -0.4542]	-0.4199	[-0.5815, -0.2638]	65	-0.5585	[-0.7311, -0.3812]	-0.4153	[-0.6015, -0.2367]
LOCI	29	-0.5112	[-0.6688, -0.3053]	-0.2549	[-0.4105, -0.0901]	30	-0.4100	[-0.6356, -0.1711]	-0.2163	[-0.3908, -0.0368]
HBOS	85	-0.5591	[-0.6997, -0.4159]	-0.3512	[-0.5053, -0.2105]	85	-0.4759	[-0.6506, -0.2950]	-0.3072	[-0.4604, -0.1612]
HDBSCAN	85	-0.2994	[-0.4117, -0.1952]	-0.0367	[-0.0807, -0.0000]	85	-0.2214	[-0.3390, -0.0943]	-0.0110	[-0.0537, +0.0286]
KNN	85	-0.6106	[-0.7514, -0.4658]	-0.4316	[-0.5871, -0.2768]	85	-0.5015	[-0.6739, -0.3086]	-0.3721	[-0.5396, -0.2101]
SOD	58	-0.4807	[-0.6098, -0.3406]	-0.2141	[-0.3166, -0.1152]	55	-0.3947	[-0.5557, -0.2181]	-0.1670	[-0.2332, -0.0933]
ROD	39	-0.5261	[-0.7414, -0.2927]	-0.3318	[-0.5691, -0.1080]	42	-0.4224	[-0.6855, -0.1658]	-0.2955	[-0.5126, -0.0851]
IForest	85	-0.5569	[-0.6862, -0.4224]	-0.2925	[-0.4217, -0.1708]	85	-0.4607	[-0.6241, -0.2982]	-0.2324	[-0.3481, -0.1235]
INNE	85	-0.5927	[-0.7236, -0.4495]	-0.3502	[-0.4971, -0.2083]	85	-0.4904	[-0.6485, -0.3221]	-0.2652	[-0.4057, -0.1366]
DIF	85	-0.5184	[-0.6529, -0.3793]	-0.2543	[-0.3670, -0.1502]	85	-0.4244	[-0.5836, -0.2600]	-0.1917	[-0.2944, -0.0985]
FeatureBagging	85	-0.5550	[-0.7034, -0.3928]	-0.4068	[-0.5685, -0.2523]	85	-0.4720	[-0.6667, -0.2691]	-0.3681	[-0.5345, -0.1990]
LSCP	50	-0.6178	[-0.7899, -0.4409]	-0.4462	[-0.6526, -0.2335]	50	-0.5495	[-0.7728, -0.3086]	-0.4211	[-0.6500, -0.2045]

Continued on next page

Table 133 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
LODA	85	-0.4168	[-0.5340, -0.3011]	-0.0925	[-0.1394, -0.0513]	85	-0.1779	[-0.2846, -0.0714]	-0.0120	[-0.0377, +0.0114]
SUOD	85	-0.5738	[-0.7139, -0.4225]	-0.3622	[-0.5112, -0.2186]	84	-0.4895	[-0.6679, -0.3077]	-0.3140	[-0.4653, -0.1741]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	-0.5801	[-0.7745, -0.3803]	-0.4354	[-0.6476, -0.2308]	0	—	[—, —]	—	[—, —]
VAE	50	-0.5648	[-0.7712, -0.3604]	-0.4188	[-0.6379, -0.2056]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	-0.2143	[-0.3287, -0.0928]	-0.1026	[-0.1563, -0.0489]	85	-0.1891	[-0.3603, -0.0157]	-0.0896	[-0.1609, -0.0215]
MO_GAAL	85	-0.2427	[-0.3583, -0.1278]	-0.0981	[-0.1525, -0.0454]	85	-0.1800	[-0.3443, -0.0145]	-0.0895	[-0.1649, -0.0191]
DeepSVDD	85	-0.5638	[-0.7079, -0.4004]	-0.4038	[-0.5544, -0.2558]	85	-0.4837	[-0.6609, -0.2835]	-0.3490	[-0.5050, -0.1921]
AnoGAN	85	-0.5509	[-0.7036, -0.3864]	-0.3597	[-0.4936, -0.2219]	85	-0.4805	[-0.6655, -0.2857]	-0.3562	[-0.5126, -0.2002]
ALAD	85	-0.3009	[-0.3915, -0.2079]	-0.0513	[-0.0879, -0.0171]	85	-0.2369	[-0.3565, -0.1138]	-0.0358	[-0.0765, +0.0048]
AE1SVM	85	-0.5691	[-0.7176, -0.4094]	-0.4070	[-0.5712, -0.2442]	85	-0.4831	[-0.6714, -0.2843]	-0.3731	[-0.5446, -0.2061]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	35	-0.3201	[-0.5351, -0.0919]	-0.0990	[-0.2493, +0.0337]	54	-0.2656	[-0.4372, -0.1014]	-0.0931	[-0.2371, +0.0142]
LUNAR	85	-0.6251	[-0.7530, -0.4803]	-0.4533	[-0.6028, -0.3028]	85	-0.5254	[-0.6977, -0.3518]	-0.3829	[-0.5441, -0.2216]

Table 134: NSA/voronoi versus all 43 other registered detectors, Original features, 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
ABOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
COPOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
QMCD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
KDE	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
Sampling	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
GMM	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
PCA	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
KPCA	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
MCD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
CD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
OCSVM	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LMDD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LOF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
COF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
CBLOF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LOCI	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
HBOS	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
HDBSCAN	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
KNN	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
ROD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
IForest	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
INNE	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
DIF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
FeatureBagging	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LSCP	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LODA	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SUOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]

Continued on next page

Table 134 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
AutoEncoder	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
VAE	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SO_GAAL	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
MO_GAAL	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
DeepSVDD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AnoGAN	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
ALAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AE1SVM	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LUNAR	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]

Table 135: NSA/deterministic versus all 43 other registered detectors, Original features, 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
ABOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
COPOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
QMCD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
KDE	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
Sampling	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
GMM	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
PCA	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
KPCA	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
MCD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
CD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
OCSVM	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LMDD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LOF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
COF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
CBLOF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LOCI	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
HBOS	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
HDBSCAN	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
KNN	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
ROD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
IForest	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
INNE	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
DIF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
FeatureBagging	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LSCP	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LODA	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SUOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]

Continued on next page

Table 135 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
VAE	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SO_GAAL	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
MO_GAAL	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
DeepSVDD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AnoGAN	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
ALAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AE1SVM	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LUNAR	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]

Table 136: NSA/suppressed versus all 43 other registered detectors, Original features, 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	77	-0.3404	[-0.5285, -0.1474]	-0.2743	[-0.4327, -0.1284]	85	-0.3883	[-0.5869, -0.1939]	-0.2785	[-0.4411, -0.1243]
ABOD	85	-0.4170	[-0.5871, -0.2412]	-0.3738	[-0.5431, -0.2146]	85	-0.4423	[-0.6286, -0.2536]	-0.3511	[-0.5187, -0.1910]
COPOD	85	-0.3410	[-0.5312, -0.1492]	-0.2861	[-0.4554, -0.1264]	85	-0.3959	[-0.5917, -0.1911]	-0.2953	[-0.4747, -0.1253]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	84	-0.1509	[-0.2978, -0.0086]	-0.0575	[-0.1564, +0.0268]	85	-0.2124	[-0.3608, -0.0648]	-0.0725	[-0.1746, +0.0106]
QMCD	85	-0.0137	[-0.1433, +0.1063]	-0.0199	[-0.0890, +0.0417]	85	-0.0018	[-0.1338, +0.0949]	-0.0002	[-0.0546, +0.0506]
KDE	85	-0.4415	[-0.6058, -0.2902]	-0.4006	[-0.5564, -0.2457]	85	-0.4774	[-0.6550, -0.3089]	-0.3755	[-0.5427, -0.2072]
Sampling	65	-0.4193	[-0.6101, -0.2280]	-0.3849	[-0.5766, -0.2022]	50	-0.4671	[-0.6910, -0.2355]	-0.3998	[-0.6369, -0.1741]
GMM	85	-0.4055	[-0.5721, -0.2329]	-0.3454	[-0.4904, -0.1994]	85	-0.4441	[-0.6271, -0.2606]	-0.3165	[-0.4735, -0.1621]
PCA	85	-0.3695	[-0.5665, -0.1732]	-0.3454	[-0.5138, -0.1759]	85	-0.4104	[-0.6135, -0.1972]	-0.3378	[-0.5098, -0.1623]
KPCA	85	-0.4400	[-0.5977, -0.2777]	-0.3965	[-0.5601, -0.2412]	85	-0.4741	[-0.6475, -0.2949]	-0.3731	[-0.5419, -0.2040]
MCD	85	-0.3812	[-0.5618, -0.2124]	-0.2847	[-0.4352, -0.1434]	85	-0.4059	[-0.5893, -0.2236]	-0.2569	[-0.4025, -0.1203]
CD	78	-0.3723	[-0.5609, -0.1927]	-0.3124	[-0.4670, -0.1656]	79	-0.4268	[-0.6271, -0.2298]	-0.3067	[-0.4661, -0.1521]
OCSVM	85	-0.4041	[-0.5838, -0.2217]	-0.3739	[-0.5513, -0.2051]	85	-0.4455	[-0.6339, -0.2595]	-0.3630	[-0.5398, -0.1802]
LMDD	52	-0.3073	[-0.5495, -0.0693]	-0.2739	[-0.4729, -0.0786]	55	-0.3365	[-0.5701, -0.1062]	-0.2583	[-0.4495, -0.0695]
LOF	85	-0.3565	[-0.5517, -0.1584]	-0.3449	[-0.5162, -0.1774]	85	-0.3936	[-0.6045, -0.1696]	-0.3392	[-0.5184, -0.1604]
COF	85	-0.2291	[-0.3882, -0.0705]	-0.1588	[-0.2698, -0.0560]	85	-0.2697	[-0.4318, -0.1046]	-0.1502	[-0.2433, -0.0614]
CBLOF	80	-0.4008	[-0.5712, -0.2370]	-0.3578	[-0.5274, -0.1917]	65	-0.4686	[-0.6643, -0.2742]	-0.3839	[-0.5803, -0.1972]
LOCI	29	-0.3819	[-0.5846, -0.1730]	-0.2379	[-0.3986, -0.0745]	30	-0.4044	[-0.6452, -0.1517]	-0.2025	[-0.3853, +0.0035]
HBOS	85	-0.3677	[-0.5472, -0.2002]	-0.2929	[-0.4607, -0.1419]	85	-0.4045	[-0.5884, -0.2216]	-0.2782	[-0.4400, -0.1242]
HDBSCAN	85	-0.1081	[-0.2485, +0.0285]	+0.0216	[-0.0295, +0.0680]	85	-0.1500	[-0.2841, -0.0124]	+0.0180	[-0.0290, +0.0650]
KNN	85	-0.4192	[-0.5942, -0.2501]	-0.3734	[-0.5461, -0.2028]	85	-0.4301	[-0.6272, -0.2346]	-0.3431	[-0.5214, -0.1638]
SOD	58	-0.2837	[-0.4473, -0.1235]	-0.1585	[-0.2668, -0.0617]	55	-0.2874	[-0.4604, -0.1137]	-0.1297	[-0.1943, -0.0589]
ROD	39	-0.2467	[-0.4985, -0.0196]	-0.2465	[-0.4964, -0.0354]	42	-0.3331	[-0.6209, -0.0803]	-0.2639	[-0.5024, -0.0407]
IForest	85	-0.3655	[-0.5204, -0.2047]	-0.2343	[-0.3644, -0.1093]	85	-0.3893	[-0.5605, -0.2128]	-0.2034	[-0.3327, -0.0850]
INNE	85	-0.4013	[-0.5673, -0.2293]	-0.2920	[-0.4408, -0.1512]	85	-0.4190	[-0.5945, -0.2468]	-0.2362	[-0.3823, -0.1028]
DIF	85	-0.3270	[-0.5024, -0.1545]	-0.1960	[-0.3164, -0.0821]	85	-0.3530	[-0.5243, -0.1799]	-0.1627	[-0.2701, -0.0630]
FeatureBagging	85	-0.3637	[-0.5482, -0.1725]	-0.3485	[-0.5222, -0.1850]	85	-0.4006	[-0.6115, -0.1850]	-0.3391	[-0.5227, -0.1634]

Continued on next page

Table 136 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
LSCP	50	-0.3927	[-0.6283, -0.1632]	-0.3717	[-0.5961, -0.1390]	50	-0.4308	[-0.6817, -0.1658]	-0.3779	[-0.6107, -0.1346]
LODA	85	-0.2255	[-0.3681, -0.0849]	-0.0342	[-0.0822, +0.0113]	85	-0.1065	[-0.2152, +0.0041]	+0.0170	[-0.0121, +0.0486]
SUOD	85	-0.3824	[-0.5615, -0.2027]	-0.3040	[-0.4590, -0.1495]	84	-0.4178	[-0.6045, -0.2279]	-0.2849	[-0.4414, -0.1318]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	-0.3550	[-0.6026, -0.1048]	-0.3609	[-0.6026, -0.1229]	0	—	[—, —]	—	[—, —]
VAE	50	-0.3397	[-0.5948, -0.0858]	-0.3444	[-0.5924, -0.1073]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	-0.0230	[-0.1890, +0.1426]	-0.0444	[-0.1115, +0.0242]	85	-0.1177	[-0.2957, +0.0673]	-0.0606	[-0.1327, +0.0130]
MO_GAAL	85	-0.0513	[-0.2162, +0.1053]	-0.0399	[-0.1098, +0.0303]	85	-0.1086	[-0.2834, +0.0779]	-0.0605	[-0.1382, +0.0161]
DeepSVDD	85	-0.3724	[-0.5514, -0.1908]	-0.3456	[-0.5084, -0.1882]	85	-0.4123	[-0.6053, -0.2125]	-0.3200	[-0.4842, -0.1538]
AnoGAN	85	-0.3595	[-0.5460, -0.1670]	-0.3014	[-0.4496, -0.1525]	85	-0.4091	[-0.6074, -0.2035]	-0.3272	[-0.4860, -0.1605]
ALAD	85	-0.1095	[-0.2380, +0.0161]	+0.0069	[-0.0407, +0.0529]	85	-0.1655	[-0.3018, -0.0257]	-0.0068	[-0.0547, +0.0423]
AE1SVM	85	-0.3777	[-0.5631, -0.1917]	-0.3488	[-0.5162, -0.1807]	85	-0.4117	[-0.6122, -0.2008]	-0.3441	[-0.5218, -0.1633]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	35	-0.1769	[-0.4412, +0.0906]	-0.0639	[-0.2278, +0.0869]	54	-0.1992	[-0.3930, -0.0222]	-0.0631	[-0.2098, +0.0515]
LUNAR	85	-0.4338	[-0.5910, -0.2714]	-0.3950	[-0.5543, -0.2377]	85	-0.4540	[-0.6297, -0.2677]	-0.3539	[-0.5221, -0.1894]

Table 137: NSA/dual versus all 43 other registered detectors, Original features, 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	77	+0.0109	[-0.0477, +0.0663]	+0.0811	[+0.0051, +0.1635]	85	+0.0213	[-0.0242, +0.0686]	+0.0798	[+0.0138, +0.1529]
ABOD	85	-0.0744	[-0.1247, -0.0342]	-0.0239	[-0.0497, +0.0012]	85	-0.0327	[-0.0592, -0.0067]	+0.0073	[-0.0186, +0.0436]
COPOD	85	+0.0016	[-0.0638, +0.0612]	+0.0638	[-0.0062, +0.1470]	85	+0.0138	[-0.0373, +0.0662]	+0.0630	[+0.0007, +0.1370]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	84	+0.1876	[+0.0869, +0.2836]	+0.2902	[+0.1301, +0.4598]	85	+0.1972	[+0.1014, +0.2909]	+0.2858	[+0.1319, +0.4491]
QMCD	85	+0.3290	[+0.1537, +0.5085]	+0.3300	[+0.1597, +0.5069]	85	+0.4079	[+0.2203, +0.5947]	+0.3581	[+0.1975, +0.5151]
KDE	85	-0.0989	[-0.1687, -0.0433]	-0.0508	[-0.0955, -0.0066]	85	-0.0677	[-0.1155, -0.0296]	-0.0171	[-0.0440, +0.0163]
Sampling	65	-0.0935	[-0.1623, -0.0393]	-0.0313	[-0.0715, +0.0044]	50	-0.0780	[-0.1343, -0.0306]	-0.0081	[-0.0268, +0.0107]
GMM	85	-0.0628	[-0.1063, -0.0254]	+0.0045	[-0.0692, +0.0868]	85	-0.0345	[-0.0679, -0.0071]	+0.0418	[-0.0122, +0.1138]
PCA	85	-0.0268	[-0.0765, +0.0207]	+0.0044	[-0.0203, +0.0314]	85	-0.0007	[-0.0419, +0.0441]	+0.0205	[-0.0046, +0.0506]
KPCA	85	-0.0973	[-0.1598, -0.0476]	-0.0467	[-0.0853, -0.0109]	85	-0.0644	[-0.1090, -0.0296]	-0.0148	[-0.0339, +0.0058]
MCD	85	-0.0386	[-0.0877, +0.0043]	+0.0651	[-0.0016, +0.1462]	85	+0.0037	[-0.0524, +0.0556]	+0.1015	[+0.0267, +0.1970]
CD	78	-0.0400	[-0.0846, -0.0011]	+0.0385	[-0.0372, +0.1267]	79	-0.0264	[-0.0600, +0.0007]	+0.0503	[-0.0109, +0.1232]
OCSVM	85	-0.0615	[-0.1073, -0.0255]	-0.0241	[-0.0429, -0.0080]	85	-0.0358	[-0.0685, -0.0100]	-0.0047	[-0.0131, +0.0044]
LMDD	52	+0.0485	[+0.0063, +0.0953]	+0.0519	[+0.0044, +0.1028]	55	+0.0718	[+0.0340, +0.1142]	+0.0602	[+0.0175, +0.1031]
LOF	85	-0.0139	[-0.0597, +0.0300]	+0.0049	[-0.0148, +0.0225]	85	+0.0161	[-0.0295, +0.0718]	+0.0192	[-0.0020, +0.0467]
COF	85	+0.1135	[+0.0106, +0.2208]	+0.1910	[+0.0397, +0.3586]	85	+0.1400	[+0.0414, +0.2414]	+0.2081	[+0.0624, +0.3710]

Continued on next page

Table 137 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
CBLOF	80	-0.0917	[-0.1470, -0.0463]	-0.0350	[-0.0560, -0.0158]	65	-0.0638	[-0.1126, -0.0272]	-0.0068	[-0.0258, +0.0119]
LOCI	29	+0.0820	[-0.0283, +0.1899]	+0.1434	[+0.0197, +0.2721]	30	+0.1028	[-0.0129, +0.2184]	+0.1540	[+0.0402, +0.2678]
HBOS	85	-0.0250	[-0.1069, +0.0489]	+0.0569	[-0.0138, +0.1306]	85	+0.0052	[-0.0693, +0.0689]	+0.0801	[+0.0091, +0.1553]
HDBSCAN	85	+0.2346	[+0.1354, +0.3405]	+0.3714	[+0.2232, +0.5230]	85	+0.2597	[+0.1627, +0.3526]	+0.3763	[+0.2338, +0.5241]
KNN	85	-0.0766	[-0.1300, -0.0326]	-0.0235	[-0.0447, -0.0041]	85	-0.0204	[-0.0591, +0.0156]	+0.0153	[-0.0131, +0.0533]
SOD	58	+0.0372	[-0.0340, +0.1049]	+0.1731	[+0.0335, +0.3320]	55	+0.0691	[-0.0340, +0.1611]	+0.2297	[+0.0307, +0.4398]
ROD	39	-0.0235	[-0.1062, +0.0540]	-0.0104	[-0.0744, +0.0606]	42	-0.0066	[-0.0576, +0.0568]	+0.0235	[-0.0280, +0.0786]
IForest	85	-0.0229	[-0.1001, +0.0437]	+0.1156	[+0.0173, +0.2496]	85	+0.0204	[-0.0468, +0.0906]	+0.1550	[+0.0579, +0.2756]
INNE	85	-0.0587	[-0.1188, -0.0087]	+0.0578	[-0.0138, +0.1528]	85	-0.0094	[-0.0672, +0.0450]	+0.1222	[+0.0357, +0.2295]
DIF	85	+0.0156	[-0.0484, +0.0778]	+0.1538	[+0.0539, +0.2695]	85	+0.0566	[-0.0109, +0.1270]	+0.1957	[+0.0860, +0.3188]
FeatureBagging	85	-0.0210	[-0.0670, +0.0251]	+0.0013	[-0.0182, +0.0207]	85	+0.0091	[-0.0365, +0.0632]	+0.0193	[-0.0050, +0.0483]
LSCP	50	-0.0901	[-0.1610, -0.0282]	-0.0097	[-0.0416, +0.0173]	50	-0.0416	[-0.0983, +0.0216]	+0.0138	[-0.0199, +0.0555]
LODA	85	+0.1172	[+0.0163, +0.2244]	+0.3156	[+0.1660, +0.4759]	85	+0.3032	[+0.1636, +0.4480]	+0.3754	[+0.2111, +0.5351]
SUOD	85	-0.0398	[-0.1122, +0.0185]	+0.0458	[-0.0097, +0.1085]	84	-0.0081	[-0.0680, +0.0485]	+0.0734	[+0.0151, +0.1392]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	-0.0524	[-0.1343, +0.0167]	+0.0011	[-0.0247, +0.0271]	0	—	[—, —]	—	[—, —]
VAE	50	-0.0372	[-0.1133, +0.0365]	+0.0176	[-0.0135, +0.0573]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	+0.3197	[+0.2072, +0.4337]	+0.3055	[+0.1731, +0.4418]	85	+0.2920	[+0.1780, +0.4106]	+0.2978	[+0.1623, +0.4453]
MO_GAAL	85	+0.2913	[+0.1727, +0.4173]	+0.3099	[+0.1782, +0.4489]	85	+0.3011	[+0.2026, +0.4032]	+0.2978	[+0.1791, +0.4385]

Continued on next page

Table 137 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
DeepSVDD	85	-0.0297	[-0.0687, +0.0110]	+0.0043	[-0.0357, +0.0531]	85	-0.0027	[-0.0306, +0.0304]	+0.0383	[+0.0010, +0.0815]
AnoGAN	85	-0.0168	[-0.0802, +0.0407]	+0.0484	[-0.0121, +0.1226]	85	+0.0006	[-0.0496, +0.0515]	+0.0312	[-0.0227, +0.0965]
ALAD	85	+0.2331	[+0.1429, +0.3205]	+0.3568	[+0.2106, +0.5122]	85	+0.2442	[+0.1616, +0.3198]	+0.3515	[+0.2072, +0.5057]
AE1SVM	85	-0.0351	[-0.0841, +0.0087]	+0.0011	[-0.0206, +0.0247]	85	-0.0021	[-0.0414, +0.0390]	+0.0142	[-0.0064, +0.0405]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	35	+0.2230	[+0.1000, +0.3509]	+0.2686	[+0.1174, +0.4322]	54	+0.2569	[+0.1467, +0.3713]	+0.3341	[+0.1960, +0.4854]
LUNAR	85	-0.0911	[-0.1618, -0.0354]	-0.0452	[-0.0768, -0.0185]	85	-0.0443	[-0.0970, -0.0019]	+0.0045	[-0.0210, +0.0437]

Table 138: NSA/annealed versus all 43 other registered detectors, Original features, 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	77	-0.5056	[-0.6661, -0.3350]	-0.3256	[-0.4768, -0.1842]	85	-0.5122	[-0.6766, -0.3458]	-0.3344	[-0.4782, -0.1959]
ABOD	85	-0.5895	[-0.7337, -0.4411]	-0.4312	[-0.5872, -0.2773]	85	-0.5663	[-0.7192, -0.3992]	-0.4069	[-0.5567, -0.2597]
COPOD	85	-0.5135	[-0.6812, -0.3368]	-0.3434	[-0.5088, -0.1858]	85	-0.5198	[-0.6838, -0.3452]	-0.3512	[-0.5114, -0.2044]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	84	-0.3275	[-0.4364, -0.2246]	-0.1168	[-0.2058, -0.0424]	85	-0.3364	[-0.4498, -0.2279]	-0.1284	[-0.2206, -0.0540]
QMCD	85	-0.1862	[-0.2835, -0.0975]	-0.0773	[-0.1312, -0.0305]	85	-0.1257	[-0.2406, -0.0436]	-0.0561	[-0.0988, -0.0201]
KDE	85	-0.6140	[-0.7459, -0.4760]	-0.4580	[-0.6060, -0.3081]	85	-0.6013	[-0.7422, -0.4533]	-0.4313	[-0.5795, -0.2874]
Sampling	65	-0.6133	[-0.7570, -0.4590]	-0.4626	[-0.6325, -0.3029]	50	-0.6394	[-0.8071, -0.4756]	-0.4578	[-0.6626, -0.2605]
GMM	85	-0.5780	[-0.7192, -0.4303]	-0.4027	[-0.5452, -0.2573]	85	-0.5680	[-0.7185, -0.4087]	-0.3724	[-0.5082, -0.2411]
PCA	85	-0.5420	[-0.7049, -0.3693]	-0.4028	[-0.5698, -0.2480]	85	-0.5343	[-0.7000, -0.3621]	-0.3937	[-0.5472, -0.2412]
KPCA	85	-0.6125	[-0.7459, -0.4723]	-0.4539	[-0.6027, -0.3038]	85	-0.5980	[-0.7479, -0.4468]	-0.4290	[-0.5795, -0.2790]
MCD	85	-0.5537	[-0.6947, -0.4096]	-0.3421	[-0.4883, -0.2054]	85	-0.5298	[-0.6795, -0.3776]	-0.3127	[-0.4397, -0.1887]
CD	78	-0.5532	[-0.7052, -0.3944]	-0.3729	[-0.5219, -0.2320]	79	-0.5586	[-0.7216, -0.3904]	-0.3653	[-0.5028, -0.2320]
OCSVM	85	-0.5766	[-0.7276, -0.4222]	-0.4313	[-0.5886, -0.2750]	85	-0.5694	[-0.7252, -0.4048]	-0.4189	[-0.5743, -0.2648]
LMDD	52	-0.4660	[-0.6815, -0.2523]	-0.3261	[-0.5156, -0.1415]	55	-0.4576	[-0.6550, -0.2497]	-0.3189	[-0.4943, -0.1584]
LOF	85	-0.5290	[-0.6959, -0.3537]	-0.4023	[-0.5708, -0.2389]	85	-0.5175	[-0.6902, -0.3365]	-0.3950	[-0.5489, -0.2308]
COF	85	-0.4016	[-0.5251, -0.2807]	-0.2162	[-0.3125, -0.1245]	85	-0.3936	[-0.5242, -0.2635]	-0.2061	[-0.2933, -0.1268]
CBLOF	80	-0.5808	[-0.7221, -0.4350]	-0.4185	[-0.5817, -0.2556]	65	-0.6176	[-0.7590, -0.4688]	-0.4389	[-0.6106, -0.2689]
LOCI	29	-0.4696	[-0.6434, -0.2857]	-0.2344	[-0.4158, -0.0467]	30	-0.4385	[-0.6424, -0.2152]	-0.2438	[-0.3910, -0.0917]
HBOS	85	-0.5402	[-0.6841, -0.3988]	-0.3503	[-0.5037, -0.2029]	85	-0.5284	[-0.6725, -0.3756]	-0.3341	[-0.4770, -0.2026]
HDBSCAN	85	-0.2806	[-0.3917, -0.1793]	-0.0358	[-0.0782, +0.0065]	85	-0.2739	[-0.3632, -0.1805]	-0.0379	[-0.0639, -0.0136]
KNN	85	-0.5917	[-0.7325, -0.4455]	-0.4308	[-0.5929, -0.2678]	85	-0.5540	[-0.7118, -0.3868]	-0.3990	[-0.5473, -0.2494]
SOD	58	-0.4449	[-0.5773, -0.3072]	-0.2060	[-0.3115, -0.0974]	55	-0.4603	[-0.5935, -0.3254]	-0.1935	[-0.2576, -0.1282]
ROD	39	-0.4889	[-0.7304, -0.2542]	-0.3265	[-0.5684, -0.0950]	42	-0.4895	[-0.7159, -0.2568]	-0.3368	[-0.5447, -0.1382]
IForest	85	-0.5380	[-0.6703, -0.4020]	-0.2917	[-0.4246, -0.1647]	85	-0.5132	[-0.6502, -0.3708]	-0.2592	[-0.3650, -0.1618]
INNE	85	-0.5738	[-0.7165, -0.4305]	-0.3494	[-0.4997, -0.2075]	85	-0.5429	[-0.6866, -0.3946]	-0.2921	[-0.4231, -0.1718]
DIF	85	-0.4995	[-0.6412, -0.3531]	-0.2534	[-0.3676, -0.1433]	85	-0.4770	[-0.6175, -0.3374]	-0.2185	[-0.3107, -0.1370]
FeatureBagging	85	-0.5362	[-0.7022, -0.3697]	-0.4059	[-0.5733, -0.2426]	85	-0.5245	[-0.6908, -0.3422]	-0.3949	[-0.5566, -0.2404]
LSCP	50	-0.6156	[-0.7920, -0.4321]	-0.4585	[-0.6801, -0.2581]	50	-0.6031	[-0.7871, -0.4099]	-0.4358	[-0.6531, -0.2174]
LODA	85	-0.3980	[-0.5185, -0.2808]	-0.0916	[-0.1387, -0.0482]	85	-0.2304	[-0.3205, -0.1445]	-0.0388	[-0.0573, -0.0219]
SUOD	85	-0.5549	[-0.7054, -0.4031]	-0.3614	[-0.5130, -0.2105]	84	-0.5416	[-0.6905, -0.3777]	-0.3410	[-0.4791, -0.2072]

Continued on next page

Table 138 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	-0.5779	[-0.7791, -0.3692]	-0.4477	[-0.6590, -0.2359]	0	—	[—, —]	—	[—, —]
VAE	50	-0.5627	[-0.7765, -0.3454]	-0.4312	[-0.6514, -0.2199]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	-0.1955	[-0.3203, -0.0730]	-0.1017	[-0.1547, -0.0448]	85	-0.2416	[-0.3829, -0.1105]	-0.1164	[-0.1707, -0.0644]
MO_GAAL	85	-0.2238	[-0.3458, -0.0993]	-0.0973	[-0.1532, -0.0387]	85	-0.2325	[-0.3612, -0.1049]	-0.1164	[-0.1747, -0.0587]
DeepSVDD	85	-0.5449	[-0.6948, -0.3890]	-0.4029	[-0.5568, -0.2535]	85	-0.5363	[-0.6980, -0.3688]	-0.3759	[-0.5169, -0.2406]
AnoGAN	85	-0.5320	[-0.6942, -0.3733]	-0.3588	[-0.4937, -0.2169]	85	-0.5330	[-0.6924, -0.3556]	-0.3830	[-0.5256, -0.2371]
ALAD	85	-0.2820	[-0.3759, -0.1912]	-0.0505	[-0.0884, -0.0140]	85	-0.2894	[-0.3815, -0.1942]	-0.0627	[-0.0923, -0.0349]
AE1SVM	85	-0.5502	[-0.7106, -0.3891]	-0.4061	[-0.5664, -0.2426]	85	-0.5357	[-0.7007, -0.3625]	-0.4000	[-0.5605, -0.2406]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	35	-0.2773	[-0.5374, -0.0377]	-0.0792	[-0.2328, +0.0658]	54	-0.3231	[-0.4686, -0.1862]	-0.1253	[-0.2519, -0.0322]
LUNAR	85	-0.6063	[-0.7378, -0.4696]	-0.4524	[-0.6035, -0.3009]	85	-0.5779	[-0.7193, -0.4195]	-0.4098	[-0.5605, -0.2599]

Table 139: NSA/binary/hamming versus all 43 other registered detectors, Original features, 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	77	-0.2735	[-0.3676, -0.1820]	-0.2973	[-0.4220, -0.1819]	85	-0.2650	[-0.3507, -0.1789]	-0.3033	[-0.4337, -0.1844]
ABOD	85	-0.3512	[-0.4271, -0.2755]	-0.3990	[-0.5321, -0.2709]	85	-0.3190	[-0.4001, -0.2407]	-0.3759	[-0.5134, -0.2458]
COPOD	85	-0.2752	[-0.3666, -0.1825]	-0.3113	[-0.4517, -0.1789]	85	-0.2726	[-0.3626, -0.1778]	-0.3202	[-0.4665, -0.1832]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	84	-0.0898	[-0.1697, -0.0083]	-0.0844	[-0.1700, -0.0077]	85	-0.0891	[-0.1626, -0.0211]	-0.0974	[-0.1842, -0.0281]
QMCD	85	+0.0521	[-0.0805, +0.2072]	-0.0451	[-0.1074, +0.0217]	85	+0.1216	[-0.0076, +0.2560]	-0.0250	[-0.0752, +0.0204]
KDE	85	-0.3758	[-0.4467, -0.3035]	-0.4259	[-0.5603, -0.3061]	85	-0.3541	[-0.4264, -0.2771]	-0.4003	[-0.5348, -0.2703]
Sampling	65	-0.3498	[-0.4346, -0.2739]	-0.4170	[-0.5665, -0.2754]	50	-0.3756	[-0.4705, -0.3016]	-0.4279	[-0.6262, -0.2404]
GMM	85	-0.3397	[-0.4186, -0.2610]	-0.3706	[-0.4880, -0.2519]	85	-0.3208	[-0.3985, -0.2336]	-0.3414	[-0.4624, -0.2214]
PCA	85	-0.3037	[-0.3949, -0.2123]	-0.3706	[-0.5154, -0.2359]	85	-0.2871	[-0.3764, -0.1948]	-0.3627	[-0.5056, -0.2251]
KPCA	85	-0.3742	[-0.4459, -0.3001]	-0.4217	[-0.5550, -0.2973]	85	-0.3507	[-0.4210, -0.2725]	-0.3980	[-0.5336, -0.2657]
MCD	85	-0.3154	[-0.3949, -0.2405]	-0.3100	[-0.4251, -0.2034]	85	-0.2826	[-0.3624, -0.2059]	-0.2817	[-0.3977, -0.1763]
CD	78	-0.3217	[-0.4082, -0.2366]	-0.3437	[-0.4655, -0.2239]	79	-0.3117	[-0.3984, -0.2271]	-0.3362	[-0.4621, -0.2151]
OCSVM	85	-0.3383	[-0.4167, -0.2549]	-0.3991	[-0.5389, -0.2608]	85	-0.3221	[-0.3991, -0.2396]	-0.3878	[-0.5345, -0.2494]
LMDD	52	-0.2187	[-0.3129, -0.1215]	-0.2947	[-0.4412, -0.1553]	55	-0.1879	[-0.2790, -0.0971]	-0.2798	[-0.4252, -0.1410]
LOF	85	-0.2907	[-0.3803, -0.2057]	-0.3702	[-0.5177, -0.2308]	85	-0.2702	[-0.3665, -0.1752]	-0.3640	[-0.5122, -0.2195]
COF	85	-0.1633	[-0.2543, -0.0633]	-0.1841	[-0.2786, -0.0907]	85	-0.1464	[-0.2291, -0.0629]	-0.1750	[-0.2541, -0.1031]
CBLOF	80	-0.3545	[-0.4320, -0.2784]	-0.3868	[-0.5296, -0.2539]	65	-0.3530	[-0.4306, -0.2894]	-0.4058	[-0.5748, -0.2533]
LOCI	29	-0.2447	[-0.3517, -0.1510]	-0.2259	[-0.3558, -0.1196]	30	-0.1873	[-0.2702, -0.1146]	-0.2083	[-0.3376, -0.0948]
HBOS	85	-0.3019	[-0.3863, -0.2218]	-0.3182	[-0.4498, -0.1941]	85	-0.2812	[-0.3657, -0.1979]	-0.3031	[-0.4373, -0.1794]
HDBSCAN	85	-0.0423	[-0.1388, +0.0412]	-0.0037	[-0.0458, +0.0372]	85	-0.0267	[-0.1003, +0.0354]	-0.0069	[-0.0314, +0.0148]
KNN	85	-0.3535	[-0.4290, -0.2776]	-0.3986	[-0.5378, -0.2664]	85	-0.3068	[-0.3941, -0.2150]	-0.3679	[-0.5123, -0.2286]
SOD	58	-0.2228	[-0.3007, -0.1401]	-0.1833	[-0.2689, -0.1040]	55	-0.2152	[-0.3016, -0.1348]	-0.1649	[-0.2335, -0.1001]
ROD	39	-0.2755	[-0.3938, -0.1581]	-0.3089	[-0.4859, -0.1421]	42	-0.2516	[-0.3502, -0.1484]	-0.3168	[-0.4992, -0.1453]
IForest	85	-0.2997	[-0.3744, -0.2262]	-0.2595	[-0.3667, -0.1605]	85	-0.2660	[-0.3414, -0.1978]	-0.2282	[-0.3218, -0.1407]
INNE	85	-0.3355	[-0.4087, -0.2656]	-0.3172	[-0.4469, -0.1971]	85	-0.2957	[-0.3732, -0.2207]	-0.2610	[-0.3847, -0.1484]
DIF	85	-0.2612	[-0.3455, -0.1863]	-0.2213	[-0.3047, -0.1425]	85	-0.2297	[-0.3113, -0.1604]	-0.1875	[-0.2639, -0.1166]
FeatureBagging	85	-0.2979	[-0.3850, -0.2143]	-0.3738	[-0.5215, -0.2378]	85	-0.2773	[-0.3713, -0.1827]	-0.3639	[-0.5101, -0.2209]
LSCP	50	-0.3512	[-0.4621, -0.2516]	-0.4088	[-0.6032, -0.2212]	50	-0.3393	[-0.4475, -0.2372]	-0.4060	[-0.6125, -0.2032]

Continued on next page

Table 139 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
LODA	85	-0.1597	[-0.2524, -0.0717]	-0.0595	[-0.1042, -0.0134]	85	+0.0169	[-0.0683, +0.1085]	-0.0078	[-0.0343, +0.0199]
SUOD	85	-0.3166	[-0.3971, -0.2348]	-0.3293	[-0.4572, -0.2067]	84	-0.2957	[-0.3750, -0.2114]	-0.3103	[-0.4353, -0.1843]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	-0.3135	[-0.4387, -0.1951]	-0.3980	[-0.5987, -0.2121]	0	—	[—, —]	—	[—, —]
VAE	50	-0.2982	[-0.4287, -0.1775]	-0.3815	[-0.5836, -0.1974]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	+0.0428	[-0.0668, +0.1560]	-0.0696	[-0.1154, -0.0237]	85	+0.0056	[-0.1081, +0.1199]	-0.0854	[-0.1321, -0.0399]
MO_GAAL	85	+0.0145	[-0.1005, +0.1359]	-0.0651	[-0.1144, -0.0147]	85	+0.0148	[-0.0734, +0.1063]	-0.0854	[-0.1301, -0.0425]
DeepSVDD	85	-0.3066	[-0.3919, -0.2234]	-0.3708	[-0.5037, -0.2416]	85	-0.2890	[-0.3744, -0.2034]	-0.3449	[-0.4755, -0.2161]
AnoGAN	85	-0.2937	[-0.3829, -0.2014]	-0.3267	[-0.4355, -0.2177]	85	-0.2858	[-0.3739, -0.1955]	-0.3520	[-0.4845, -0.2240]
ALAD	85	-0.0438	[-0.1031, +0.0148]	-0.0183	[-0.0427, +0.0145]	85	-0.0422	[-0.0882, -0.0029]	-0.0316	[-0.0469, -0.0160]
AE1SVM	85	-0.3120	[-0.4023, -0.2219]	-0.3740	[-0.5171, -0.2328]	85	-0.2884	[-0.3781, -0.1973]	-0.3690	[-0.5127, -0.2294]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	35	-0.0765	[-0.2127, +0.0655]	-0.0722	[-0.1716, +0.0169]	54	-0.0410	[-0.1438, +0.0606]	-0.0877	[-0.2082, -0.0036]
LUNAR	85	-0.3680	[-0.4400, -0.2915]	-0.4203	[-0.5539, -0.2920]	85	-0.3306	[-0.4071, -0.2495]	-0.3787	[-0.5180, -0.2448]

Table 140: NSA/binary/rcontiguous versus all 43 other registered detectors, Original features, 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	67	-0.2843	[-0.3624, -0.1976]	-0.3203	[-0.4559, -0.1894]	75	-0.2592	[-0.3449, -0.1709]	-0.3105	[-0.4411, -0.1809]
ABOD	75	-0.3556	[-0.4051, -0.3082]	-0.4305	[-0.5680, -0.2987]	75	-0.3149	[-0.3864, -0.2483]	-0.3902	[-0.5376, -0.2541]
COPOD	75	-0.2840	[-0.3607, -0.2015]	-0.3340	[-0.4960, -0.1871]	75	-0.2705	[-0.3577, -0.1766]	-0.3280	[-0.4781, -0.1874]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	74	-0.0694	[-0.1442, +0.0002]	-0.0812	[-0.1634, -0.0102]	75	-0.0579	[-0.1297, +0.0091]	-0.0787	[-0.1664, -0.0093]
QMCD	75	+0.1052	[-0.0327, +0.2493]	-0.0250	[-0.0921, +0.0315]	75	+0.1870	[+0.0332, +0.3347]	+0.0070	[-0.0561, +0.0623]
KDE	75	-0.3804	[-0.4266, -0.3375]	-0.4582	[-0.5888, -0.3320]	75	-0.3520	[-0.4087, -0.2967]	-0.4164	[-0.5600, -0.2775]
Sampling	60	-0.3449	[-0.4075, -0.2868]	-0.4330	[-0.5983, -0.2723]	50	-0.3411	[-0.4237, -0.2668]	-0.4138	[-0.6115, -0.2151]
GMM	75	-0.3433	[-0.3963, -0.2877]	-0.3975	[-0.5237, -0.2767]	75	-0.3189	[-0.3862, -0.2577]	-0.3515	[-0.4803, -0.2279]
PCA	75	-0.3070	[-0.3830, -0.2223]	-0.3997	[-0.5562, -0.2450]	75	-0.2814	[-0.3736, -0.1870]	-0.3759	[-0.5288, -0.2267]
KPCA	75	-0.3795	[-0.4229, -0.3372]	-0.4541	[-0.5854, -0.3219]	75	-0.3498	[-0.4097, -0.2964]	-0.4144	[-0.5601, -0.2745]
MCD	75	-0.3174	[-0.3790, -0.2568]	-0.3306	[-0.4567, -0.2112]	75	-0.2778	[-0.3617, -0.1942]	-0.2851	[-0.4110, -0.1603]
CD	68	-0.3188	[-0.3888, -0.2494]	-0.3675	[-0.4978, -0.2371]	69	-0.3022	[-0.3840, -0.2246]	-0.3416	[-0.4725, -0.2138]
OCSVM	75	-0.3428	[-0.4039, -0.2768]	-0.4310	[-0.5808, -0.2840]	75	-0.3190	[-0.3956, -0.2459]	-0.4041	[-0.5597, -0.2583]
LMDD	42	-0.2508	[-0.3411, -0.1399]	-0.3393	[-0.5036, -0.1716]	45	-0.1849	[-0.2885, -0.0772]	-0.2960	[-0.4468, -0.1493]
LOF	75	-0.2936	[-0.3699, -0.2192]	-0.3983	[-0.5595, -0.2442]	75	-0.2616	[-0.3618, -0.1520]	-0.3785	[-0.5342, -0.2208]
COF	75	-0.1472	[-0.2419, -0.0482]	-0.1893	[-0.2813, -0.1029]	75	-0.1204	[-0.2081, -0.0362]	-0.1638	[-0.2428, -0.0912]
CBLOF	70	-0.3680	[-0.4189, -0.3175]	-0.4239	[-0.5717, -0.2748]	60	-0.3425	[-0.4140, -0.2761]	-0.4100	[-0.5877, -0.2437]
LOCI	24	-0.2496	[-0.3327, -0.1619]	-0.2333	[-0.3454, -0.1389]	25	-0.1622	[-0.2505, -0.0703]	-0.1936	[-0.2863, -0.0880]
HBOS	75	-0.2999	[-0.3752, -0.2195]	-0.3355	[-0.4861, -0.1955]	75	-0.2708	[-0.3598, -0.1767]	-0.3049	[-0.4524, -0.1673]
HDBSCAN	75	-0.0188	[-0.1094, +0.0610]	+0.0058	[-0.0350, +0.0384]	75	+0.0114	[-0.0605, +0.0773]	+0.0188	[-0.0126, +0.0512]
KNN	75	-0.3575	[-0.4057, -0.3093]	-0.4288	[-0.5772, -0.2863]	75	-0.3029	[-0.3853, -0.2181]	-0.3806	[-0.5358, -0.2323]
SOD	50	-0.2564	[-0.3224, -0.1899]	-0.1990	[-0.2800, -0.1242]	50	-0.1927	[-0.2674, -0.1230]	-0.1559	[-0.2282, -0.0859]
ROD	29	-0.2862	[-0.3799, -0.1553]	-0.3664	[-0.5983, -0.1419]	32	-0.2316	[-0.3508, -0.1017]	-0.3362	[-0.5468, -0.1347]
IForest	75	-0.2969	[-0.3588, -0.2308]	-0.2693	[-0.3937, -0.1520]	75	-0.2474	[-0.3196, -0.1702]	-0.2156	[-0.3213, -0.1202]
INNE	75	-0.3388	[-0.3923, -0.2909]	-0.3393	[-0.4886, -0.2056]	75	-0.2896	[-0.3630, -0.2197]	-0.2619	[-0.3996, -0.1308]
DIF	75	-0.2555	[-0.3267, -0.1841]	-0.2335	[-0.3320, -0.1398]	75	-0.2104	[-0.2929, -0.1281]	-0.1786	[-0.2667, -0.0998]
FeatureBagging	75	-0.2976	[-0.3684, -0.2200]	-0.4015	[-0.5654, -0.2450]	75	-0.2707	[-0.3661, -0.1653]	-0.3780	[-0.5291, -0.2317]
LSCP	50	-0.3187	[-0.4074, -0.2323]	-0.4071	[-0.6151, -0.2089]	50	-0.3048	[-0.4078, -0.1932]	-0.3919	[-0.6025, -0.1842]

Continued on next page

Table 140 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
LODA	75	-0.1355	[-0.2277, -0.0365]	-0.0442	[-0.0913, -0.0000]	75	+0.0696	[-0.0244, +0.1737]	+0.0266	[-0.0033, +0.0597]
SUOD	75	-0.3188	[-0.3852, -0.2457]	-0.3483	[-0.4922, -0.2076]	74	-0.2868	[-0.3677, -0.2007]	-0.3110	[-0.4464, -0.1851]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	-0.2810	[-0.3866, -0.1783]	-0.3963	[-0.6110, -0.1949]	0	—	[—, —]	—	[—, —]
VAE	50	-0.2657	[-0.3782, -0.1503]	-0.3798	[-0.5977, -0.1724]	0	—	[—, —]	—	[—, —]
SO_GAAL	75	+0.0751	[-0.0222, +0.1825]	-0.0701	[-0.1119, -0.0275]	75	+0.0429	[-0.0761, +0.1653]	-0.0681	[-0.1234, -0.0184]
MO_GAAL	75	+0.0502	[-0.0594, +0.1632]	-0.0620	[-0.1094, -0.0152]	75	+0.0518	[-0.0429, +0.1541]	-0.0687	[-0.1211, -0.0179]
DeepSVDD	75	-0.3090	[-0.3792, -0.2337]	-0.3991	[-0.5420, -0.2601]	75	-0.2815	[-0.3687, -0.1907]	-0.3556	[-0.5001, -0.2202]
AnoGAN	75	-0.2977	[-0.3734, -0.2104]	-0.3503	[-0.4695, -0.2273]	75	-0.2783	[-0.3671, -0.1798]	-0.3612	[-0.4970, -0.2268]
ALAD	75	-0.0171	[-0.0781, +0.0417]	-0.0065	[-0.0294, +0.0179]	75	-0.0040	[-0.0557, +0.0485]	-0.0036	[-0.0262, +0.0205]
AE1SVM	75	-0.3146	[-0.3862, -0.2355]	-0.4033	[-0.5601, -0.2515]	75	-0.2817	[-0.3699, -0.1918]	-0.3827	[-0.5352, -0.2290]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	25	-0.1182	[-0.2371, +0.0268]	-0.0908	[-0.2053, +0.0160]	44	-0.0298	[-0.1663, +0.0995]	-0.0829	[-0.2210, +0.0203]
LUNAR	75	-0.3746	[-0.4187, -0.3308]	-0.4533	[-0.5883, -0.3197]	75	-0.3321	[-0.3946, -0.2667]	-0.3956	[-0.5465, -0.2546]

Table 141: NSA/binary/rchunk versus all 43 other registered detectors, Original features, 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	70	-0.2310	[-0.3244, -0.1344]	-0.3025	[-0.4288, -0.1808]	84	-0.2453	[-0.3300, -0.1610]	-0.3076	[-0.4496, -0.1856]
ABOD	78	-0.3129	[-0.3895, -0.2318]	-0.4050	[-0.5480, -0.2729]	84	-0.2987	[-0.3750, -0.2158]	-0.3803	[-0.5241, -0.2500]
COPOD	78	-0.2368	[-0.3361, -0.1347]	-0.3146	[-0.4592, -0.1729]	84	-0.2518	[-0.3427, -0.1490]	-0.3245	[-0.4713, -0.1871]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	77	-0.0533	[-0.1292, +0.0187]	-0.0916	[-0.1769, -0.0192]	84	-0.0702	[-0.1393, -0.0065]	-0.1022	[-0.1907, -0.0323]
QMCD	78	+0.0879	[-0.0213, +0.2108]	-0.0466	[-0.0967, +0.0013]	84	+0.1399	[+0.0103, +0.2680]	-0.0293	[-0.0757, +0.0114]
KDE	78	-0.3381	[-0.4078, -0.2620]	-0.4315	[-0.5657, -0.3026]	84	-0.3352	[-0.4002, -0.2602]	-0.4051	[-0.5438, -0.2772]
Sampling	62	-0.3277	[-0.4014, -0.2596]	-0.4247	[-0.5858, -0.2788]	50	-0.3627	[-0.4372, -0.2943]	-0.4325	[-0.6353, -0.2503]
GMM	78	-0.3007	[-0.3790, -0.2154]	-0.3765	[-0.5042, -0.2587]	84	-0.3014	[-0.3736, -0.2229]	-0.3461	[-0.4739, -0.2308]
PCA	78	-0.2611	[-0.3611, -0.1533]	-0.3761	[-0.5227, -0.2308]	84	-0.2678	[-0.3607, -0.1704]	-0.3673	[-0.5128, -0.2283]
KPCA	78	-0.3361	[-0.4099, -0.2563]	-0.4273	[-0.5679, -0.2918]	84	-0.3321	[-0.3979, -0.2592]	-0.4028	[-0.5444, -0.2684]
MCD	78	-0.2760	[-0.3557, -0.1924]	-0.3170	[-0.4380, -0.2032]	84	-0.2632	[-0.3408, -0.1805]	-0.2864	[-0.3999, -0.1776]
CD	71	-0.2832	[-0.3705, -0.1950]	-0.3506	[-0.4774, -0.2277]	78	-0.2903	[-0.3719, -0.2054]	-0.3397	[-0.4681, -0.2168]
OCSVM	78	-0.3048	[-0.3858, -0.2220]	-0.4056	[-0.5481, -0.2670]	84	-0.3041	[-0.3798, -0.2215]	-0.3927	[-0.5383, -0.2519]
LMDD	45	-0.1712	[-0.2905, -0.0449]	-0.2956	[-0.4518, -0.1451]	54	-0.1727	[-0.2680, -0.0758]	-0.2889	[-0.4409, -0.1428]
LOF	78	-0.2556	[-0.3493, -0.1552]	-0.3768	[-0.5304, -0.2353]	84	-0.2523	[-0.3505, -0.1507]	-0.3689	[-0.5180, -0.2233]
COF	78	-0.1255	[-0.2115, -0.0379]	-0.1887	[-0.2809, -0.1023]	84	-0.1275	[-0.2044, -0.0526]	-0.1788	[-0.2591, -0.1043]
CBLOF	73	-0.3201	[-0.3932, -0.2456]	-0.3929	[-0.5422, -0.2569]	65	-0.3446	[-0.4091, -0.2860]	-0.4104	[-0.5772, -0.2544]
LOCI	25	-0.1872	[-0.2734, -0.0967]	-0.2355	[-0.3646, -0.1238]	29	-0.1590	[-0.2291, -0.0896]	-0.2147	[-0.3374, -0.1000]
HBOS	78	-0.2638	[-0.3451, -0.1788]	-0.3228	[-0.4606, -0.1915]	84	-0.2589	[-0.3391, -0.1726]	-0.3059	[-0.4432, -0.1848]
HDBSCAN	78	-0.0036	[-0.0868, +0.0642]	-0.0092	[-0.0391, +0.0157]	84	-0.0073	[-0.0695, +0.0510]	-0.0113	[-0.0293, +0.0050]
KNN	78	-0.3151	[-0.3936, -0.2287]	-0.4047	[-0.5500, -0.2614]	84	-0.2873	[-0.3700, -0.1949]	-0.3725	[-0.5211, -0.2326]
SOD	52	-0.1980	[-0.2788, -0.1185]	-0.1874	[-0.2731, -0.1051]	55	-0.2044	[-0.2757, -0.1356]	-0.1694	[-0.2358, -0.1084]
ROD	32	-0.1871	[-0.3262, -0.0248]	-0.3113	[-0.5186, -0.1180]	41	-0.2209	[-0.3409, -0.0987]	-0.3228	[-0.5129, -0.1408]
IForest	78	-0.2559	[-0.3362, -0.1649]	-0.2626	[-0.3765, -0.1561]	84	-0.2453	[-0.3187, -0.1685]	-0.2310	[-0.3306, -0.1405]
INNE	78	-0.2927	[-0.3706, -0.2033]	-0.3214	[-0.4590, -0.1950]	84	-0.2795	[-0.3528, -0.2090]	-0.2661	[-0.3894, -0.1558]
DIF	78	-0.2218	[-0.3064, -0.1376]	-0.2272	[-0.3190, -0.1357]	84	-0.2103	[-0.2876, -0.1366]	-0.1906	[-0.2683, -0.1201]
FeatureBagging	78	-0.2606	[-0.3562, -0.1619]	-0.3804	[-0.5362, -0.2349]	84	-0.2588	[-0.3510, -0.1594]	-0.3685	[-0.5172, -0.2294]
LSCP	50	-0.3297	[-0.4281, -0.2357]	-0.4218	[-0.6293, -0.2347]	50	-0.3263	[-0.4245, -0.2201]	-0.4106	[-0.6154, -0.2060]

Continued on next page

Table 141 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
LODA	78	-0.1240	[-0.2014, -0.0411]	-0.0630	[-0.0995, -0.0315]	84	+0.0359	[-0.0541, +0.1322]	-0.0124	[-0.0379, +0.0146]
SUOD	78	-0.2778	[-0.3617, -0.1828]	-0.3342	[-0.4679, -0.2073]	83	-0.2741	[-0.3554, -0.1854]	-0.3141	[-0.4446, -0.1941]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	-0.2920	[-0.4096, -0.1753]	-0.4110	[-0.6179, -0.2083]	0	—	[—, —]	—	[—, —]
VAE	50	-0.2768	[-0.3996, -0.1500]	-0.3944	[-0.6086, -0.1949]	0	—	[—, —]	—	[—, —]
SO_GAAL	78	+0.0908	[-0.0024, +0.1945]	-0.0717	[-0.1154, -0.0248]	84	+0.0249	[-0.0789, +0.1348]	-0.0905	[-0.1360, -0.0477]
MO_GAAL	78	+0.0417	[-0.0616, +0.1518]	-0.0740	[-0.1166, -0.0314]	84	+0.0351	[-0.0466, +0.1252]	-0.0900	[-0.1335, -0.0472]
DeepSVDD	78	-0.2663	[-0.3548, -0.1675]	-0.3767	[-0.5168, -0.2400]	84	-0.2717	[-0.3557, -0.1859]	-0.3501	[-0.4816, -0.2232]
AnoGAN	78	-0.2625	[-0.3507, -0.1659]	-0.3358	[-0.4542, -0.2158]	84	-0.2678	[-0.3577, -0.1731]	-0.3569	[-0.4914, -0.2240]
ALAD	78	-0.0178	[-0.0663, +0.0313]	-0.0292	[-0.0502, -0.0088]	84	-0.0245	[-0.0655, +0.0146]	-0.0365	[-0.0520, -0.0219]
AE1SVM	78	-0.2690	[-0.3666, -0.1592]	-0.3801	[-0.5340, -0.2337]	84	-0.2690	[-0.3609, -0.1722]	-0.3735	[-0.5209, -0.2289]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	28	-0.0364	[-0.1961, +0.1163]	-0.0728	[-0.1858, +0.0319]	53	-0.0308	[-0.1463, +0.0811]	-0.0925	[-0.2099, -0.0057]
LUNAR	78	-0.3294	[-0.4049, -0.2465]	-0.4250	[-0.5633, -0.2903]	84	-0.3130	[-0.3838, -0.2338]	-0.3836	[-0.5250, -0.2441]

Table 142: NSA/fixed versus all 43 other registered detectors, PCA(2), 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	85	-0.0031	[-0.0998, +0.0934]	+0.0377	[-0.0542, +0.1364]	85	-0.0256	[-0.1364, +0.0756]	-0.0307	[-0.1126, +0.0439]
ABOD	85	-0.1619	[-0.2420, -0.0866]	-0.1432	[-0.2472, -0.0576]	85	-0.1520	[-0.2350, -0.0756]	-0.1592	[-0.2582, -0.0724]
COPOD	85	-0.0479	[-0.1474, +0.0528]	-0.0095	[-0.1143, +0.0992]	85	-0.0612	[-0.1648, +0.0330]	-0.0699	[-0.1549, +0.0047]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	85	+0.0880	[-0.0112, +0.1880]	+0.1316	[+0.0449, +0.2289]	85	+0.0622	[-0.0305, +0.1517]	+0.0671	[+0.0120, +0.1253]
QMCD	85	+0.0073	[-0.0738, +0.0941]	+0.0207	[-0.0470, +0.1018]	85	+0.0692	[-0.0021, +0.1447]	+0.0190	[-0.0348, +0.0755]
KDE	85	-0.1638	[-0.2435, -0.0937]	-0.1585	[-0.2611, -0.0754]	85	-0.1579	[-0.2339, -0.0897]	-0.1750	[-0.2818, -0.0839]
Sampling	65	-0.1687	[-0.2716, -0.0743]	-0.1534	[-0.2721, -0.0584]	50	-0.1362	[-0.2364, -0.0399]	-0.1884	[-0.3409, -0.0601]
GMM	85	-0.0286	[-0.1280, +0.0744]	-0.0698	[-0.2076, +0.0473]	85	-0.0486	[-0.1613, +0.0584]	-0.1279	[-0.2477, -0.0255]
PCA	85	-0.0373	[-0.1462, +0.0751]	-0.0625	[-0.1981, +0.0730]	85	-0.0741	[-0.1690, +0.0127]	-0.1363	[-0.2608, -0.0371]
KPCA	85	-0.1633	[-0.2428, -0.0926]	-0.1501	[-0.2511, -0.0702]	85	-0.1524	[-0.2300, -0.0814]	-0.1638	[-0.2658, -0.0764]
MCD	85	-0.0922	[-0.1853, -0.0006]	-0.0673	[-0.2056, +0.0624]	85	-0.0889	[-0.1932, +0.0094]	-0.1042	[-0.2271, -0.0028]
CD	85	+0.0254	[-0.0729, +0.1418]	+0.0166	[-0.0791, +0.1220]	85	-0.0058	[-0.1199, +0.1116]	-0.0443	[-0.1326, +0.0341]
OCSVM	85	-0.1532	[-0.2311, -0.0824]	-0.1507	[-0.2490, -0.0716]	85	-0.1528	[-0.2309, -0.0817]	-0.1656	[-0.2705, -0.0792]
LMDD	85	+0.0257	[-0.0745, +0.1314]	+0.0242	[-0.0904, +0.1450]	85	+0.0117	[-0.1028, +0.1245]	-0.0185	[-0.1025, +0.0584]
LOF	85	-0.0597	[-0.1487, +0.0191]	-0.0787	[-0.1935, +0.0144]	85	-0.0291	[-0.1414, +0.0906]	-0.0831	[-0.1795, +0.0100]
COF	85	-0.0008	[-0.0995, +0.0953]	+0.0624	[-0.0415, +0.1621]	85	-0.0106	[-0.0947, +0.0687]	+0.0159	[-0.0508, +0.0760]
CBLOF	80	-0.1524	[-0.2346, -0.0766]	-0.1448	[-0.2517, -0.0600]	65	-0.1577	[-0.2576, -0.0663]	-0.2003	[-0.3302, -0.0830]
LOCI	28	+0.0414	[-0.1155, +0.2012]	-0.0234	[-0.1260, +0.0869]	30	-0.0014	[-0.1642, +0.1615]	-0.0304	[-0.1285, +0.0769]
HBOS	85	-0.1200	[-0.1922, -0.0526]	-0.0180	[-0.0668, +0.0266]	85	-0.0803	[-0.1506, -0.0210]	-0.0031	[-0.0379, +0.0349]

Continued on next page

Table 142 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
HDBSCAN	85	+0.0203	[-0.0761, +0.1128]	+0.1284	[+0.0589, +0.2093]	85	+0.0266	[-0.0478, +0.0991]	+0.0849	[+0.0400, +0.1354]
KNN	85	-0.1574	[-0.2437, -0.0816]	-0.1443	[-0.2526, -0.0566]	85	-0.1166	[-0.2159, -0.0181]	-0.1517	[-0.2643, -0.0527]
SOD	63	+0.0459	[-0.0464, +0.1281]	+0.0808	[+0.0028, +0.1650]	55	+0.0436	[-0.0437, +0.1288]	+0.0255	[-0.0481, +0.0913]
ROD	85	-0.0502	[-0.1509, +0.0644]	-0.0179	[-0.1299, +0.0999]	85	-0.0332	[-0.1361, +0.0721]	-0.0597	[-0.1491, +0.0223]
IForest	85	-0.1064	[-0.1792, -0.0439]	-0.0267	[-0.0795, +0.0299]	85	-0.0954	[-0.1745, -0.0239]	-0.0609	[-0.1088, -0.0178]
INNE	85	-0.1321	[-0.2173, -0.0572]	-0.1273	[-0.2319, -0.0453]	85	-0.1222	[-0.2005, -0.0509]	-0.1270	[-0.2286, -0.0383]
DIF	83	-0.1180	[-0.1961, -0.0468]	-0.0876	[-0.2035, +0.0053]	85	-0.1025	[-0.1804, -0.0355]	-0.1125	[-0.2149, -0.0348]
FeatureBagging	85	-0.0816	[-0.1666, -0.0015]	-0.0914	[-0.1889, -0.0088]	85	-0.0438	[-0.1506, +0.0640]	-0.1039	[-0.2103, -0.0098]
LSCP	50	-0.1812	[-0.2823, -0.0933]	-0.1451	[-0.3021, -0.0316]	50	-0.1161	[-0.2138, -0.0311]	-0.1353	[-0.2665, -0.0299]
LODA	85	-0.0514	[-0.0982, -0.0025]	+0.0227	[-0.0240, +0.0781]	85	+0.0146	[-0.0319, +0.0641]	+0.0327	[-0.0046, +0.0771]
SUOD	84	-0.1264	[-0.2017, -0.0604]	-0.0696	[-0.1444, -0.0015]	85	-0.0999	[-0.1763, -0.0324]	-0.0767	[-0.1422, -0.0229]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	-0.0097	[-0.1401, +0.1269]	-0.0089	[-0.1740, +0.1340]	0	—	[—, —]	—	[—, —]
VAE	50	-0.0099	[-0.1519, +0.1389]	-0.0288	[-0.2320, +0.1426]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	+0.0444	[-0.0621, +0.1728]	+0.0419	[-0.0399, +0.1329]	85	+0.0346	[-0.0679, +0.1572]	+0.0042	[-0.0451, +0.0603]
MO_GAAL	85	+0.0521	[-0.0676, +0.1891]	+0.0248	[-0.0620, +0.1278]	85	+0.0483	[-0.0473, +0.1539]	-0.0027	[-0.0622, +0.0575]
DeepSVDD	85	-0.0879	[-0.1756, -0.0062]	-0.0644	[-0.1912, +0.0433]	85	-0.0750	[-0.1602, -0.0021]	-0.0867	[-0.1907, +0.0011]
AnoGAN	85	-0.0293	[-0.1212, +0.0704]	+0.0058	[-0.0951, +0.1109]	85	-0.0576	[-0.1696, +0.0626]	-0.0750	[-0.1696, +0.0108]
ALAD	85	+0.1087	[+0.0183, +0.1939]	+0.1513	[+0.0717, +0.2473]	85	+0.0784	[+0.0047, +0.1551]	+0.0827	[+0.0320, +0.1396]
AE1SVM	85	-0.0416	[-0.1391, +0.0629]	-0.0682	[-0.2032, +0.0674]	85	-0.0641	[-0.1765, +0.0400]	-0.1338	[-0.2507, -0.0323]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]

Continued on next page

Table 142 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
RGraph	53	+0.2189	[+0.0653, +0.3678]	+0.1572	[+0.0659, +0.2599]	85	+0.2045	[+0.0885, +0.3254]	+0.1218	[+0.0678, +0.1815]
LUNAR	85	-0.1328	[-0.2194, -0.0530]	-0.1249	[-0.2399, -0.0292]	85	-0.1189	[-0.2093, -0.0361]	-0.1467	[-0.2641, -0.0509]

Table 143: NSA/variable versus all 43 other registered detectors, PCA(2), 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	85	+0.1111	[+0.0088, +0.2094]	+0.1226	[+0.0330, +0.2243]	85	+0.0511	[-0.0585, +0.1584]	+0.0372	[-0.0416, +0.1192]
ABOD	85	-0.0478	[-0.1039, -0.0064]	-0.0583	[-0.1422, +0.0042]	85	-0.0754	[-0.1389, -0.0239]	-0.0912	[-0.1787, -0.0193]
COPOD	85	+0.0662	[-0.0292, +0.1603]	+0.0754	[-0.0264, +0.1800]	85	+0.0154	[-0.0841, +0.1064]	-0.0020	[-0.0785, +0.0730]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	85	+0.2021	[+0.1291, +0.2774]	+0.2166	[+0.1215, +0.3251]	85	+0.1388	[+0.0666, +0.2108]	+0.1350	[+0.0719, +0.2057]
QMCD	85	+0.1215	[+0.0528, +0.2034]	+0.1056	[+0.0262, +0.2043]	85	+0.1459	[+0.0617, +0.2356]	+0.0870	[+0.0269, +0.1571]
KDE	85	-0.0496	[-0.1037, -0.0049]	-0.0736	[-0.1618, -0.0025]	85	-0.0813	[-0.1431, -0.0311]	-0.1070	[-0.1971, -0.0346]
Sampling	65	-0.0462	[-0.1161, +0.0137]	-0.0702	[-0.1716, +0.0091]	50	-0.0280	[-0.0943, +0.0288]	-0.1121	[-0.2338, -0.0197]
GMM	85	+0.0855	[-0.0156, +0.1871]	+0.0151	[-0.1087, +0.1343]	85	+0.0280	[-0.0822, +0.1421]	-0.0600	[-0.1735, +0.0330]
PCA	85	+0.0768	[-0.0243, +0.1788]	+0.0224	[-0.1017, +0.1492]	85	+0.0026	[-0.0809, +0.0840]	-0.0683	[-0.1781, +0.0243]
KPCA	85	-0.0492	[-0.1020, -0.0079]	-0.0652	[-0.1494, +0.0032]	85	-0.0758	[-0.1399, -0.0259]	-0.0958	[-0.1757, -0.0292]
MCD	85	+0.0219	[-0.0639, +0.1097]	+0.0176	[-0.1138, +0.1468]	85	-0.0122	[-0.1069, +0.0816]	-0.0363	[-0.1545, +0.0614]
CD	85	+0.1395	[+0.0445, +0.2483]	+0.1015	[+0.0078, +0.2104]	85	+0.0708	[-0.0431, +0.1916]	+0.0236	[-0.0623, +0.1102]
OCSVM	85	-0.0390	[-0.0933, +0.0064]	-0.0658	[-0.1517, +0.0011]	85	-0.0762	[-0.1400, -0.0251]	-0.0976	[-0.1795, -0.0293]
LMDD	85	+0.1399	[+0.0481, +0.2353]	+0.1091	[-0.0036, +0.2309]	85	+0.0883	[-0.0242, +0.2036]	+0.0495	[-0.0372, +0.1291]
LOF	85	+0.0544	[-0.0300, +0.1338]	+0.0062	[-0.0975, +0.0950]	85	+0.0476	[-0.0629, +0.1682]	-0.0151	[-0.1025, +0.0762]
COF	85	+0.1134	[+0.0392, +0.1792]	+0.1473	[+0.0405, +0.2584]	85	+0.0660	[-0.0090, +0.1275]	+0.0839	[+0.0156, +0.1527]
CBLOF	80	-0.0456	[-0.1006, -0.0012]	-0.0712	[-0.1613, -0.0021]	65	-0.0734	[-0.1528, -0.0095]	-0.1280	[-0.2407, -0.0371]

Continued on next page

Table 143 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
LOCI	28	+0.1249	[-0.0276, +0.2790]	+0.0717	[-0.0084, +0.1558]	30	+0.0344	[-0.1090, +0.1836]	+0.0286	[-0.0536, +0.1200]
HBOS	85	-0.0058	[-0.0605, +0.0356]	+0.0669	[+0.0177, +0.1213]	85	-0.0037	[-0.0667, +0.0505]	+0.0649	[+0.0365, +0.0969]
HDBSCAN	85	+0.1345	[+0.0727, +0.2004]	+0.2133	[+0.1289, +0.3129]	85	+0.1033	[+0.0530, +0.1535]	+0.1528	[+0.0961, +0.2158]
KNN	85	-0.0433	[-0.1039, +0.0060]	-0.0594	[-0.1570, +0.0160]	85	-0.0399	[-0.1187, +0.0407]	-0.0838	[-0.1778, -0.0032]
SOD	63	+0.1630	[+0.1092, +0.2123]	+0.1565	[+0.0714, +0.2528]	55	+0.1425	[+0.1068, +0.1813]	+0.0983	[+0.0264, +0.1687]
ROD	85	+0.0639	[-0.0350, +0.1740]	+0.0670	[-0.0365, +0.1909]	85	+0.0434	[-0.0562, +0.1493]	+0.0082	[-0.0739, +0.0924]
IForest	85	+0.0077	[-0.0530, +0.0620]	+0.0582	[+0.0095, +0.1175]	85	-0.0187	[-0.0936, +0.0546]	+0.0070	[-0.0342, +0.0492]
INNE	85	-0.0180	[-0.0793, +0.0389]	-0.0424	[-0.1342, +0.0337]	85	-0.0456	[-0.1062, +0.0079]	-0.0591	[-0.1454, +0.0135]
DIF	83	+0.0049	[-0.0600, +0.0622]	-0.0012	[-0.0994, +0.0805]	85	-0.0259	[-0.0894, +0.0399]	-0.0446	[-0.1320, +0.0245]
FeatureBagging	85	+0.0325	[-0.0490, +0.1131]	-0.0065	[-0.0920, +0.0658]	85	+0.0328	[-0.0713, +0.1369]	-0.0360	[-0.1356, +0.0516]
LSCP	50	-0.0397	[-0.1086, +0.0050]	-0.0590	[-0.1934, +0.0283]	50	-0.0079	[-0.0903, +0.0778]	-0.0590	[-0.1636, +0.0330]
LODA	85	+0.0627	[+0.0250, +0.1081]	+0.1076	[+0.0429, +0.1873]	85	+0.0912	[+0.0422, +0.1386]	+0.1006	[+0.0513, +0.1514]
SUOD	84	-0.0126	[-0.0780, +0.0408]	+0.0155	[-0.0539, +0.0841]	85	-0.0233	[-0.0971, +0.0504]	-0.0087	[-0.0608, +0.0398]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	+0.1318	[+0.0008, +0.2674]	+0.0773	[-0.0669, +0.2289]	0	—	[—, —]	—	[—, —]
VAE	50	+0.1316	[-0.0023, +0.2746]	+0.0573	[-0.1340, +0.2300]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	+0.1585	[+0.0618, +0.2714]	+0.1268	[+0.0568, +0.2211]	85	+0.1112	[+0.0158, +0.2284]	+0.0722	[+0.0219, +0.1368]
MO_GAAL	85	+0.1663	[+0.0594, +0.2840]	+0.1097	[+0.0289, +0.2122]	85	+0.1249	[+0.0398, +0.2222]	+0.0652	[+0.0123, +0.1304]
DeepSVDD	85	+0.0262	[-0.0450, +0.0875]	+0.0205	[-0.0937, +0.1196]	85	+0.0016	[-0.0765, +0.0698]	-0.0188	[-0.1158, +0.0638]

Continued on next page

Table 143 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
AnoGAN	85	+0.0848	[+0.0009, +0.1768]	+0.0907	[-0.0044, +0.1993]	85	+0.0190	[-0.0900, +0.1391]	-0.0071	[-0.0944, +0.0773]
ALAD	85	+0.2228	[+0.1561, +0.2882]	+0.2362	[+0.1432, +0.3477]	85	+0.1550	[+0.0949, +0.2165]	+0.1506	[+0.0874, +0.2236]
AE1SVM	85	+0.0725	[-0.0164, +0.1697]	+0.0167	[-0.1035, +0.1425]	85	+0.0125	[-0.0890, +0.1147]	-0.0658	[-0.1748, +0.0332]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	53	+0.3270	[+0.1913, +0.4499]	+0.2505	[+0.1433, +0.3766]	85	+0.2811	[+0.1793, +0.3896]	+0.1897	[+0.1212, +0.2664]
LUNAR	85	-0.0186	[-0.0806, +0.0363]	-0.0400	[-0.1439, +0.0406]	85	-0.0423	[-0.1142, +0.0196]	-0.0787	[-0.1839, +0.0040]

Table 144: NSA/coverage versus all 43 other registered detectors, PCA(2), 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	85	+0.0948	[-0.0010, +0.1924]	+0.1158	[+0.0281, +0.2178]	85	+0.0497	[-0.0583, +0.1563]	+0.0331	[-0.0442, +0.1139]
ABOD	85	-0.0640	[-0.1174, -0.0206]	-0.0650	[-0.1489, -0.0021]	85	-0.0768	[-0.1438, -0.0224]	-0.0954	[-0.1883, -0.0189]
COPOD	85	+0.0500	[-0.0398, +0.1476]	+0.0686	[-0.0317, +0.1740]	85	+0.0140	[-0.0871, +0.1093]	-0.0061	[-0.0838, +0.0711]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	85	+0.1859	[+0.1091, +0.2613]	+0.2098	[+0.1151, +0.3184]	85	+0.1374	[+0.0630, +0.2100]	+0.1309	[+0.0681, +0.1998]
QMCD	85	+0.1052	[+0.0357, +0.1856]	+0.0989	[+0.0186, +0.1964]	85	+0.1445	[+0.0622, +0.2336]	+0.0829	[+0.0235, +0.1480]
KDE	85	-0.0659	[-0.1212, -0.0179]	-0.0803	[-0.1690, -0.0111]	85	-0.0827	[-0.1502, -0.0301]	-0.1112	[-0.2066, -0.0332]
Sampling	65	-0.0623	[-0.1364, -0.0008]	-0.0765	[-0.1732, +0.0034]	50	-0.0293	[-0.0988, +0.0290]	-0.1207	[-0.2565, -0.0186]
GMM	85	+0.0693	[-0.0298, +0.1721]	+0.0083	[-0.1192, +0.1263]	85	+0.0266	[-0.0847, +0.1340]	-0.0641	[-0.1870, +0.0326]
PCA	85	+0.0606	[-0.0397, +0.1712]	+0.0157	[-0.1064, +0.1451]	85	+0.0012	[-0.0835, +0.0801]	-0.0725	[-0.1885, +0.0199]
KPCA	85	-0.0654	[-0.1183, -0.0218]	-0.0719	[-0.1591, -0.0078]	85	-0.0772	[-0.1411, -0.0252]	-0.1000	[-0.1872, -0.0269]
MCD	85	+0.0057	[-0.0816, +0.0955]	+0.0108	[-0.1247, +0.1454]	85	-0.0136	[-0.1083, +0.0811]	-0.0404	[-0.1639, +0.0566]
CD	85	+0.1233	[+0.0257, +0.2379]	+0.0948	[+0.0009, +0.2064]	85	+0.0694	[-0.0463, +0.1926]	+0.0195	[-0.0639, +0.1037]
OCSVM	85	-0.0553	[-0.1130, -0.0088]	-0.0725	[-0.1597, -0.0067]	85	-0.0776	[-0.1415, -0.0230]	-0.1018	[-0.1919, -0.0290]
LMDD	85	+0.1236	[+0.0327, +0.2201]	+0.1023	[-0.0109, +0.2265]	85	+0.0869	[-0.0272, +0.2012]	+0.0454	[-0.0445, +0.1255]
LOF	85	+0.0381	[-0.0421, +0.1168]	-0.0005	[-0.1085, +0.0830]	85	+0.0462	[-0.0633, +0.1629]	-0.0193	[-0.1142, +0.0752]
COF	85	+0.0971	[+0.0217, +0.1641]	+0.1406	[+0.0383, +0.2517]	85	+0.0646	[-0.0125, +0.1287]	+0.0797	[+0.0107, +0.1451]
CBLOF	80	-0.0586	[-0.1181, -0.0116]	-0.0760	[-0.1692, -0.0073]	65	-0.0765	[-0.1603, -0.0096]	-0.1373	[-0.2630, -0.0427]
LOCI	28	+0.1108	[-0.0395, +0.2702]	+0.0643	[-0.0178, +0.1527]	30	+0.0336	[-0.1078, +0.1885]	+0.0356	[-0.0524, +0.1300]
HBOS	85	-0.0221	[-0.0756, +0.0198]	+0.0601	[+0.0129, +0.1141]	85	-0.0051	[-0.0676, +0.0506]	+0.0607	[+0.0316, +0.0945]

Continued on next page

Table 144 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
HDBSCAN	85	+0.1182	[+0.0516, +0.1886]	+0.2066	[+0.1225, +0.3059]	85	+0.1019	[+0.0519, +0.1518]	+0.1487	[+0.0920, +0.2116]
KNN	85	-0.0596	[-0.1190, -0.0067]	-0.0661	[-0.1609, +0.0110]	85	-0.0413	[-0.1271, +0.0388]	-0.0879	[-0.1941, -0.0035]
SOD	63	+0.1475	[+0.0841, +0.2030]	+0.1530	[+0.0669, +0.2492]	55	+0.1408	[+0.1029, +0.1812]	+0.0881	[+0.0066, +0.1623]
ROD	85	+0.0476	[-0.0481, +0.1546]	+0.0603	[-0.0425, +0.1780]	85	+0.0420	[-0.0612, +0.1495]	+0.0041	[-0.0822, +0.0928]
IForest	85	-0.0085	[-0.0687, +0.0436]	+0.0515	[+0.0037, +0.1093]	85	-0.0201	[-0.1012, +0.0524]	+0.0029	[-0.0422, +0.0479]
INNE	85	-0.0342	[-0.0978, +0.0207]	-0.0492	[-0.1402, +0.0236]	85	-0.0470	[-0.1122, +0.0086]	-0.0632	[-0.1578, +0.0151]
DIF	83	-0.0137	[-0.0773, +0.0421]	-0.0080	[-0.1074, +0.0695]	85	-0.0273	[-0.0968, +0.0390]	-0.0487	[-0.1429, +0.0240]
FeatureBagging	85	+0.0162	[-0.0650, +0.0925]	-0.0132	[-0.0972, +0.0563]	85	+0.0314	[-0.0759, +0.1375]	-0.0401	[-0.1452, +0.0527]
LSCP	50	-0.0586	[-0.1251, -0.0161]	-0.0669	[-0.1994, +0.0183]	50	-0.0092	[-0.0918, +0.0730]	-0.0677	[-0.1851, +0.0332]
LODA	85	+0.0465	[+0.0115, +0.0873]	+0.1008	[+0.0380, +0.1772]	85	+0.0898	[+0.0424, +0.1352]	+0.0965	[+0.0513, +0.1464]
SUOD	84	-0.0288	[-0.0925, +0.0236]	+0.0088	[-0.0580, +0.0770]	85	-0.0247	[-0.1016, +0.0477]	-0.0129	[-0.0684, +0.0382]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	+0.1129	[-0.0115, +0.2517]	+0.0693	[-0.0803, +0.2211]	0	—	[—, —]	—	[—, —]
VAE	50	+0.1127	[-0.0216, +0.2527]	+0.0494	[-0.1397, +0.2205]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	+0.1422	[+0.0440, +0.2591]	+0.1201	[+0.0494, +0.2151]	85	+0.1098	[+0.0129, +0.2221]	+0.0680	[+0.0187, +0.1339]
MO_GAAL	85	+0.1500	[+0.0415, +0.2700]	+0.1029	[+0.0261, +0.2049]	85	+0.1235	[+0.0373, +0.2220]	+0.0611	[+0.0020, +0.1279]
DeepSVDD	85	+0.0100	[-0.0595, +0.0714]	+0.0138	[-0.0971, +0.1117]	85	+0.0002	[-0.0789, +0.0672]	-0.0229	[-0.1246, +0.0615]
AnoGAN	85	+0.0686	[-0.0161, +0.1627]	+0.0839	[-0.0101, +0.1909]	85	+0.0176	[-0.0985, +0.1322]	-0.0112	[-0.1002, +0.0781]
ALAD	85	+0.2065	[+0.1384, +0.2756]	+0.2295	[+0.1360, +0.3379]	85	+0.1536	[+0.0947, +0.2155]	+0.1465	[+0.0859, +0.2166]

Continued on next page

Table 144 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
AE1SVM	85	+0.0562	[-0.0328, +0.1540]	+0.0100	[-0.1120, +0.1338]	85	+0.0111	[-0.0931, +0.1174]	-0.0699	[-0.1885, +0.0291]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	53	+0.3144	[+0.1825, +0.4430]	+0.2416	[+0.1374, +0.3683]	85	+0.2797	[+0.1744, +0.3887]	+0.1856	[+0.1173, +0.2590]
LUNAR	85	-0.0349	[-0.1005, +0.0205]	-0.0468	[-0.1496, +0.0322]	85	-0.0437	[-0.1219, +0.0205]	-0.0829	[-0.1930, +0.0016]

Table 145: NSA/grid versus all 43 other registered detectors, PCA(2), 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	85	+0.1081	[+0.0098, +0.2072]	+0.1256	[+0.0322, +0.2336]	85	+0.0603	[-0.0462, +0.1608]	+0.0495	[-0.0291, +0.1300]
ABOD	85	-0.0508	[-0.1048, -0.0092]	-0.0553	[-0.1373, +0.0077]	85	-0.0662	[-0.1229, -0.0191]	-0.0789	[-0.1603, -0.0143]
COPOD	85	+0.0632	[-0.0350, +0.1579]	+0.0784	[-0.0226, +0.1901]	85	+0.0247	[-0.0632, +0.1133]	+0.0103	[-0.0628, +0.0836]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	85	+0.1991	[+0.1273, +0.2703]	+0.2195	[+0.1244, +0.3331]	85	+0.1480	[+0.0660, +0.2262]	+0.1473	[+0.0739, +0.2289]
QMCD	85	+0.1185	[+0.0499, +0.1980]	+0.1086	[+0.0321, +0.2033]	85	+0.1551	[+0.0663, +0.2483]	+0.0993	[+0.0307, +0.1765]
KDE	85	-0.0526	[-0.1059, -0.0071]	-0.0706	[-0.1597, -0.0020]	85	-0.0720	[-0.1242, -0.0299]	-0.0947	[-0.1776, -0.0267]
Sampling	65	-0.0496	[-0.1192, +0.0088]	-0.0711	[-0.1640, +0.0032]	50	-0.0173	[-0.0743, +0.0292]	-0.0912	[-0.2031, -0.0069]
GMM	85	+0.0825	[-0.0210, +0.1853]	+0.0181	[-0.1053, +0.1354]	85	+0.0372	[-0.0687, +0.1419]	-0.0477	[-0.1584, +0.0458]
PCA	85	+0.0739	[-0.0256, +0.1784]	+0.0254	[-0.0954, +0.1550]	85	+0.0118	[-0.0639, +0.0935]	-0.0560	[-0.1614, +0.0400]
KPCA	85	-0.0522	[-0.1059, -0.0101]	-0.0622	[-0.1430, +0.0025]	85	-0.0665	[-0.1135, -0.0253]	-0.0835	[-0.1615, -0.0227]
MCD	85	+0.0190	[-0.0651, +0.1052]	+0.0206	[-0.1079, +0.1502]	85	-0.0030	[-0.0954, +0.0887]	-0.0240	[-0.1411, +0.0741]
CD	85	+0.1365	[+0.0423, +0.2451]	+0.1045	[+0.0103, +0.2150]	85	+0.0801	[-0.0304, +0.1955]	+0.0359	[-0.0466, +0.1210]
OCSVM	85	-0.0420	[-0.0972, +0.0032]	-0.0628	[-0.1453, +0.0038]	85	-0.0669	[-0.1169, -0.0263]	-0.0853	[-0.1588, -0.0252]
LMDD	85	+0.1369	[+0.0455, +0.2351]	+0.1121	[-0.0048, +0.2319]	85	+0.0976	[-0.0141, +0.2149]	+0.0618	[-0.0209, +0.1467]
LOF	85	+0.0514	[-0.0316, +0.1281]	+0.0092	[-0.0941, +0.0997]	85	+0.0568	[-0.0474, +0.1697]	-0.0028	[-0.0912, +0.0890]
COF	85	+0.1104	[+0.0366, +0.1776]	+0.1503	[+0.0431, +0.2664]	85	+0.0753	[+0.0021, +0.1391]	+0.0962	[+0.0237, +0.1717]
CBLOF	80	-0.0473	[-0.1019, -0.0019]	-0.0676	[-0.1535, -0.0004]	65	-0.0579	[-0.1212, -0.0052]	-0.1099	[-0.2173, -0.0221]

Continued on next page

Table 145 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
LOCI	28	+0.1198	[-0.0336, +0.2733]	+0.0786	[-0.0167, +0.1755]	30	+0.0420	[-0.1136, +0.1985]	+0.0274	[-0.0827, +0.1447]
HBOS	85	-0.0088	[-0.0602, +0.0336]	+0.0699	[+0.0176, +0.1254]	85	+0.0055	[-0.0419, +0.0507]	+0.0772	[+0.0477, +0.1106]
HDBSCAN	85	+0.1315	[+0.0709, +0.1980]	+0.2163	[+0.1334, +0.3141]	85	+0.1125	[+0.0519, +0.1754]	+0.1651	[+0.0988, +0.2442]
KNN	85	-0.0463	[-0.1050, +0.0035]	-0.0564	[-0.1494, +0.0232]	85	-0.0307	[-0.1129, +0.0519]	-0.0715	[-0.1676, +0.0092]
SOD	63	+0.1646	[+0.1066, +0.2172]	+0.1662	[+0.0776, +0.2575]	55	+0.1527	[+0.1166, +0.1887]	+0.1180	[+0.0392, +0.2006]
ROD	85	+0.0609	[-0.0363, +0.1696]	+0.0700	[-0.0358, +0.1901]	85	+0.0527	[-0.0375, +0.1517]	+0.0205	[-0.0605, +0.1097]
IForest	85	+0.0047	[-0.0569, +0.0589]	+0.0612	[+0.0097, +0.1248]	85	-0.0095	[-0.0823, +0.0637]	+0.0193	[-0.0291, +0.0678]
INNE	85	-0.0210	[-0.0820, +0.0357]	-0.0394	[-0.1277, +0.0365]	85	-0.0363	[-0.0897, +0.0114]	-0.0468	[-0.1261, +0.0186]
DIF	83	+0.0020	[-0.0630, +0.0579]	+0.0019	[-0.0976, +0.0830]	85	-0.0166	[-0.0788, +0.0424]	-0.0323	[-0.1181, +0.0385]
FeatureBagging	85	+0.0295	[-0.0504, +0.1076]	-0.0035	[-0.0902, +0.0707]	85	+0.0420	[-0.0599, +0.1445]	-0.0237	[-0.1232, +0.0692]
LSCP	50	-0.0417	[-0.1080, +0.0029]	-0.0582	[-0.1766, +0.0229]	50	+0.0028	[-0.0710, +0.0785]	-0.0382	[-0.1448, +0.0482]
LODA	85	+0.0597	[+0.0217, +0.1040]	+0.1106	[+0.0478, +0.1824]	85	+0.1004	[+0.0510, +0.1438]	+0.1129	[+0.0585, +0.1739]
SUOD	84	-0.0155	[-0.0796, +0.0394]	+0.0187	[-0.0493, +0.0873]	85	-0.0140	[-0.0795, +0.0522]	+0.0036	[-0.0440, +0.0501]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	+0.1298	[+0.0024, +0.2610]	+0.0780	[-0.0642, +0.2262]	0	—	[—, —]	—	[—, —]
VAE	50	+0.1296	[-0.0041, +0.2613]	+0.0581	[-0.1215, +0.2354]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	+0.1555	[+0.0577, +0.2703]	+0.1298	[+0.0582, +0.2229]	85	+0.1205	[+0.0198, +0.2341]	+0.0845	[+0.0189, +0.1566]
MO_GAAL	85	+0.1633	[+0.0538, +0.2858]	+0.1127	[+0.0334, +0.2179]	85	+0.1342	[+0.0415, +0.2361]	+0.0775	[+0.0155, +0.1461]
DeepSVDD	85	+0.0232	[-0.0470, +0.0863]	+0.0235	[-0.0887, +0.1251]	85	+0.0109	[-0.0538, +0.0715]	-0.0065	[-0.1062, +0.0769]

Continued on next page

Table 145 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
AnoGAN	85	+0.0818	[-0.0034, +0.1758]	+0.0937	[-0.0094, +0.2018]	85	+0.0283	[-0.0756, +0.1414]	+0.0052	[-0.0829, +0.0915]
ALAD	85	+0.2198	[+0.1512, +0.2869]	+0.2392	[+0.1460, +0.3466]	85	+0.1643	[+0.1005, +0.2293]	+0.1629	[+0.0881, +0.2423]
AE1SVM	85	+0.0695	[-0.0185, +0.1648]	+0.0197	[-0.1023, +0.1433]	85	+0.0218	[-0.0757, +0.1259]	-0.0535	[-0.1644, +0.0430]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	53	+0.3236	[+0.1953, +0.4478]	+0.2553	[+0.1431, +0.3864]	85	+0.2904	[+0.1829, +0.4041]	+0.2020	[+0.1237, +0.2871]
LUNAR	85	-0.0216	[-0.0847, +0.0341]	-0.0370	[-0.1388, +0.0431]	85	-0.0330	[-0.1023, +0.0255]	-0.0664	[-0.1641, +0.0135]

Table 146: NSA/hierarchical versus all 43 other registered detectors, PCA(2), 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	85	+0.0729	[-0.0408, +0.1825]	+0.0904	[-0.0017, +0.1890]	85	+0.0407	[-0.0665, +0.1427]	+0.0140	[-0.0585, +0.0854]
ABOD	85	-0.0859	[-0.1566, -0.0282]	-0.0904	[-0.1831, -0.0169]	85	-0.0858	[-0.1585, -0.0256]	-0.1144	[-0.2048, -0.0338]
COPOD	85	+0.0280	[-0.0802, +0.1324]	+0.0432	[-0.0570, +0.1452]	85	+0.0050	[-0.0970, +0.1017]	-0.0252	[-0.1021, +0.0417]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	85	+0.1639	[+0.0820, +0.2461]	+0.1844	[+0.0962, +0.2840]	85	+0.1284	[+0.0692, +0.1923]	+0.1118	[+0.0588, +0.1688]
QMCD	85	+0.0833	[+0.0200, +0.1571]	+0.0735	[+0.0076, +0.1572]	85	+0.1355	[+0.0507, +0.2225]	+0.0638	[-0.0034, +0.1312]
KDE	85	-0.0878	[-0.1615, -0.0274]	-0.1057	[-0.1945, -0.0302]	85	-0.0917	[-0.1635, -0.0258]	-0.1302	[-0.2238, -0.0504]
Sampling	65	-0.0827	[-0.1727, -0.0064]	-0.1084	[-0.2100, -0.0256]	50	-0.0635	[-0.1522, +0.0089]	-0.1448	[-0.2636, -0.0407]
GMM	85	+0.0473	[-0.0682, +0.1617]	-0.0171	[-0.1440, +0.0987]	85	+0.0176	[-0.0935, +0.1179]	-0.0832	[-0.1902, +0.0070]
PCA	85	+0.0387	[-0.0757, +0.1598]	-0.0097	[-0.1361, +0.1197]	85	-0.0078	[-0.0980, +0.0773]	-0.0916	[-0.1939, -0.0040]
KPCA	85	-0.0873	[-0.1564, -0.0306]	-0.0973	[-0.1826, -0.0277]	85	-0.0862	[-0.1618, -0.0227]	-0.1190	[-0.2066, -0.0399]
MCD	85	-0.0162	[-0.1171, +0.0789]	-0.0146	[-0.1458, +0.1093]	85	-0.0226	[-0.1176, +0.0690]	-0.0595	[-0.1699, +0.0253]
CD	85	+0.1014	[-0.0078, +0.2165]	+0.0693	[-0.0260, +0.1697]	85	+0.0604	[-0.0459, +0.1724]	+0.0004	[-0.0823, +0.0741]
OCSVM	85	-0.0772	[-0.1459, -0.0186]	-0.0979	[-0.1858, -0.0274]	85	-0.0866	[-0.1635, -0.0174]	-0.1208	[-0.2094, -0.0443]
LMDD	85	+0.1017	[-0.0068, +0.2084]	+0.0769	[-0.0373, +0.1934]	85	+0.0779	[-0.0336, +0.1876]	+0.0263	[-0.0554, +0.0957]
LOF	85	+0.0162	[-0.0773, +0.1014]	-0.0260	[-0.1342, +0.0615]	85	+0.0372	[-0.0717, +0.1511]	-0.0383	[-0.1267, +0.0440]
COF	85	+0.0752	[-0.0113, +0.1464]	+0.1152	[+0.0136, +0.2157]	85	+0.0556	[-0.0150, +0.1146]	+0.0607	[-0.0075, +0.1190]
CBLOF	80	-0.0748	[-0.1468, -0.0136]	-0.0985	[-0.1950, -0.0236]	65	-0.1017	[-0.1896, -0.0275]	-0.1594	[-0.2714, -0.0607]
LOCI	28	+0.0786	[-0.0906, +0.2545]	+0.0477	[-0.0541, +0.1569]	30	+0.0639	[-0.0757, +0.2025]	+0.0205	[-0.0555, +0.1025]
HBOS	85	-0.0440	[-0.1104, +0.0102]	+0.0347	[-0.0135, +0.0804]	85	-0.0141	[-0.0896, +0.0593]	+0.0417	[+0.0086, +0.0816]

Continued on next page

Table 146 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
HDBSCAN	85	+0.0963	[+0.0249, +0.1728]	+0.1812	[+0.1096, +0.2738]	85	+0.0929	[+0.0398, +0.1435]	+0.1296	[+0.0804, +0.1835]
KNN	85	-0.0815	[-0.1556, -0.0182]	-0.0915	[-0.1909, -0.0078]	85	-0.0503	[-0.1384, +0.0303]	-0.1070	[-0.2074, -0.0214]
SOD	63	+0.1330	[+0.0657, +0.1912]	+0.1361	[+0.0566, +0.2190]	55	+0.1104	[+0.0544, +0.1603]	+0.0669	[+0.0145, +0.1179]
ROD	85	+0.0257	[-0.0835, +0.1403]	+0.0349	[-0.0688, +0.1482]	85	+0.0330	[-0.0726, +0.1439]	-0.0150	[-0.0913, +0.0608]
IForest	85	-0.0305	[-0.1049, +0.0395]	+0.0260	[-0.0224, +0.0822]	85	-0.0291	[-0.1078, +0.0392]	-0.0162	[-0.0582, +0.0206]
INNE	85	-0.0561	[-0.1329, +0.0088]	-0.0746	[-0.1651, +0.0054]	85	-0.0560	[-0.1314, +0.0097]	-0.0823	[-0.1714, +0.0018]
DIF	83	-0.0357	[-0.1168, +0.0299]	-0.0327	[-0.1330, +0.0506]	85	-0.0363	[-0.1126, +0.0326]	-0.0678	[-0.1544, -0.0008]
FeatureBagging	85	-0.0057	[-0.0959, +0.0747]	-0.0386	[-0.1279, +0.0401]	85	+0.0224	[-0.0846, +0.1235]	-0.0592	[-0.1529, +0.0213]
LSCP	50	-0.0742	[-0.1645, -0.0150]	-0.0951	[-0.2282, +0.0007]	50	-0.0434	[-0.1396, +0.0445]	-0.0918	[-0.1926, -0.0123]
LODA	85	+0.0246	[-0.0147, +0.0641]	+0.0754	[+0.0268, +0.1339]	85	+0.0808	[+0.0227, +0.1471]	+0.0774	[+0.0277, +0.1303]
SUOD	84	-0.0505	[-0.1279, +0.0142]	-0.0164	[-0.0868, +0.0496]	85	-0.0337	[-0.1145, +0.0425]	-0.0319	[-0.0891, +0.0195]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	+0.0973	[-0.0502, +0.2446]	+0.0412	[-0.1043, +0.1769]	0	—	[—, —]	—	[—, —]
VAE	50	+0.0971	[-0.0500, +0.2455]	+0.0212	[-0.1548, +0.1847]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	+0.1203	[+0.0124, +0.2371]	+0.0946	[+0.0238, +0.1831]	85	+0.1008	[+0.0125, +0.2096]	+0.0490	[+0.0076, +0.0981]
MO_GAAL	85	+0.1281	[+0.0127, +0.2543]	+0.0775	[+0.0003, +0.1752]	85	+0.1145	[+0.0270, +0.2146]	+0.0420	[-0.0018, +0.0931]
DeepSVDD	85	-0.0119	[-0.0974, +0.0603]	-0.0116	[-0.1257, +0.0901]	85	-0.0088	[-0.0979, +0.0714]	-0.0420	[-0.1382, +0.0400]
AnoGAN	85	+0.0466	[-0.0525, +0.1480]	+0.0585	[-0.0388, +0.1654]	85	+0.0086	[-0.0960, +0.1192]	-0.0303	[-0.1114, +0.0459]
ALAD	85	+0.1846	[+0.1113, +0.2587]	+0.2041	[+0.1219, +0.2985]	85	+0.1446	[+0.0875, +0.2036]	+0.1274	[+0.0740, +0.1822]

Continued on next page

Table 146 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
AE1SVM	85	+0.0343	[-0.0692, +0.1452]	-0.0154	[-0.1423, +0.1088]	85	+0.0021	[-0.0994, +0.1054]	-0.0890	[-0.1971, +0.0023]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	53	+0.2950	[+0.1640, +0.4199]	+0.2186	[+0.1218, +0.3376]	85	+0.2707	[+0.1604, +0.3802]	+0.1665	[+0.1090, +0.2302]
LUNAR	85	-0.0568	[-0.1330, +0.0097]	-0.0722	[-0.1820, +0.0130]	85	-0.0527	[-0.1339, +0.0165]	-0.1020	[-0.2072, -0.0206]

Table 147: NSA/voronoi versus all 43 other registered detectors, PCA(2), 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	85	+0.1199	[+0.0274, +0.2131]	+0.1276	[+0.0302, +0.2390]	85	+0.0710	[-0.0271, +0.1795]	+0.0646	[-0.0258, +0.1609]
ABOD	85	-0.0390	[-0.0790, -0.0078]	-0.0532	[-0.1208, +0.0047]	85	-0.0555	[-0.1089, -0.0135]	-0.0638	[-0.1365, -0.0030]
COPOD	85	+0.0750	[-0.0102, +0.1606]	+0.0805	[-0.0161, +0.1870]	85	+0.0354	[-0.0588, +0.1233]	+0.0255	[-0.0483, +0.1125]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	85	+0.2109	[+0.1424, +0.2817]	+0.2216	[+0.1245, +0.3343]	85	+0.1587	[+0.0873, +0.2274]	+0.1625	[+0.0840, +0.2476]
QMCD	85	+0.1303	[+0.0495, +0.2290]	+0.1107	[+0.0370, +0.2024]	85	+0.1658	[+0.0771, +0.2707]	+0.1144	[+0.0473, +0.1881]
KDE	85	-0.0408	[-0.0827, -0.0052]	-0.0685	[-0.1375, -0.0115]	85	-0.0613	[-0.1145, -0.0200]	-0.0796	[-0.1452, -0.0219]
Sampling	65	-0.0330	[-0.0853, +0.0087]	-0.0617	[-0.1316, -0.0042]	50	-0.0114	[-0.0558, +0.0252]	-0.0696	[-0.1570, -0.0030]
GMM	85	+0.0943	[+0.0010, +0.1926]	+0.0202	[-0.0886, +0.1323]	85	+0.0479	[-0.0560, +0.1545]	-0.0326	[-0.1333, +0.0667]
PCA	85	+0.0857	[-0.0033, +0.1875]	+0.0275	[-0.0844, +0.1541]	85	+0.0225	[-0.0527, +0.0996]	-0.0409	[-0.1373, +0.0561]
KPCA	85	-0.0403	[-0.0801, -0.0104]	-0.0601	[-0.1207, -0.0100]	85	-0.0558	[-0.1055, -0.0186]	-0.0684	[-0.1293, -0.0191]
MCD	85	+0.0308	[-0.0423, +0.1129]	+0.0227	[-0.0932, +0.1480]	85	+0.0077	[-0.0785, +0.0960]	-0.0088	[-0.1156, +0.0978]
CD	85	+0.1484	[+0.0548, +0.2567]	+0.1066	[+0.0103, +0.2186]	85	+0.0908	[-0.0215, +0.2066]	+0.0510	[-0.0418, +0.1531]
OCSVM	85	-0.0302	[-0.0710, +0.0026]	-0.0607	[-0.1232, -0.0094]	85	-0.0562	[-0.1084, -0.0169]	-0.0702	[-0.1324, -0.0184]
LMDD	85	+0.1487	[+0.0667, +0.2379]	+0.1142	[+0.0083, +0.2399]	85	+0.1083	[+0.0030, +0.2165]	+0.0769	[-0.0089, +0.1671]
LOF	85	+0.0632	[-0.0087, +0.1349]	+0.0113	[-0.0736, +0.0881]	85	+0.0675	[-0.0361, +0.1794]	+0.0123	[-0.0789, +0.1143]
COF	85	+0.1222	[+0.0524, +0.1817]	+0.1524	[+0.0490, +0.2625]	85	+0.0859	[+0.0158, +0.1472]	+0.1113	[+0.0334, +0.1909]
CBLOF	80	-0.0315	[-0.0718, +0.0010]	-0.0558	[-0.1256, +0.0001]	65	-0.0582	[-0.1194, -0.0114]	-0.0943	[-0.1793, -0.0271]

Continued on next page

Table 147 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
LOCI	28	+0.1222	[-0.0307, +0.2844]	+0.0653	[-0.0384, +0.1734]	30	+0.0652	[-0.0759, +0.2064]	+0.0377	[-0.0738, +0.1530]
HBOS	85	+0.0030	[-0.0343, +0.0355]	+0.0720	[+0.0263, +0.1230]	85	+0.0162	[-0.0319, +0.0606]	+0.0923	[+0.0548, +0.1330]
HDBSCAN	85	+0.1433	[+0.0917, +0.2057]	+0.2184	[+0.1324, +0.3143]	85	+0.1232	[+0.0711, +0.1753]	+0.1802	[+0.1091, +0.2573]
KNN	85	-0.0345	[-0.0810, +0.0067]	-0.0543	[-0.1287, +0.0163]	85	-0.0200	[-0.0988, +0.0584]	-0.0564	[-0.1355, +0.0195]
SOD	63	+0.1803	[+0.1218, +0.2376]	+0.1769	[+0.0917, +0.2687]	55	+0.1565	[+0.1237, +0.1890]	+0.1358	[+0.0670, +0.2124]
ROD	85	+0.0727	[-0.0174, +0.1749]	+0.0721	[-0.0311, +0.1960]	85	+0.0634	[-0.0265, +0.1646]	+0.0357	[-0.0406, +0.1323]
IForest	85	+0.0165	[-0.0372, +0.0666]	+0.0633	[+0.0025, +0.1298]	85	+0.0012	[-0.0698, +0.0703]	+0.0345	[-0.0220, +0.0943]
INNE	85	-0.0091	[-0.0586, +0.0362]	-0.0373	[-0.1063, +0.0256]	85	-0.0256	[-0.0787, +0.0194]	-0.0316	[-0.0973, +0.0257]
DIF	83	+0.0142	[-0.0358, +0.0624]	+0.0046	[-0.0813, +0.0863]	85	-0.0059	[-0.0628, +0.0506]	-0.0171	[-0.0944, +0.0586]
FeatureBagging	85	+0.0413	[-0.0270, +0.1108]	-0.0014	[-0.0687, +0.0617]	85	+0.0527	[-0.0490, +0.1535]	-0.0085	[-0.1013, +0.0897]
LSCP	50	-0.0235	[-0.0597, +0.0083]	-0.0462	[-0.1383, +0.0151]	50	+0.0087	[-0.0595, +0.0809]	-0.0166	[-0.1124, +0.0846]
LODA	85	+0.0716	[+0.0330, +0.1143]	+0.1127	[+0.0532, +0.1807]	85	+0.1111	[+0.0612, +0.1586]	+0.1280	[+0.0727, +0.1865]
SUOD	84	-0.0037	[-0.0554, +0.0447]	+0.0204	[-0.0386, +0.0840]	85	-0.0033	[-0.0668, +0.0610]	+0.0187	[-0.0340, +0.0782]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	+0.1480	[+0.0338, +0.2734]	+0.0900	[-0.0229, +0.2383]	0	—	[—, —]	—	[—, —]
VAE	50	+0.1478	[+0.0328, +0.2780]	+0.0701	[-0.0836, +0.2424]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	+0.1673	[+0.0755, +0.2740]	+0.1319	[+0.0466, +0.2375]	85	+0.1312	[+0.0345, +0.2423]	+0.0996	[+0.0303, +0.1858]
MO_GAAL	85	+0.1751	[+0.0765, +0.2952]	+0.1147	[+0.0331, +0.2277]	85	+0.1449	[+0.0615, +0.2424]	+0.0927	[+0.0268, +0.1736]
DeepSVDD	85	+0.0351	[-0.0204, +0.0882]	+0.0256	[-0.0700, +0.1222]	85	+0.0216	[-0.0471, +0.0817]	+0.0086	[-0.0787, +0.0917]

Continued on next page

Table 147 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
AnoGAN	85	+0.0936	[+0.0144, +0.1834]	+0.0957	[-0.0085, +0.2102]	85	+0.0390	[-0.0659, +0.1515]	+0.0204	[-0.0710, +0.1180]
ALAD	85	+0.2316	[+0.1709, +0.2925]	+0.2413	[+0.1484, +0.3453]	85	+0.1750	[+0.1197, +0.2341]	+0.1780	[+0.1013, +0.2633]
AE1SVM	85	+0.0813	[+0.0004, +0.1744]	+0.0218	[-0.0900, +0.1512]	85	+0.0325	[-0.0633, +0.1341]	-0.0384	[-0.1416, +0.0668]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	53	+0.3232	[+0.1930, +0.4427]	+0.2482	[+0.1335, +0.3829]	85	+0.3011	[+0.1993, +0.4066]	+0.2171	[+0.1365, +0.3030]
LUNAR	85	-0.0098	[-0.0580, +0.0339]	-0.0349	[-0.1179, +0.0350]	85	-0.0223	[-0.0860, +0.0336]	-0.0513	[-0.1387, +0.0255]

Table 148: NSA/deterministic versus all 43 other registered detectors, PCA(2), 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	85	+0.0116	[-0.0938, +0.1149]	+0.0480	[-0.0439, +0.1529]	85	-0.0106	[-0.1289, +0.0972]	-0.0309	[-0.1197, +0.0516]
ABOD	85	-0.1472	[-0.2284, -0.0768]	-0.1328	[-0.2378, -0.0487]	85	-0.1371	[-0.2302, -0.0552]	-0.1593	[-0.2693, -0.0631]
COPOD	85	-0.0333	[-0.1357, +0.0687]	+0.0009	[-0.1053, +0.1154]	85	-0.0462	[-0.1576, +0.0621]	-0.0700	[-0.1614, +0.0159]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	85	+0.1026	[+0.0054, +0.1994]	+0.1420	[+0.0613, +0.2416]	85	+0.0771	[+0.0084, +0.1554]	+0.0670	[+0.0205, +0.1185]
QMCD	85	+0.0220	[-0.0510, +0.1003]	+0.0311	[-0.0273, +0.1035]	85	+0.0842	[+0.0093, +0.1678]	+0.0189	[-0.0336, +0.0685]
KDE	85	-0.1491	[-0.2249, -0.0840]	-0.1481	[-0.2540, -0.0636]	85	-0.1430	[-0.2340, -0.0616]	-0.1751	[-0.2796, -0.0812]
Sampling	65	-0.1507	[-0.2507, -0.0585]	-0.1510	[-0.2714, -0.0575]	50	-0.1273	[-0.2387, -0.0344]	-0.1956	[-0.3450, -0.0651]
GMM	85	-0.0140	[-0.1155, +0.0936]	-0.0594	[-0.1965, +0.0717]	85	-0.0337	[-0.1500, +0.0793]	-0.1281	[-0.2588, -0.0222]
PCA	85	-0.0226	[-0.1357, +0.0958]	-0.0521	[-0.2032, +0.0874]	85	-0.0591	[-0.1604, +0.0368]	-0.1364	[-0.2600, -0.0299]
KPCA	85	-0.1486	[-0.2267, -0.0786]	-0.1397	[-0.2459, -0.0605]	85	-0.1374	[-0.2286, -0.0515]	-0.1639	[-0.2680, -0.0729]
MCD	85	-0.0775	[-0.1692, +0.0157]	-0.0569	[-0.2046, +0.0790]	85	-0.0739	[-0.1809, +0.0262]	-0.1043	[-0.2310, -0.0003]
CD	85	+0.0401	[-0.0580, +0.1600]	+0.0270	[-0.0671, +0.1447]	85	+0.0092	[-0.1050, +0.1290]	-0.0444	[-0.1404, +0.0425]
OCSVM	85	-0.1385	[-0.2140, -0.0686]	-0.1403	[-0.2490, -0.0574]	85	-0.1378	[-0.2297, -0.0561]	-0.1657	[-0.2666, -0.0751]
LMDD	85	+0.0404	[-0.0652, +0.1489]	+0.0346	[-0.0859, +0.1568]	85	+0.0266	[-0.1018, +0.1440]	-0.0186	[-0.1119, +0.0669]
LOF	85	-0.0451	[-0.1347, +0.0333]	-0.0683	[-0.1910, +0.0305]	85	-0.0141	[-0.1303, +0.1090]	-0.0832	[-0.1899, +0.0180]
COF	85	+0.0139	[-0.0769, +0.1053]	+0.0728	[-0.0287, +0.1718]	85	+0.0043	[-0.0813, +0.0743]	+0.0158	[-0.0486, +0.0701]
CBLOF	80	-0.1402	[-0.2202, -0.0703]	-0.1390	[-0.2525, -0.0525]	65	-0.1614	[-0.2698, -0.0685]	-0.2107	[-0.3414, -0.0968]
LOCI	28	+0.0550	[-0.0922, +0.2079]	+0.0100	[-0.0898, +0.1120]	30	+0.0257	[-0.1309, +0.1535]	-0.0195	[-0.0999, +0.0573]
HBOS	85	-0.1053	[-0.1787, -0.0399]	-0.0076	[-0.0528, +0.0342]	85	-0.0654	[-0.1539, +0.0168]	-0.0032	[-0.0427, +0.0374]

Continued on next page

Table 148 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
HDBSCAN	85	+0.0350	[-0.0586, +0.1290]	+0.1388	[+0.0689, +0.2239]	85	+0.0416	[-0.0226, +0.1028]	+0.0848	[+0.0437, +0.1295]
KNN	85	-0.1428	[-0.2249, -0.0720]	-0.1339	[-0.2436, -0.0424]	85	-0.1016	[-0.2036, -0.0125]	-0.1518	[-0.2710, -0.0536]
SOD	63	+0.0600	[-0.0202, +0.1362]	+0.0925	[+0.0160, +0.1780]	55	+0.0519	[-0.0333, +0.1293]	+0.0194	[-0.0482, +0.0688]
ROD	85	-0.0356	[-0.1371, +0.0819]	-0.0075	[-0.1208, +0.1158]	85	-0.0182	[-0.1351, +0.0978]	-0.0598	[-0.1528, +0.0340]
IForest	85	-0.0917	[-0.1658, -0.0281]	-0.0163	[-0.0712, +0.0480]	85	-0.0804	[-0.1674, -0.0039]	-0.0610	[-0.1157, -0.0094]
INNE	85	-0.1174	[-0.2001, -0.0461]	-0.1169	[-0.2282, -0.0283]	85	-0.1073	[-0.1995, -0.0306]	-0.1271	[-0.2340, -0.0303]
DIF	83	-0.0970	[-0.1776, -0.0302]	-0.0755	[-0.1985, +0.0233]	85	-0.0875	[-0.1770, -0.0104]	-0.1126	[-0.2199, -0.0272]
FeatureBagging	85	-0.0670	[-0.1546, +0.0156]	-0.0810	[-0.1814, +0.0054]	85	-0.0289	[-0.1425, +0.0789]	-0.1040	[-0.2168, -0.0018]
LSCP	50	-0.1638	[-0.2708, -0.0767]	-0.1461	[-0.3073, -0.0313]	50	-0.1072	[-0.2219, -0.0070]	-0.1425	[-0.2818, -0.0202]
LODA	85	-0.0367	[-0.0812, +0.0111]	+0.0331	[-0.0062, +0.0822]	85	+0.0295	[-0.0293, +0.1004]	+0.0325	[-0.0025, +0.0691]
SUOD	84	-0.1122	[-0.1862, -0.0513]	-0.0593	[-0.1429, +0.0134]	85	-0.0849	[-0.1760, -0.0025]	-0.0768	[-0.1505, -0.0127]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	+0.0077	[-0.1317, +0.1564]	-0.0099	[-0.1864, +0.1584]	0	—	[—, —]	—	[—, —]
VAE	50	+0.0075	[-0.1519, +0.1631]	-0.0298	[-0.2484, +0.1651]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	+0.0590	[-0.0557, +0.1919]	+0.0523	[-0.0282, +0.1493]	85	+0.0496	[-0.0502, +0.1708]	+0.0041	[-0.0424, +0.0645]
MO_GAAL	85	+0.0668	[-0.0571, +0.2031]	+0.0351	[-0.0559, +0.1455]	85	+0.0632	[-0.0312, +0.1686]	-0.0028	[-0.0671, +0.0657]
DeepSVDD	85	-0.0732	[-0.1643, +0.0062]	-0.0540	[-0.1898, +0.0588]	85	-0.0601	[-0.1623, +0.0285]	-0.0869	[-0.2047, +0.0118]
AnoGAN	85	-0.0147	[-0.1078, +0.0897]	+0.0161	[-0.0830, +0.1312]	85	-0.0427	[-0.1602, +0.0800]	-0.0751	[-0.1718, +0.0165]
ALAD	85	+0.1233	[+0.0321, +0.2101]	+0.1617	[+0.0815, +0.2547]	85	+0.0933	[+0.0226, +0.1648]	+0.0826	[+0.0396, +0.1345]
AE1SVM	85	-0.0270	[-0.1280, +0.0845]	-0.0578	[-0.2019, +0.0835]	85	-0.0492	[-0.1672, +0.0679]	-0.1339	[-0.2613, -0.0222]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]

Continued on next page

Table 148 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
RGraph	53	+0.2416	[+0.0900, +0.3937]	+0.1904	[+0.0913, +0.3121]	85	+0.2194	[+0.1098, +0.3387]	+0.1216	[+0.0718, +0.1760]
LUNAR	85	-0.1181	[-0.2011, -0.0414]	-0.1145	[-0.2419, -0.0172]	85	-0.1039	[-0.2007, -0.0216]	-0.1468	[-0.2730, -0.0456]

Table 149: NSA/suppressed versus all 43 other registered detectors, PCA(2), 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	85	+0.1011	[+0.0015, +0.1996]	+0.1135	[+0.0212, +0.2217]	85	+0.0478	[-0.0558, +0.1478]	+0.0448	[-0.0388, +0.1384]
ABOD	85	-0.0578	[-0.1053, -0.0186]	-0.0674	[-0.1508, +0.0019]	85	-0.0787	[-0.1384, -0.0306]	-0.0837	[-0.1731, -0.0098]
COPOD	85	+0.0562	[-0.0379, +0.1536]	+0.0663	[-0.0375, +0.1807]	85	+0.0122	[-0.0855, +0.1049]	+0.0056	[-0.0792, +0.0962]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	85	+0.1921	[+0.1248, +0.2639]	+0.2075	[+0.1117, +0.3194]	85	+0.1355	[+0.0677, +0.2039]	+0.1426	[+0.0699, +0.2255]
QMCD	85	+0.1115	[+0.0380, +0.2018]	+0.0965	[+0.0232, +0.1914]	85	+0.1426	[+0.0558, +0.2371]	+0.0945	[+0.0292, +0.1652]
KDE	85	-0.0596	[-0.1097, -0.0150]	-0.0826	[-0.1737, -0.0143]	85	-0.0845	[-0.1450, -0.0363]	-0.0995	[-0.1879, -0.0293]
Sampling	65	-0.0542	[-0.1162, -0.0066]	-0.0749	[-0.1739, +0.0003]	50	-0.0432	[-0.1010, +0.0068]	-0.0974	[-0.2136, -0.0126]
GMM	85	+0.0755	[-0.0272, +0.1819]	+0.0060	[-0.1269, +0.1303]	85	+0.0247	[-0.0775, +0.1314]	-0.0524	[-0.1663, +0.0532]
PCA	85	+0.0669	[-0.0287, +0.1664]	+0.0134	[-0.1216, +0.1464]	85	-0.0007	[-0.0824, +0.0785]	-0.0608	[-0.1724, +0.0418]
KPCA	85	-0.0591	[-0.1076, -0.0203]	-0.0742	[-0.1620, -0.0111]	85	-0.0790	[-0.1395, -0.0327]	-0.0883	[-0.1722, -0.0238]
MCD	85	+0.0120	[-0.0691, +0.0976]	+0.0085	[-0.1301, +0.1432]	85	-0.0155	[-0.1028, +0.0770]	-0.0287	[-0.1579, +0.0791]
CD	85	+0.1296	[+0.0330, +0.2413]	+0.0924	[-0.0014, +0.2047]	85	+0.0676	[-0.0448, +0.1814]	+0.0312	[-0.0603, +0.1287]
OCSVM	85	-0.0490	[-0.0997, -0.0078]	-0.0748	[-0.1570, -0.0116]	85	-0.0794	[-0.1396, -0.0319]	-0.0901	[-0.1732, -0.0238]
LMDD	85	+0.1299	[+0.0400, +0.2267]	+0.1000	[-0.0181, +0.2253]	85	+0.0851	[-0.0253, +0.1946]	+0.0570	[-0.0386, +0.1508]
LOF	85	+0.0444	[-0.0365, +0.1225]	-0.0029	[-0.1087, +0.0839]	85	+0.0443	[-0.0607, +0.1587]	-0.0076	[-0.1037, +0.0969]
COF	85	+0.1034	[+0.0336, +0.1623]	+0.1382	[+0.0351, +0.2530]	85	+0.0627	[-0.0079, +0.1201]	+0.0914	[+0.0142, +0.1705]
CBLOF	80	-0.0497	[-0.0977, -0.0089]	-0.0703	[-0.1681, -0.0009]	65	-0.0838	[-0.1569, -0.0234]	-0.1205	[-0.2328, -0.0303]
LOCI	28	+0.1108	[-0.0405, +0.2729]	+0.0547	[-0.0404, +0.1549]	30	+0.0521	[-0.0839, +0.2020]	+0.0300	[-0.0596, +0.1269]

Continued on next page

Table 149 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
HBOS	85	-0.0158	[-0.0629, +0.0252]	+0.0578	[+0.0154, +0.1059]	85	-0.0070	[-0.0660, +0.0440]	+0.0724	[+0.0362, +0.1141]
HDBSCAN	85	+0.1245	[+0.0711, +0.1890]	+0.2042	[+0.1203, +0.3053]	85	+0.1000	[+0.0497, +0.1532]	+0.1604	[+0.0950, +0.2362]
KNN	85	-0.0533	[-0.1076, -0.0021]	-0.0684	[-0.1621, +0.0108]	85	-0.0432	[-0.1231, +0.0359]	-0.0762	[-0.1797, +0.0068]
SOD	63	+0.1673	[+0.1130, +0.2167]	+0.1596	[+0.0746, +0.2583]	55	+0.1278	[+0.0928, +0.1657]	+0.1092	[+0.0274, +0.1922]
ROD	85	+0.0539	[-0.0380, +0.1637]	+0.0579	[-0.0503, +0.1827]	85	+0.0402	[-0.0539, +0.1410]	+0.0158	[-0.0737, +0.1167]
IForest	85	-0.0023	[-0.0621, +0.0543]	+0.0491	[-0.0066, +0.1149]	85	-0.0220	[-0.0936, +0.0459]	+0.0146	[-0.0371, +0.0717]
INNE	85	-0.0279	[-0.0850, +0.0248]	-0.0515	[-0.1427, +0.0177]	85	-0.0488	[-0.1068, +0.0035]	-0.0515	[-0.1342, +0.0185]
DIF	83	-0.0042	[-0.0639, +0.0524]	-0.0099	[-0.1166, +0.0776]	85	-0.0291	[-0.0923, +0.0313]	-0.0370	[-0.1280, +0.0435]
FeatureBagging	85	+0.0225	[-0.0553, +0.0979]	-0.0156	[-0.0981, +0.0554]	85	+0.0295	[-0.0673, +0.1304]	-0.0284	[-0.1348, +0.0696]
LSCP	50	-0.0444	[-0.1053, +0.0074]	-0.0626	[-0.1982, +0.0277]	50	-0.0231	[-0.1022, +0.0601]	-0.0444	[-0.1689, +0.0646]
LODA	85	+0.0528	[+0.0150, +0.0961]	+0.0985	[+0.0407, +0.1689]	85	+0.0879	[+0.0355, +0.1371]	+0.1082	[+0.0548, +0.1629]
SUOD	84	-0.0225	[-0.0820, +0.0340]	+0.0059	[-0.0637, +0.0786]	85	-0.0265	[-0.0939, +0.0409]	-0.0012	[-0.0614, +0.0601]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	+0.1271	[-0.0014, +0.2601]	+0.0736	[-0.0870, +0.2317]	0	—	[—, —]	—	[—, —]
VAE	50	+0.1269	[-0.0100, +0.2743]	+0.0537	[-0.1412, +0.2419]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	+0.1485	[+0.0530, +0.2587]	+0.1177	[+0.0341, +0.2208]	85	+0.1080	[+0.0134, +0.2249]	+0.0797	[+0.0208, +0.1610]
MO_GAAL	85	+0.1563	[+0.0580, +0.2725]	+0.1006	[+0.0148, +0.2076]	85	+0.1217	[+0.0372, +0.2182]	+0.0728	[+0.0100, +0.1530]
DeepSVDD	85	+0.0163	[-0.0468, +0.0739]	+0.0114	[-0.1061, +0.1166]	85	-0.0016	[-0.0763, +0.0614]	-0.0112	[-0.1153, +0.0854]
AnoGAN	85	+0.0748	[-0.0126, +0.1641]	+0.0816	[-0.0139, +0.1929]	85	+0.0158	[-0.0872, +0.1345]	+0.0005	[-0.0948, +0.1005]

Continued on next page

Table 149 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ALAD	85	+0.2128	[+0.1528, +0.2710]	+0.2271	[+0.1334, +0.3381]	85	+0.1518	[+0.0977, +0.2124]	+0.1582	[+0.0877, +0.2414]
AE1SVM	85	+0.0625	[-0.0240, +0.1594]	+0.0076	[-0.1215, +0.1389]	85	+0.0093	[-0.0906, +0.1094]	-0.0583	[-0.1722, +0.0508]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	53	+0.3096	[+0.1840, +0.4286]	+0.2383	[+0.1301, +0.3794]	85	+0.2778	[+0.1726, +0.3859]	+0.1973	[+0.1206, +0.2824]
LUNAR	85	-0.0286	[-0.0841, +0.0228]	-0.0491	[-0.1520, +0.0325]	85	-0.0455	[-0.1178, +0.0144]	-0.0712	[-0.1771, +0.0171]

Table 150: NSA/dual versus all 43 other registered detectors, PCA(2), 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	85	+0.1464	[+0.0751, +0.2271]	+0.1794	[+0.0806, +0.2980]	85	+0.1092	[+0.0262, +0.1945]	+0.1248	[+0.0273, +0.2398]
ABOD	85	-0.0125	[-0.0262, -0.0011]	-0.0014	[-0.0147, +0.0117]	85	-0.0173	[-0.0471, +0.0032]	-0.0036	[-0.0219, +0.0161]
COPOD	85	+0.1015	[+0.0367, +0.1777]	+0.1322	[+0.0552, +0.2252]	85	+0.0735	[+0.0107, +0.1433]	+0.0857	[+0.0226, +0.1591]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	85	+0.2374	[+0.1615, +0.3112]	+0.2734	[+0.1576, +0.3991]	85	+0.1969	[+0.0988, +0.2858]	+0.2227	[+0.1059, +0.3488]
QMCD	85	+0.1568	[+0.0572, +0.2812]	+0.1625	[+0.0394, +0.3021]	85	+0.2040	[+0.0846, +0.3356]	+0.1746	[+0.0516, +0.3112]
KDE	85	-0.0143	[-0.0324, +0.0044]	-0.0167	[-0.0401, +0.0033]	85	-0.0232	[-0.0454, -0.0017]	-0.0194	[-0.0468, +0.0050]
Sampling	65	+0.0095	[-0.0177, +0.0302]	+0.0115	[-0.0200, +0.0447]	50	+0.0283	[+0.0143, +0.0435]	+0.0087	[-0.0292, +0.0359]
GMM	85	+0.1208	[+0.0481, +0.2043]	+0.0719	[+0.0123, +0.1588]	85	+0.0861	[-0.0050, +0.1796]	+0.0276	[-0.0250, +0.0880]
PCA	85	+0.1122	[+0.0379, +0.1973]	+0.0793	[+0.0060, +0.1796]	85	+0.0607	[-0.0022, +0.1262]	+0.0193	[-0.0374, +0.0808]
KPCA	85	-0.0138	[-0.0257, -0.0029]	-0.0083	[-0.0284, +0.0098]	85	-0.0176	[-0.0348, -0.0042]	-0.0082	[-0.0380, +0.0201]
MCD	85	+0.0573	[-0.0012, +0.1282]	+0.0744	[+0.0031, +0.1737]	85	+0.0459	[-0.0273, +0.1221]	+0.0514	[-0.0099, +0.1206]
CD	85	+0.1749	[+0.0936, +0.2699]	+0.1583	[+0.0614, +0.2770]	85	+0.1289	[+0.0258, +0.2423]	+0.1112	[+0.0160, +0.2233]
OCSVM	85	-0.0037	[-0.0183, +0.0113]	-0.0089	[-0.0333, +0.0143]	85	-0.0180	[-0.0362, -0.0024]	-0.0100	[-0.0411, +0.0211]
LMDD	85	+0.1752	[+0.1091, +0.2548]	+0.1659	[+0.0789, +0.2756]	85	+0.1464	[+0.0516, +0.2449]	+0.1371	[+0.0594, +0.2244]
LOF	85	+0.0897	[+0.0395, +0.1436]	+0.0630	[+0.0201, +0.1083]	85	+0.1057	[+0.0126, +0.2065]	+0.0725	[+0.0045, +0.1566]
COF	85	+0.1487	[+0.0904, +0.2085]	+0.2041	[+0.0960, +0.3235]	85	+0.1241	[+0.0512, +0.1945]	+0.1715	[+0.0729, +0.2801]

Continued on next page

Table 150 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
CBLOF	80	-0.0012	[-0.0180, +0.0142]	-0.0068	[-0.0261, +0.0124]	65	+0.0033	[-0.0103, +0.0165]	+0.0001	[-0.0395, +0.0323]
LOCI	28	+0.1532	[+0.0207, +0.2938]	+0.1204	[+0.0104, +0.2429]	30	+0.1069	[-0.0586, +0.2709]	+0.0788	[-0.0672, +0.2326]
HBOS	85	+0.0295	[+0.0155, +0.0437]	+0.1237	[+0.0516, +0.2174]	85	+0.0544	[+0.0216, +0.0848]	+0.1525	[+0.0726, +0.2523]
HDBSCAN	85	+0.1698	[+0.1137, +0.2315]	+0.2701	[+0.1589, +0.3973]	85	+0.1614	[+0.0830, +0.2397]	+0.2405	[+0.1257, +0.3634]
KNN	85	-0.0080	[-0.0294, +0.0151]	-0.0025	[-0.0297, +0.0256]	85	+0.0182	[-0.0496, +0.0903]	+0.0039	[-0.0343, +0.0383]
SOD	63	+0.2068	[+0.1549, +0.2658]	+0.2232	[+0.1232, +0.3324]	55	+0.1928	[+0.1456, +0.2408]	+0.2064	[+0.1049, +0.3180]
ROD	85	+0.0992	[+0.0263, +0.1937]	+0.1238	[+0.0390, +0.2322]	85	+0.1016	[+0.0304, +0.1835]	+0.0959	[+0.0284, +0.1795]
IForest	85	+0.0430	[+0.0077, +0.0771]	+0.1150	[+0.0364, +0.2053]	85	+0.0394	[-0.0207, +0.0961]	+0.0947	[+0.0166, +0.1804]
INNE	85	+0.0174	[-0.0057, +0.0435]	+0.0144	[-0.0185, +0.0529]	85	+0.0125	[-0.0219, +0.0456]	+0.0286	[-0.0051, +0.0668]
DIF	83	+0.0402	[+0.0112, +0.0692]	+0.0549	[-0.0028, +0.1195]	85	+0.0323	[-0.0169, +0.0817]	+0.0431	[-0.0186, +0.1200]
FeatureBagging	85	+0.0678	[+0.0227, +0.1190]	+0.0503	[+0.0113, +0.0907]	85	+0.0909	[+0.0040, +0.1804]	+0.0517	[-0.0097, +0.1186]
LSCP	50	+0.0037	[-0.0107, +0.0187]	+0.0090	[-0.0175, +0.0353]	50	+0.0484	[-0.0028, +0.1021]	+0.0617	[+0.0109, +0.1250]
LODA	85	+0.0981	[+0.0496, +0.1496]	+0.1644	[+0.0599, +0.2878]	85	+0.1493	[+0.0729, +0.2329]	+0.1882	[+0.0837, +0.3076]
SUOD	84	+0.0226	[-0.0040, +0.0520]	+0.0719	[+0.0188, +0.1364]	85	+0.0349	[-0.0117, +0.0862]	+0.0789	[+0.0190, +0.1582]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	+0.1752	[+0.0761, +0.2852]	+0.1452	[+0.0635, +0.2569]	0	—	[—, —]	—	[—, —]
VAE	50	+0.1750	[+0.0780, +0.2921]	+0.1253	[+0.0402, +0.2593]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	+0.1938	[+0.1005, +0.3005]	+0.1836	[+0.0822, +0.2985]	85	+0.1693	[+0.0667, +0.2777]	+0.1598	[+0.0572, +0.2726]
MO_GAAL	85	+0.2016	[+0.1059, +0.3151]	+0.1665	[+0.0749, +0.2749]	85	+0.1830	[+0.0870, +0.2831]	+0.1529	[+0.0682, +0.2512]

Continued on next page

Table 150 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
DeepSVDD	85	+0.0616	[+0.0318, +0.0939]	+0.0774	[+0.0323, +0.1457]	85	+0.0597	[+0.0230, +0.0995]	+0.0688	[+0.0267, +0.1237]
AnoGAN	85	+0.1201	[+0.0593, +0.1960]	+0.1475	[+0.0520, +0.2682]	85	+0.0771	[-0.0119, +0.1783]	+0.0806	[-0.0090, +0.1877]
ALAD	85	+0.2581	[+0.1958, +0.3186]	+0.2931	[+0.1748, +0.4238]	85	+0.2131	[+0.1375, +0.2846]	+0.2383	[+0.1209, +0.3687]
AE1SVM	85	+0.1078	[+0.0403, +0.1927]	+0.0735	[-0.0058, +0.1765]	85	+0.0706	[-0.0106, +0.1619]	+0.0218	[-0.0421, +0.0893]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	53	+0.3398	[+0.1965, +0.4878]	+0.2762	[+0.1430, +0.4204]	85	+0.3392	[+0.2128, +0.4693]	+0.2773	[+0.1566, +0.4036]
LUNAR	85	+0.0167	[-0.0073, +0.0412]	+0.0168	[-0.0089, +0.0429]	85	+0.0159	[-0.0257, +0.0588]	+0.0089	[-0.0198, +0.0392]

Table 151: NSA/annealed versus all 43 other registered detectors, PCA(2), 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	85	+0.0079	[-0.0885, +0.1016]	+0.0457	[-0.0473, +0.1434]	85	-0.0206	[-0.1251, +0.0805]	-0.0320	[-0.1162, +0.0473]
ABOD	85	-0.1509	[-0.2254, -0.0815]	-0.1352	[-0.2395, -0.0528]	85	-0.1471	[-0.2344, -0.0694]	-0.1604	[-0.2656, -0.0679]
COPOD	85	-0.0369	[-0.1344, +0.0617]	-0.0015	[-0.1073, +0.0986]	85	-0.0563	[-0.1645, +0.0453]	-0.0711	[-0.1614, +0.0089]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	85	+0.0990	[+0.0028, +0.2006]	+0.1397	[+0.0575, +0.2303]	85	+0.0671	[-0.0081, +0.1500]	+0.0658	[+0.0204, +0.1193]
QMCD	85	+0.0183	[-0.0607, +0.1016]	+0.0288	[-0.0318, +0.1017]	85	+0.0742	[-0.0075, +0.1616]	+0.0178	[-0.0261, +0.0671]
KDE	85	-0.1528	[-0.2284, -0.0886]	-0.1504	[-0.2558, -0.0662]	85	-0.1530	[-0.2385, -0.0787]	-0.1762	[-0.2864, -0.0812]
Sampling	65	-0.1566	[-0.2509, -0.0664]	-0.1540	[-0.2718, -0.0615]	50	-0.1433	[-0.2461, -0.0500]	-0.1944	[-0.3429, -0.0637]
GMM	85	-0.0176	[-0.1143, +0.0786]	-0.0618	[-0.2021, +0.0635]	85	-0.0437	[-0.1521, +0.0599]	-0.1292	[-0.2552, -0.0252]
PCA	85	-0.0263	[-0.1313, +0.0828]	-0.0544	[-0.1903, +0.0763]	85	-0.0691	[-0.1654, +0.0144]	-0.1375	[-0.2670, -0.0336]
KPCA	85	-0.1523	[-0.2275, -0.0867]	-0.1420	[-0.2454, -0.0628]	85	-0.1475	[-0.2332, -0.0686]	-0.1650	[-0.2695, -0.0760]
MCD	85	-0.0812	[-0.1709, +0.0028]	-0.0593	[-0.2030, +0.0709]	85	-0.0839	[-0.1841, +0.0120]	-0.1055	[-0.2335, -0.0064]
CD	85	+0.0364	[-0.0577, +0.1476]	+0.0246	[-0.0702, +0.1277]	85	-0.0009	[-0.1082, +0.1183]	-0.0456	[-0.1327, +0.0388]
OCSVM	85	-0.1422	[-0.2158, -0.0757]	-0.1426	[-0.2458, -0.0630]	85	-0.1479	[-0.2317, -0.0708]	-0.1668	[-0.2730, -0.0776]
LMDD	85	+0.0367	[-0.0620, +0.1390]	+0.0322	[-0.0802, +0.1487]	85	+0.0166	[-0.0936, +0.1251]	-0.0197	[-0.1128, +0.0583]
LOF	85	-0.0488	[-0.1335, +0.0259]	-0.0707	[-0.1880, +0.0214]	85	-0.0241	[-0.1357, +0.0900]	-0.0843	[-0.1861, +0.0124]
COF	85	+0.0102	[-0.0834, +0.1002]	+0.0704	[-0.0263, +0.1656]	85	-0.0057	[-0.0900, +0.0695]	+0.0147	[-0.0504, +0.0711]
CBLOF	80	-0.1411	[-0.2199, -0.0720]	-0.1379	[-0.2410, -0.0540]	65	-0.1718	[-0.2740, -0.0812]	-0.2112	[-0.3353, -0.0990]
LOCI	28	+0.0597	[-0.0907, +0.2175]	+0.0169	[-0.0916, +0.1381]	30	+0.0266	[-0.1162, +0.1525]	-0.0218	[-0.1067, +0.0609]
HBOS	85	-0.1090	[-0.1795, -0.0464]	-0.0100	[-0.0514, +0.0271]	85	-0.0754	[-0.1551, -0.0012]	-0.0043	[-0.0430, +0.0319]

Continued on next page

Table 151 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
HDBSCAN	85	+0.0313	[-0.0628, +0.1200]	+0.1364	[+0.0696, +0.2177]	85	+0.0316	[-0.0334, +0.0992]	+0.0836	[+0.0446, +0.1283]
KNN	85	-0.1465	[-0.2285, -0.0730]	-0.1362	[-0.2498, -0.0447]	85	-0.1116	[-0.2067, -0.0182]	-0.1530	[-0.2680, -0.0524]
SOD	63	+0.0590	[-0.0233, +0.1369]	+0.0901	[+0.0146, +0.1675]	55	+0.0359	[-0.0538, +0.1253]	+0.0189	[-0.0518, +0.0743]
ROD	85	-0.0393	[-0.1384, +0.0692]	-0.0099	[-0.1201, +0.1036]	85	-0.0282	[-0.1380, +0.0850]	-0.0609	[-0.1522, +0.0254]
IForest	85	-0.0954	[-0.1604, -0.0347]	-0.0187	[-0.0711, +0.0363]	85	-0.0904	[-0.1688, -0.0207]	-0.0621	[-0.1141, -0.0154]
INNE	85	-0.1211	[-0.2035, -0.0487]	-0.1193	[-0.2279, -0.0342]	85	-0.1173	[-0.2003, -0.0464]	-0.1283	[-0.2336, -0.0345]
DIF	83	-0.1014	[-0.1784, -0.0354]	-0.0787	[-0.2013, +0.0147]	85	-0.0975	[-0.1760, -0.0313]	-0.1137	[-0.2217, -0.0314]
FeatureBagging	85	-0.0707	[-0.1576, +0.0078]	-0.0834	[-0.1809, +0.0029]	85	-0.0389	[-0.1448, +0.0620]	-0.1052	[-0.2159, -0.0095]
LSCP	50	-0.1720	[-0.2738, -0.0902]	-0.1531	[-0.3153, -0.0379]	50	-0.1232	[-0.2214, -0.0376]	-0.1413	[-0.2743, -0.0309]
LODA	85	-0.0404	[-0.0826, +0.0042]	+0.0307	[-0.0086, +0.0799]	85	+0.0195	[-0.0373, +0.0876]	+0.0314	[+0.0005, +0.0669]
SUOD	84	-0.1161	[-0.1912, -0.0568]	-0.0621	[-0.1318, +0.0043]	85	-0.0949	[-0.1793, -0.0236]	-0.0779	[-0.1470, -0.0167]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AutoEncoder	50	-0.0005	[-0.1328, +0.1354]	-0.0169	[-0.1808, +0.1284]	0	—	[—, —]	—	[—, —]
VAE	50	-0.0007	[-0.1449, +0.1447]	-0.0368	[-0.2463, +0.1355]	0	—	[—, —]	—	[—, —]
SO_GAAL	85	+0.0554	[-0.0529, +0.1795]	+0.0499	[-0.0285, +0.1415]	85	+0.0395	[-0.0560, +0.1548]	+0.0030	[-0.0399, +0.0569]
MO_GAAL	85	+0.0631	[-0.0546, +0.1968]	+0.0328	[-0.0544, +0.1379]	85	+0.0532	[-0.0383, +0.1579]	-0.0039	[-0.0634, +0.0589]
DeepSVDD	85	-0.0769	[-0.1612, -0.0006]	-0.0564	[-0.1833, +0.0472]	85	-0.0701	[-0.1636, +0.0104]	-0.0880	[-0.1960, +0.0060]
AnoGAN	85	-0.0183	[-0.1054, +0.0755]	+0.0138	[-0.0814, +0.1202]	85	-0.0527	[-0.1581, +0.0650]	-0.0762	[-0.1694, +0.0128]
ALAD	85	+0.1196	[+0.0348, +0.2053]	+0.1594	[+0.0819, +0.2482]	85	+0.0833	[+0.0119, +0.1615]	+0.0814	[+0.0404, +0.1313]
AE1SVM	85	-0.0307	[-0.1268, +0.0734]	-0.0602	[-0.1977, +0.0705]	85	-0.0592	[-0.1710, +0.0484]	-0.1350	[-0.2619, -0.0279]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]

Continued on next page

Table 151 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
RGraph	53	+0.2332	[+0.0778, +0.3895]	+0.1869	[+0.0862, +0.3002]	85	+0.2094	[+0.0913, +0.3356]	+0.1205	[+0.0714, +0.1749]
LUNAR	85	-0.1218	[-0.2051, -0.0452]	-0.1169	[-0.2379, -0.0227]	85	-0.1140	[-0.2117, -0.0346]	-0.1479	[-0.2661, -0.0487]

Table 152: NSA/binary/hamming versus all 43 other registered detectors, PCA(2), 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
ABOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
COPOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
QMCD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
KDE	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
Sampling	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
GMM	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
PCA	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
KPCA	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
MCD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
CD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
OCSVM	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LMDD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LOF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
COF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
CBLOF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LOCI	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
HBOS	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
HDBSCAN	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
KNN	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
ROD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
IForest	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
INNE	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
DIF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
FeatureBagging	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LSCP	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LODA	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SUOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]

Continued on next page

Table 152 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
AutoEncoder	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
VAE	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SO_GAAL	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
MO_GAAL	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
DeepSVDD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AnoGAN	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
ALAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AE1SVM	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LUNAR	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]

Table 153: NSA/binary/rcontiguous versus all 43 other registered detectors, PCA(2), 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
ABOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
COPOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
QMCD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
KDE	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
Sampling	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
GMM	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
PCA	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
KPCA	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
MCD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
CD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
OCSVM	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LMDD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LOF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
COF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
CBLOF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LOCI	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
HBOS	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
HDBSCAN	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
KNN	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
ROD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
IForest	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
INNE	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
DIF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
FeatureBagging	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LSCP	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LODA	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SUOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]

Continued on next page

Table 153 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
AutoEncoder	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
VAE	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SO_GAAL	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
MO_GAAL	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
DeepSVDD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AnoGAN	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
ALAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AE1SVM	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LUNAR	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]

Table 154: NSA/binary/rchunk versus all 43 other registered detectors, PCA(2), 10 and 5 percent retained normals. Each paired difference has a dataset-cluster 95 percent interval; P is matched cases out of 85.

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
ECOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
ABOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
COPOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
MAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOS	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
QMCD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
KDE	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
Sampling	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
GMM	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
PCA	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
KPCA	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
MCD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
CD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
OCSVM	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LMDD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LOF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
COF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
CBLOF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LOCI	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
HBOS	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
HDBSCAN	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
KNN	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
ROD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
IForest	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
INNE	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
DIF	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
FeatureBagging	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LSCP	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LODA	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SUOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
XGBOD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]

Continued on next page

Table 154 continued

Comparator	10% P	10% ΔAUC	10% CI	10% ΔAP	10% CI	5% P	5% ΔAUC	5% CI	5% ΔAP	5% CI
AutoEncoder	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
VAE	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
SO_GAAL	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
MO_GAAL	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
DeepSVDD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AnoGAN	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
ALAD	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
AE1SVM	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
DevNet	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
RGraph	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]
LUNAR	0	—	[—, —]	—	[—, —]	0	—	[—, —]	—	[—, —]

Q Relative gains with the same comparator held fixed

The following contrasts use four matched successful outcomes per dataset/seed: the NSA and comparator at the smaller size and the same two configurations at full retention. A positive gap change favors NSA's relative standing under reduced training availability. This distinguishes closing a deficit from becoming the best detector. All 2,236 comparator contrasts are available in the accompanying machine-readable results.

Table 155: All NSA relative-gap changes versus IForest and KNN, Original features. Positive means NSA gains relative to the same comparator as training data shrink. All intervals are descriptive dataset-cluster 95 percent intervals.

NSA configuration	Comparator	Retained	Matched	Δ AUC gap	95% CI	Δ AP gap	95% CI
NSA fixed	KNN	10%	85/85	-0.0200	[-0.0696, +0.0309]	+0.0621	[+0.0108, +0.1241]
NSA fixed	IForest	10%	85/85	-0.0103	[-0.0545, +0.0350]	+0.1108	[+0.0483, +0.1891]
NSA variable	KNN	10%	85/85	-0.1167	[-0.2445, -0.0038]	-0.0603	[-0.1804, +0.0455]
NSA variable	IForest	10%	85/85	-0.1069	[-0.2219, -0.0067]	-0.0116	[-0.1147, +0.0747]
NSA coverage	KNN	10%	85/85	-0.1184	[-0.2448, -0.0065]	-0.0605	[-0.1884, +0.0415]
NSA coverage	IForest	10%	85/85	-0.1086	[-0.2233, -0.0054]	-0.0118	[-0.1177, +0.0747]
NSA grid	KNN	10%	85/85	-0.1234	[-0.2467, -0.0144]	-0.0637	[-0.1895, +0.0407]
NSA grid	IForest	10%	85/85	-0.1137	[-0.2278, -0.0133]	-0.0151	[-0.1268, +0.0754]
NSA hierarchical	KNN	10%	85/85	-0.0102	[-0.0665, +0.0424]	+0.0673	[+0.0112, +0.1332]
NSA hierarchical	IForest	10%	85/85	-0.0004	[-0.0673, +0.0604]	+0.1160	[+0.0481, +0.1978]
NSA voronoi	KNN	10%	0/85	—	[—, —]	—	[—, —]
NSA voronoi	IForest	10%	0/85	—	[—, —]	—	[—, —]
NSA deterministic	KNN	10%	0/85	—	[—, —]	—	[—, —]
NSA deterministic	IForest	10%	0/85	—	[—, —]	—	[—, —]
NSA suppressed	KNN	10%	85/85	-0.1103	[-0.2294, -0.0044]	-0.0554	[-0.1685, +0.0471]
NSA suppressed	IForest	10%	85/85	-0.1006	[-0.2082, -0.0019]	-0.0067	[-0.1020, +0.0725]
NSA dual	KNN	10%	81/85	+0.1175	[+0.0727, +0.1634]	+0.1315	[+0.0753, +0.1957]
NSA dual	IForest	10%	81/85	+0.1270	[+0.0728, +0.1807]	+0.1777	[+0.0946, +0.2736]
NSA annealed	KNN	10%	85/85	-0.0105	[-0.0639, +0.0422]	+0.0735	[+0.0218, +0.1357]
NSA annealed	IForest	10%	85/85	-0.0008	[-0.0446, +0.0420]	+0.1222	[+0.0626, +0.1913]
NSA binary Hamming	KNN	10%	76/85	+0.0193	[-0.0147, +0.0574]	+0.0546	[-0.0330, +0.1321]
NSA binary Hamming	IForest	10%	76/85	+0.0300	[-0.0019, +0.0668]	+0.1050	[+0.0138, +0.1933]
NSA binary contiguous	KNN	10%	59/85	+0.0293	[-0.0395, +0.1012]	+0.0469	[-0.0314, +0.1334]

Continued on next page

Table 155 continued

NSA configuration	Comparator	Retained	Matched	Δ AUC gap	95% CI	Δ AP gap	95% CI
NSA binary contiguous	IForest	10%	59/85	+0.0464	[-0.0121, +0.1130]	+0.1425	[+0.0765, +0.2135]
NSA binary chunk	KNN	10%	75/85	+0.0517	[+0.0084, +0.0974]	+0.0754	[+0.0167, +0.1483]
NSA binary chunk	IForest	10%	75/85	+0.0641	[+0.0264, +0.1025]	+0.1330	[+0.0679, +0.2104]
NSA fixed	KNN	5%	85/85	+0.0223	[-0.0535, +0.1084]	+0.1136	[+0.0450, +0.1932]
NSA fixed	IForest	5%	85/85	+0.0191	[-0.0411, +0.0816]	+0.1629	[+0.0936, +0.2432]
NSA variable	KNN	5%	85/85	-0.1218	[-0.2855, +0.0314]	-0.0280	[-0.1702, +0.0964]
NSA variable	IForest	5%	85/85	-0.1250	[-0.2614, +0.0038]	+0.0213	[-0.0961, +0.1248]
NSA coverage	KNN	5%	85/85	-0.1243	[-0.2865, +0.0317]	-0.0328	[-0.1728, +0.0911]
NSA coverage	IForest	5%	85/85	-0.1276	[-0.2664, +0.0046]	+0.0165	[-0.1000, +0.1187]
NSA grid	KNN	5%	85/85	-0.1281	[-0.2881, +0.0179]	-0.0360	[-0.1806, +0.0795]
NSA grid	IForest	5%	85/85	-0.1313	[-0.2742, -0.0070]	+0.0133	[-0.1065, +0.1217]
NSA hierarchical	KNN	5%	85/85	+0.0990	[+0.0156, +0.1891]	+0.1268	[+0.0549, +0.2064]
NSA hierarchical	IForest	5%	85/85	+0.0958	[+0.0229, +0.1731]	+0.1761	[+0.1068, +0.2502]
NSA voronoi	KNN	5%	0/85	—	[—, —]	—	[—, —]
NSA voronoi	IForest	5%	0/85	—	[—, —]	—	[—, —]
NSA deterministic	KNN	5%	0/85	—	[—, —]	—	[—, —]
NSA deterministic	IForest	5%	0/85	—	[—, —]	—	[—, —]
NSA suppressed	KNN	5%	85/85	-0.1211	[-0.2814, +0.0249]	-0.0251	[-0.1652, +0.1013]
NSA suppressed	IForest	5%	85/85	-0.1244	[-0.2604, +0.0025]	+0.0242	[-0.0932, +0.1235]
NSA dual	KNN	5%	81/85	+0.1759	[+0.1201, +0.2296]	+0.1721	[+0.0991, +0.2581]
NSA dual	IForest	5%	81/85	+0.1723	[+0.1053, +0.2361]	+0.2202	[+0.1317, +0.3221]
NSA annealed	KNN	5%	85/85	+0.0272	[-0.0402, +0.0909]	+0.1053	[+0.0423, +0.1809]
NSA annealed	IForest	5%	85/85	+0.0240	[-0.0211, +0.0739]	+0.1546	[+0.0804, +0.2292]
NSA binary Hamming	KNN	5%	76/85	+0.0605	[-0.0033, +0.1236]	+0.0861	[-0.0304, +0.1862]
NSA binary Hamming	IForest	5%	76/85	+0.0658	[+0.0097, +0.1210]	+0.1408	[+0.0328, +0.2344]
NSA binary contiguous	KNN	5%	59/85	+0.0524	[-0.0265, +0.1349]	+0.0928	[+0.0318, +0.1715]
NSA binary contiguous	IForest	5%	59/85	+0.0781	[+0.0093, +0.1499]	+0.1985	[+0.1262, +0.2746]
NSA binary chunk	KNN	5%	75/85	+0.0873	[+0.0238, +0.1610]	+0.1136	[+0.0349, +0.1973]
NSA binary chunk	IForest	5%	75/85	+0.0945	[+0.0420, +0.1483]	+0.1766	[+0.1044, +0.2603]

Table 156: All NSA relative-gap changes versus IForest and KNN, PCA(2). Positive means NSA gains relative to the same comparator as training data shrink. All intervals are descriptive dataset-cluster 95 percent intervals.

NSA configuration	Comparator	Retained	Matched	Δ AUC gap	95% CI	Δ AP gap	95% CI
NSA fixed	KNN	10%	85/85	+0.0557	[-0.0304, +0.1548]	+0.0003	[-0.0900, +0.0901]
NSA fixed	IForest	10%	85/85	+0.0702	[-0.0034, +0.1494]	+0.0329	[-0.0248, +0.0940]
NSA variable	KNN	10%	85/85	+0.0598	[-0.0136, +0.1413]	-0.0163	[-0.0938, +0.0492]
NSA variable	IForest	10%	85/85	+0.0742	[+0.0097, +0.1490]	+0.0163	[-0.0102, +0.0434]
NSA coverage	KNN	10%	85/85	+0.0669	[-0.0149, +0.1564]	-0.0042	[-0.0784, +0.0636]
NSA coverage	IForest	10%	85/85	+0.0814	[+0.0113, +0.1679]	+0.0284	[+0.0007, +0.0541]
NSA grid	KNN	10%	85/85	+0.0610	[-0.0113, +0.1397]	-0.0052	[-0.0708, +0.0587]
NSA grid	IForest	10%	85/85	+0.0754	[+0.0127, +0.1463]	+0.0274	[+0.0000, +0.0550]
NSA hierarchical	KNN	10%	85/85	+0.0390	[-0.0423, +0.1270]	-0.0261	[-0.0941, +0.0392]
NSA hierarchical	IForest	10%	85/85	+0.0534	[-0.0189, +0.1376]	+0.0065	[-0.0291, +0.0389]
NSA voronoi	KNN	10%	85/85	+0.0571	[-0.0146, +0.1370]	-0.0192	[-0.0844, +0.0491]
NSA voronoi	IForest	10%	85/85	+0.0715	[+0.0103, +0.1440]	+0.0134	[-0.0320, +0.0553]
NSA deterministic	KNN	10%	85/85	+0.0499	[-0.0325, +0.1463]	-0.0061	[-0.0994, +0.0845]
NSA deterministic	IForest	10%	85/85	+0.0644	[-0.0066, +0.1407]	+0.0265	[-0.0312, +0.0829]
NSA suppressed	KNN	10%	85/85	+0.0687	[-0.0060, +0.1559]	-0.0026	[-0.0865, +0.0751]
NSA suppressed	IForest	10%	85/85	+0.0832	[+0.0177, +0.1595]	+0.0300	[-0.0166, +0.0712]
NSA dual	KNN	10%	85/85	+0.0707	[+0.0132, +0.1355]	+0.0129	[-0.0306, +0.0566]
NSA dual	IForest	10%	85/85	+0.0851	[+0.0349, +0.1461]	+0.0455	[-0.0086, +0.1091]
NSA annealed	KNN	10%	85/85	+0.0496	[-0.0397, +0.1573]	-0.0162	[-0.1110, +0.0812]
NSA annealed	IForest	10%	85/85	+0.0641	[-0.0129, +0.1550]	+0.0164	[-0.0430, +0.0821]
NSA binary Hamming	KNN	10%	0/85	—	[—, —]	—	[—, —]
NSA binary Hamming	IForest	10%	0/85	—	[—, —]	—	[—, —]
NSA binary contiguous	KNN	10%	0/85	—	[—, —]	—	[—, —]
NSA binary contiguous	IForest	10%	0/85	—	[—, —]	—	[—, —]
NSA binary chunk	KNN	10%	0/85	—	[—, —]	—	[—, —]
NSA binary chunk	IForest	10%	0/85	—	[—, —]	—	[—, —]
NSA fixed	KNN	5%	85/85	+0.0966	[-0.0285, +0.2359]	-0.0072	[-0.1034, +0.0962]

Continued on next page

Table 156 continued

NSA configuration	Comparator	Retained	Matched	Δ AUC gap	95% CI	Δ AP gap	95% CI
NSA fixed	IForest	5%	85/85	+0.0813	[-0.0095, +0.1698]	-0.0013	[-0.0652, +0.0578]
NSA variable	KNN	5%	85/85	+0.0631	[-0.0395, +0.1744]	-0.0407	[-0.1239, +0.0356]
NSA variable	IForest	5%	85/85	+0.0478	[-0.0361, +0.1415]	-0.0348	[-0.0753, +0.0023]
NSA coverage	KNN	5%	85/85	+0.0851	[-0.0267, +0.2104]	-0.0260	[-0.1183, +0.0606]
NSA coverage	IForest	5%	85/85	+0.0698	[-0.0206, +0.1711]	-0.0202	[-0.0679, +0.0226]
NSA grid	KNN	5%	85/85	+0.0766	[-0.0334, +0.1917]	-0.0203	[-0.1019, +0.0620]
NSA grid	IForest	5%	85/85	+0.0612	[-0.0188, +0.1536]	-0.0145	[-0.0557, +0.0214]
NSA hierarchical	KNN	5%	85/85	+0.0701	[-0.0278, +0.1729]	-0.0416	[-0.1175, +0.0315]
NSA hierarchical	IForest	5%	85/85	+0.0548	[-0.0197, +0.1456]	-0.0357	[-0.0834, +0.0081]
NSA voronoi	KNN	5%	85/85	+0.0716	[-0.0314, +0.1774]	-0.0213	[-0.0997, +0.0566]
NSA voronoi	IForest	5%	85/85	+0.0562	[-0.0222, +0.1461]	-0.0154	[-0.0655, +0.0285]
NSA deterministic	KNN	5%	85/85	+0.0911	[-0.0211, +0.2085]	-0.0241	[-0.1239, +0.0806]
NSA deterministic	IForest	5%	85/85	+0.0757	[-0.0051, +0.1582]	-0.0182	[-0.0920, +0.0479]
NSA suppressed	KNN	5%	85/85	+0.0788	[-0.0194, +0.1830]	-0.0104	[-0.0967, +0.0732]
NSA suppressed	IForest	5%	85/85	+0.0635	[-0.0124, +0.1560]	-0.0045	[-0.0440, +0.0349]
NSA dual	KNN	5%	85/85	+0.0968	[-0.0005, +0.1970]	+0.0193	[-0.0465, +0.0769]
NSA dual	IForest	5%	85/85	+0.0815	[+0.0097, +0.1640]	+0.0252	[-0.0414, +0.0897]
NSA annealed	KNN	5%	85/85	+0.0844	[-0.0334, +0.2221]	-0.0330	[-0.1354, +0.0701]
NSA annealed	IForest	5%	85/85	+0.0691	[-0.0142, +0.1586]	-0.0271	[-0.0975, +0.0348]
NSA binary Hamming	KNN	5%	0/85	—	[—, —]	—	[—, —]
NSA binary Hamming	IForest	5%	0/85	—	[—, —]	—	[—, —]
NSA binary contiguous	KNN	5%	0/85	—	[—, —]	—	[—, —]
NSA binary contiguous	IForest	5%	0/85	—	[—, —]	—	[—, —]
NSA binary chunk	KNN	5%	0/85	—	[—, —]	—	[—, —]
NSA binary chunk	IForest	5%	0/85	—	[—, —]	—	[—, —]

R New-run failure and timing accounting

Table 157: Outcome totals for the 19,040 additional detector records after measured-time qualification.

Outcome	Records	Share
success	15739	82.66%
overbudget success	26	0.14%
timeout	511	2.68%
error	1334	7.01%
build error	70	0.37%
requires labeled outliers	680	3.57%
requires 1d	340	1.79%
requires 2d or 3d	170	0.89%
requires 2d	170	0.89%

Table 158: New-run outcomes by configuration, 340 expected records each. S=qualified success, O=over-budget success, T=timeout, E=error, B=build error, L=labeled-outlier requirement; final columns are dimension restrictions.

Configuration	S	O	T	E	B	L	1D	2/3D	2D
NSA fixed	340	0	0	0	0	0	0	0	0
NSA variable	340	0	0	0	0	0	0	0	0
NSA coverage	340	0	0	0	0	0	0	0	0
NSA grid	340	0	0	0	0	0	0	0	0
NSA hierarchical	340	0	0	0	0	0	0	0	0
NSA voronoi	170	0	0	0	0	0	0	170	0
NSA deterministic	170	0	0	0	0	0	0	0	170
NSA suppressed	340	0	0	0	0	0	0	0	0
NSA dual	340	0	0	0	0	0	0	0	0
NSA annealed	340	0	0	0	0	0	0	0	0
NSA binary Hamming	170	0	0	170	0	0	0	0	0
NSA binary contiguous	150	0	0	190	0	0	0	0	0
NSA binary chunk	162	0	0	178	0	0	0	0	0
ECOD	332	8	0	0	0	0	0	0	0
ABOD	340	0	0	0	0	0	0	0	0
COPOD	340	0	0	0	0	0	0	0	0
MAD	0	0	0	0	0	0	340	0	0
SOS	339	0	0	1	0	0	0	0	0
QMCD	340	0	0	0	0	0	0	0	0
KDE	340	0	0	0	0	0	0	0	0
Sampling	230	0	0	110	0	0	0	0	0
GMM	340	0	0	0	0	0	0	0	0
PCA	340	0	0	0	0	0	0	0	0
KPCA	340	0	0	0	0	0	0	0	0
MCD	340	0	0	0	0	0	0	0	0
CD	327	0	1	12	0	0	0	0	0
OCSVM	340	0	0	0	0	0	0	0	0
LMDD	277	0	63	0	0	0	0	0	0
LOF	340	0	0	0	0	0	0	0	0

Continued on next page

Table 158 continued

Configuration	S	O	T	E	B	L	1D	2/3D	2D
COF	340	0	0	0	0	0	0	0	0
CBLOF	290	0	0	50	0	0	0	0	0
LOCI	117	0	223	0	0	0	0	0	0
HBOS	340	0	0	0	0	0	0	0	0
HDBSCAN	340	0	0	0	0	0	0	0	0
KNN	340	0	0	0	0	0	0	0	0
SOD	231	14	25	0	70	0	0	0	0
ROD	251	2	86	1	0	0	0	0	0
IForest	340	0	0	0	0	0	0	0	0
INNE	340	0	0	0	0	0	0	0	0
DIF	338	2	0	0	0	0	0	0	0
FeatureBagging	340	0	0	0	0	0	0	0	0
LSCP	200	0	0	140	0	0	0	0	0
LODA	340	0	0	0	0	0	0	0	0
SUOD	338	0	0	2	0	0	0	0	0
XGBOD	0	0	0	0	0	340	0	0	0
AutoEncoder	100	0	0	240	0	0	0	0	0
VAE	100	0	0	240	0	0	0	0	0
SO_GAAL	340	0	0	0	0	0	0	0	0
MO_GAAL	340	0	0	0	0	0	0	0	0
DeepSVDD	340	0	0	0	0	0	0	0	0
AnoGAN	340	0	0	0	0	0	0	0	0
ALAD	340	0	0	0	0	0	0	0	0
AE1SVM	340	0	0	0	0	0	0	0	0
DevNet	0	0	0	0	0	340	0	0	0
RGraph	227	0	113	0	0	0	0	0	0
LUNAR	340	0	0	0	0	0	0	0	0

Table 159: Raw versus strict sensitivity, Original features. First line is all successful AUROC/AP (coverage); second line is time-qualified AUROC/AP (coverage). Available-case means can reflect different successful subsets.

Configuration	10% retained	5% retained
NSA fixed	0.2982/0.1093 (85/85)	0.2951/0.1173 (85/85)
	0.2982/0.1093 (85/85)	0.2951/0.1173 (85/85)
NSA variable	0.4350/0.1620 (85/85)	0.3844/0.1509 (85/85)
	0.4350/0.1620 (85/85)	0.3844/0.1509 (85/85)
NSA coverage	0.4333/0.1620 (85/85)	0.3819/0.1463 (85/85)
	0.4333/0.1620 (85/85)	0.3819/0.1463 (85/85)
NSA grid	0.4323/0.1573 (85/85)	0.3823/0.1417 (85/85)
	0.4323/0.1573 (85/85)	0.3823/0.1417 (85/85)
NSA hierarchical	0.2452/0.1031 (85/85)	0.3089/0.1193 (85/85)
	0.2452/0.1031 (85/85)	0.3089/0.1193 (85/85)
NSA voronoi	—/— (0/85)	—/— (0/85)
	—/— (0/85)	—/— (0/85)
NSA deterministic	—/— (0/85)	—/— (0/85)
	—/— (0/85)	—/— (0/85)
NSA suppressed	0.4366/0.1614 (85/85)	0.3803/0.1483 (85/85)
	0.4366/0.1614 (85/85)	0.3803/0.1483 (85/85)
NSA dual	0.7792/0.5112 (85/85)	0.7900/0.5066 (85/85)
	0.7792/0.5112 (85/85)	0.7900/0.5066 (85/85)
NSA annealed	0.2641/0.1040 (85/85)	0.2564/0.0924 (85/85)
	0.2641/0.1040 (85/85)	0.2564/0.0924 (85/85)
NSA binary Hamming	0.5024/0.1361 (85/85)	0.5036/0.1234 (85/85)
	0.5024/0.1361 (85/85)	0.5036/0.1234 (85/85)
NSA binary contiguous	0.5317/0.1321 (75/85)	0.5437/0.1339 (75/85)
	0.5317/0.1321 (75/85)	0.5437/0.1339 (75/85)
NSA binary chunk	0.5350/0.1290 (78/85)	0.5223/0.1186 (84/85)
	0.5350/0.1290 (78/85)	0.5223/0.1186 (84/85)
ECOD	0.7703/0.4309 (85/85)	0.7686/0.4268 (85/85)
	0.7729/0.4329 (77/85)	0.7686/0.4268 (85/85)
ABOD	0.8536/0.5352 (85/85)	0.8227/0.4993 (85/85)
	0.8536/0.5352 (85/85)	0.8227/0.4993 (85/85)
COPOD	0.7776/0.4474 (85/85)	0.7762/0.4436 (85/85)
	0.7776/0.4474 (85/85)	0.7762/0.4436 (85/85)
MAD	—/— (0/85)	—/— (0/85)
	—/— (0/85)	—/— (0/85)
SOS	0.5916/0.2208 (84/85)	0.5928/0.2208 (85/85)
	0.5916/0.2208 (84/85)	0.5928/0.2208 (85/85)
QMCD	0.4502/0.1812 (85/85)	0.3821/0.1485 (85/85)
	0.4502/0.1812 (85/85)	0.3821/0.1485 (85/85)
KDE	0.8781/0.5620 (85/85)	0.8577/0.5237 (85/85)
	0.8781/0.5620 (85/85)	0.8577/0.5237 (85/85)
Sampling	0.8732/0.5472 (65/85)	0.8850/0.5191 (50/85)
	0.8732/0.5472 (65/85)	0.8850/0.5191 (50/85)
GMM	0.8420/0.5067 (85/85)	0.8244/0.4648 (85/85)
	0.8420/0.5067 (85/85)	0.8244/0.4648 (85/85)
PCA	0.8061/0.5068 (85/85)	0.7907/0.4861 (85/85)
	0.8061/0.5068 (85/85)	0.7907/0.4861 (85/85)
KPCA	0.8766/0.5579 (85/85)	0.8544/0.5214 (85/85)
	0.8766/0.5579 (85/85)	0.8544/0.5214 (85/85)
MCD	0.8178/0.4461 (85/85)	0.7862/0.4051 (85/85)
	0.8178/0.4461 (85/85)	0.7862/0.4051 (85/85)

Continued on next page

Table 159 continued

Configuration	10% retained	5% retained
CD	0.8216/0.4776 (78/85)	0.8162/0.4575 (79/85)
	0.8216/0.4776 (78/85)	0.8162/0.4575 (79/85)
OCSVM	0.8407/0.5353 (85/85)	0.8258/0.5112 (85/85)
	0.8407/0.5353 (85/85)	0.8258/0.5112 (85/85)
LMDD	0.7443/0.4517 (52/85)	0.7284/0.4163 (55/85)
	0.7443/0.4517 (52/85)	0.7284/0.4163 (55/85)
LOF	0.7931/0.5063 (85/85)	0.7739/0.4874 (85/85)
	0.7931/0.5063 (85/85)	0.7739/0.4874 (85/85)
COF	0.6657/0.3202 (85/85)	0.6500/0.2984 (85/85)
	0.6657/0.3202 (85/85)	0.6500/0.2984 (85/85)
CBLOF	0.8580/0.5274 (80/85)	0.8652/0.5216 (65/85)
	0.8580/0.5274 (80/85)	0.8652/0.5216 (65/85)
LOCI	0.7087/0.3674 (29/85)	0.6819/0.3476 (30/85)
	0.7087/0.3674 (29/85)	0.6819/0.3476 (30/85)
HBOS	0.8043/0.4543 (85/85)	0.7848/0.4265 (85/85)
	0.8043/0.4543 (85/85)	0.7848/0.4265 (85/85)
HDBSCAN	0.5446/0.1398 (85/85)	0.5303/0.1303 (85/85)
	0.5446/0.1398 (85/85)	0.5303/0.1303 (85/85)
KNN	0.8558/0.5347 (85/85)	0.8104/0.4913 (85/85)
	0.8558/0.5347 (85/85)	0.8104/0.4913 (85/85)
SOD	0.7340/0.3233 (67/85)	0.7238/0.2796 (55/85)
	0.7325/0.3226 (58/85)	0.7238/0.2796 (55/85)
ROD	0.8199/0.4720 (40/85)	0.7856/0.4115 (43/85)
	0.8054/0.4483 (39/85)	0.7826/0.4245 (42/85)
IForest	0.8021/0.3957 (85/85)	0.7696/0.3516 (85/85)
	0.8021/0.3957 (85/85)	0.7696/0.3516 (85/85)
INNE	0.8379/0.4534 (85/85)	0.7993/0.3844 (85/85)
	0.8379/0.4534 (85/85)	0.7993/0.3844 (85/85)
DIF	0.7636/0.3574 (85/85)	0.7334/0.3109 (85/85)
	0.7636/0.3574 (85/85)	0.7334/0.3109 (85/85)
FeatureBagging	0.8002/0.5099 (85/85)	0.7809/0.4873 (85/85)
	0.8002/0.5099 (85/85)	0.7809/0.4873 (85/85)
LSCP	0.8724/0.5188 (50/85)	0.8487/0.4971 (50/85)
	0.8724/0.5188 (50/85)	0.8487/0.4971 (50/85)
LODA	0.6620/0.1956 (85/85)	0.4868/0.1312 (85/85)
	0.6620/0.1956 (85/85)	0.4868/0.1312 (85/85)
SUOD	0.8190/0.4654 (85/85)	0.7981/0.4334 (84/85)
	0.8190/0.4654 (85/85)	0.7981/0.4334 (84/85)
XGBOD	—/— (0/85)	—/— (0/85)
	—/— (0/85)	—/— (0/85)
AutoEncoder	0.8347/0.5080 (50/85)	—/— (0/85)
	0.8347/0.5080 (50/85)	—/— (0/85)
VAE	0.8194/0.4915 (50/85)	—/— (0/85)
	0.8194/0.4915 (50/85)	—/— (0/85)
SO_GAAL	0.4595/0.2057 (85/85)	0.4980/0.2088 (85/85)
	0.4595/0.2057 (85/85)	0.4980/0.2088 (85/85)
MO_GAAL	0.4879/0.2013 (85/85)	0.4889/0.2088 (85/85)
	0.4879/0.2013 (85/85)	0.4889/0.2088 (85/85)
DeepSVDD	0.8089/0.5069 (85/85)	0.7927/0.4683 (85/85)
	0.8089/0.5069 (85/85)	0.7927/0.4683 (85/85)
AnoGAN	0.7961/0.4628 (85/85)	0.7894/0.4754 (85/85)
	0.7961/0.4628 (85/85)	0.7894/0.4754 (85/85)

Continued on next page

Table 159 continued

Configuration	10% retained	5% retained
ALAD	0.5461/0.1544 (85/85)	0.5458/0.1551 (85/85)
	0.5461/0.1544 (85/85)	0.5458/0.1551 (85/85)
AE1SVM	0.8143/0.5101 (85/85)	0.7921/0.4924 (85/85)
	0.8143/0.5101 (85/85)	0.7921/0.4924 (85/85)
DevNet	—/— (0/85)	—/— (0/85)
	—/— (0/85)	—/— (0/85)
RGraph	0.5518/0.2457 (35/85)	0.5585/0.2186 (54/85)
	0.5518/0.2457 (35/85)	0.5585/0.2186 (54/85)
LUNAR	0.8703/0.5564 (85/85)	0.8343/0.5021 (85/85)
	0.8703/0.5564 (85/85)	0.8343/0.5021 (85/85)

Table 160: Raw versus strict sensitivity, PCA(2). First line is all successful AUROC/AP (coverage); second line is time-qualified AUROC/AP (coverage). Available-case means can reflect different successful subsets.

Configuration	10% retained	5% retained
NSA fixed	0.6132/0.2789 (85/85)	0.5917/0.2200 (85/85)
	0.6132/0.2789 (85/85)	0.5917/0.2200 (85/85)
NSA variable	0.7273/0.3638 (85/85)	0.6683/0.2879 (85/85)
	0.7273/0.3638 (85/85)	0.6683/0.2879 (85/85)
NSA coverage	0.7110/0.3571 (85/85)	0.6669/0.2838 (85/85)
	0.7110/0.3571 (85/85)	0.6669/0.2838 (85/85)
NSA grid	0.7243/0.3668 (85/85)	0.6775/0.3002 (85/85)
	0.7243/0.3668 (85/85)	0.6775/0.3002 (85/85)
NSA hierarchical	0.6891/0.3316 (85/85)	0.6579/0.2647 (85/85)
	0.6891/0.3316 (85/85)	0.6579/0.2647 (85/85)
NSA voronoi	0.7361/0.3689 (85/85)	0.6882/0.3153 (85/85)
	0.7361/0.3689 (85/85)	0.6882/0.3153 (85/85)
NSA deterministic	0.6278/0.2893 (85/85)	0.6066/0.2198 (85/85)
	0.6278/0.2893 (85/85)	0.6066/0.2198 (85/85)
NSA suppressed	0.7173/0.3547 (85/85)	0.6650/0.2955 (85/85)
	0.7173/0.3547 (85/85)	0.6650/0.2955 (85/85)
NSA dual	0.7626/0.4206 (85/85)	0.7264/0.3755 (85/85)
	0.7626/0.4206 (85/85)	0.7264/0.3755 (85/85)
NSA annealed	0.6241/0.2869 (85/85)	0.5966/0.2187 (85/85)
	0.6241/0.2869 (85/85)	0.5966/0.2187 (85/85)
NSA binary Hamming	—/— (0/85)	—/— (0/85)
	—/— (0/85)	—/— (0/85)
NSA binary contiguous	—/— (0/85)	—/— (0/85)
	—/— (0/85)	—/— (0/85)
NSA binary chunk	—/— (0/85)	—/— (0/85)
	—/— (0/85)	—/— (0/85)
ECOD	0.6162/0.2412 (85/85)	0.6172/0.2507 (85/85)
	0.6162/0.2412 (85/85)	0.6172/0.2507 (85/85)
ABOD	0.7751/0.4221 (85/85)	0.7437/0.3791 (85/85)
	0.7751/0.4221 (85/85)	0.7437/0.3791 (85/85)
COPOD	0.6611/0.2884 (85/85)	0.6529/0.2899 (85/85)
	0.6611/0.2884 (85/85)	0.6529/0.2899 (85/85)
MAD	—/— (0/85)	—/— (0/85)
	—/— (0/85)	—/— (0/85)
SOS	0.5252/0.1472 (85/85)	0.5295/0.1529 (85/85)
	0.5252/0.1472 (85/85)	0.5295/0.1529 (85/85)

Continued on next page

Table 160 continued

Configuration	10% retained	5% retained
QMCD	0.6058/0.2582 (85/85)	0.5224/0.2009 (85/85)
	0.6058/0.2582 (85/85)	0.5224/0.2009 (85/85)
KDE	0.7769/0.4373 (85/85)	0.7496/0.3949 (85/85)
	0.7769/0.4373 (85/85)	0.7496/0.3949 (85/85)
Sampling	0.7742/0.4442 (65/85)	0.7676/0.3969 (50/85)
	0.7742/0.4442 (65/85)	0.7676/0.3969 (50/85)
GMM	0.6418/0.3487 (85/85)	0.6403/0.3479 (85/85)
	0.6418/0.3487 (85/85)	0.6403/0.3479 (85/85)
PCA	0.6504/0.3413 (85/85)	0.6657/0.3562 (85/85)
	0.6504/0.3413 (85/85)	0.6657/0.3562 (85/85)
KPCA	0.7764/0.4290 (85/85)	0.7440/0.3837 (85/85)
	0.7764/0.4290 (85/85)	0.7440/0.3837 (85/85)
MCD	0.7053/0.3462 (85/85)	0.6805/0.3242 (85/85)
	0.7053/0.3462 (85/85)	0.6805/0.3242 (85/85)
CD	0.5877/0.2623 (85/85)	0.5975/0.2643 (85/85)
	0.5877/0.2623 (85/85)	0.5975/0.2643 (85/85)
OCSVM	0.7663/0.4295 (85/85)	0.7445/0.3855 (85/85)
	0.7663/0.4295 (85/85)	0.7445/0.3855 (85/85)
LMDD	0.5874/0.2547 (85/85)	0.5800/0.2384 (85/85)
	0.5874/0.2547 (85/85)	0.5800/0.2384 (85/85)
LOF	0.6729/0.3576 (85/85)	0.6207/0.3030 (85/85)
	0.6729/0.3576 (85/85)	0.6207/0.3030 (85/85)
COF	0.6139/0.2165 (85/85)	0.6023/0.2040 (85/85)
	0.6139/0.2165 (85/85)	0.6023/0.2040 (85/85)
CBLOF	0.7665/0.4357 (80/85)	0.7703/0.4210 (65/85)
	0.7665/0.4357 (80/85)	0.7703/0.4210 (65/85)
LOCI	0.5645/0.2412 (28/85)	0.5263/0.2354 (30/85)
	0.5645/0.2412 (28/85)	0.5263/0.2354 (30/85)
HBOS	0.7331/0.2969 (85/85)	0.6720/0.2230 (85/85)
	0.7331/0.2969 (85/85)	0.6720/0.2230 (85/85)
HDBSCAN	0.5928/0.1505 (85/85)	0.5650/0.1351 (85/85)
	0.5928/0.1505 (85/85)	0.5650/0.1351 (85/85)
KNN	0.7706/0.4232 (85/85)	0.7082/0.3717 (85/85)
	0.7706/0.4232 (85/85)	0.7082/0.3717 (85/85)
SOD	0.5607/0.2068 (68/85)	0.5845/0.2026 (55/85)
	0.5624/0.2075 (63/85)	0.5845/0.2026 (55/85)
ROD	0.6634/0.2968 (85/85)	0.6248/0.2797 (85/85)
	0.6634/0.2968 (85/85)	0.6248/0.2797 (85/85)
IForest	0.7196/0.3056 (85/85)	0.6870/0.2809 (85/85)
	0.7196/0.3056 (85/85)	0.6870/0.2809 (85/85)
INNE	0.7452/0.4062 (85/85)	0.7139/0.3470 (85/85)
	0.7452/0.4062 (85/85)	0.7139/0.3470 (85/85)
DIF	0.7238/0.3667 (85/85)	0.6941/0.3325 (85/85)
	0.7250/0.3661 (83/85)	0.6941/0.3325 (85/85)
FeatureBagging	0.6948/0.3703 (85/85)	0.6355/0.3239 (85/85)
	0.6948/0.3703 (85/85)	0.6355/0.3239 (85/85)
LSCP	0.7968/0.4374 (50/85)	0.7475/0.3439 (50/85)
	0.7968/0.4374 (50/85)	0.7475/0.3439 (50/85)
LODA	0.6645/0.2562 (85/85)	0.5771/0.1873 (85/85)
	0.6645/0.2562 (85/85)	0.5771/0.1873 (85/85)
SUOD	0.7402/0.3490 (84/85)	0.6915/0.2966 (85/85)
	0.7402/0.3490 (84/85)	0.6915/0.2966 (85/85)

Continued on next page

Table 160 continued

Configuration	10% retained	5% retained
XGBOD	—/— (0/85)	—/— (0/85)
	—/— (0/85)	—/— (0/85)
AutoEncoder	0.6253/0.3012 (50/85)	—/— (0/85)
	0.6253/0.3012 (50/85)	—/— (0/85)
VAE	0.6255/0.3211 (50/85)	—/— (0/85)
	0.6255/0.3211 (50/85)	—/— (0/85)
SO_GAAL	0.5688/0.2370 (85/85)	0.5571/0.2157 (85/85)
	0.5688/0.2370 (85/85)	0.5571/0.2157 (85/85)
MO_GAAL	0.5610/0.2541 (85/85)	0.5434/0.2227 (85/85)
	0.5610/0.2541 (85/85)	0.5434/0.2227 (85/85)
DeepSVDD	0.7010/0.3433 (85/85)	0.6667/0.3067 (85/85)
	0.7010/0.3433 (85/85)	0.6667/0.3067 (85/85)
AnoGAN	0.6425/0.2731 (85/85)	0.6493/0.2950 (85/85)
	0.6425/0.2731 (85/85)	0.6493/0.2950 (85/85)
ALAD	0.5045/0.1276 (85/85)	0.5133/0.1373 (85/85)
	0.5045/0.1276 (85/85)	0.5133/0.1373 (85/85)
AE1SVM	0.6548/0.3471 (85/85)	0.6558/0.3537 (85/85)
	0.6548/0.3471 (85/85)	0.6558/0.3537 (85/85)
DevNet	—/— (0/85)	—/— (0/85)
	—/— (0/85)	—/— (0/85)
RGraph	0.4038/0.1112 (53/85)	0.3872/0.0982 (85/85)
	0.4038/0.1112 (53/85)	0.3872/0.0982 (85/85)
LUNAR	0.7459/0.4038 (85/85)	0.7106/0.3666 (85/85)
	0.7459/0.4038 (85/85)	0.7106/0.3666 (85/85)

The six-size analysis contains 6,708 same-size NSA-versus-other-detector comparisons. All 2,236 comparisons at the new sizes, all 224 new-size method changes, and all 2,236 matched relative-gap changes are checked against the aggregate raw records, including paired coverage, dataset macro means, and both bootstrap intervals. Raw files contain AUROC/AP aggregates, not per-test decision vectors; this reconstruction validates the reporting arithmetic and cannot independently recompute each metric from predictions.

S Training-batch sensitivity: complete outcomes and rank checks

The following tables disclose the separate training-batch intervention. Overall score columns are dataset-macro available-case means; the later dataset tables average the qualified splits within each named dataset. An incomplete mean is not ranked against a complete field. The machine-readable cell comparison retains all 740 supplemental records, their original counterparts, timings, training counts, and source hashes.

Table 161: All six AutoEncoder/VAE panels under the separate training-batch sensitivity. Fresh/reused counts describe the selected 85 rows per panel.

Features	Retained	Method	Fresh / reused	Original runs	Sensitivity runs	Original AUROC	Sensitivity AUROC	Original AP	Sensitivity AP
Original features	100%	AutoEncoder	0 / 85	85/85	85/85	0.8684	0.8684	0.5702	0.5702
Original features	100%	VAE	0 / 85	85/85	85/85	0.8299	0.8299	0.5466	0.5466
Original features	75%	AutoEncoder	0 / 85	85/85	85/85	0.8627	0.8627	0.5625	0.5625
Original features	75%	VAE	0 / 85	85/85	85/85	0.8234	0.8234	0.5340	0.5340
Original features	50%	AutoEncoder	0 / 85	85/85	85/85	0.8517	0.8517	0.5511	0.5511
Original features	50%	VAE	0 / 85	85/85	85/85	0.8141	0.8141	0.5253	0.5253
Original features	25%	AutoEncoder	15 / 70	57/85	72/85	0.8564	0.8309	0.5614	0.5239
Original features	25%	VAE	15 / 70	69/85	84/85	0.8317	0.8084	0.5436	0.5115
Original features	10%	AutoEncoder	85 / 0	50/85	85/85	0.8347	0.8119	0.5080	0.5071
Original features	10%	VAE	85 / 0	50/85	85/85	0.8194	0.8021	0.4915	0.4992
Original features	5%	AutoEncoder	85 / 0	0/85	85/85	—	0.7952	—	0.4945
Original features	5%	VAE	85 / 0	0/85	85/85	—	0.7863	—	0.4860
PCA(2)	100%	AutoEncoder	0 / 85	85/85	85/85	0.6830	0.6830	0.3647	0.3647
PCA(2)	100%	VAE	0 / 85	85/85	85/85	0.6718	0.6718	0.3697	0.3697
PCA(2)	75%	AutoEncoder	0 / 85	85/85	85/85	0.6767	0.6767	0.3495	0.3495
PCA(2)	75%	VAE	0 / 85	85/85	85/85	0.6690	0.6690	0.3697	0.3697

Continued on next page

Table 161 continued

Features	Retained	Method	Fresh / reused	Original runs	Sensitivity runs	Original AUROC	Sensitivity AUROC	Original AP	Sensitivity AP
PCA(2)	50%	AutoEncoder	0 / 85	85/85	85/85	0.6591	0.6591	0.3211	0.3211
PCA(2)	50%	VAE	0 / 85	85/85	85/85	0.6623	0.6623	0.3577	0.3577
PCA(2)	25%	AutoEncoder	15 / 70	60/85	75/85	0.6651	0.6531	0.3316	0.3081
PCA(2)	25%	VAE	15 / 70	70/85	85/85	0.6461	0.6453	0.3866	0.3524
PCA(2)	10%	AutoEncoder	85 / 0	50/85	85/85	0.6253	0.6306	0.3012	0.3183
PCA(2)	10%	VAE	85 / 0	50/85	85/85	0.6255	0.6297	0.3211	0.3295
PCA(2)	5%	AutoEncoder	85 / 0	0/85	85/85	—	0.6400	—	0.3151
PCA(2)	5%	VAE	85 / 0	0/85	85/85	—	0.6243	—	0.3205

S.1 Original features

Original fixed baselines: ABOD, COPOD, KDE, GMM, KPCA, MCD, OCSVM, LOF, HDBSCAN, KNN, IForest, INNE, FeatureBagging, LODA, SO_GAAL, MO_GAAL, DeepSVDD, ALAD, LUNAR. Sensitivity fixed baselines: ABOD, COPOD, KDE, GMM, KPCA, MCD, OCSVM, LOF, HDBSCAN, KNN, IForest, INNE, FeatureBagging, LODA, SO_GAAL, MO_GAAL, DeepSVDD, ALAD, LUNAR.

Table 162: Original features: all NSA AUROC ranks before and after the training-batch sensitivity. Only a focal NSA with 85/85 qualified runs receives a rank.

NSA method	Retained	Runs	Score	Original field rank	Adapted field rank
NSA annealed	100%	85/85	0.3161	20/20	20/20
NSA annealed	75%	85/85	0.3112	20/20	20/20
NSA annealed	50%	85/85	0.3023	20/20	20/20
NSA annealed	25%	85/85	0.2789	20/20	20/20
NSA annealed	10%	85/85	0.2641	20/20	20/20
NSA annealed	5%	85/85	0.2564	20/20	20/20
NSA binary Hamming	100%	76/85	0.5259	—	—
NSA binary Hamming	75%	83/85	0.5226	—	—
NSA binary Hamming	50%	85/85	0.5288	19/20	19/20
NSA binary Hamming	25%	85/85	0.5172	20/20	20/20
NSA binary Hamming	10%	85/85	0.5024	18/20	18/20
NSA binary Hamming	5%	85/85	0.5036	17/20	17/20
NSA binary chunk	100%	75/85	0.5269	—	—
NSA binary chunk	75%	75/85	0.5292	—	—
NSA binary chunk	50%	75/85	0.5328	—	—
NSA binary chunk	25%	75/85	0.5191	—	—
NSA binary chunk	10%	78/85	0.5350	—	—
NSA binary chunk	5%	84/85	0.5223	—	—
NSA binary contiguous	100%	59/85	0.5350	—	—
NSA binary contiguous	75%	60/85	0.5404	—	—
NSA binary contiguous	50%	60/85	0.5335	—	—
NSA binary contiguous	25%	63/85	0.5218	—	—
NSA binary contiguous	10%	75/85	0.5317	—	—
NSA binary contiguous	5%	75/85	0.5437	—	—
NSA coverage	100%	85/85	0.5931	17/20	17/20
NSA coverage	75%	85/85	0.5814	17/20	17/20
NSA coverage	50%	85/85	0.5534	17/20	17/20
NSA coverage	25%	85/85	0.5028	20/20	20/20
NSA coverage	10%	85/85	0.4333	20/20	20/20
NSA coverage	5%	85/85	0.3819	20/20	20/20
NSA deterministic	100%	0/85	—	—	—
NSA deterministic	75%	0/85	—	—	—
NSA deterministic	50%	0/85	—	—	—
NSA deterministic	25%	0/85	—	—	—
NSA deterministic	10%	0/85	—	—	—

Continued on next page

Table 162 continued

NSA method	Retained	Runs	Score	Original field rank	Adapted field rank
NSA deterministic	5%	0/85	—	—	—
NSA dual	100%	81/85	0.7047	—	—
NSA dual	75%	80/85	0.7352	—	—
NSA dual	50%	81/85	0.7318	—	—
NSA dual	25%	83/85	0.7488	—	—
NSA dual	10%	85/85	0.7792	14/20	14/20
NSA dual	5%	85/85	0.7900	10/20	10/20
NSA fixed	100%	85/85	0.3597	20/20	20/20
NSA fixed	75%	85/85	0.3504	20/20	20/20
NSA fixed	50%	85/85	0.3460	20/20	20/20
NSA fixed	25%	85/85	0.3297	20/20	20/20
NSA fixed	10%	85/85	0.2982	20/20	20/20
NSA fixed	5%	85/85	0.2951	20/20	20/20
NSA grid	100%	85/85	0.5972	17/20	17/20
NSA grid	75%	85/85	0.5880	17/20	17/20
NSA grid	50%	85/85	0.5569	17/20	17/20
NSA grid	25%	84/85	0.5025	—	—
NSA grid	10%	85/85	0.4323	20/20	20/20
NSA grid	5%	85/85	0.3823	20/20	20/20
NSA hierarchical	100%	85/85	0.2968	20/20	20/20
NSA hierarchical	75%	85/85	0.3078	20/20	20/20
NSA hierarchical	50%	85/85	0.2753	20/20	20/20
NSA hierarchical	25%	85/85	0.2712	20/20	20/20
NSA hierarchical	10%	85/85	0.2452	20/20	20/20
NSA hierarchical	5%	85/85	0.3089	20/20	20/20
NSA suppressed	100%	85/85	0.5883	17/20	17/20
NSA suppressed	75%	85/85	0.5656	17/20	17/20
NSA suppressed	50%	85/85	0.5516	17/20	17/20
NSA suppressed	25%	85/85	0.4920	20/20	20/20
NSA suppressed	10%	85/85	0.4366	20/20	20/20
NSA suppressed	5%	85/85	0.3803	20/20	20/20
NSA variable	100%	85/85	0.5931	17/20	17/20
NSA variable	75%	85/85	0.5821	17/20	17/20
NSA variable	50%	85/85	0.5531	17/20	17/20
NSA variable	25%	85/85	0.5025	20/20	20/20
NSA variable	10%	85/85	0.4350	20/20	20/20
NSA variable	5%	85/85	0.3844	20/20	20/20
NSA voronoi	100%	0/85	—	—	—
NSA voronoi	75%	0/85	—	—	—
NSA voronoi	50%	0/85	—	—	—
NSA voronoi	25%	0/85	—	—	—
NSA voronoi	10%	0/85	—	—	—
NSA voronoi	5%	0/85	—	—	—

Table 163: Original features: all NSA AP ranks before and after the training-batch sensitivity. Only a focal NSA with 85/85 qualified runs receives a rank.

NSA method	Retained	Runs	Score	Original field rank	Adapted field rank
NSA annealed	100%	85/85	0.1120	20/20	20/20
NSA annealed	75%	85/85	0.1058	20/20	20/20
NSA annealed	50%	85/85	0.1070	20/20	20/20
NSA annealed	25%	85/85	0.0988	20/20	20/20
NSA annealed	10%	85/85	0.1040	20/20	20/20
NSA annealed	5%	85/85	0.0924	20/20	20/20
NSA binary Hamming	100%	76/85	0.1681	—	—
NSA binary Hamming	75%	83/85	0.1643	—	—
NSA binary Hamming	50%	85/85	0.1573	20/20	20/20
NSA binary Hamming	25%	85/85	0.1449	20/20	20/20
NSA binary Hamming	10%	85/85	0.1361	20/20	20/20
NSA binary Hamming	5%	85/85	0.1234	20/20	20/20
NSA binary chunk	100%	75/85	0.1261	—	—
NSA binary chunk	75%	75/85	0.1258	—	—
NSA binary chunk	50%	75/85	0.1211	—	—
NSA binary chunk	25%	75/85	0.1082	—	—
NSA binary chunk	10%	78/85	0.1290	—	—
NSA binary chunk	5%	84/85	0.1186	—	—
NSA binary contiguous	100%	59/85	0.1578	—	—
NSA binary contiguous	75%	60/85	0.1498	—	—
NSA binary contiguous	50%	60/85	0.1511	—	—
NSA binary contiguous	25%	63/85	0.1203	—	—
NSA binary contiguous	10%	75/85	0.1321	—	—
NSA binary contiguous	5%	75/85	0.1339	—	—
NSA coverage	100%	85/85	0.3040	16/20	16/20
NSA coverage	75%	85/85	0.2855	16/20	16/20
NSA coverage	50%	85/85	0.2455	16/20	16/20
NSA coverage	25%	85/85	0.2053	18/20	18/20
NSA coverage	10%	85/85	0.1620	18/20	18/20
NSA coverage	5%	85/85	0.1463	18/20	18/20
NSA deterministic	100%	0/85	—	—	—
NSA deterministic	75%	0/85	—	—	—
NSA deterministic	50%	0/85	—	—	—
NSA deterministic	25%	0/85	—	—	—
NSA deterministic	10%	0/85	—	—	—
NSA deterministic	5%	0/85	—	—	—
NSA dual	100%	81/85	0.4609	—	—
NSA dual	75%	80/85	0.5006	—	—
NSA dual	50%	81/85	0.4966	—	—
NSA dual	25%	83/85	0.5142	—	—
NSA dual	10%	85/85	0.5112	7/20	7/20
NSA dual	5%	85/85	0.5066	4/20	4/20

Continued on next page

Table 163 continued

NSA method	Retained	Runs	Score	Original field rank	Adapted field rank
NSA fixed	100%	85/85	0.1287	20/20	20/20
NSA fixed	75%	85/85	0.1272	20/20	20/20
NSA fixed	50%	85/85	0.1232	20/20	20/20
NSA fixed	25%	85/85	0.1211	20/20	20/20
NSA fixed	10%	85/85	0.1093	20/20	20/20
NSA fixed	5%	85/85	0.1173	20/20	20/20
NSA grid	100%	85/85	0.3026	16/20	16/20
NSA grid	75%	85/85	0.2869	16/20	16/20
NSA grid	50%	85/85	0.2510	16/20	16/20
NSA grid	25%	84/85	0.1994	—	—
NSA grid	10%	85/85	0.1573	18/20	18/20
NSA grid	5%	85/85	0.1417	18/20	18/20
NSA hierarchical	100%	85/85	0.1173	20/20	20/20
NSA hierarchical	75%	85/85	0.1145	20/20	20/20
NSA hierarchical	50%	85/85	0.0938	20/20	20/20
NSA hierarchical	25%	85/85	0.1092	20/20	20/20
NSA hierarchical	10%	85/85	0.1031	20/20	20/20
NSA hierarchical	5%	85/85	0.1193	20/20	20/20
NSA suppressed	100%	85/85	0.2982	16/20	16/20
NSA suppressed	75%	85/85	0.2705	16/20	16/20
NSA suppressed	50%	85/85	0.2440	16/20	16/20
NSA suppressed	25%	85/85	0.1989	18/20	18/20
NSA suppressed	10%	85/85	0.1614	18/20	18/20
NSA suppressed	5%	85/85	0.1483	18/20	18/20
NSA variable	100%	85/85	0.3038	16/20	16/20
NSA variable	75%	85/85	0.2865	16/20	16/20
NSA variable	50%	85/85	0.2456	16/20	16/20
NSA variable	25%	85/85	0.2054	18/20	18/20
NSA variable	10%	85/85	0.1620	18/20	18/20
NSA variable	5%	85/85	0.1509	18/20	18/20
NSA voronoi	100%	0/85	—	—	—
NSA voronoi	75%	0/85	—	—	—
NSA voronoi	50%	0/85	—	—	—
NSA voronoi	25%	0/85	—	—	—
NSA voronoi	10%	0/85	—	—	—
NSA voronoi	5%	0/85	—	—	—

S.2 PCA(2)

Original fixed baselines: ECOD, ABOD, COPOD, SOS, QMCD, KDE, GMM, PCA, KPCA, MCD, CD, OCSVM, LOF, COF, HDBSCAN, KNN, IForest, INNE, FeatureBagging, LODA, SO_GAAL, MO_GAAL, DeepSVDD, ALAD, LUNAR. Sensitivity fixed baselines: ECOD, ABOD, COPOD, SOS, QMCD, KDE, GMM, PCA, KPCA, MCD, CD, OCSVM, LOF, COF, HDBSCAN, KNN, IForest, INNE, FeatureBagging, LODA, VAE, SO_GAAL, MO_GAAL, DeepSVDD, ALAD, LUNAR.

Table 164: PCA(2): all NSA AUROC ranks before and after the training-batch sensitivity. Only a focal NSA with 85/85 qualified runs receives a rank.

NSA method	Retained	Runs	Score	Original field rank	Adapted field rank
NSA annealed	100%	85/85	0.5998	21/26	22/27
NSA annealed	75%	85/85	0.6183	21/26	22/27
NSA annealed	50%	85/85	0.6401	18/26	19/27
NSA annealed	25%	85/85	0.6286	19/26	20/27
NSA annealed	10%	85/85	0.6241	17/26	18/27
NSA annealed	5%	85/85	0.5966	19/26	20/27
NSA binary Hamming	100%	0/85	—	—	—
NSA binary Hamming	75%	0/85	—	—	—
NSA binary Hamming	50%	0/85	—	—	—
NSA binary Hamming	25%	0/85	—	—	—
NSA binary Hamming	10%	0/85	—	—	—
NSA binary Hamming	5%	0/85	—	—	—
NSA binary chunk	100%	0/85	—	—	—
NSA binary chunk	75%	0/85	—	—	—
NSA binary chunk	50%	0/85	—	—	—
NSA binary chunk	25%	0/85	—	—	—
NSA binary chunk	10%	0/85	—	—	—
NSA binary chunk	5%	0/85	—	—	—
NSA binary contiguous	100%	0/85	—	—	—
NSA binary contiguous	75%	0/85	—	—	—
NSA binary contiguous	50%	0/85	—	—	—
NSA binary contiguous	25%	0/85	—	—	—
NSA binary contiguous	10%	0/85	—	—	—
NSA binary contiguous	5%	0/85	—	—	—
NSA coverage	100%	85/85	0.6694	15/26	16/27
NSA coverage	75%	85/85	0.6763	15/26	15/27
NSA coverage	50%	85/85	0.7039	13/26	13/27
NSA coverage	25%	85/85	0.6954	14/26	14/27
NSA coverage	10%	85/85	0.7110	9/26	9/27
NSA coverage	5%	85/85	0.6669	10/26	10/27
NSA deterministic	100%	85/85	0.6032	21/26	22/27
NSA deterministic	75%	85/85	0.6149	21/26	22/27
NSA deterministic	50%	85/85	0.6459	18/26	19/27
NSA deterministic	25%	85/85	0.6420	17/26	18/27
NSA deterministic	10%	85/85	0.6278	17/26	18/27
NSA deterministic	5%	85/85	0.6066	17/26	18/27
NSA dual	100%	85/85	0.7172	13/26	13/27
NSA dual	75%	85/85	0.7268	12/26	12/27
NSA dual	50%	85/85	0.7419	12/26	12/27
NSA dual	25%	85/85	0.7603	10/26	10/27
NSA dual	10%	85/85	0.7626	6/26	6/27
NSA dual	5%	85/85	0.7264	5/26	5/27

Continued on next page

Table 164 continued

NSA method	Retained	Runs	Score	Original field rank	Adapted field rank
NSA fixed	100%	85/85	0.5827	22/26	23/27
NSA fixed	75%	85/85	0.5923	21/26	22/27
NSA fixed	50%	85/85	0.6202	20/26	21/27
NSA fixed	25%	85/85	0.6299	19/26	20/27
NSA fixed	10%	85/85	0.6132	19/26	20/27
NSA fixed	5%	85/85	0.5917	19/26	20/27
NSA grid	100%	85/85	0.6886	15/26	15/27
NSA grid	75%	85/85	0.6942	15/26	15/27
NSA grid	50%	85/85	0.7105	13/26	13/27
NSA grid	25%	85/85	0.7261	11/26	11/27
NSA grid	10%	85/85	0.7243	8/26	8/27
NSA grid	5%	85/85	0.6775	10/26	10/27
NSA hierarchical	100%	85/85	0.6754	15/26	15/27
NSA hierarchical	75%	85/85	0.6676	15/26	16/27
NSA hierarchical	50%	85/85	0.6921	15/26	15/27
NSA hierarchical	25%	85/85	0.6836	14/26	14/27
NSA hierarchical	10%	85/85	0.6891	12/26	12/27
NSA hierarchical	5%	85/85	0.6579	12/26	12/27
NSA suppressed	100%	85/85	0.6738	15/26	15/27
NSA suppressed	75%	85/85	0.6841	15/26	15/27
NSA suppressed	50%	85/85	0.7063	13/26	13/27
NSA suppressed	25%	85/85	0.7120	12/26	12/27
NSA suppressed	10%	85/85	0.7173	9/26	9/27
NSA suppressed	5%	85/85	0.6650	12/26	12/27
NSA variable	100%	85/85	0.6928	15/26	15/27
NSA variable	75%	85/85	0.6982	14/26	14/27
NSA variable	50%	85/85	0.7139	13/26	13/27
NSA variable	25%	85/85	0.7240	11/26	11/27
NSA variable	10%	85/85	0.7273	8/26	8/27
NSA variable	5%	85/85	0.6683	10/26	10/27
NSA voronoi	100%	85/85	0.7043	15/26	15/27
NSA voronoi	75%	85/85	0.7164	14/26	14/27
NSA voronoi	50%	85/85	0.7315	12/26	12/27
NSA voronoi	25%	85/85	0.7432	11/26	11/27
NSA voronoi	10%	85/85	0.7361	8/26	8/27
NSA voronoi	5%	85/85	0.6882	8/26	8/27

Table 165: PCA(2): all NSA AP ranks before and after the training-batch sensitivity. Only a focal NSA with 85/85 qualified runs receives a rank.

NSA method	Retained	Runs	Score	Original field rank	Adapted field rank
NSA annealed	100%	85/85	0.3672	13/26	14/27
NSA annealed	75%	85/85	0.3715	12/26	12/27
NSA annealed	50%	85/85	0.3835	12/26	12/27
NSA annealed	25%	85/85	0.3403	15/26	16/27
NSA annealed	10%	85/85	0.2869	16/26	17/27
NSA annealed	5%	85/85	0.2187	19/26	20/27
NSA binary Hamming	100%	0/85	—	—	—

Continued on next page

Table 165 continued

NSA method	Retained	Runs	Score	Original field rank	Adapted field rank
NSA binary Hamming	75%	0/85	—	—	—
NSA binary Hamming	50%	0/85	—	—	—
NSA binary Hamming	25%	0/85	—	—	—
NSA binary Hamming	10%	0/85	—	—	—
NSA binary Hamming	5%	0/85	—	—	—
NSA binary chunk	100%	0/85	—	—	—
NSA binary chunk	75%	0/85	—	—	—
NSA binary chunk	50%	0/85	—	—	—
NSA binary chunk	25%	0/85	—	—	—
NSA binary chunk	10%	0/85	—	—	—
NSA binary chunk	5%	0/85	—	—	—
NSA binary contiguous	100%	0/85	—	—	—
NSA binary contiguous	75%	0/85	—	—	—
NSA binary contiguous	50%	0/85	—	—	—
NSA binary contiguous	25%	0/85	—	—	—
NSA binary contiguous	10%	0/85	—	—	—
NSA binary contiguous	5%	0/85	—	—	—
NSA coverage	100%	85/85	0.4253	10/26	10/27
NSA coverage	75%	85/85	0.4310	10/26	10/27
NSA coverage	50%	85/85	0.4422	10/26	10/27
NSA coverage	25%	85/85	0.4102	10/26	10/27
NSA coverage	10%	85/85	0.3571	10/26	10/27
NSA coverage	5%	85/85	0.2838	15/26	16/27
NSA deterministic	100%	85/85	0.3594	14/26	15/27
NSA deterministic	75%	85/85	0.3586	13/26	14/27
NSA deterministic	50%	85/85	0.3928	12/26	12/27
NSA deterministic	25%	85/85	0.3450	14/26	15/27
NSA deterministic	10%	85/85	0.2893	15/26	16/27
NSA deterministic	5%	85/85	0.2198	19/26	20/27
NSA dual	100%	85/85	0.4718	6/26	6/27
NSA dual	75%	85/85	0.4777	6/26	6/27
NSA dual	50%	85/85	0.4801	6/26	6/27
NSA dual	25%	85/85	0.4699	3/26	3/27
NSA dual	10%	85/85	0.4206	6/26	6/27
NSA dual	5%	85/85	0.3755	5/26	5/27
NSA fixed	100%	85/85	0.3427	16/26	17/27
NSA fixed	75%	85/85	0.3328	17/26	18/27
NSA fixed	50%	85/85	0.3564	14/26	15/27
NSA fixed	25%	85/85	0.3474	14/26	15/27
NSA fixed	10%	85/85	0.2789	16/26	17/27
NSA fixed	5%	85/85	0.2200	19/26	20/27
NSA grid	100%	85/85	0.4361	10/26	10/27
NSA grid	75%	85/85	0.4385	10/26	10/27
NSA grid	50%	85/85	0.4454	10/26	10/27

Continued on next page

Table 165 continued

NSA method	Retained	Runs	Score	Original field rank	Adapted field rank
NSA grid	25%	85/85	0.4208	8/26	8/27
NSA grid	10%	85/85	0.3668	9/26	9/27
NSA grid	5%	85/85	0.3002	14/26	15/27
NSA hierarchical	100%	85/85	0.4218	10/26	10/27
NSA hierarchical	75%	85/85	0.4090	10/26	10/27
NSA hierarchical	50%	85/85	0.4266	10/26	10/27
NSA hierarchical	25%	85/85	0.3857	11/26	11/27
NSA hierarchical	10%	85/85	0.3316	14/26	14/27
NSA hierarchical	5%	85/85	0.2647	16/26	17/27
NSA suppressed	100%	85/85	0.4214	10/26	10/27
NSA suppressed	75%	85/85	0.4288	10/26	10/27
NSA suppressed	50%	85/85	0.4402	10/26	10/27
NSA suppressed	25%	85/85	0.4186	8/26	8/27
NSA suppressed	10%	85/85	0.3547	10/26	10/27
NSA suppressed	5%	85/85	0.2955	14/26	15/27
NSA variable	100%	85/85	0.4441	10/26	10/27
NSA variable	75%	85/85	0.4435	10/26	10/27
NSA variable	50%	85/85	0.4466	10/26	10/27
NSA variable	25%	85/85	0.4204	8/26	8/27
NSA variable	10%	85/85	0.3638	9/26	9/27
NSA variable	5%	85/85	0.2879	15/26	16/27
NSA voronoi	100%	85/85	0.4521	9/26	9/27
NSA voronoi	75%	85/85	0.4590	8/26	8/27
NSA voronoi	50%	85/85	0.4550	9/26	9/27
NSA voronoi	25%	85/85	0.4347	8/26	8/27
NSA voronoi	10%	85/85	0.3689	9/26	9/27
NSA voronoi	5%	85/85	0.3153	12/26	13/27

Table 166: Original features: every dataset panel rerun in the training-batch sensitivity. AUROC/AP average the qualified splits shown in the coverage column.

Dataset	Retained	Method	Normals	Original runs	Sensitivity runs	AUROC	AP	Sensitivity outcomes
glass	25%	AutoEncoder	30	0/5	5/5	0.6915	0.1197	success:5
glass	25%	VAE	30	0/5	5/5	0.7140	0.1295	success:5
lympho	25%	AutoEncoder	21	0/5	5/5	0.9871	0.7967	success:5
lympho	25%	VAE	21	0/5	5/5	0.9879	0.8400	success:5
vertebral	25%	AutoEncoder	31	0/5	5/5	0.4562	0.1296	success:5
vertebral	25%	VAE	31	0/5	5/5	0.3984	0.1149	success:5
arrhythmia	10%	AutoEncoder	23	0/5	5/5	0.7820	0.4413	success:5
arrhythmia	10%	VAE	23	0/5	5/5	0.7847	0.4381	success:5
cardio	10%	AutoEncoder	60	5/5	5/5	0.9583	0.6821	success:5
cardio	10%	VAE	60	5/5	5/5	0.9610	0.6959	success:5
glass	10%	AutoEncoder	12	0/5	5/5	0.7085	0.1205	success:5
glass	10%	VAE	12	0/5	5/5	0.6988	0.1291	success:5
ionosphere	10%	AutoEncoder	13	0/5	5/5	0.9247	0.8952	success:5
ionosphere	10%	VAE	13	0/5	5/5	0.9150	0.8825	success:5
letter	10%	AutoEncoder	60	5/5	5/5	0.5528	0.0939	success:5
letter	10%	VAE	60	5/5	5/5	0.5361	0.0870	success:5
lympho	10%	AutoEncoder	8	0/5	5/5	0.9741	0.8015	success:5
lympho	10%	VAE	8	0/5	5/5	0.9759	0.8033	success:5
mnist	10%	AutoEncoder	60	5/5	5/5	0.8898	0.4316	success:5
mnist	10%	VAE	60	5/5	5/5	0.8829	0.4256	success:5
musk	10%	AutoEncoder	60	5/5	5/5	0.9997	0.9916	success:5
musk	10%	VAE	60	5/5	5/5	0.9998	0.9933	success:5
optdigits	10%	AutoEncoder	60	5/5	5/5	0.6600	0.0399	success:5
optdigits	10%	VAE	60	5/5	5/5	0.6254	0.0359	success:5
pendigits	10%	AutoEncoder	60	5/5	5/5	0.9261	0.1962	success:5
pendigits	10%	VAE	60	5/5	5/5	0.9013	0.1594	success:5
pima	10%	AutoEncoder	29	0/5	5/5	0.6922	0.5048	success:5
pima	10%	VAE	29	0/5	5/5	0.7052	0.5267	success:5
satellite	10%	AutoEncoder	60	5/5	5/5	0.6870	0.7075	success:5
satellite	10%	VAE	60	5/5	5/5	0.6818	0.6991	success:5
satimage-2	10%	AutoEncoder	60	5/5	5/5	0.9836	0.9126	success:5
satimage-2	10%	VAE	60	5/5	5/5	0.9773	0.8703	success:5

Continued on next page

Table 166 continued

Dataset	Retained	Method	Normals	Original runs	Sensitivity runs	AUROC	AP	Sensitivity outcomes
shuttle	10%	AutoEncoder	60	5/5	5/5	0.9951	0.8515	success:5
shuttle	10%	VAE	60	5/5	5/5	0.9954	0.8555	success:5
vertebral	10%	AutoEncoder	12	0/5	5/5	0.4476	0.1187	success:5
vertebral	10%	VAE	12	0/5	5/5	0.4337	0.1216	success:5
vowels	10%	AutoEncoder	60	5/5	5/5	0.6947	0.1731	success:5
vowels	10%	VAE	60	5/5	5/5	0.6335	0.0926	success:5
wbc	10%	AutoEncoder	21	0/5	5/5	0.9252	0.6581	success:5
wbc	10%	VAE	21	0/5	5/5	0.9278	0.6699	success:5
arrhythmia	5%	AutoEncoder	11	0/5	5/5	0.7659	0.4664	success:5
arrhythmia	5%	VAE	11	0/5	5/5	0.7630	0.4619	success:5
cardio	5%	AutoEncoder	30	0/5	5/5	0.9589	0.6976	success:5
cardio	5%	VAE	30	0/5	5/5	0.9593	0.7025	success:5
glass	5%	AutoEncoder	6	0/5	5/5	0.6152	0.1019	success:5
glass	5%	VAE	6	0/5	5/5	0.6470	0.1095	success:5
ionosphere	5%	AutoEncoder	6	0/5	5/5	0.9139	0.8756	success:5
ionosphere	5%	VAE	6	0/5	5/5	0.9082	0.8717	success:5
letter	5%	AutoEncoder	30	0/5	5/5	0.5465	0.0911	success:5
letter	5%	VAE	30	0/5	5/5	0.5281	0.0849	success:5
lympho	5%	AutoEncoder	4	0/5	5/5	0.9466	0.7482	success:5
lympho	5%	VAE	4	0/5	5/5	0.9328	0.7451	success:5
mnist	5%	AutoEncoder	30	0/5	5/5	0.8737	0.4023	success:5
mnist	5%	VAE	30	0/5	5/5	0.8670	0.3973	success:5
musk	5%	AutoEncoder	30	0/5	5/5	0.9983	0.9313	success:5
musk	5%	VAE	30	0/5	5/5	0.9981	0.9264	success:5
optdigits	5%	AutoEncoder	30	0/5	5/5	0.6510	0.0385	success:5
optdigits	5%	VAE	30	0/5	5/5	0.6201	0.0351	success:5
pendigits	5%	AutoEncoder	30	0/5	5/5	0.9204	0.1898	success:5
pendigits	5%	VAE	30	0/5	5/5	0.8962	0.1601	success:5
pima	5%	AutoEncoder	14	0/5	5/5	0.6840	0.5011	success:5
pima	5%	VAE	14	0/5	5/5	0.6955	0.5197	success:5
satellite	5%	AutoEncoder	30	0/5	5/5	0.6904	0.7071	success:5
satellite	5%	VAE	30	0/5	5/5	0.6849	0.6954	success:5
satimage-2	5%	AutoEncoder	30	0/5	5/5	0.9813	0.9108	success:5
satimage-2	5%	VAE	30	0/5	5/5	0.9706	0.8457	success:5
shuttle	5%	AutoEncoder	30	0/5	5/5	0.9957	0.8690	success:5

Continued on next page

Table 166 continued

Dataset	Retained	Method	Normals	Original runs	Sensitivity runs	AUROC	AP	Sensitivity outcomes
shuttle	5%	VAE	30	0/5	5/5	0.9959	0.8745	success:5
vertebral	5%	AutoEncoder	6	0/5	5/5	0.3913	0.1074	success:5
vertebral	5%	VAE	6	0/5	5/5	0.3710	0.1041	success:5
vowels	5%	AutoEncoder	30	0/5	5/5	0.6734	0.1483	success:5
vowels	5%	VAE	30	0/5	5/5	0.6198	0.1114	success:5
wbc	5%	AutoEncoder	10	0/5	5/5	0.9122	0.6205	success:5
wbc	5%	VAE	10	0/5	5/5	0.9104	0.6170	success:5

Table 167: PCA(2): every dataset panel rerun in the training-batch sensitivity. AUROC/AP average the qualified splits shown in the coverage column.

Dataset	Retained	Method	Normals	Original runs	Sensitivity runs	AUROC	AP	Sensitivity outcomes
glass	25%	AutoEncoder	30	0/5	5/5	0.5628	0.0741	success:5
glass	25%	VAE	30	0/5	5/5	0.6841	0.0987	success:5
lympho	25%	AutoEncoder	21	0/5	5/5	0.8414	0.4103	success:5
lympho	25%	VAE	21	0/5	5/5	0.8241	0.3680	success:5
vertebral	25%	AutoEncoder	31	0/5	5/5	0.3869	0.1101	success:5
vertebral	25%	VAE	31	0/5	5/5	0.4173	0.1124	success:5
arrhythmia	10%	AutoEncoder	23	0/5	5/5	0.6747	0.3523	success:5
arrhythmia	10%	VAE	23	0/5	5/5	0.6429	0.3563	success:5
cardio	10%	AutoEncoder	60	5/5	5/5	0.8804	0.5328	success:5
cardio	10%	VAE	60	5/5	5/5	0.9015	0.5682	success:5
glass	10%	AutoEncoder	12	0/5	5/5	0.5726	0.0963	success:5
glass	10%	VAE	12	0/5	5/5	0.6348	0.1212	success:5
ionosphere	10%	AutoEncoder	13	0/5	5/5	0.5946	0.4521	success:5
ionosphere	10%	VAE	13	0/5	5/5	0.5385	0.4151	success:5
letter	10%	AutoEncoder	60	5/5	5/5	0.3799	0.0523	success:5
letter	10%	VAE	60	5/5	5/5	0.4033	0.0551	success:5
lympho	10%	AutoEncoder	8	0/5	5/5	0.7155	0.2559	success:5
lympho	10%	VAE	8	0/5	5/5	0.7172	0.2190	success:5
mnist	10%	AutoEncoder	60	5/5	5/5	0.5884	0.1772	success:5
mnist	10%	VAE	60	5/5	5/5	0.6334	0.2020	success:5
musk	10%	AutoEncoder	60	5/5	5/5	0.4522	0.2048	success:5
musk	10%	VAE	60	5/5	5/5	0.4258	0.1572	success:5
optdigits	10%	AutoEncoder	60	5/5	5/5	0.2706	0.0190	success:5
optdigits	10%	VAE	60	5/5	5/5	0.2820	0.0186	success:5
pendigits	10%	AutoEncoder	60	5/5	5/5	0.8804	0.1831	success:5
pendigits	10%	VAE	60	5/5	5/5	0.7786	0.1018	success:5
pima	10%	AutoEncoder	29	0/5	5/5	0.6291	0.4628	success:5
pima	10%	VAE	29	0/5	5/5	0.6626	0.5014	success:5
satellite	10%	AutoEncoder	60	5/5	5/5	0.6223	0.5880	success:5
satellite	10%	VAE	60	5/5	5/5	0.6106	0.6089	success:5
satimage-2	10%	AutoEncoder	60	5/5	5/5	0.9019	0.6613	success:5
satimage-2	10%	VAE	60	5/5	5/5	0.9341	0.8431	success:5

Continued on next page

Table 167 continued

Dataset	Retained	Method	Normals	Original runs	Sensitivity runs	AUROC	AP	Sensitivity outcomes
shuttle	10%	AutoEncoder	60	5/5	5/5	0.9282	0.5661	success:5
shuttle	10%	VAE	60	5/5	5/5	0.9508	0.6282	success:5
vertebral	10%	AutoEncoder	12	0/5	5/5	0.3806	0.1139	success:5
vertebral	10%	VAE	12	0/5	5/5	0.3550	0.1017	success:5
vowels	10%	AutoEncoder	60	5/5	5/5	0.3488	0.0275	success:5
vowels	10%	VAE	60	5/5	5/5	0.3350	0.0279	success:5
wbc	10%	AutoEncoder	21	0/5	5/5	0.8997	0.6666	success:5
wbc	10%	VAE	21	0/5	5/5	0.8991	0.6751	success:5
arrhythmia	5%	AutoEncoder	11	0/5	5/5	0.6836	0.4092	success:5
arrhythmia	5%	VAE	11	0/5	5/5	0.6500	0.3422	success:5
cardio	5%	AutoEncoder	30	0/5	5/5	0.8580	0.4943	success:5
cardio	5%	VAE	30	0/5	5/5	0.8957	0.5414	success:5
glass	5%	AutoEncoder	6	0/5	5/5	0.5445	0.1469	success:5
glass	5%	VAE	6	0/5	5/5	0.4793	0.1327	success:5
ionosphere	5%	AutoEncoder	6	0/5	5/5	0.5711	0.4732	success:5
ionosphere	5%	VAE	6	0/5	5/5	0.5242	0.4123	success:5
letter	5%	AutoEncoder	30	0/5	5/5	0.3532	0.0485	success:5
letter	5%	VAE	30	0/5	5/5	0.4025	0.0516	success:5
lympho	5%	AutoEncoder	4	0/5	5/5	0.7448	0.3023	success:5
lympho	5%	VAE	4	0/5	5/5	0.6017	0.1365	success:5
mnist	5%	AutoEncoder	30	0/5	5/5	0.6667	0.1856	success:5
mnist	5%	VAE	30	0/5	5/5	0.6956	0.2246	success:5
musk	5%	AutoEncoder	30	0/5	5/5	0.4815	0.2214	success:5
musk	5%	VAE	30	0/5	5/5	0.5336	0.2191	success:5
optdigits	5%	AutoEncoder	30	0/5	5/5	0.4236	0.0257	success:5
optdigits	5%	VAE	30	0/5	5/5	0.3745	0.0218	success:5
pendigits	5%	AutoEncoder	30	0/5	5/5	0.8529	0.1728	success:5
pendigits	5%	VAE	30	0/5	5/5	0.7978	0.1147	success:5
pima	5%	AutoEncoder	14	0/5	5/5	0.5914	0.4227	success:5
pima	5%	VAE	14	0/5	5/5	0.6110	0.4459	success:5
satellite	5%	AutoEncoder	30	0/5	5/5	0.6396	0.6067	success:5
satellite	5%	VAE	30	0/5	5/5	0.6260	0.6195	success:5
satimage-2	5%	AutoEncoder	30	0/5	5/5	0.8836	0.4909	success:5
satimage-2	5%	VAE	30	0/5	5/5	0.9148	0.8067	success:5
shuttle	5%	AutoEncoder	30	0/5	5/5	0.9477	0.6348	success:5

Continued on next page

Table 167 continued

Dataset	Retained	Method	Normals	Original runs	Sensitivity runs	AUROC	AP	Sensitivity outcomes
shuttle	5%	VAE	30	0/5	5/5	0.9467	0.6623	success:5
vertebral	5%	AutoEncoder	6	0/5	5/5	0.4056	0.1165	success:5
vertebral	5%	VAE	6	0/5	5/5	0.3460	0.0995	success:5
vowels	5%	AutoEncoder	30	0/5	5/5	0.3293	0.0279	success:5
vowels	5%	VAE	30	0/5	5/5	0.3234	0.0266	success:5
wbc	5%	AutoEncoder	10	0/5	5/5	0.9033	0.5766	success:5
wbc	5%	VAE	10	0/5	5/5	0.8899	0.5907	success:5